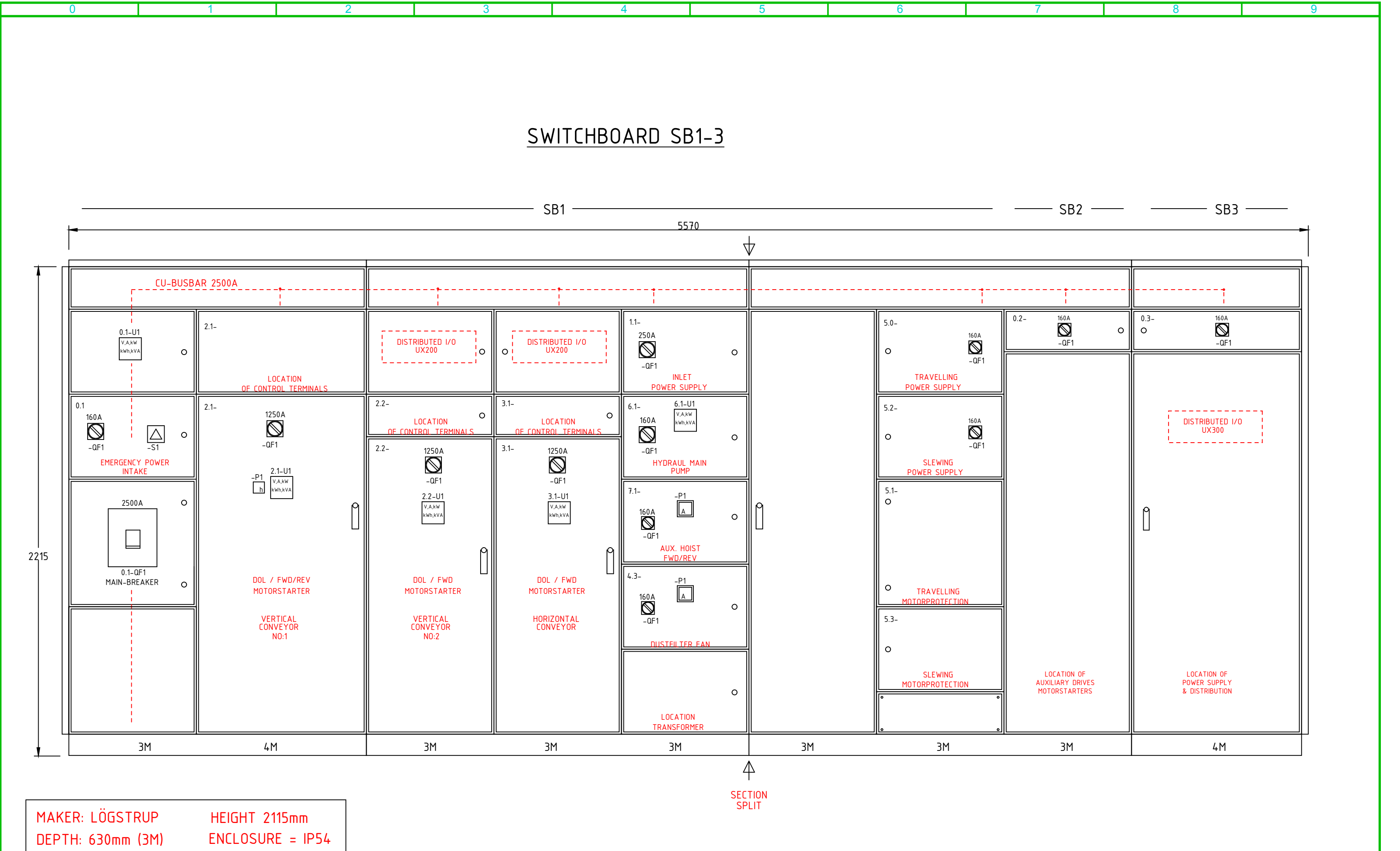


0	1	2	3	4	5	6	7	8	9		
DRAWING AND DOCUMENT LIST					PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ENGLAND					BMH_DRAWING_LIST	
COVER No.	DOCUMENT No.	No. OF SHEETS	DESCRIPTION								
A1	A524	1	DRAWING LIST								
A1	A524 A	49	SWITCHBOARD LAYOUT								
A1	A524 B	9	SINGLE LINE DIAGRAM								
A1	A524 C	52	PARTS LIST INTERNAL								
A1	A524 D	6	PARTS LIST EXTERNAL								
A1	A524 E	108	CIRCUIT DIAGRAM								
A1	A524 F	30	SWITCHBOARD CONNECTION TABLE SB1								
A1	A524 G	65	CABLE LIST								
A1	A524 H	3	LOCATION OF EXTERNAL EQUIPMENT								
A1	A524 I	47	PLC INPUT/OUTPUT LIST								
A1	A524 K	5	COMMUNICATION SYSTEMS								
A1	A524 L	2	LIGHTNING, GROUNDING AND EARTHING								
A1	A524 O	26	LOCAL CONTROL BOXES & JUNCTION BOXES								
A2	A524 R	28	NAME PLATE LIST								
A2	A524 T	44	FACTORY ACCEPTANCE TEST (FAT-TEST)								
A2	A524 U	88	SITE ACCEPTANCE TEST (SAT-TEST)								
A2	A524 V	2	FAT INSPECTION PROCEDURE								
A2	A524 X	2	SAT INSPECTION PROCEDURE								
		AS BUILT				PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT DRAWING LIST		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-18	
REV.	MODIFICATION			DATE				CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 -	
								APPR.	FILE NAME A524.DB	SHEET 1	CONT. SHEET 1

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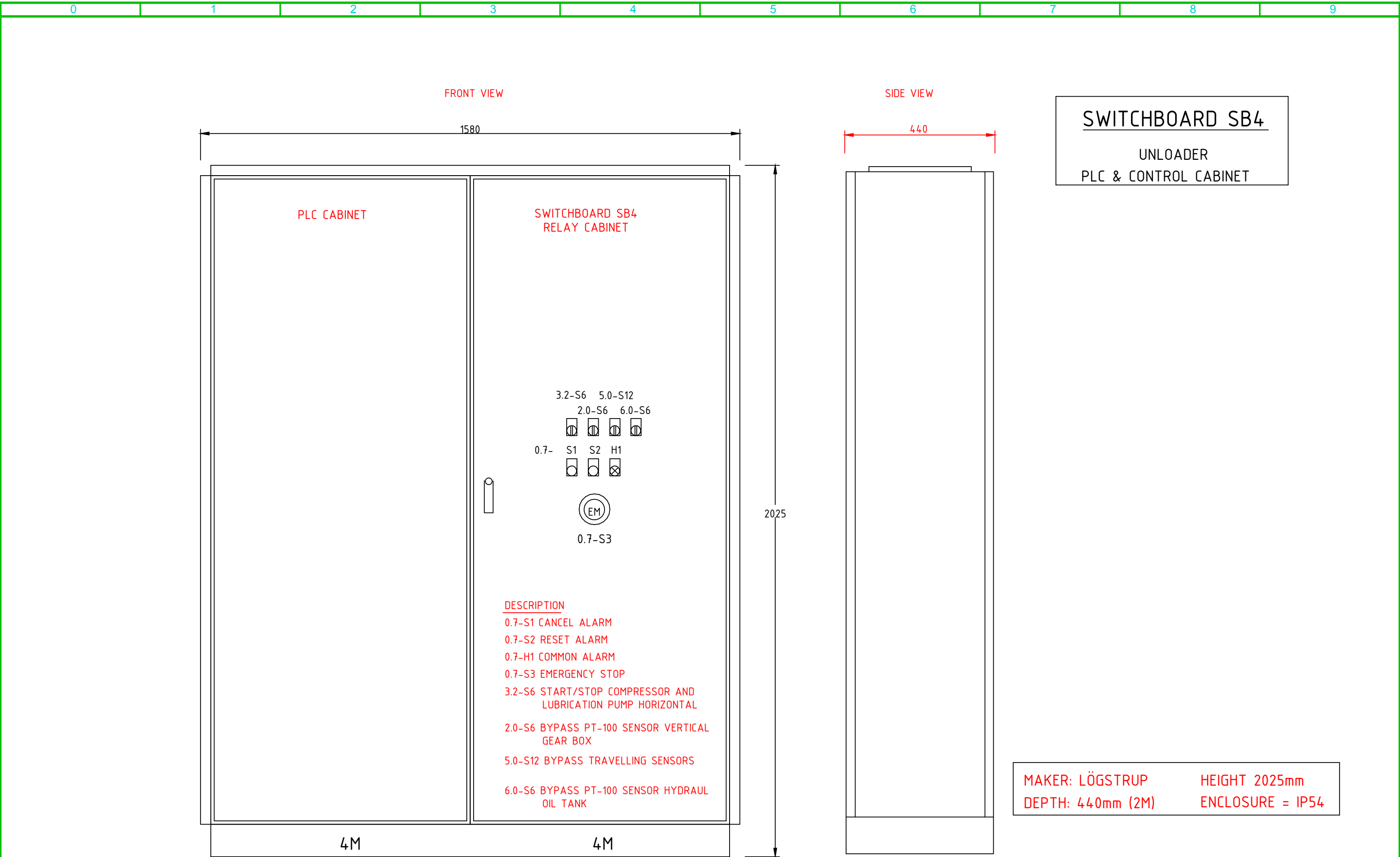
SB1

- SB2

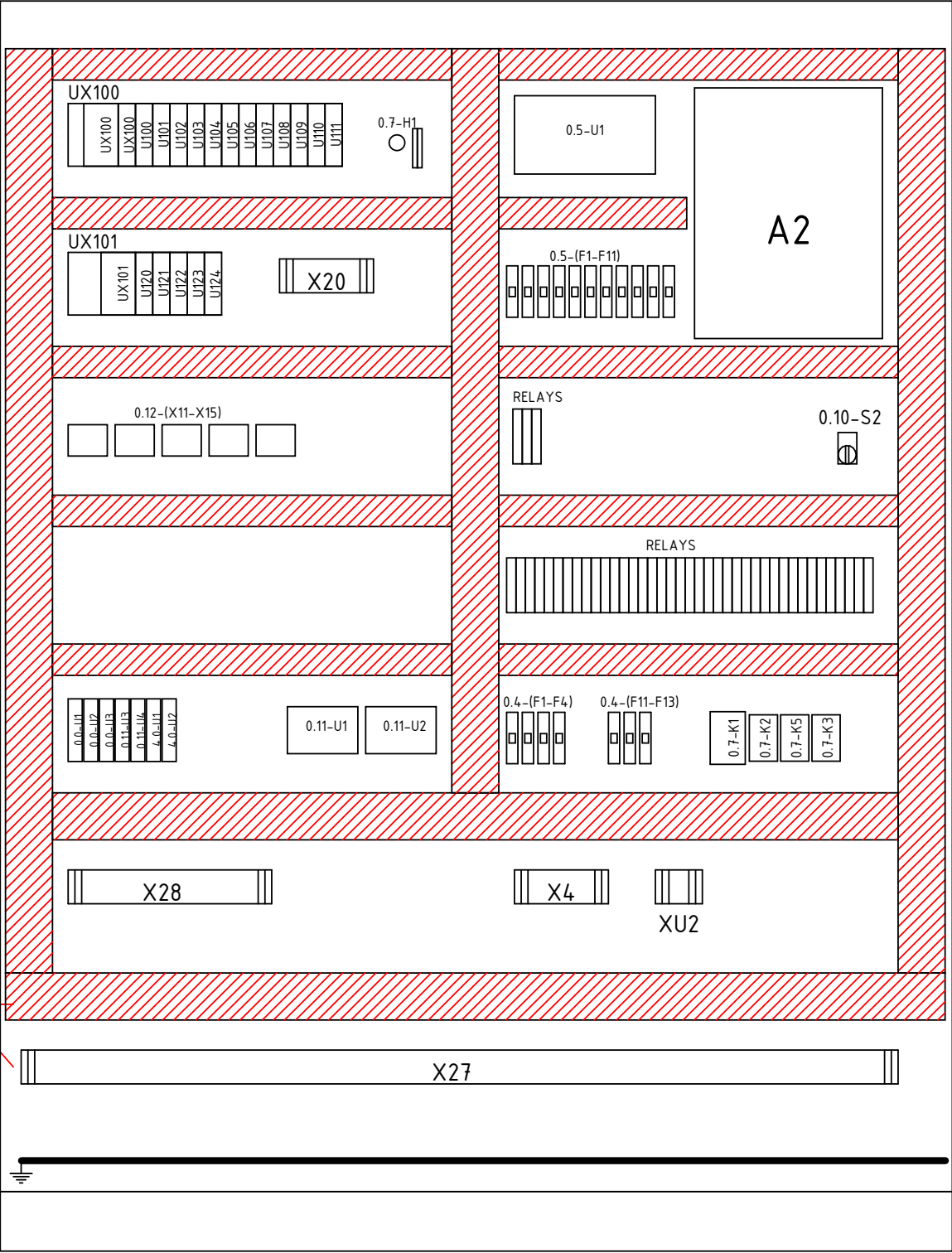
SB3



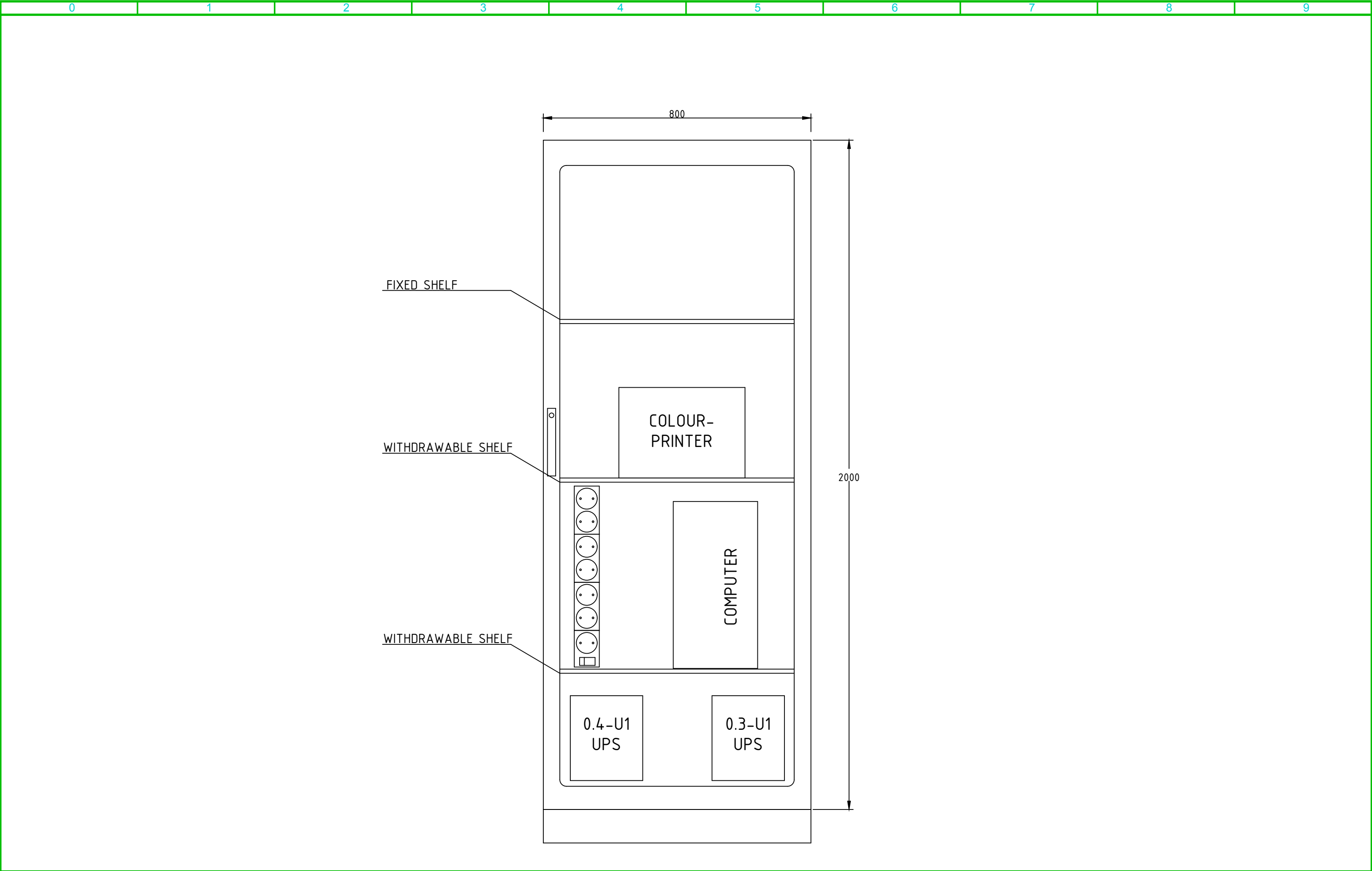
ON DISTANCE



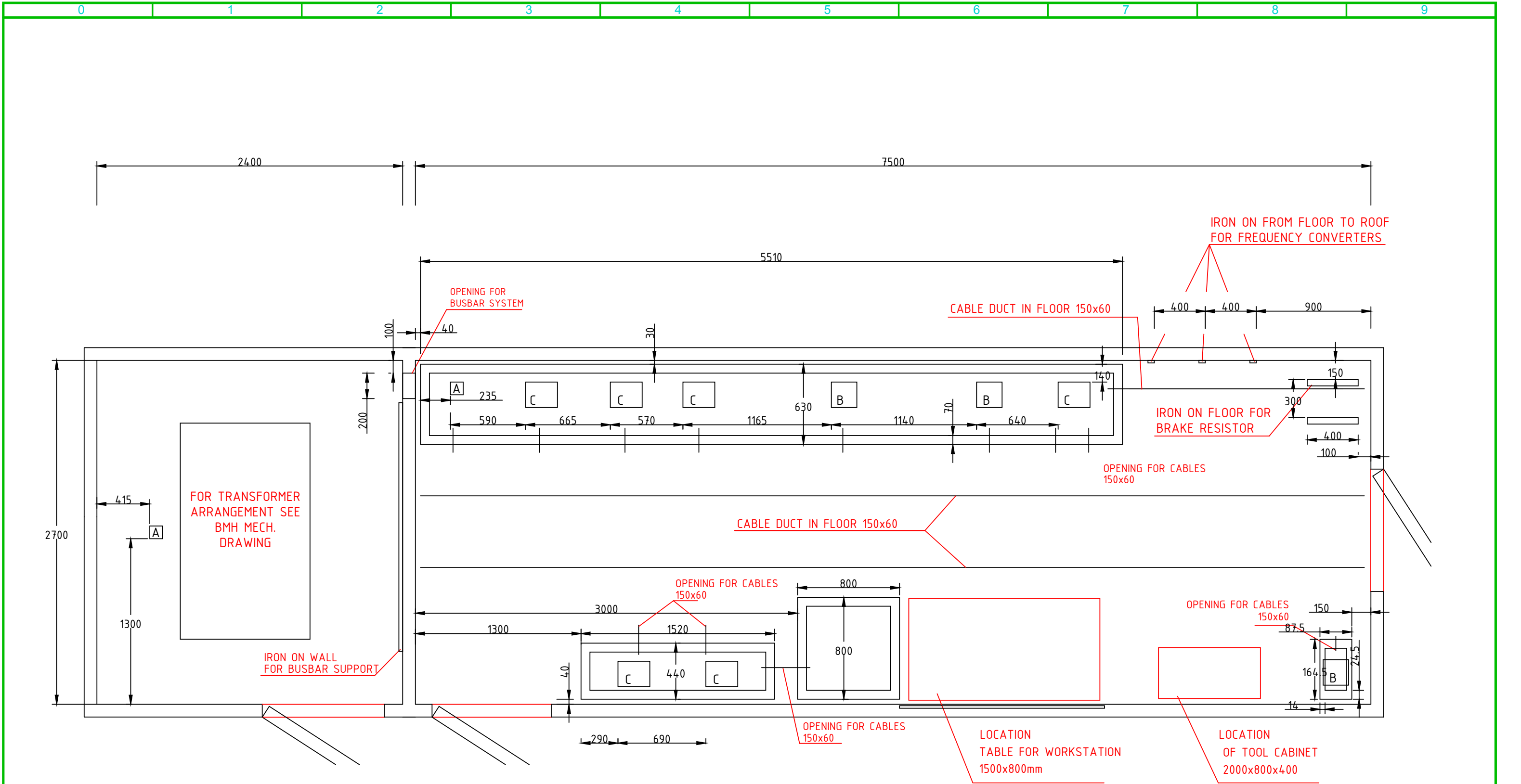
LAST 0-No. = 0114



ON DISTANCE

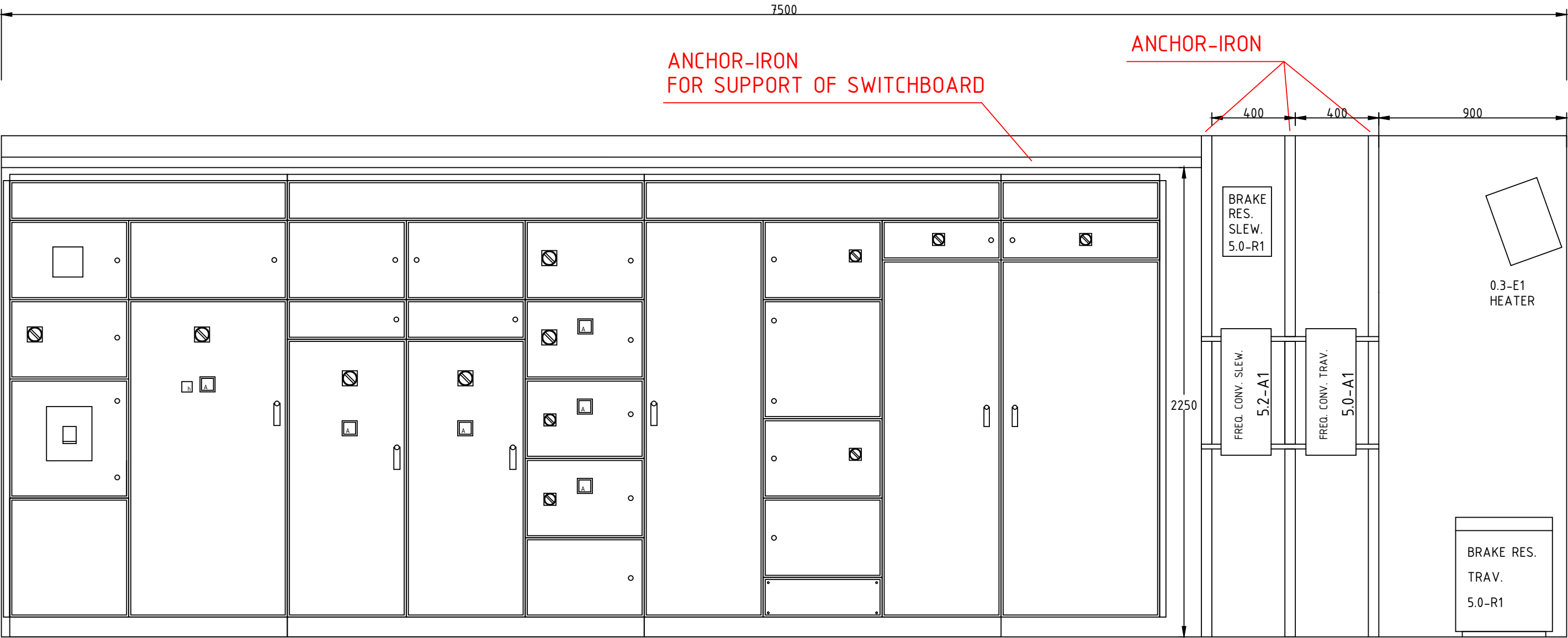


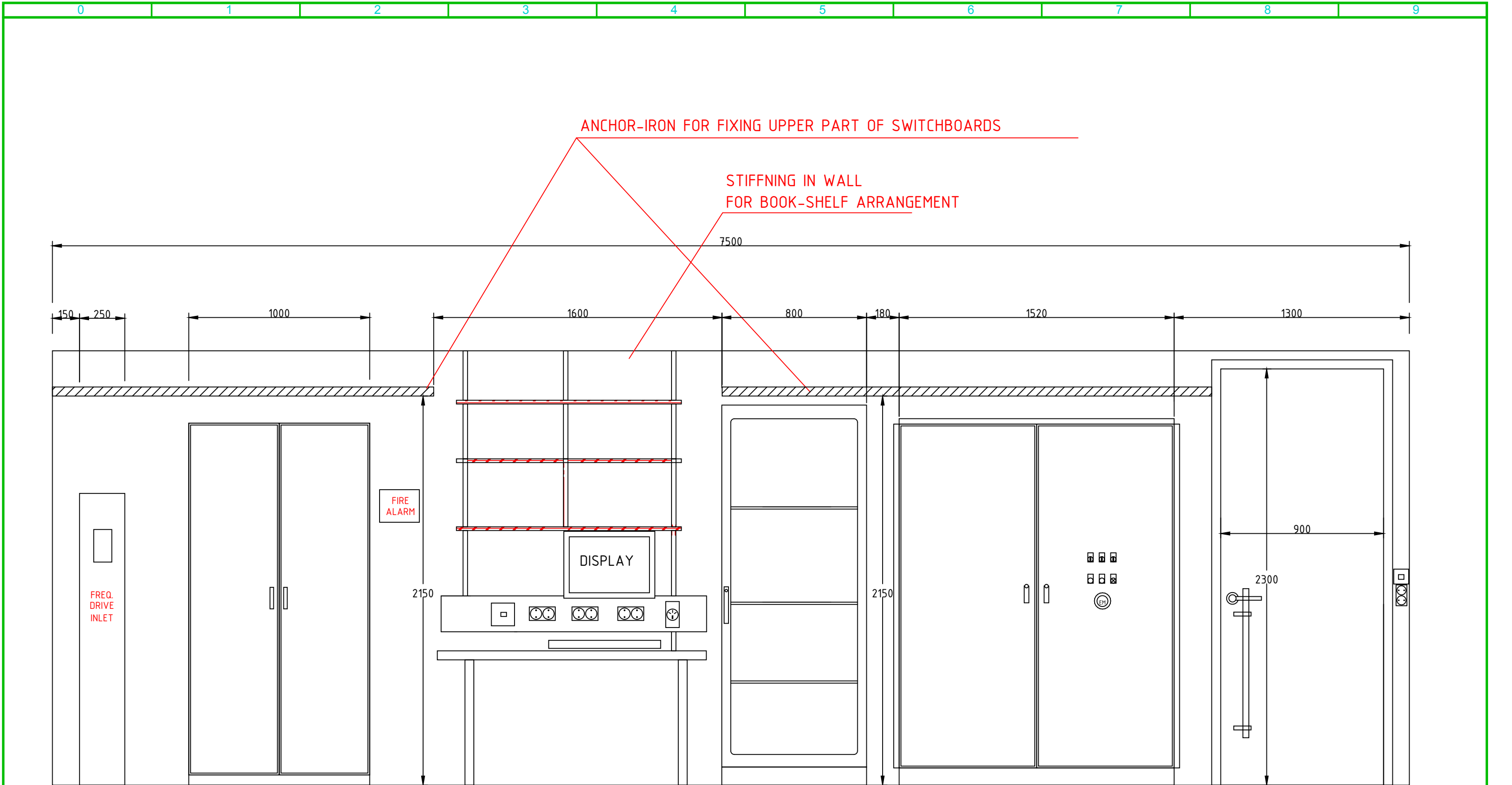


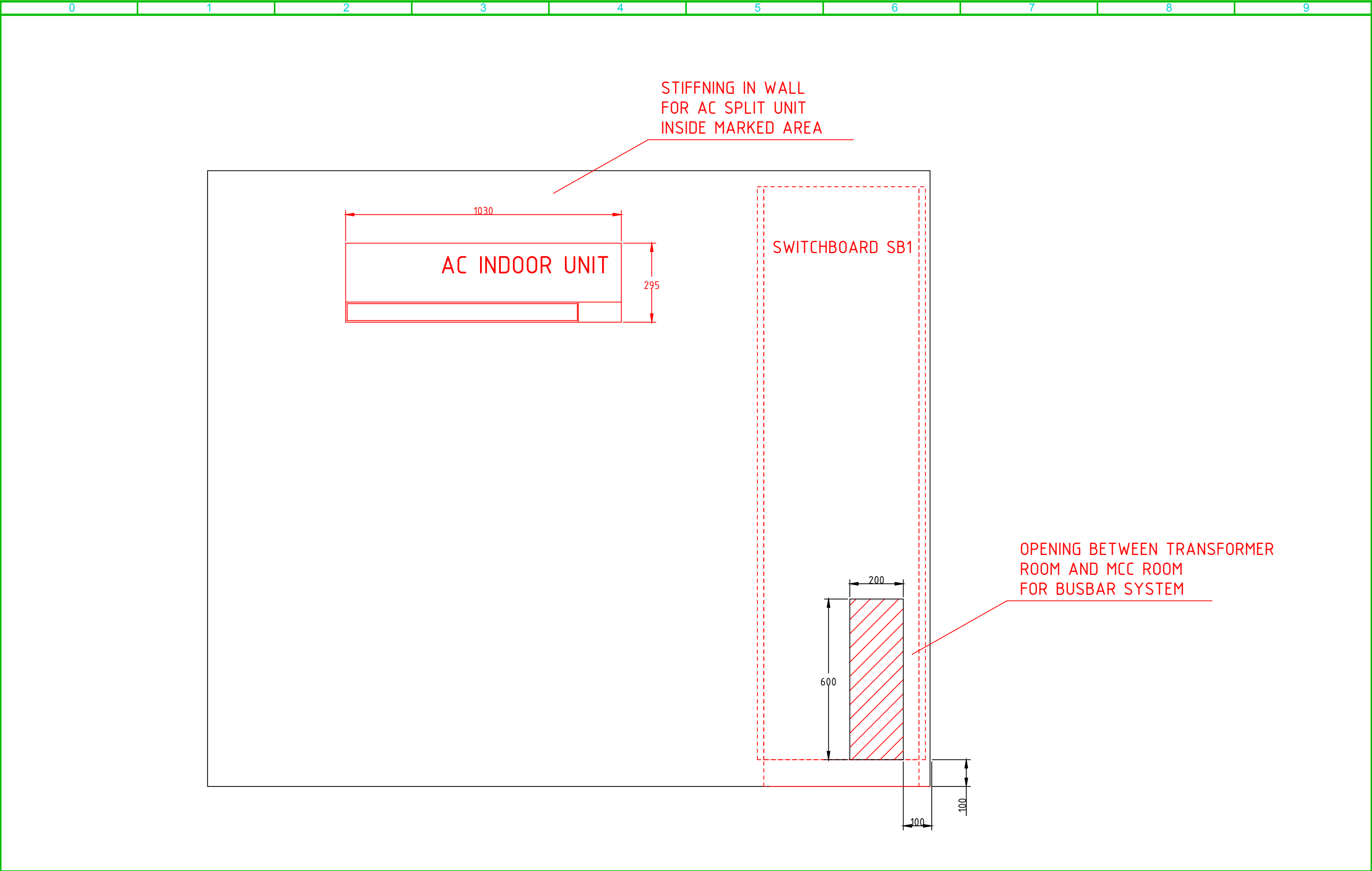


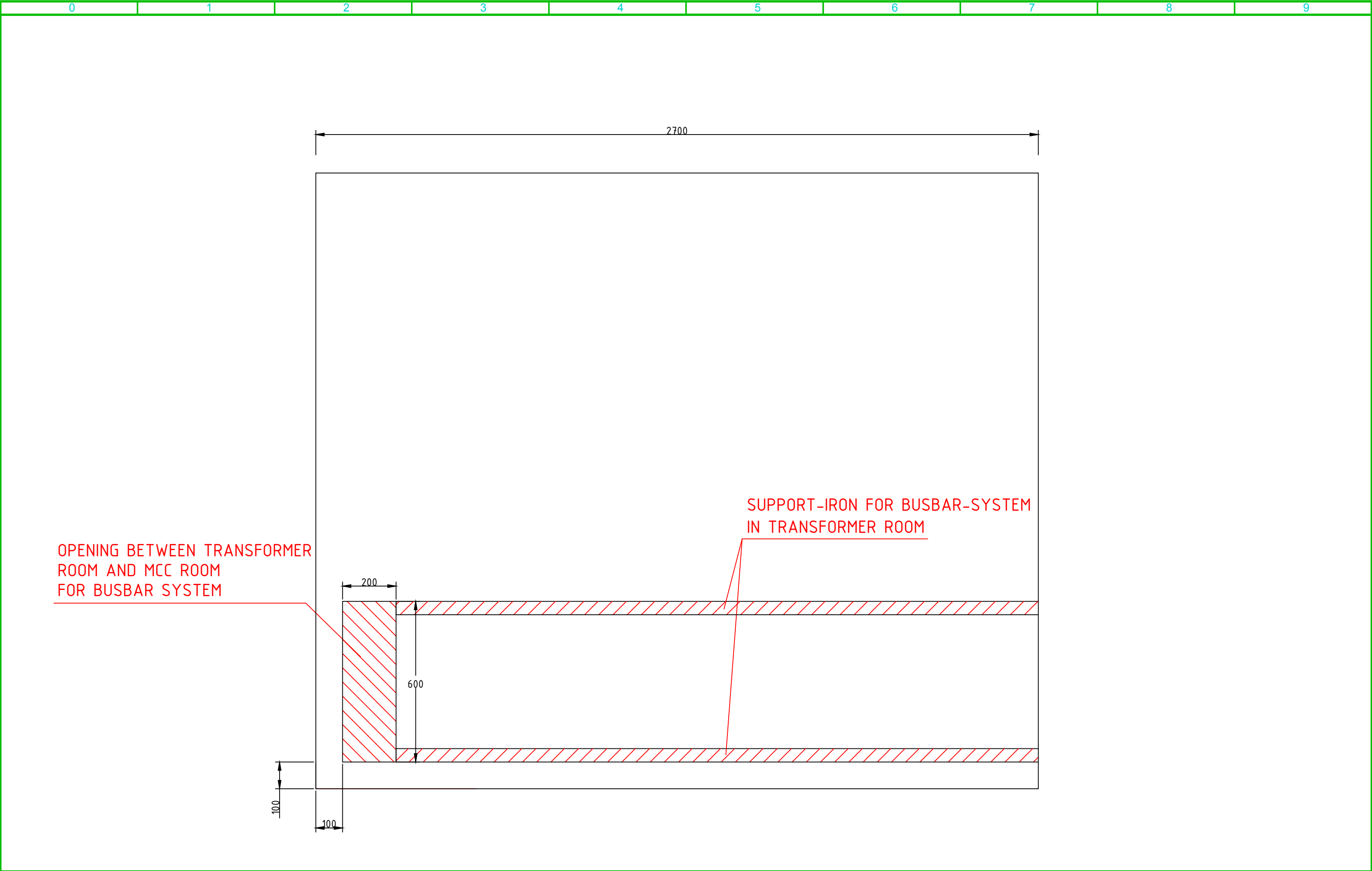
NOTE:
CABLE DUCT TROUGH
BOTTOM FOR OUTGOING
CABLES

A=100x100
B=200X200
C=250X200

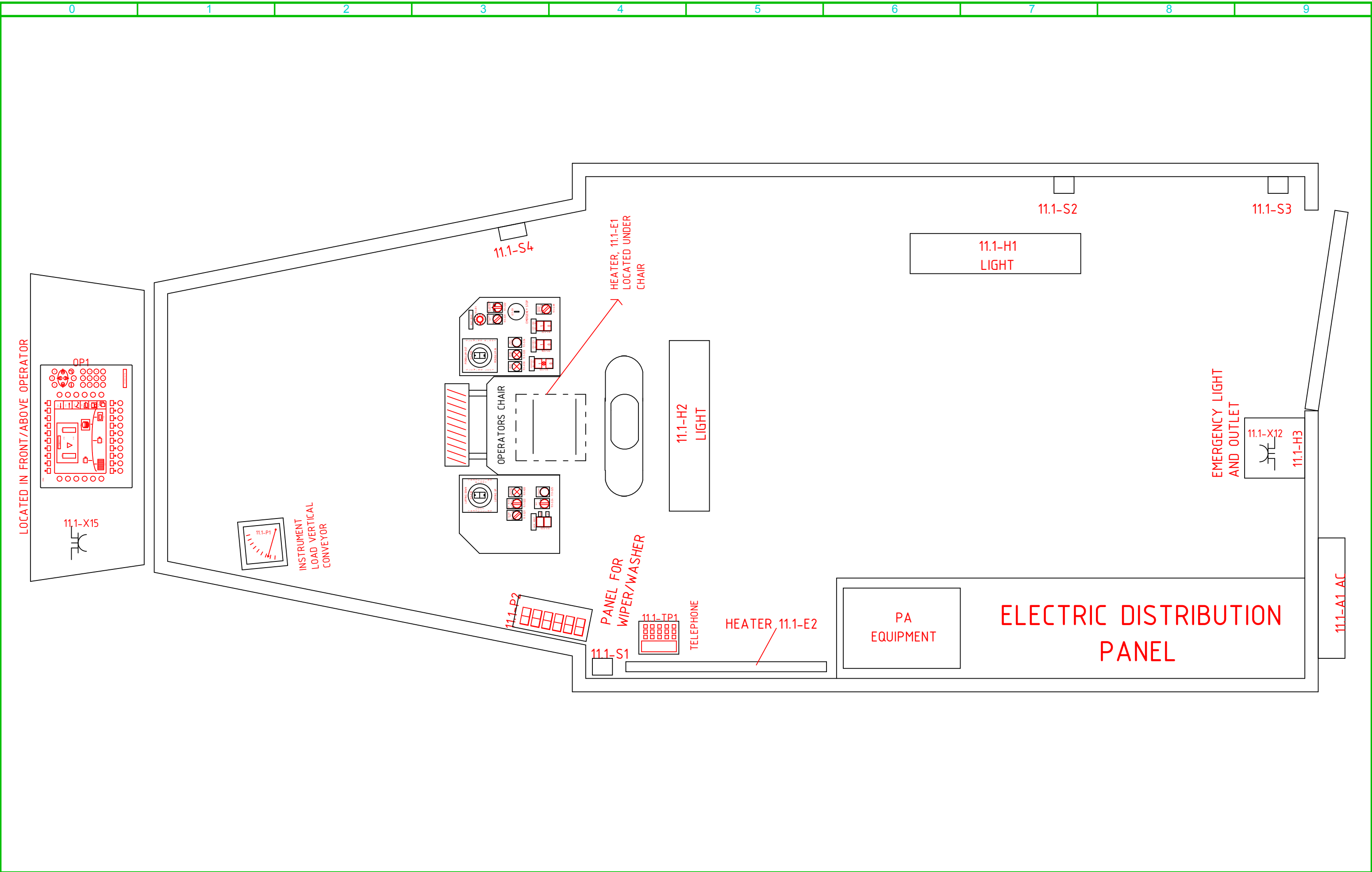




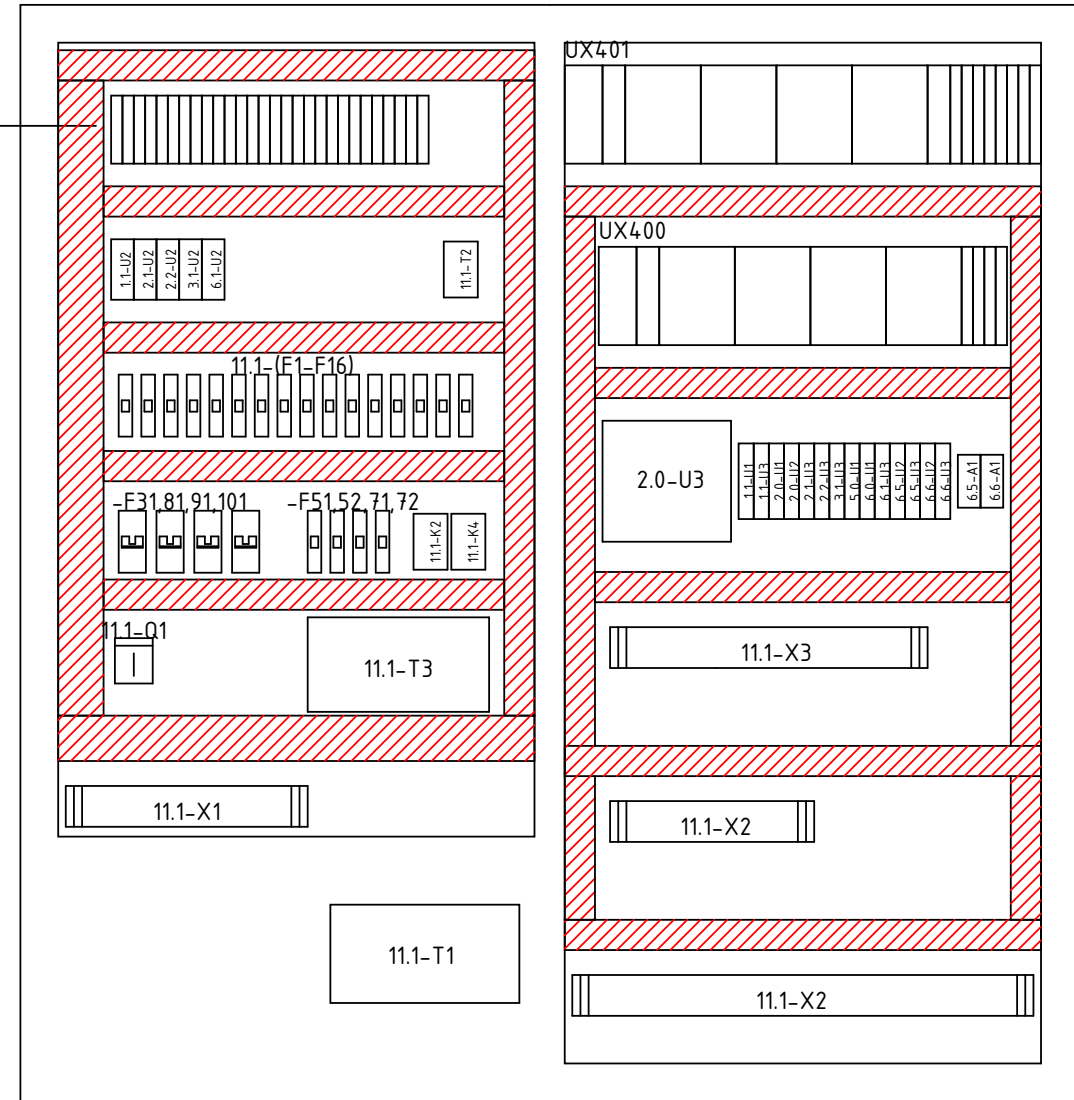




AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN			AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT ELHOUSE LAYOUT WALL VIEW, TRANSFORMER ROOM		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
REV.	MODIFICATION				DATE		CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - A		
							APPR.	FILE NAME A524.DB		SHEET 12	CONT. SHEET 15	



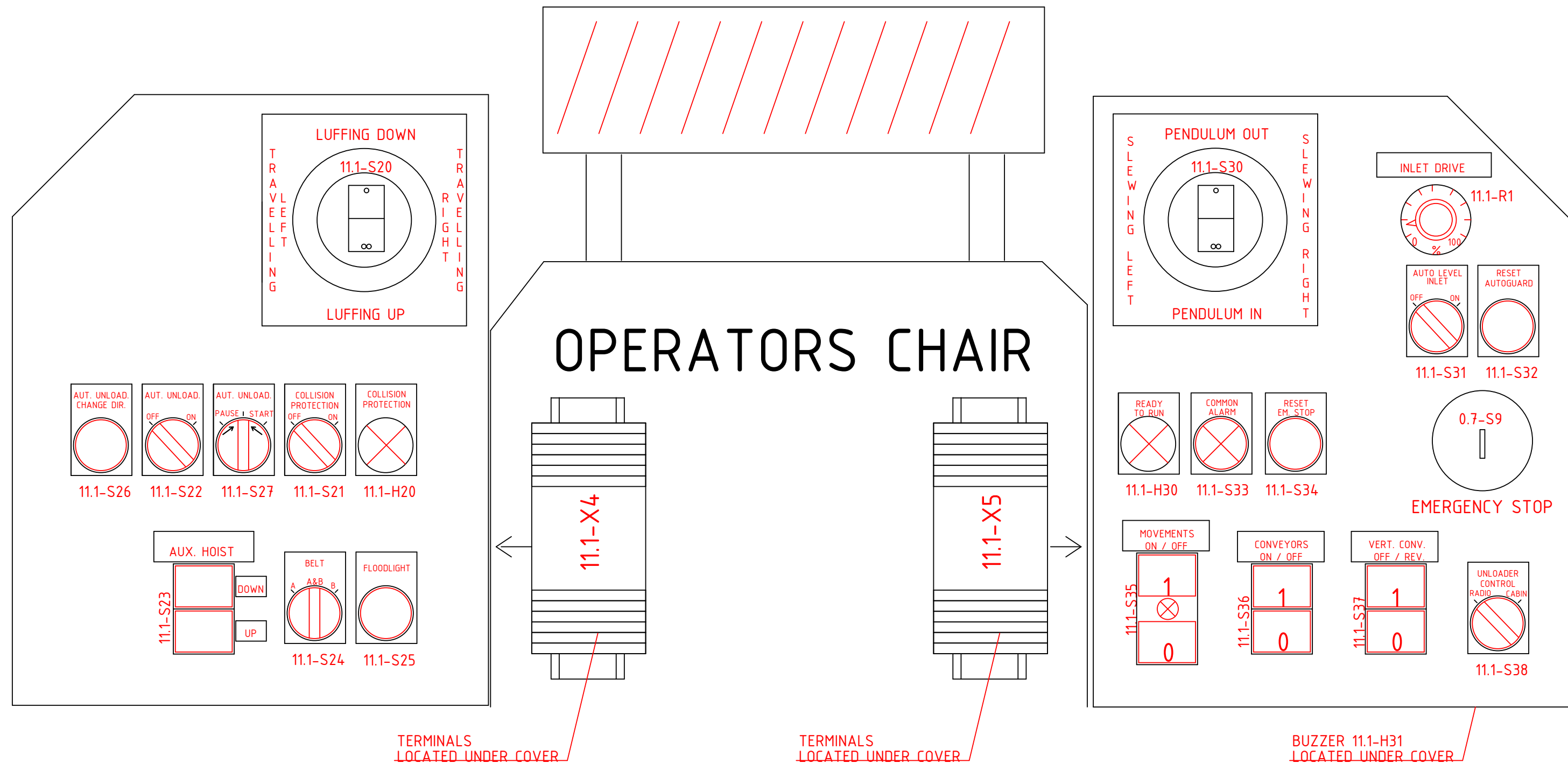
0.10-K2
0.12-K1
0.12-K2
0.12-K3
0.12-K4
0.12-K5
0.12-K6
0.12-K7
0.12-K8
0.12-K9
0.12-K10
0.12-K11
0.12-K12
0.12-K13
0.12-K14
0.12-K15
0.12-K16
0.12-K17
1.1-K3
2.1-K5
2.2-K3
3.1-K3
6.1-K3
6.5-K1
6.6-K1
7.1-K7
11.1-K1
11.1-K3
11.1-K5



ON DISTANCE

DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - A	
APPR.	FILE NAME A524.DB	SHEET 16	CONT. SHEET 17

reproduced, or exploited in any other unauthorized way.



NUMBER OF TERMINALS: 12

0.1-X1

1	1.7
2	1.7
3	1.7
4	1.7
5	1.8
6	1.8
11	1.3
12	1.4
21	1.0
22	1.0
23	1.1
24	1.1

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

TERMINAL CONFIGURATION

SWITCHBOARD SB1

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-16

DRAWING NO.
A524 - A

SHEET
20

CONT. SHEET
21

reproduced, or exploited in any other unauthorized way.

NUMBER OF TERMINALS: 11

0.1-X2

L1	2.8
L2	2.8
L3	2.8
1	2.4
2	2.4
3	2.4
4	2.5
11	2.2
12	2.2
13	2.2
14	2.2



AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007
SHIPUNLOADER MERSEY DOCKS
ELECTRICAL EQUIPMENT
TERMINAL CONFIGURATION
SWITCHBOARD SB1

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-16

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - A

APPR.

FILE NAME
A524.DB

SHEET
21

CONT. SHEET
22

NUMBER OF TERMINALS: 7

1.1-X1

3	4	11	12	13	14	15
106.3	106.3	106.3	106.3	106.3	106.3	106.3

NUMBER OF TERMINALS: 18

2.1-X1

1	100.3
2	100.3
3	101.3
4	101.3
11	101.3
12	101.3
13	101.3
14	101.3
15	101.3
16	101.3
17	101.3
18	101.3
21	100.3
22	100.3
23	100.3
24	100.3
25	100.3
26	100.3

NUMBER OF TERMINALS: 14

2.2-X1

1	103.3
2	103.3
3	103.3
4	103.3
11	103.3
12	103.3
13	103.3
14	103.3
15	103.3
16	103.3
21	103.3
22	103.3
23	103.3
24	103.3

reproduced, or exploited in any other unauthorized way.

NUMBER OF TERMINALS: 8

4.3-X1

3	80.3
4	80.3
11	80.3
12	80.3
13	80.3
14	80.3
15	80.3
16	80.3

NUMBER OF TERMINALS: 43

5.1-X1

L1	56.0
L2	56.0
L3	56.1
1	58.3
2	58.3
3	56.1
4	56.1
5	58.7
6	58.7
11	56.2
12	58.3
13	56.3
14	56.4
15	56.6
16	56.7
17	56.8
18	57.3
19	57.4
20	57.6
21	57.7
22	57.8
23	58.3
24	58.3
31	58.3
32	58.3
33	58.3
34	58.3
35	58.4
36	58.4
37	58.5
38	58.5
39	58.5
40	58.6
41	58.6
42	58.6
43	58.7
44	58.7
45	58.7
46	58.8
47	58.8
48	58.8
49	58.9
50	58.9

NUMBER OF TERMINALS: 19

5.3-X1

L1	63.0
L2	63.0
L3	63.1
1	64.3
2	64.3
3	63.1
4	63.1
5	64.7
6	64.7
11	63.2
12	64.3
13	63.3
14	63.5
15	64.3
16	64.3
31	64.6
32	64.6
33	64.7
34	64.7

NUMBER OF TERMINALS: 21

7.1-X1

1	76.2
2	76.2
3	75.2
4	75.2
11	75.2
12	75.2
13	75.2
14	75.2
15	75.2
16	75.2
17	76.2
18	76.2
19	75.2
20	75.2
21	76.2
31	75.2
32	75.2
41	76.8
42	76.9
43	76.7
44	76.7

NUMBER OF TERMINALS: 6

X11

19.219.219.319.321.221.2

1	2	3	4	11	12
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NUMBER OF TERMINALS: 90

X3

101	5.1
102	5.1
103	5.1
104	5.2
105	5.4
106	5.4
107	5.4
108	5.4
109	5.6
110	5.6
111	5.7
112	5.7
113	6.1
114	6.1
115	6.1
116	6.2
117	4.1
118	4.1
119	4.3
120	4.4
121	6.3
122	6.3
123	6.3
124	6.3
125	6.5
126	6.5
127	6.5
128	6.5
129	6.7
130	6.7
131	6.7
132	6.7
133	7.4
134	7.4
135	7.5
136	7.5
137	7.6
138	7.7
139	7.7
140	7.8
141	8.1
142	8.1
143	8.2
144	8.3
145	8.3
146	8.4
147	8.5
148	8.6
149	8.7
150	8.7
151	9.1
152	9.1
153	10.1
154	10.1
155	10.2
156	10.3
157	10.3
158	10.4
159	10.5
160	10.5
161	10.7
162	10.7
163	11.1
164	11.1
165	11.2
166	11.2
167	11.4
168	11.4
169	11.5
170	11.5

171	12.1
172	12.1
173	12.2
174	12.3
175	12.3
176	12.4
177	12.5
178	12.5
179	12.6
180	12.7
181	7.8
182	7.9
183	
184	
185	
186	
187	
188	
189	
190	

NUMBER OF TERMINALS: 6

X12

19.419.419.519.521.321.3

1	2	3	4	11	12
---	---	---	---	----	----

reproduced, or exploited in any other unauthorized way.

NUMBER OF TERMINALS: 14

X31

201	13.0
202	13.0
203	13.2
204	13.2
205	13.3
206	13.3
207	13.4
208	13.5
209	13.5
210	13.6
211	13.7
212	13.7
213	
214	

NUMBER OF TERMINALS: 15

XU1

301	9.2
302	9.2
303	9.3
304	9.4
305	9.4
306	9.5
307	9.6
308	9.7
309	9.8
310	9.8
311	
312	
313	
314	
315	

reproduced, or exploited in any other unauthorized way.

0	1	2	3	4	5	6	7	8	9
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NUMBER OF TERMINALS: 35

X4

1	19.0
2	19.1
3	19.2
4	19.2
5	19.3
6	19.3
7	19.4
8	19.4
9	19.5
10	19.5
11	19.6
12	19.6
13	38.7
14	38.7
15	82.5
16	82.5
17	86.4
18	86.4
19	86.4
20	86.4
21	86.4
22	86.4
23	86.4
24	86.4
25	86.4
26	86.4
27	86.4
28	86.4
29	86.4
30	
31	
32	
33	
34	
35	

NUMBER OF TERMINALS: 17

X20

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5

108.5



AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007
SHIPUNLOADER MERSEY DOCKS
ELECTRICAL EQUIPMENT
TERMINAL CONFIGURATION
SWITCHBOARD SB4

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-16

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - A

APPR.

FILE NAME
A524.DB

SHEET
39

CONT. SHEET
40

NUMBER OF TERMINALS: 230

X27

101	7.1
102	7.1
103	7.1
104	7.1
105	7.1
106	7.1
107	21.2
108	21.2
109	21.3
110	21.3
111	21.4
112	21.4
113	21.6
114	21.6
115	32.8
116	32.8
117	32.2
118	32.2
119	32.2
120	32.2
121	32.2
122	32.2
123	32.2
124	32.2
125	32.2
126	32.2
127	32.2
128	32.2
129	32.2
130	32.2
131	32.2
132	32.2
133	32.2
134	32.2
135	78.2
136	78.2
137	78.2
138	78.2
139	78.2
140	78.2
141	78.2
142	78.2
143	78.2
144	78.2
145	78.2
146	38.2
147	38.2
148	40.2
149	40.2
150	40.2
151	40.2
152	40.2
153	40.2
154	42.2
155	42.2
156	42.2
157	42.2
158	42.2
159	42.2
160	42.2
161	42.2
162	42.2
163	42.2
164	42.2
165	42.2
166	42.2
167	50.2
168	50.2
169	50.2
170	50.2

171	50.2
172	50.2
173	50.2
174	50.2
175	50.2
176	50.2
177	50.2
178	50.2
179	50.2
180	50.2
181	50.2
182	50.2
183	50.2
184	50.7
185	50.7
186	51.2
187	51.2
188	51.2
189	51.2
190	51.2
191	51.2
192	51.2
193	52.2
194	52.2
195	52.2
196	52.2
197	52.2
198	52.2
199	52.2
200	53.7
201	53.7
202	53.7
203	53.7
204	53.7
205	53.7
206	53.7
207	53.7
208	55.2
209	55.2
210	59.2
211	59.2
212	59.2
213	59.2
214	59.2
215	59.2
216	59.2
217	59.2
218	59.2
219	59.2
220	59.2
221	59.2
222	90.2
223	59.7
224	59.7
225	62.2
226	62.2
227	79.2
228	79.2
229	79.2
230	79.2
231	79.2
232	79.2
233	79.2
234	79.2
235	83.2
236	83.2
237	83.2
238	83.2
239	83.2
240	83.2

241	83.8
242	83.8
243	83.8
244	83.8
245	83.8
246	83.8
247	84.2
248	84.2
249	84.2
250	84.2
251	84.2
252	84.2
253	84.8
254	84.8
255	84.8
256	84.8
257	84.8
258	84.8
259	81.7
260	81.7
261	81.7
262	81.7
263	81.7
264	81.7
265	81.7
266	81.7
267	81.7
268	81.7
269	81.7
270	82.5
271	82.5
272	82.5
273	82.5
274	85.3
275	85.3
276	85.3
277	85.3
278	85.3
279	85.3
280	85.3
281	85.3
282	85.3
283	85.3
284	85.3
285	85.3
286	85.3
287	85.3
288	85.3
289	85.3
290	
291	
292	
293	
294	
295	
296	
297	
298	
299	90.2
300	90.2
301	90.8
302	90.8
303	107.2
304	107.2
305	22.5
306	22.6
307	22.6
308	22.6
309	77B.2
310	77B.2

311	77B.2
312	77B.7
313	77B.7
314	77B.7
315	77B.7
316	40.6
317	40.6
318	40.6
319	40.6
320	
321	
322	
323	
324	
325	
326	
327	
328	
329	
330	

NUMBER OF TERMINALS: 50

X28

401	38.2
402	38.2
403	38.2
404	38.2
405	38.2
406	38.2
407	38.2
408	38.2
409	38.2
410	48.6
411	48.6
412	48.6
413	48.6
414	48.6
415	48.6
416	48.6
417	48.6
418	48.6
419	48.6
420	48.6
421	48.6
422	48.6
423	48.6
424	48.6
425	48.6
426	48.6
427	48.6
428	48.6
429	48.6
430	83.2
431	83.2
432	83.2
433	83.2
434	83.2
435	83.2
436	83.2
437	83.2
438	83.2
439	84.2
440	84.2
441	84.2
442	84.2
443	84.2
444	84.2
445	84.2
446	84.2
447	84.2
448	
449	
450	

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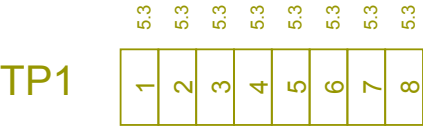
NUMBER OF TERMINALS: 10

XU2

20.020.020.220.220.320.4

51	52	53	54	55	56	57	58	59	60
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NUMBER OF TERMINALS: 8



DATE

2006-10-16

DRAWING NO.

A524 - A

SHEET

43

CONT. SHEET

44

reproduced, or exploited in any other unauthorized way.

NUMBER OF TERMINALS: 50

11.1-X1

1	14.0
2	14.0
3	14.1
4	14.1
5	14.1
6	14.2
7	14.2
8	14.2
9	14.3
10	14.3
11	14.5
12	14.5
13	14.6
14	14.6
15	14.7
16	14.7
17	14.7
18	14.8
19	14.9
20	14.9
21	16.5
22	16.5
23	16.7
24	16.7
25	16.8
26	16.9
27	17.4
28	17.4
29	17.6
30	17.6
31	18.1
32	18.1
33	68.3
34	68.3
35	72.2
36	72.2
37	93.3
38	93.3
39	99.3
40	99.3
41	102.3
42	102.3
43	105.3
44	105.3
45	
46	
101	20.3
102	20.4
103	20.3
104	20.4

NUMBER OF TERMINALS: 140

11.1-X2

201	45.5
202	45.5
203	21.4
204	21.4
205	21.6
206	21.6
207	32.8
208	32.8
209	33.2
210	33.3
211	7.1
212	7.1
213	104.7
214	104.7
215	15.2
216	15.2
217	16.1
218	16.1
219	16.1
220	53.2
221	53.2
222	72.2
223	72.2
224	74.3
225	74.3
226	77.2
227	77.2
228	77.2
229	77.2
230	77.2
231	77.2
232	34.7
233	34.7
234	34.7
235	34.7
236	34.7
237	60.2
238	60.2
239	65.2
240	65.2
241	65.2
242	73.2
243	73.2
244	77.7
245	77.7
246	77.7
247	77.7
248	77.7
249	77.7
250	104.2
251	104.2
252	104.2
253	32.8
254	32.8
255	33.2
256	33.2
257	33.2
258	33.2
259	33.2
260	33.2
261	33.2
262	33.2
263	33.2
264	33.2
265	33.2
266	33.2
267	66.2
268	66.2
269	66.2
270	66.2

271	71.2
272	71.2
273	71.2
274	45.5
275	45.5
276	45.5
277	45.5
278	45.5
279	45.5
280	45.5
281	45.5
282	45.5
283	45.5
284	46.5
285	46.5
286	46.5
287	46.5
288	46.5
289	46.5
290	47.5
291	47.5
292	47.5
293	47.5
294	41.5
295	41.5
296	49.3
297	49.3
298	49.3
299	74.3
300	74.3
301	74.3
302	74.3
303	74.3
304	90.2
305	90.2
306	90.2
307	92.2
308	93.3
309	98.3
310	105.3
311	105.3
312	77B.6
313	77B.6
314	40.7
315	40.7
316	40.7
317	40.7
318	40.8
319	40.8
320	77.2
321	77.2
322	77.2
323	47.5
324	
325	
326	
327	
328	
329	
330	
331	
332	
333	
334	
335	
336	
337	
338	
339	
340	

NUMBER OF TERMINALS: 67

11.1-X3

53.2	53.2	53.2	72.2	60.2	60.2	60.2	73.2	104.2	66.2	66.2	66.2	68.3	68.3	68.3	68.3	68.3	72.7	72.7	72.7	73.7	73.7	73.7	73.7	104.7	104.7	41.5	41.5	72.2	73.2	93.3	93.3	93.3	93.3	93.3	95.3	95.3	95.3	95.3	98.3	98.3	98.3	98.3	98.3	99.3	99.3	99.3	99.3	102.3	102.3	102.3	102.3	102.3	105.3	105.3	105.3	105.3	105.3									
401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420		421	422	423		424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465

0	1	2	3	4	5	6	7	8	9
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NUMBER OF TERMINALS: 22

11.1-X4

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	31	32	33	34
16.0	16.0	16.0	53.1	53.1	72.1	72.1	74.2	74.2	77.1	77.1	77.1	77.1	77.1	77.1	77.1	77.1	77.1	53.1	53.1	53.1	72.1

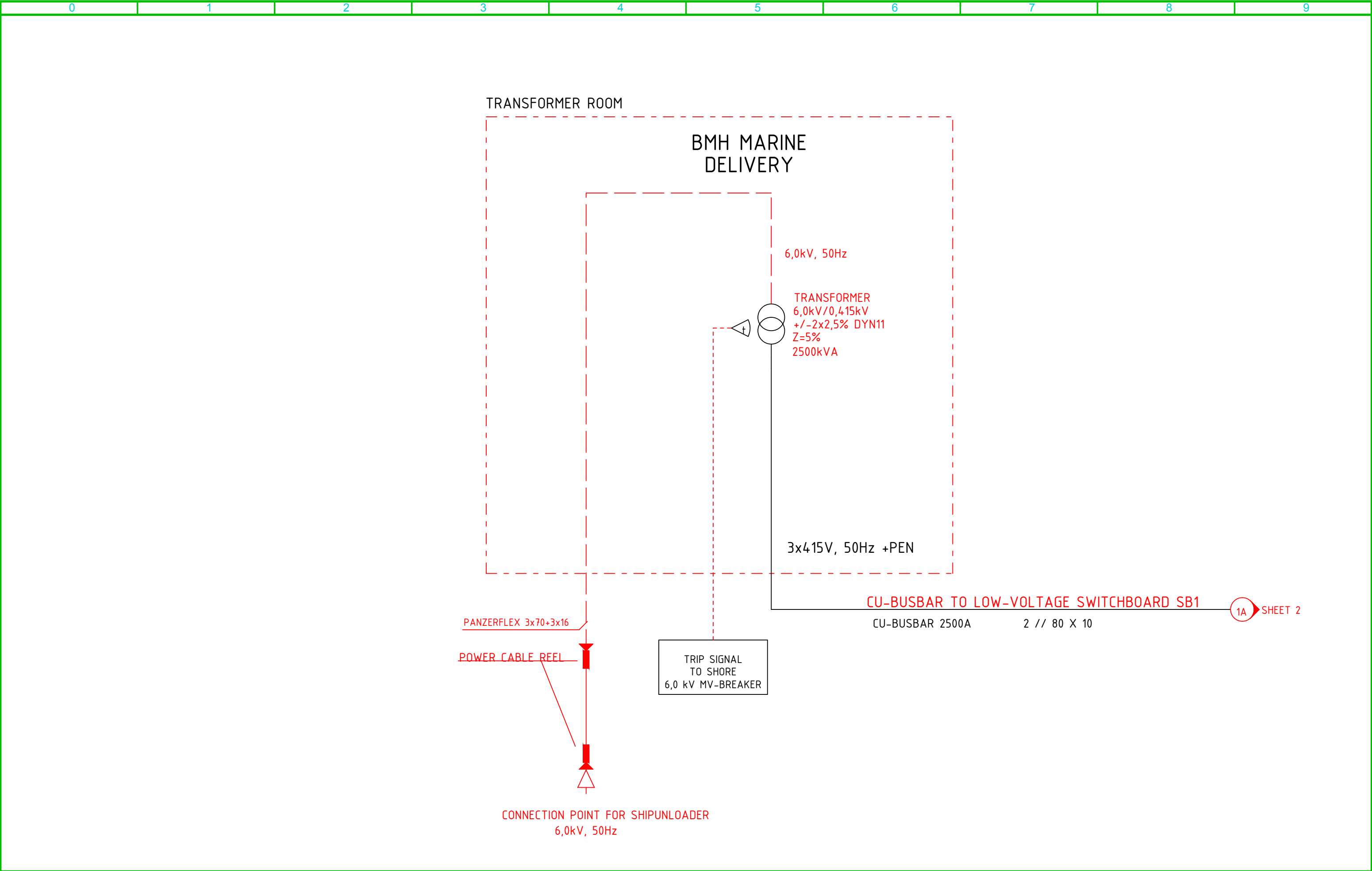
AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN			AS BUILT			BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN			PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT TERMINAL CONFIGURATION CABIN CHAIR, LEFT SIDE			DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - A		APPR.	FILE NAME A524.DB		SHEET 47	CONT. SHEET 48						
REV.	MODIFICATION				DATE										

NUMBER OF TERMINALS: 8

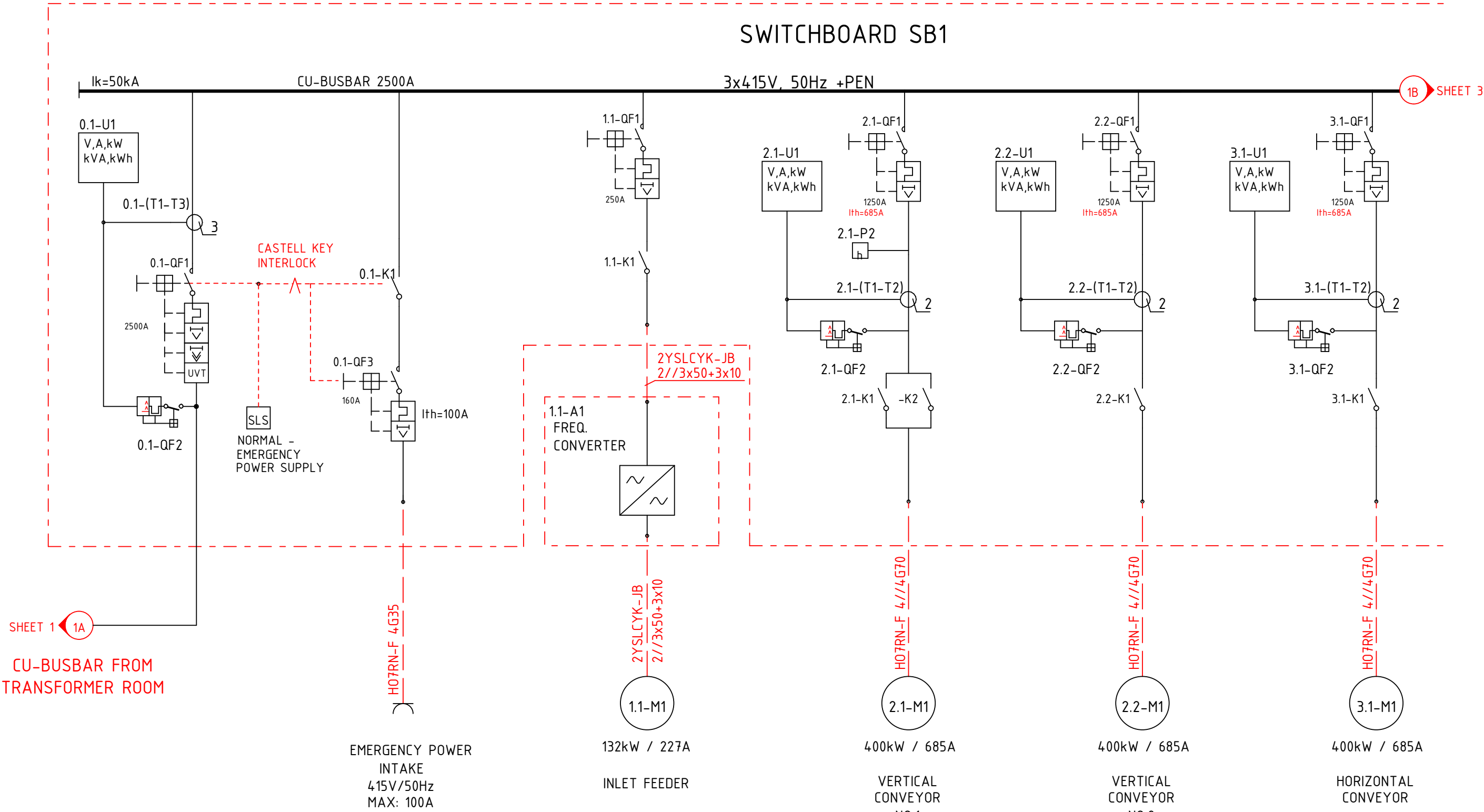
TP2

1	5,5
2	5,5
3	5,5
4	5,5
5	5,5
6	5,5
7	5,5
8	5,5

reproduced, or exploited in any other unauthorized way.



reproduced, or exploited in any other unauthorized way.



SWITCHBOARD SB1

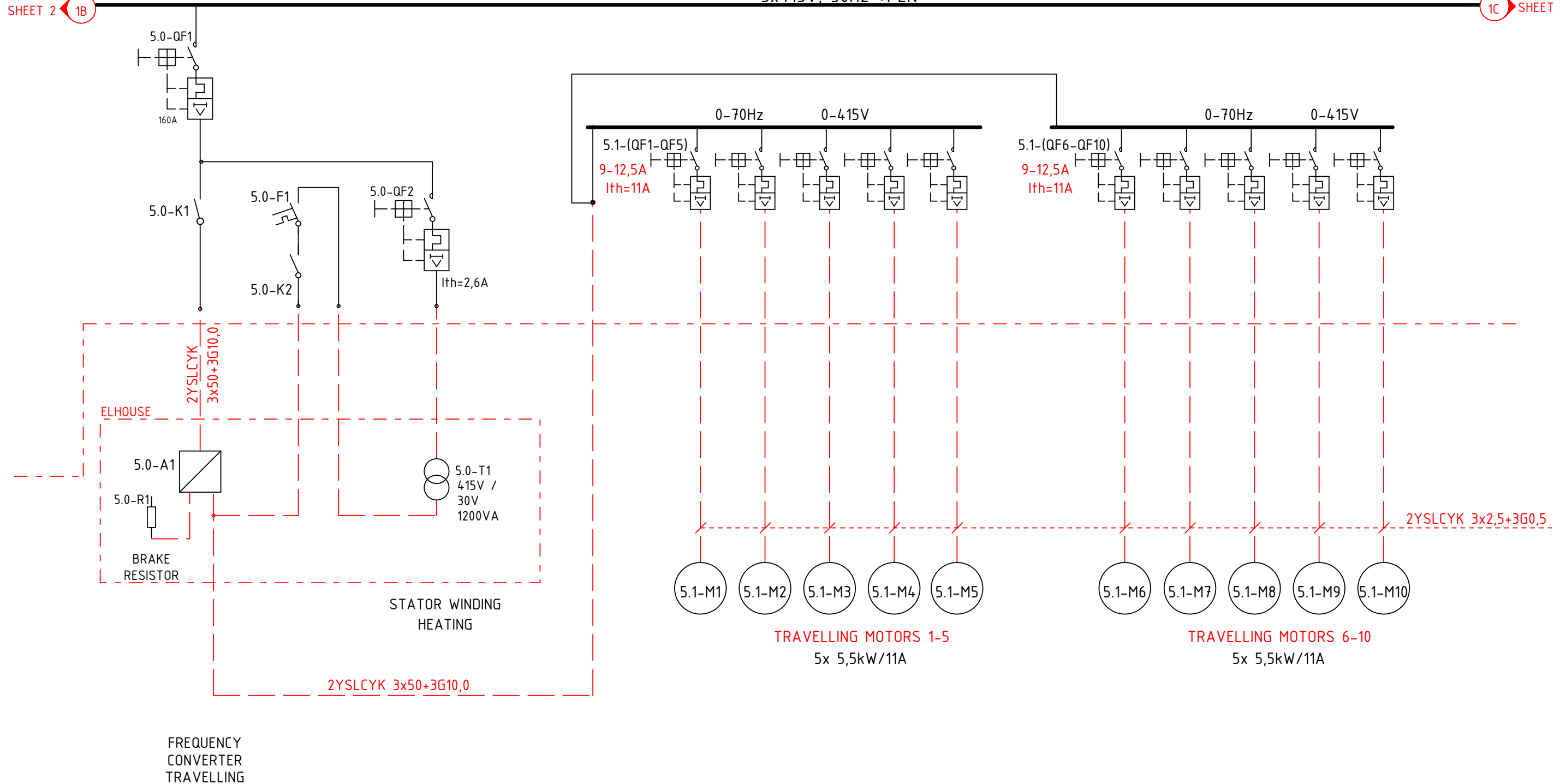
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CU-BUSBAR 2500A

3x415V, 50Hz +PEN

SHEET 2 ◀ (1B)

(1C) SHEET 4



AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

SINGLE LINE DIAGRAM

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-16

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - B

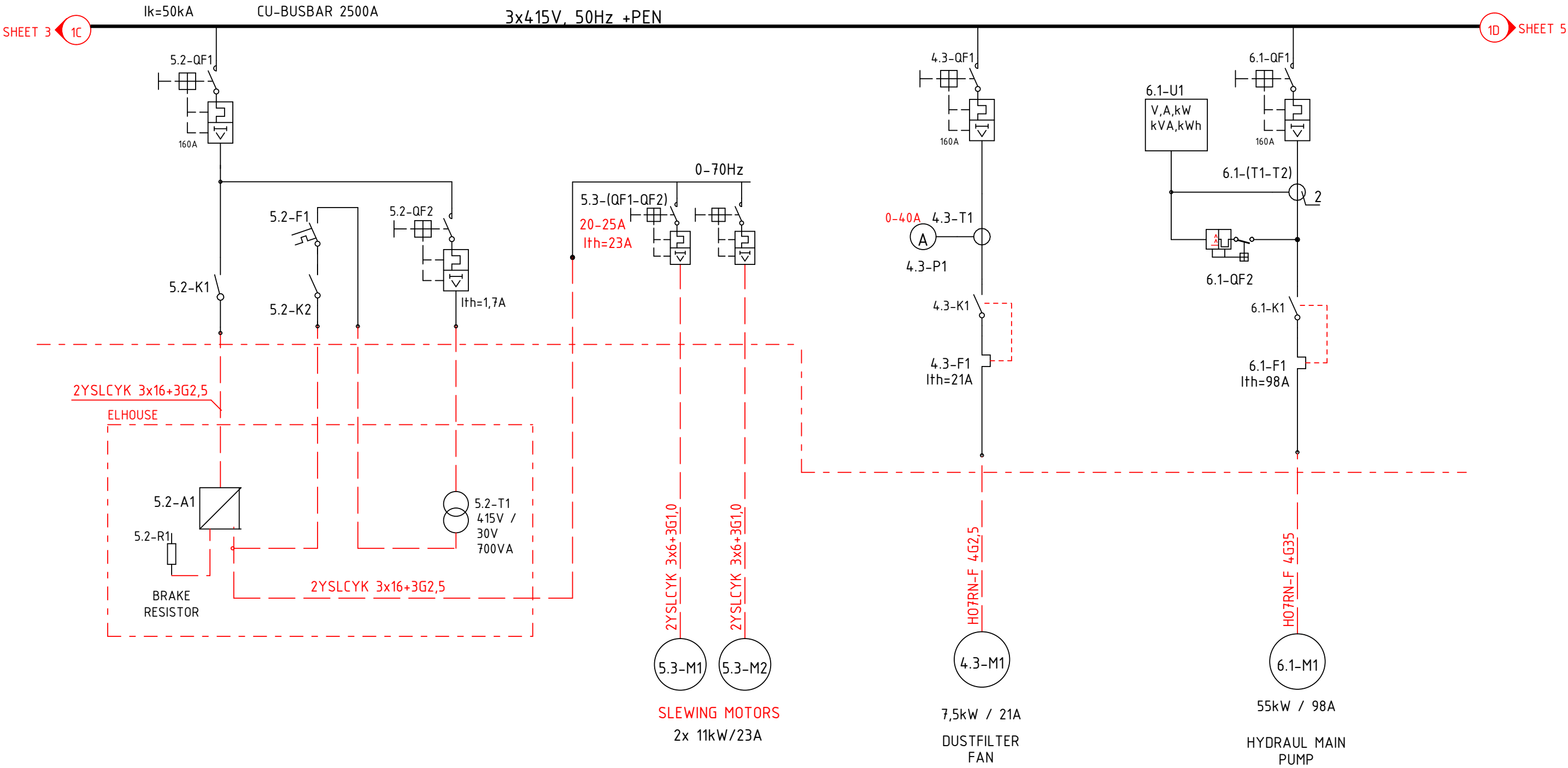
APPR.

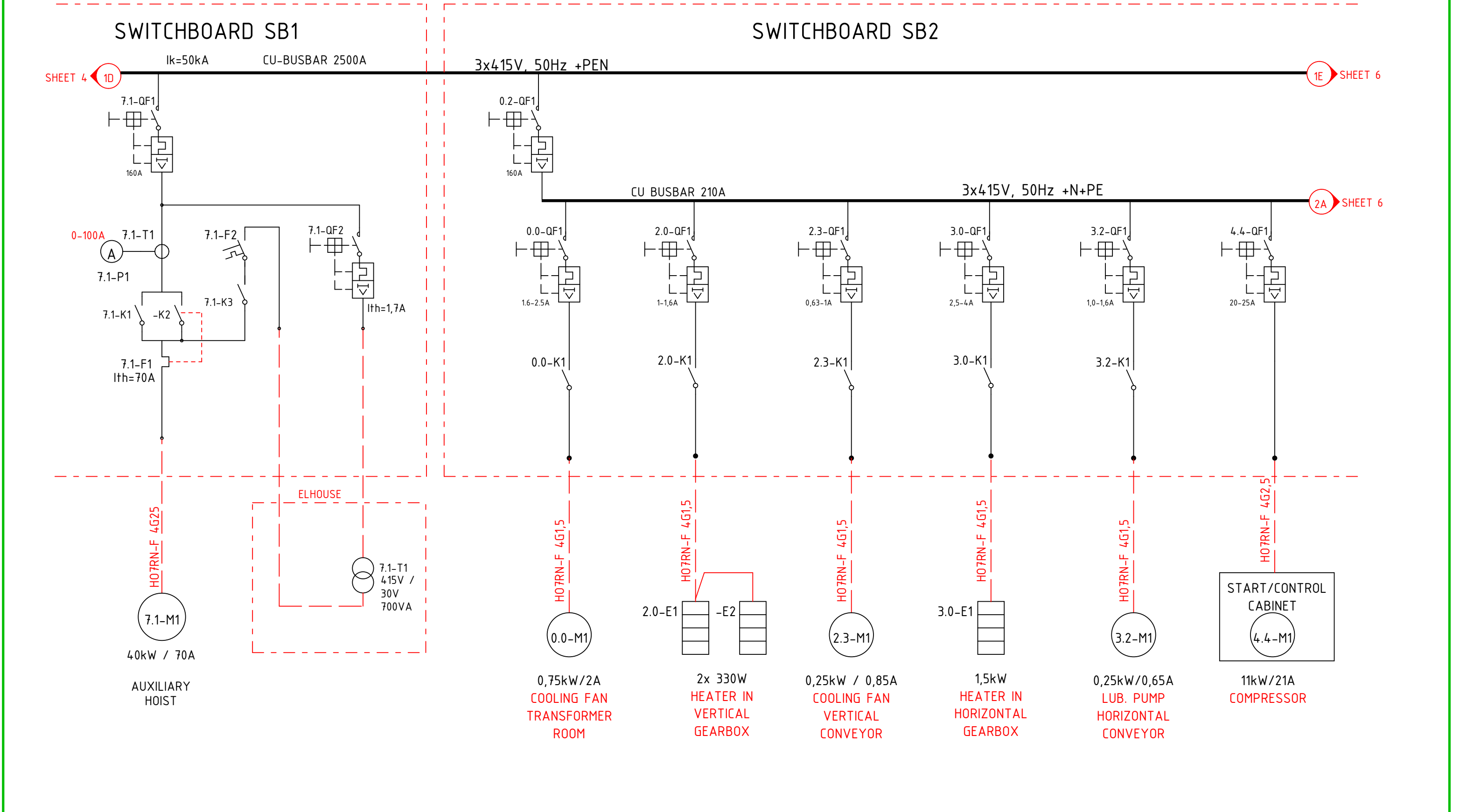
FILE NAME
A524.DB

SHEET
3

CONT. SHEET
4

SWITCHBOARD SB1





SWITCHBOARD SB2

SHEET 5

1E

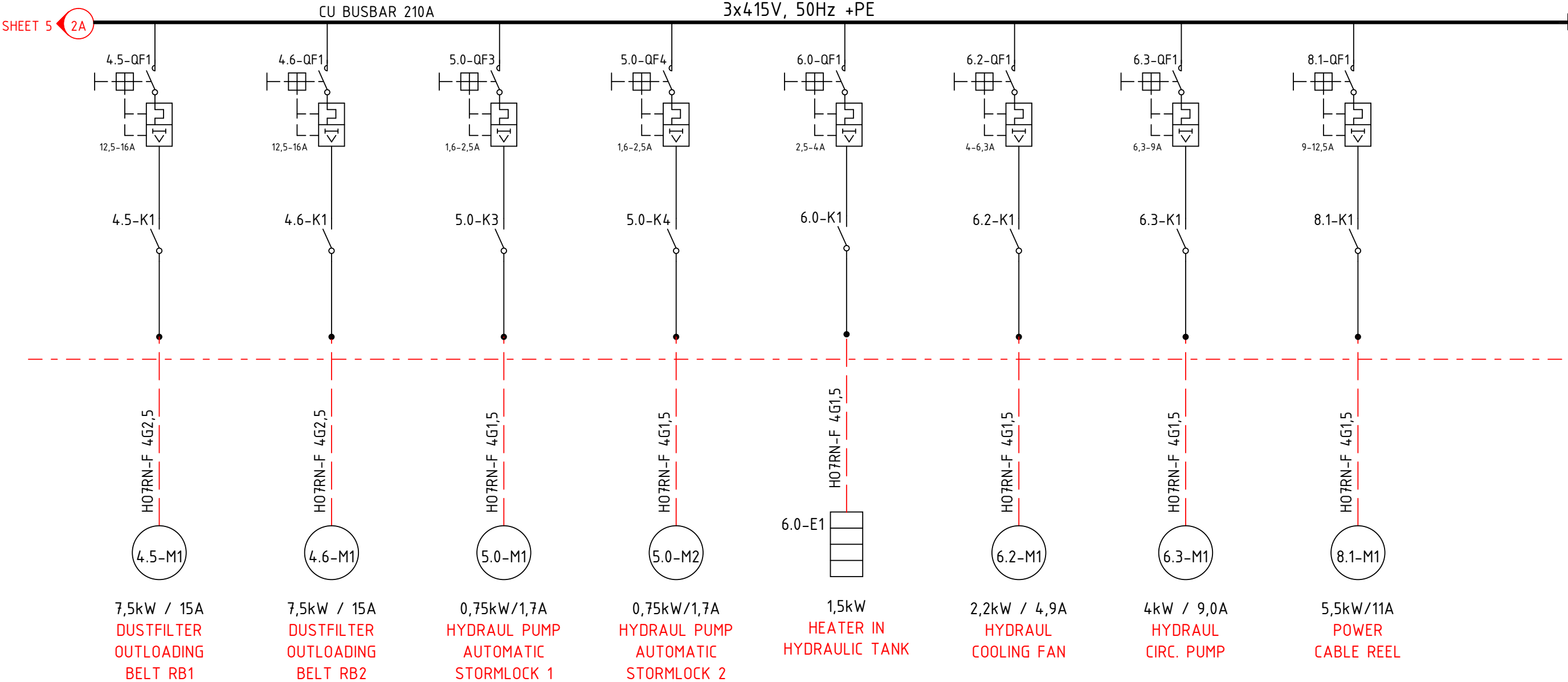
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CU-BUSBAR 2500A

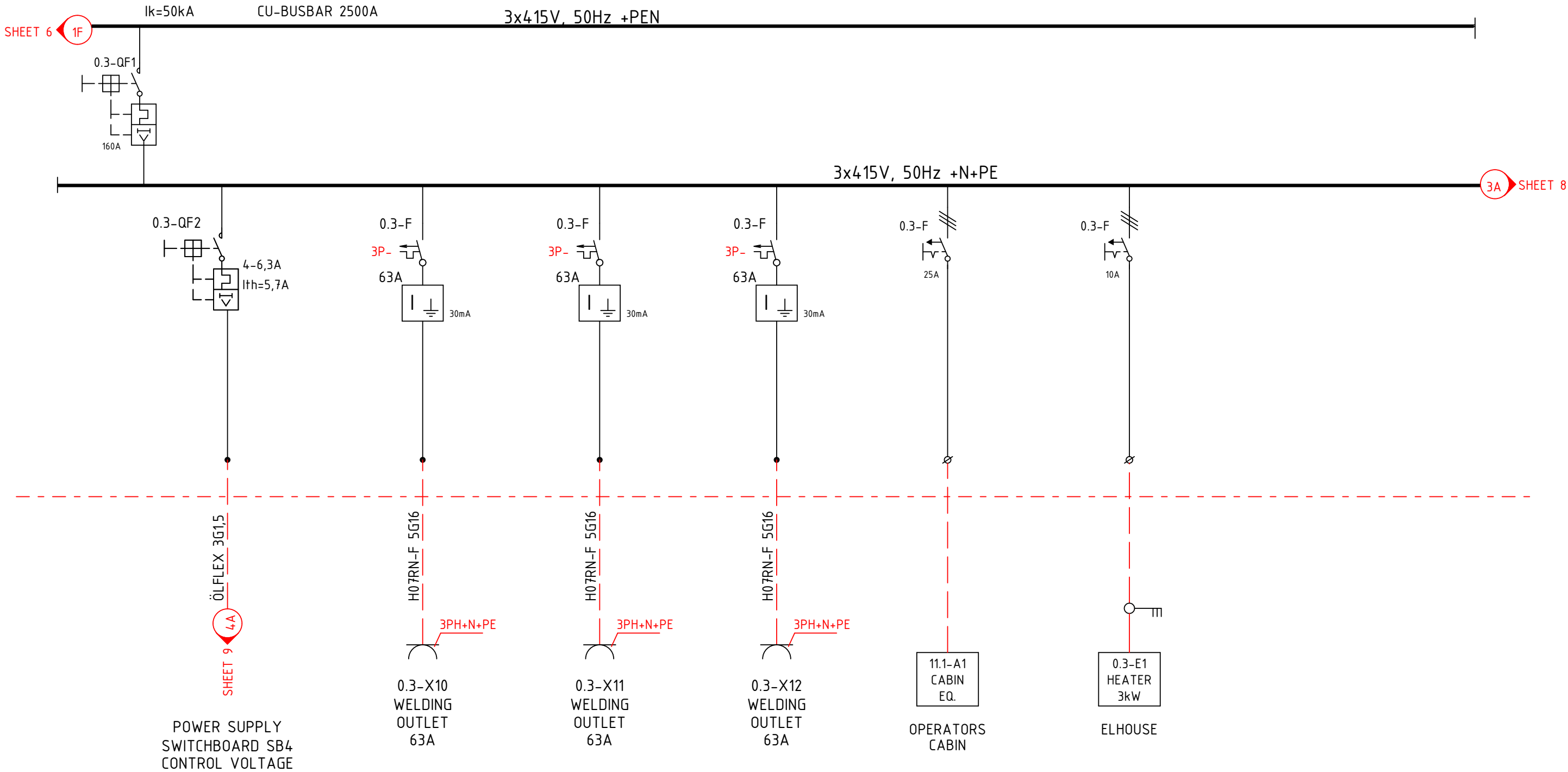
3x415V, 50Hz +PEN

1F

SHEET 7



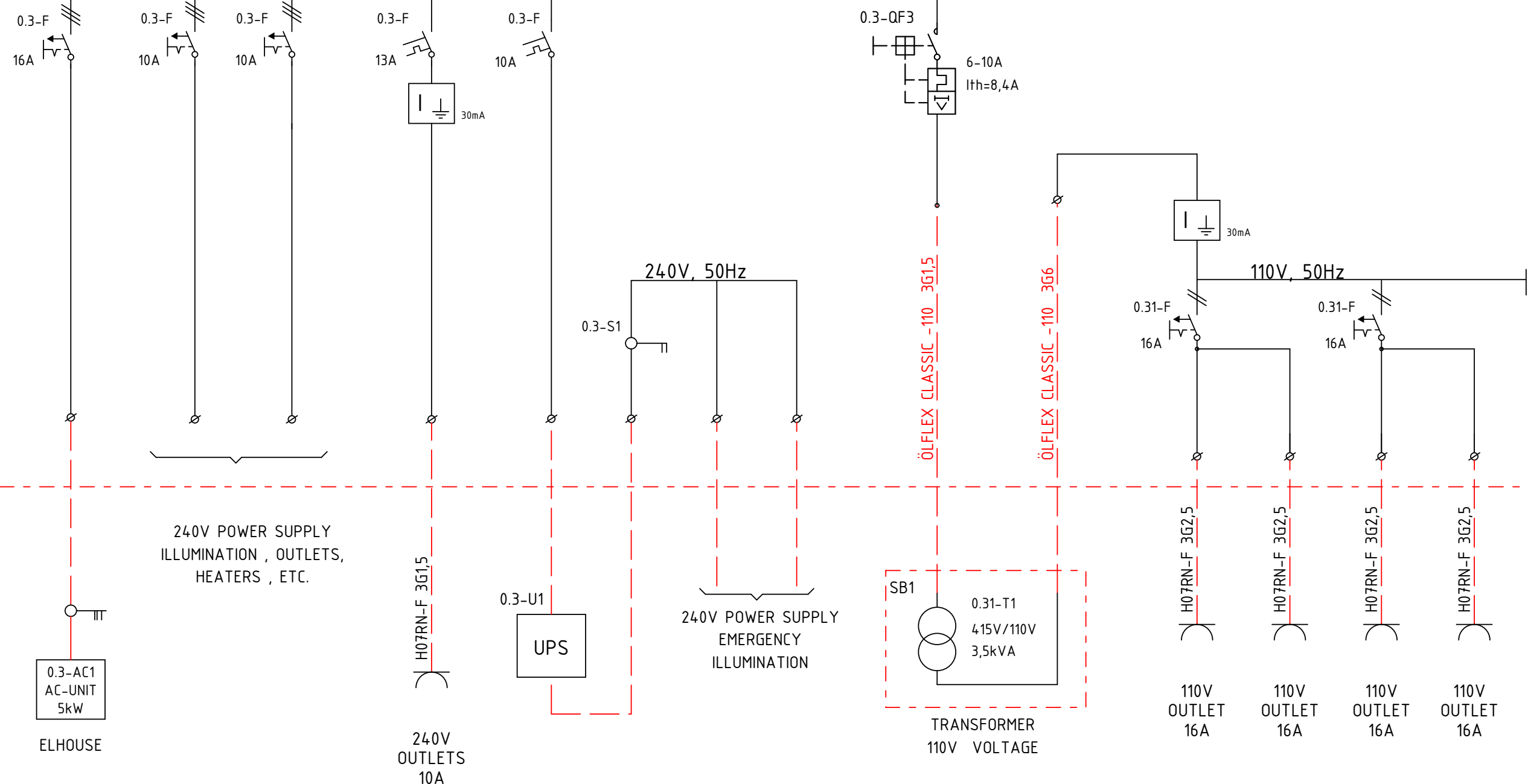
SWITCHBOARD SB3



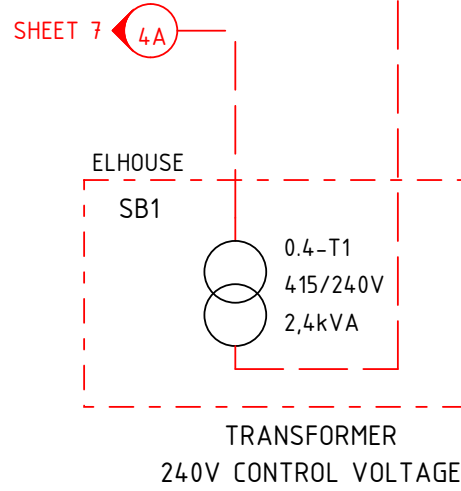
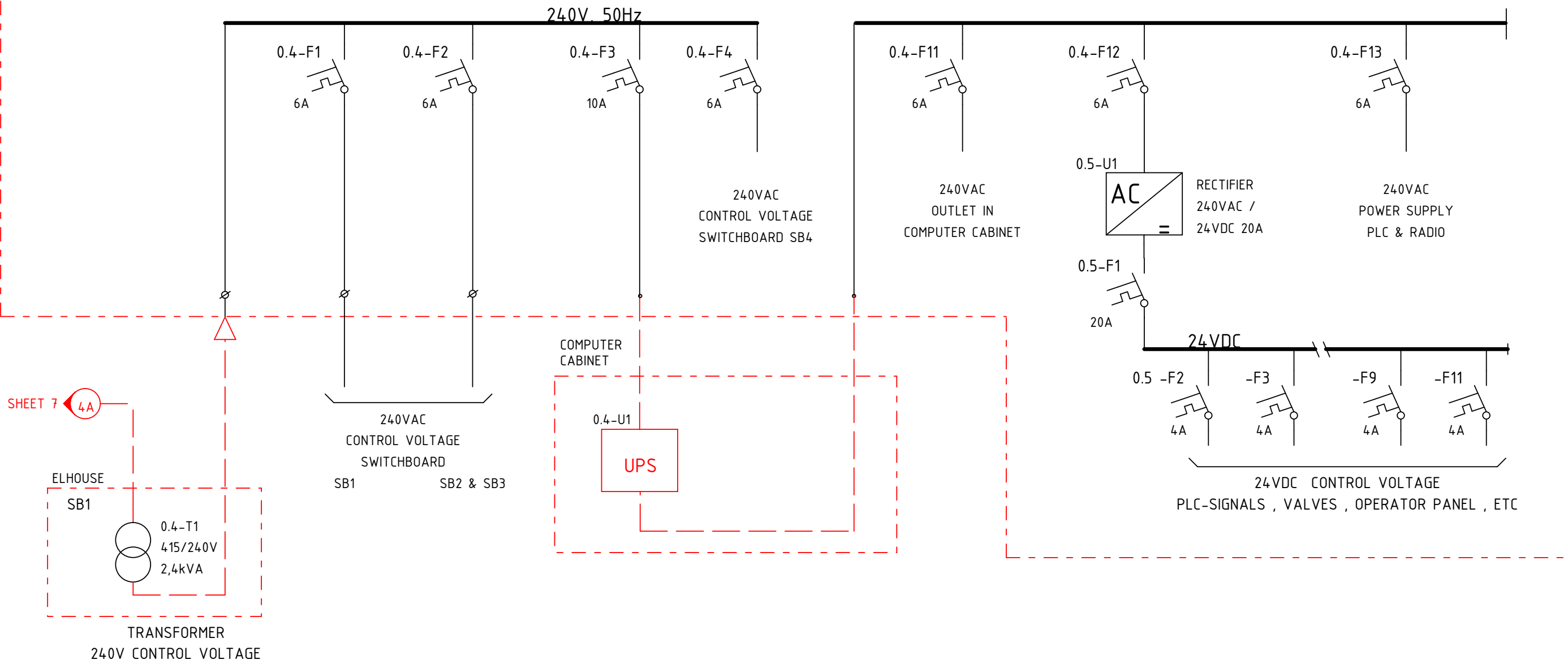
SHEET 7 3B

SWITCHBOARD SB3

3x415V, 50Hz +N+PE



SWITCHBOARD SB4



0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.1-K1	CONTACTOR A110-30-11	1SFL451001R8811	230-260V / 50-60Hz		ABB	1	2.6		
0.1-K2	TIME DELAY RELAY CT-ERD	1SVR500100R0000			ABB	1	2.5		
0.1-QF1	AUX. CONTACT	1SDA038361R1			ABB	1	1.3		
0.1-QF1	UNDervOLTAGE TRIP E1-E6	1SDA038312R1	220-230VAC		ABB	1	1.3		
0.1-QF1	TIME DELAY E1-E6 YU	1SDA038320R1	220-250VAC		ABB	1	1.3		
0.1-QF1	ACB E3N 2500 PR121/P-LSIG 3P W MP	1SDA056130R1			ABB	1	1.3		
0.1-QF1	FRAME E3 W FP 3P HR-HR	1SDA059669R1			ABB	1	1.3		
0.1-QF2	MOTOR PROTECTION BREAKER 1-1,6A	1SAM150000R1006		MS325-1,6	ABB	1	1.4		
0.1-QF3	MCCB T1B 160 R100 F FC 3P	1SDA050878R1			ABB	1	2.8		
0.1-QF3	AUX. CONTACT BLOCK AUX 3Q	1SDA051369R1			ABB	1	2.8		
0.1-QF3	EXTENDED ROTARY HANDLE	1SDA051383R1			ABB	1	2.8		
0.1-QF3	PROTECTION COVER HTC T1 3P	1SDA051415R1			ABB	1	2.8		
0.1-QF4	MOTOR PROTECTION BREAKER 1-1,6A	1SAM150000R1006		MS325-1,6	ABB	1	2.7		
0.1-S1	SELECTOR SWITCH 1-2, 3-P	CH10A222-600FT2-V			KRAUS & NAIMER	1	2.3		
0.1-T1	CURRENT TRANSFORMER 2500/5A	MAK140/100H(2500/5)			GANZ	1	1.6		
0.1-T2	CURRENT TRANSFORMER 2500/5A	MAK140/100H(2500/5)			GANZ	1	1.6		
0.1-T3	CURRENT TRANSFORMER 2500/5A	MAK140/100H(2500/5)			GANZ	1	1.6		
0.1-T4	TRANSFORMER 50VA	OFS50 (415/240)	415-240V		TRAMO	1	1.3		
0.1-T5	TRANSFORMER 200VA	OFS200 (415/240)	415-240V		TRAMO	1	2.7		
0.1-U1	MULTI INSTRUMENT 3-PHASE 4-WIRE, 2ao	INT-1534-M-5M-002-1	100-250V AC/DC		CROMPTON INSTRUMENTS	1	1.0		
0.1-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	12			
0.1-X1	END SUPPORT ZEw 35	9540000000			WEIDMULLER	2			
0.1-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
0.1-X2	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	8			
0.1-X2	END SUPPORT ZEw 35	9540000000			WEIDMULLER	2			
0.1-X2	TERMINAL WDU 35	1020500000			WEIDMULLER	3			
0.1-X2	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
0.1-X2	ENDPLATE WAP 16-35	1050100000			WEIDMULLER	1			
0.4-T1	TRANSFORMER 2,5kVA	OFS2500 (415/240)	415-240V		TRAMO	1	4.1		
0.31-T1	TRANSFORMER 3,5kVA	OFS3500 (415/110)	415-110V		TRAMO	1	4.3		
1.1-K1	CONTACTOR A260-30-22	1SFL531001R8822	230-260V / 50-60Hz		ABB	1	106.5		
1.1-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	106.4		
1.1-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	106.4		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB1

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
1

CONT. SHEET
2

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
1.1-QF1	MCCB T3S 250 R250 F F 3P	1SDA051269R1			ABB	1	106.8		
1.1-QF1	AUX. CONTACT BLOCK AUX 3Q	1SDA051369R1			ABB	1	106.8		
1.1-QF1	PROTECTION COVER HTC T3 3P	1SDA051419R1			ABB	1	106.8		
1.1-QF1	EXTENDED ROTARY HANDLE	1SDA054929R1			ABB	1	106.8		
1.1-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	7			
1.1-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
1.1-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
2.1-K1	CONTACTOR AF750-30-22	1SFL637001R7022	100-250VAC		ABB	1	101.5		
2.1-K1	FWD/REV CONNECTORS BEM750-30	1SFN086101R1000			ABB	1	101.5		
2.1-K2	CONTACTOR AF750-30-22	1SFL637001R7022	100-250VAC		ABB	1	101.6		
2.1-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	101.3		
2.1-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	101.3		
2.1-K4	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	101.3		
2.1-K4	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	101.3		
2.1-P1	WORK TIME METER	BW4018-23050	230V 50HZ		MULLER	1	101.6		
2.1-QF1	PROTECTION COVER LOCKING SKREW	1SDA013699R1			ABB	1	100.8		
2.1-QF1	AXTENDED ROTARY HANDLE S7	1SDA014213R1			ABB	1	100.8		
2.1-QF1	PROTECTION COVER S7 3P	1SDA023324R1			ABB	1	100.8		
2.1-QF1	AUX. CONTACT S6-S7 2no+2nc	1SDA023366R1			ABB	1	100.8		
2.1-QF1	MCCB S7S 1250 1000 PR212MP 3P	1SDA050144R1			ABB	1	100.8		
2.1-QF2	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	100.7		
2.1-QF2	MOTOR PROTECTION BREAKER 1-1,6A	1SAM150000R1006		MS325-1,6	ABB	1	100.7		
2.1-T1	CURRENT TRANSFORMER 800/5A	MAK86/60 800A/5A			GANZ	1	100.8		
2.1-T2	CURRENT TRANSFORMER 800/5A	MAK86/60 800A/5A			GANZ	1	100.8		
2.1-T3	CURRENT TRANSFORMER 1000/5A	MAK86/60 1000/5A			GANZ	1	100.8		
2.1-T4	CURRENT TRANSFORMER 0-10A	CTM10			EMOTRON	1	100.6		
2.1-U1	MULTI INSTRUMENT 3-PHASE 3-WIRE, 2ao	INT-1533-M-5M-002-1	100-250V AC/DC		CROMPTON INSTRUMENTS	1	100.4		
2.1-U4	LOAD CONVERTER "ELFI"	M20	100-690V 50/60Hz		EMOTRON	1	100.4		
2.1-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	18			
2.1-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
2.1-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
2.2-K1	CONTACTOR AF750-30-22	1SFL637001R7022	100-250VAC		ABB	1	103.4		
2.2-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	103.3		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB1

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
2

CONT. SHEET
3

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
2.2-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	103.3		
2.2-QF1	PROTECTION COVER LOCKING SKREW	1SDA013699R1			ABB	1	103.5		
2.2-QF1	AXTENDED ROTARY HANDLE S7	1SDA014213R1			ABB	1	103.5		
2.2-QF1	PROTECTION COVER S7 3P	1SDA023324R1			ABB	1	103.5		
2.2-QF1	AUX. CONTACT S6-S7 2no+2nc	1SDA023366R1			ABB	1	103.5		
2.2-QF1	MCCB S7S 1250 1000 PR212MP 3P	1SDA050144R1			ABB	1	103.5		
2.2-QF2	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	103.5		
2.2-QF2	MOTOR PROTECTION BREAKER 1-1,6A	1SAM150000R1006		MS325-1,6	ABB	1	103.5		
2.2-T1	CURRENT TRANSFORMER 800/5A	MAK86/60 800A/5A			GANZ	1	103.8		
2.2-T2	CURRENT TRANSFORMER 800/5A	MAK86/60 800A/5A			GANZ	1	103.8		
2.2-U1	MULTI INSTRUMENT 3-PHASE 3-WIRE, 2ao	INT-1533-M-5M-002-1	100-250V AC/DC		CROMPTON INSTRUMENTS	1	103.5		
2.2-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	14			
2.2-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
2.2-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
3.1-K1	CONTACTOR AF750-30-22	1SFL637001R7022	100-250VAC		ABB	1	94.4		
3.1-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	94.3		
3.1-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	94.3		
3.1-QF1	PROTECTION COVER LOCKING SKREW	1SDA013699R1			ABB	1	94.8		
3.1-QF1	AXTENDED ROTARY HANDLE S7	1SDA014213R1			ABB	1	94.8		
3.1-QF1	PROTECTION COVER S7 3P	1SDA023324R1			ABB	1	94.8		
3.1-QF1	AUX. CONTACT S6-S7 2no+2nc	1SDA023366R1			ABB	1	94.8		
3.1-QF1	MCCB S7S 1250 1000 PR212MP 3P	1SDA050144R1			ABB	1	94.8		
3.1-QF2	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	94.7		
3.1-QF2	MOTOR PROTECTION BREAKER 1-1,6A	1SAM150000R1006		MS325-1,6	ABB	1	94.7		
3.1-T1	CURRENT TRANSFORMER 800/5A	MAK86/60 800A/5A			GANZ	1	94.8		
3.1-T2	CURRENT TRANSFORMER 800/5A	MAK86/60 800A/5A			GANZ	1	94.8		
3.1-U1	MULTI INSTRUMENT 3-PHASE 3-WIRE, 2ao	INT-1533-M-5M-002-1	100-250V AC/DC		CROMPTON INSTRUMENTS	1	94.5		
3.1-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	14			
3.1-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
3.1-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
4.3-F1	OVERLOAD RELAY TA25 DU *18-25A	1SAZ211201R1051			ABB	1	80.5		
4.3-K1	CONTACTOR A26-30-10	1SBL241001R8810	230-260V / 50-60Hz		ABB	1	80.5		
4.3-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	80.3		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB1

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
3

CONT. SHEET
4

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
4.3-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	80.3		
4.3-P1	AMPERE METER 0-40/5A	72LA(40/5)			GANZ	1	80.7		
4.3-QF1	MCCB T2S 160 MA52 F F 3P	1SDA051218R1			ABB	1	80.8		
4.3-QF1	AUX. CONTACT BLOCK AUX 3Q	1SDA051369R1			ABB	1	80.8		
4.3-QF1	EXTENDED ROTARY HANDLE	1SDA051383R1			ABB	1	80.8		
4.3-QF1	PROTECTION COVER HTC T2 3P	1SDA051417R1			ABB	1	80.8		
4.3-T1	CURRENT TRANSFORMER 40/5A	MAK45/14(40/5)			GANZ	1	80.8		
4.3-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	8			
4.3-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
4.3-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
5.0-F1	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	54.6		
5.0-F1	AUTOMATIC FUSE S202 C40 2P	2CDS252001R0404			ABB	1	54.6		
5.0-K1	CONTACTOR A145-30-11	1SFL471001R8811	230-260V / 50-60Hz		ABB	1	54.6		
5.0-K2	CONTACTOR A40-30-10	1SBL321001R8810	230-260V / 50-60Hz		ABB	1	54.5		
5.0-K2	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	54.5		
5.0-K5	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	54.3		
5.0-K5	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	54.3		
5.0-K6	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	54.3		
5.0-K6	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	54.3		
5.0-QF1	MCCB T2S 160 R125 F F 3P	1SDA051003R1			ABB	1	54.4		
5.0-QF1	AUX. CONTACT BLOCK AUX 3Q	1SDA051369R1			ABB	1	54.4		
5.0-QF1	EXTENDED ROTARY HANDLE TMAX- RHE	1SDA051383R1			ABB	1	54.4		
5.0-QF1	PROTECTION COVER HTC T2 3P	1SDA051417R1			ABB	1	54.4		
5.0-QF2	MOTOR PROTECTION BREAKER 2,5-4A	GV2RT08			TELEMECANIQUE	1	54.7		
5.0-QF2	AUX. CONTACT BLOCK 1no + 1nc	GVAN11		3116642	TELEMECANIQUE	1	54.7		
5.0-T1	TRANSFORMER 1,2kVA	OFS1200 (415/30)	415-30V		TRAMO	1	54.7		
5.0-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	12			
5.0-X1	TERMINAL WDU 10 SL	9537480000			WEIDMULLER	6			
5.0-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
5.0-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
5.1-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	58.5		
5.1-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	58.4		
5.1-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	58.4		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB1

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
4

CONT. SHEET
5

reproduced, or exploited in any other unauthorized way.

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
5.1-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	56.3		
5.1-QF1	BUSBAR	1SAM101937R0022			ABB	1	56.3		
5.1-QF1	BUSBAR	1SAM101937R0024			ABB	1	56.3		
5.1-QF1	BUSBAR CONNECTION	1SAM101938R0002			ABB	1	56.3		
5.1-QF1	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	56.3		
5.1-QF2	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	56.4		
5.1-QF2	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	56.4		
5.1-QF3	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	56.5		
5.1-QF3	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	56.5		
5.1-QF4	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	56.7		
5.1-QF4	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	56.7		
5.1-QF5	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	56.8		
5.1-QF5	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	56.8		
5.1-QF6	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	57.3		
5.1-QF6	BUSBAR	1SAM101937R0022			ABB	1	57.3		
5.1-QF6	BUSBAR	1SAM101937R0024			ABB	1	57.3		
5.1-QF6	BUSBAR CONNECTION	1SAM101938R0002			ABB	1	57.3		
5.1-QF6	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	57.3		
5.1-QF7	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	57.4		
5.1-QF7	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	57.4		
5.1-QF8	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	57.5		
5.1-QF8	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	57.5		
5.1-QF9	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	57.7		
5.1-QF9	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	57.7		
5.1-QF10	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	57.8		
5.1-QF10	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	57.8		
5.1-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	38			
5.1-X1	TERMINAL WDU 10 SL	9537480000			WEIDMULLER	2			
5.1-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
5.1-X1	TERMINAL WDU 70	1024600000			WEIDMULLER	3			
5.1-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
5.2-F1	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	61.5		
5.2-F1	AUTOMATIC FUSE S202 C16 2P	2CDS252001R0164			ABB	1	61.5		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB1

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
5

CONT. SHEET
6

REV.	MODIFICATION	DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
5.2-K1	CONTACTOR A50-30-11	1SBL351001R8811	230-260V / 50-60Hz		ABB	1	61.6		
5.2-K2	CONTACTOR A16-30-10	1SBL181001R8810	230-260V / 50-60Hz		ABB	1	61.5		
5.2-K2	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	61.5		
5.2-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	61.3		
5.2-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	61.3		
5.2-K4	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	61.3		
5.2-K4	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	61.3		
5.2-QF1	MCCB T2S 160 R80 F F 3P	1SDA051001R1			ABB	1	61.4		
5.2-QF1	AUX. CONTACT BLOCK AUX 3Q	1SDA051369R1			ABB	1	61.4		
5.2-QF1	EXTENDED ROTARY HANDLE TMAX- RHE	1SDA051383R1			ABB	1	61.4		
5.2-QF1	PROTECTION COVER HTC T2 3P	1SDA051417R1			ABB	1	61.4		
5.2-QF2	MOTOR PROTECTION BREAKER 1,6-2,5A	GV2RT07			TELEMECANIQUE	1	61.7		
5.2-QF2	AUX. CONTACT BLOCK 1no + 1nc	GVAN11		3116642	TELEMECANIQUE	1	61.7		
5.2-T1	TRANSFORMER 0,7kVA	OFS700 (415/30)	415-30V		TRAMO	1	61.7		
5.2-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	14			
5.2-X1	TERMINAL WDU 6 SL	9537460000			WEIDMULLER	2			
5.2-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
5.2-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
5.3-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	64.5		
5.3-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	64.4		
5.3-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	64.4		
5.3-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	63.3		
5.3-QF1	BUSBAR	1SAM101937R0022			ABB	1	63.3		
5.3-QF1	BUSBAR CONNECTION	1SAM101938R0002			ABB	1	63.3		
5.3-QF1	MOTOR PROTECTION BREAKER 20-25A	1SAM150000R1014		MS325-25	ABB	1	63.3		
5.3-QF2	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	63.5		
5.3-QF2	MOTOR PROTECTION BREAKER 20-25A	1SAM150000R1014		MS325-25	ABB	1	63.5		
5.3-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	16			
5.3-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
5.3-X1	TERMINAL WDU 16	1020400000			WEIDMULLER	3			
5.3-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
5.3-X1	ENDPLATE WAP 16-35	1050100000			WEIDMULLER	1			
6.1-F1	OVERLOAD RELAY TA110 DU*80-110A	1SAZ411201R1002			ABB	1	69.8		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB1

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
6

CONT. SHEET
7

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
6.1-K1	CONTACTOR A110-30-11	1SFL451001R8811	230-260V / 50-60Hz		ABB	1	69.4		
6.1-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	69.3		
6.1-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	69.3		
6.1-QF1	MCCB T2S 160 MA100 F F 3P	1SDA051220R1			ABB	1	69.8		
6.1-QF1	AUX. CONTACT BLOCK AUX 3Q	1SDA051369R1			ABB	1	69.8		
6.1-QF1	EXTENDED ROTARY HANDLE TMAX- RHE	1SDA051383R1			ABB	1	69.8		
6.1-QF1	PROTECTION COVER HTC T3 3P	1SDA051419R1			ABB	1	69.8		
6.1-QF2	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	69.7		
6.1-QF2	MOTOR PROTECTION BREAKER 1-1,6A	1SAM150000R1006		MS325-1,6	ABB	1	69.7		
6.1-T1	CURRENT TRANSFORMER 150/5A	MAK62/30(150/5A)			GANZ	1	69.8		
6.1-T2	CURRENT TRANSFORMER 150/5A	MAK62/30(150/5A)			GANZ	1	69.8		
6.1-U1	MULTI INSTRUMENT 3-PHASE 3-WIRE, 2ao	INT-1533-M-5M-002-1	100-250V AC/DC		CROMPTON INSTRUMENTS	1	69.5		
6.1-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	15			
6.1-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
6.1-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
7.1-F1	OVERLOAD RELAY TA110 DU*80-110A	1SAZ411201R1002			ABB	1	75.4		
7.1-F2	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	76.7		
7.1-F2	AUTOMATIC FUSE S202 C25 2P	2CDS252001R0254			ABB	1	76.7		
7.1-K1	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	75.5		
7.1-K1	CONNECTION BUSBAR BEM 75-30	1SBN083501R1000			ABB	1	75.5		
7.1-K1	CONTACTOR	1SFL431001R8811	240V/60Hz		ABB	1	75.5		
7.1-K2	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	75.6		
7.1-K2	CONTACTOR	1SFL431001R8811	240V/60Hz		ABB	1	75.6		
7.1-K3	CONTACTOR A26-30-10	1SBL241001R8810	230-260V / 50-60Hz		ABB	1	76.6		
7.1-K3	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	76.6		
7.1-K4	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	75.2		
7.1-K4	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	75.2		
7.1-K5	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	75.2		
7.1-K5	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	75.2		
7.1-K6	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	76.3		
7.1-K6	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	76.3		
7.1-P1	AMPERE METER 0-100/5A	72LA(100/5A)			GANZ	1	75.7		
7.1-QF1	MCCB T2S 160 MA80 F F 3P	1SDA051219R1			ABB	1	75.3		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB1

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
7

CONT. SHEET
8

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.0-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	39.6		
0.0-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	39.4		
0.0-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	39.4		
0.0-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	39.8		
0.0-QF1	MOTOR PROTECTION BREAKER 1,6-2,5A	1SAM150000R1007		MS325-2,5	ABB	1	39.8		
0.2-QF1	MCCB T2S 160 R160 F F 3P	1SDA051004R1			ABB	1	3.1		
0.2-QF1	AUX. CONTACT BLOCK AUX 3Q	1SDA051369R1			ABB	1	3.1		
0.2-QF1	EXTENDED ROTARY HANDLE	1SDA051383R1			ABB	1	3.1		
2.0-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	96.6		
2.0-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	96.4		
2.0-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	96.4		
2.0-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	96.5		
2.0-QF1	MOTOR PROTECTION BREAKER 1-1,6A	1SAM150000R1006		MS325-1,6	ABB	1	96.5		
2.3-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	97.6		
2.3-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	97.4		
2.3-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	97.4		
2.3-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	97.5		
2.3-QF1	MOTOR PROTECTION BREAKER 0,63-1A	1SAM150000R1005		MS325-1	ABB	1	97.5		
3.0-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	92.8		
3.0-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	92.7		
3.0-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	92.7		
3.0-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	92.7		
3.0-QF1	MOTOR PROTECTION BREAKER 2,5-4A	1SAM150000R1008		MS325-4	ABB	1	92.7		
3.2-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	91.6		
3.2-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	91.4		
3.2-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	91.4		
3.2-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	91.5		
3.2-QF1	MOTOR PROTECTION BREAKER 0,63-1A	1SAM150000R1005		MS325-1	ABB	1	91.5		
4.0-K11	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	83B.6		
4.0-K11	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	83B.6		
4.0-K12	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	83B.6		
4.0-K12	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	83B.6		
4.0-K13	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	83B.4		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



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Box 566
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SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB2

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
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DRAWING NO.
A524 - C

SHEET
11

CONT. SHEET
12

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
4.0-K13	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	83B.4		
4.0-K14	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	83B.4		
4.0-K14	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	83B.4		
4.0-K15	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	84B.6		
4.0-K15	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	84B.6		
4.0-K16	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	84B.6		
4.0-K16	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	84B.6		
4.0-K17	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	84B.4		
4.0-K17	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	84B.4		
4.0-K18	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	84B.4		
4.0-K18	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	84B.4		
4.0-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	83B.3		
4.0-QF1	MOTOR PROTECTION BREAKER 1,6-2,5A	1SAM150000R1007		MS325-2,5	ABB	1	83B.3		
4.0-QF2	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	84B.3		
4.0-QF2	MOTOR PROTECTION BREAKER 1,6-2,5A	1SAM150000R1007		MS325-2,5	ABB	1	84B.3		
4.4-QF1	AUX. CONTACT BLOCK HK4-11	1SAM401901R1001			ABB	1	82.5		
4.4-QF1	MOTOR PROTECTION BREAKER 40-50A	1SAM450000R1007			ABB	1	82.5		
4.5-K1	CONTACTOR A16-30-10	1SBL181001R8810	230-260V / 50-60Hz		ABB	1	87.5		
4.5-K1	AUX. CONTACT BLOCK 1no	1SBN010010R1010			ABB	2	87.5		
4.5-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	87.3		
4.5-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	87.3		
4.5-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	87.5		
4.5-QF1	MOTOR PROTECTION BREAKER 12,5-16A	1SAM150000R1012		MS325-16	ABB	1	87.5		
4.6-K1	CONTACTOR A16-30-10	1SBL181001R8810	230-260V / 50-60Hz		ABB	1	88.5		
4.6-K1	AUX. CONTACT BLOCK 1no	1SBN010010R1010			ABB	2	88.5		
4.6-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	88.3		
4.6-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	88.3		
4.6-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	88.5		
4.6-QF1	MOTOR PROTECTION BREAKER 12,5-16A	1SAM150000R1012		MS325-16	ABB	1	88.5		
5.0-K3	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	51.8		
5.0-K4	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	52.8		
5.0-K7	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	51.7		
5.0-K7	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	51.7		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB2

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
12

CONT. SHEET
13

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
5.0-K8	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	52.7		
5.0-K8	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	52.7		
5.0-QF3	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	51.7		
5.0-QF3	MOTOR PROTECTION BREAKER 1,6-2,5A	1SAM150000R1007		MS325-2,5	ABB	1	51.7		
5.0-QF4	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	52.7		
5.0-QF4	MOTOR PROTECTION BREAKER 1,6-2,5A	1SAM150000R1007		MS325-2,5	ABB	1	52.7		
6.0-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	67.6		
6.0-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	67.4		
6.0-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	67.4		
6.0-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	67.5		
6.0-QF1	MOTOR PROTECTION BREAKER 2,5-4A	1SAM150000R1008		MS325-4	ABB	1	67.5		
6.2-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	70.6		
6.2-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	70.4		
6.2-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	70.4		
6.2-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	70.5		
6.2-QF1	MOTOR PROTECTION BREAKER 4-6,3A	1SAM150000R1009		MS325-6,3	ABB	1	70.5		
6.3-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	71.8		
6.3-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	71.7		
6.3-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	71.7		
6.3-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	71.7		
6.3-QF1	MOTOR PROTECTION BREAKER 6,3-9A	1SAM150000R1010		MS325-9	ABB	1	71.7		
8.1-K1	CONTACTOR A12-30-10	1SBL161001R8810	230-260V / 50-60Hz		ABB	1	43.6		
8.1-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	43.4		
8.1-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	43.4		
8.1-QF1	AUX. CONTACT BLOCK HK-11	1SAM101901R0001			ABB	1	43.8		
8.1-QF1	MOTOR PROTECTION BREAKER 9-12,5A	1SAM150000R1011		MS325-12,5	ABB	1	43.8		
BUSBAR	BUSBAR SUPPORT CU 30x10 4pcs	SV3000000			RITTAL	1	3.1		
BUSBAR	INSERTS FOR CU 12x5 12pcs	SV3009000			RITTAL	1	3.1		
BUSBAR	COMPONENT ADAPTOR 5/72/2/ 63A	SV3038000			RITTAL	14	3.1		
BUSBAR	COMPONENT ADAPTOR 5/108/2/ 50A	SV3042000			RITTAL	2	3.1		
BUSBAR	END COVER SUPPORT 10pcs	SV3080000			RITTAL	1	3.1		
BUSBAR	BUSBAR COVER SECTION 10x1m	SV3092000			RITTAL	1	3.1		
BUSBAR	B.B CONNECTION ADAPTOR 250A	SV3442000			RITTAL	1	3.1		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB2

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
13

CONT. SHEET
21

reproduced, or exploited in any other unauthorized way.

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.3-F1	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	5.1		
0.3-F1	AUTOMATIC FUSE S203 C63 3P	2CDS253001R0634			ABB	1	5.1		
0.3-F2	EARTH-FAULT PROTECTION F204 A-63/0,03	2CSF204101R1630			ABB	1	5.1		
0.3-F2	AUX. CONTACT S9-X	16056100			ABB	1	5.1		
0.3-F3	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	5.4		
0.3-F3	AUTOMATIC FUSE S203 C63 3P	2CDS253001R0634			ABB	1	5.4		
0.3-F4	EARTH-FAULT PROTECTION F204 A-63/0,03	2CSF204101R1630			ABB	1	5.4		
0.3-F4	AUX. CONTACT S9-X	16056100			ABB	1	5.4		
0.3-F5	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	5.6		
0.3-F5	AUTOMATIC FUSE S203 C63 3P	2CDS253001R0634			ABB	1	5.6		
0.3-F6	EARTH-FAULT PROTECTION F204 A-63/0,03	2CSF204101R1630			ABB	1	5.6		
0.3-F6	AUX. CONTACT S9-X	16056100			ABB	1	5.6		
0.3-F7	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	6.1		
0.3-F7	AUTOMATIC FUSE S203 C25 3P	2CDS253001R0254			ABB	1	6.1		
0.3-F8	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	6.3		
0.3-F8	AUTOMATIC FUSE S203 C16 3P	2CDS253001R0164			ABB	1	6.3		
0.3-F9	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	6.5		
0.3-F9	AUTOMATIC FUSE S203 C10 3P	2CDS253001R0104			ABB	1	6.5		
0.3-F10	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	6.7		
0.3-F10	AUTOMATIC FUSE S203 C10 3P	2CDS253001R0104			ABB	1	6.7		
0.3-F13	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	7.5		
0.3-F13	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	7.5		
0.3-F14	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	7.7		
0.3-F14	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	7.7		
0.3-F15	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	8.1		
0.3-F15	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	8.1		
0.3-F16	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	8.5		
0.3-F16	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	8.5		
0.3-F17	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	9.1		
0.3-F17	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	9.1		
0.3-F18	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	10.1		
0.3-F18	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	10.1		
0.3-F19	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	10.5		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB3

DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-24	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - C	
APPR.	FILE NAME A524.DB	SHEET 21	CONT. SHEET 22

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.3-F19	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	10.5		
0.3-F20	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	10.7		
0.3-F20	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	10.7		
0.3-F21	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	11.1		
0.3-F21	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	11.1		
0.3-F22	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	11.2		
0.3-F22	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	11.2		
0.3-F23	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	11.4		
0.3-F23	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	11.4		
0.3-F24	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	11.5		
0.3-F24	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	11.5		
0.3-F25	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	12.1		
0.3-F25	AUTOMATIC FUSE S201 C13 1P	2CDS251001R0134			ABB	1	12.1		
0.3-F26	EARTH-FAULT PROTECTION F202 A-25/0,03	2CSF202101R1250			ABB	1	12.1		
0.3-F26	AUX. CONTACT S9-X	16056100			ABB	1	12.1		
0.3-F27	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	12.3		
0.3-F27	AUTOMATIC FUSE S201 C13 1P	2CDS251001R0134			ABB	1	12.3		
0.3-F28	EARTH-FAULT PROTECTION F202 A-25/0,03	2CSF202101R1250			ABB	1	12.3		
0.3-F28	AUX. CONTACT S9-X	16056100			ABB	1	12.3		
0.3-F29	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	12.8		
0.3-F29	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	12.8		
0.3-K1	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	7.3		
0.3-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	7.3		
0.3-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	7.3		
0.3-K3	LIGHT SENSITIVE RELAY IC200	15284	230V / 50-60Hz	1903261	MERLIN & GERIN	1	9.7		
0.3-QF1	MCCB T2S 160 R160 F F 3P	1SDA051004R1			ABB	1	3.6		
0.3-QF1	AUX. CONTACT BLOCK AUX 3Q	1SDA051369R1			ABB	1	3.6		
0.3-QF1	EXTENDED ROTARY HANDLE	1SDA051383R1			ABB	1	3.6		
0.3-QF2	MOTOR PROTECTION BREAKER 4-6,3A	GV2RT10			TELEMECANIQUE	1	4.1		
0.3-QF2	AUX. CONTACT BLOCK 1no + 1nc	GVAN11		3116642	TELEMECANIQUE	1	4.1		
0.3-QF3	MOTOR PROTECTION BREAKER 6-10A	GV2RT14		3116670	TELEMECANIQUE	1	4.3		
0.3-QF3	AUX. CONTACT BLOCK 1no + 1nc	GVAN11		3116642	TELEMECANIQUE	1	4.3		
0.3-S1	CIRCUIT BREAKER OT 40 M3	1SCA022497R490			ABB	1	9.2		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB3

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
22

CONT. SHEET
23

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.31-F1	EARTH-FAULT PROTECTION F202 A-40/0,03	2CSF202101R1400			ABB	1	13.2		
0.31-F1	AUX. CONTACT S9-X	16056100			ABB	1	13.2		
0.31-F2	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	13.2		
0.31-F2	AUTOMATIC FUSE S201 C16 1P	2CDS251001R0164			ABB	1	13.2		
0.31-F3	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	13.4		
0.31-F3	AUTOMATIC FUSE S201 C16 1P	2CDS251001R0164			ABB	1	13.4		
0.31-F4	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	13.7		
0.31-F4	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	13.7		
4.0-K19	INTERFACE RELAY 1P	G2R-1SN(24VAC)	24V 50Hz		OMRON	1	79.3		
4.0-K19	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	79.3		
U300	POWER SUPPLY ST1PSD	313178005	24VDC		MITSUBISHI	1	28.2		
U300	SCREW CONNECTION ST1B-E4P2-H-SET	313179026			MITSUBISHI	1	28.2		
U301	INPUT MODULE 16DI ST1X16-DE1	313178013	24VDC		MITSUBISHI	1	28.3		
U301	SCREW CONNECTION ST1B-E4X16	313179032			MITSUBISHI	1	28.3		
U302	INPUT MODULE 16DI ST1X16-DE1	313178013	24VDC		MITSUBISHI	1	28.4		
U302	SCREW CONNECTION ST1B-E4X16	313179032			MITSUBISHI	1	28.4		
U303	INPUT MODULE 16DI ST1X16-DE1	313178013	24VDC		MITSUBISHI	1	28.5		
U303	SCREW CONNECTION ST1B-E4X16	313179032			MITSUBISHI	1	28.5		
U304	OUTPUT MODULE 16DO ST1Y16-TPE3	313178026	24VDC		MITSUBISHI	1	28.6		
U304	SCREW CONNECTION ST1B-E3Y16	313179038			MITSUBISHI	1	28.6		
U305	OUTPUT MODULE 16DO ST1Y16-TPE3	313178026	24VDC		MITSUBISHI	1	28.6		
U305	SCREW CONNECTION ST1B-E3Y16	313179038			MITSUBISHI	1	28.6		
UX300	PROFIBUS NODE ST1H-PB	313178000	24VDC		MITSUBISHI	1	28.1		
X3	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	74			
X3	TERMINAL WDU 6 SL	9537460000			WEIDMULLER	4			
X3	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
X3	TERMINAL WDU 16	1020400000			WEIDMULLER	12			
X3	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
X3	ENDPLATE WAP 16-35	1050100000			WEIDMULLER	1			
X12	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	6			
X12	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
X12	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
X31	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	12			



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

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SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB3

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

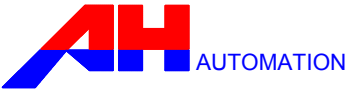
DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
23

CONT. SHEET
24

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.0-K3	INTERFACE RELAY 2P	G2R-2SN(24VDC)	24VDC		OMRON	1	38.3		
0.0-K3	RELAY SOCKET	P2RF-08E			OMRON	1	38.3		
0.0-U1	Pt-100 AMPLIFIER MCR-PT100-U-DC/3/7, 0-200C	2810311	24VDC		PHOENIX	1	38.3		
0.0-U2	Pt-100 AMPLIFIER MCR-PT100-U-DC/3/7, 0-200C	2810311	24VDC		PHOENIX	1	38.3		
0.0-U3	Pt-100 AMPLIFIER MCR-PT100-U-DC/3/7, 0-200C	2810311	24VDC		PHOENIX	1	38.3		
0.4-F1	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	19.2		
0.4-F1	AUTOMATIC FUSE S201 C6 1P	2CDS251001R0064			ABB	1	19.2		
0.4-F2	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	19.4		
0.4-F2	AUTOMATIC FUSE S201 C6 1P	2CDS251001R0064			ABB	1	19.4		
0.4-F3	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	19.6		
0.4-F3	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	19.6		
0.4-F4	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	19.8		
0.4-F4	AUTOMATIC FUSE S201 C6 1P	2CDS251001R0064			ABB	1	19.8		
0.4-F11	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	20.2		
0.4-F11	AUTOMATIC FUSE S201 C6 1P	2CDS251001R0064			ABB	1	20.2		
0.4-F12	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	20.5		
0.4-F12	AUTOMATIC FUSE S201 C6 1P	2CDS251001R0064			ABB	1	20.5		
0.4-F13	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	20.7		
0.4-F13	AUTOMATIC FUSE S201 C6 1P	2CDS251001R0064			ABB	1	20.7		
0.4-K1	INTERFACE RELAY 1P	G2R-1SN(230VAC)	230V / 50Hz		OMRON	1	19.7		
0.4-K1	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	19.7		
0.5-F1	AUTOMATIC FUSE S201 C20 1P	2CDS251001R0204			ABB	1	21.0		
0.5-F2	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	21.0		
0.5-F2	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	21.0		
0.5-F3	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	21.2		
0.5-F3	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	21.2		
0.5-F4	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	21.3		
0.5-F4	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	21.3		
0.5-F5	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	21.4		
0.5-F5	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	21.4		
0.5-F6	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	21.7		
0.5-F6	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	21.7		
0.5-F7	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	21.8		



SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007
SHIPUNLOADER MERSEY DOCKS
ELECTRICAL EQUIPMENT
PARTS LIST
SWITCHBOARD SB4

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
31

CONT. SHEET
32

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.5-F7	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	21.8		
0.5-F8	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	22.1		
0.5-F8	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	22.1		
0.5-F9	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	22.2		
0.5-F9	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	22.2		
0.5-F10	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	22.3		
0.5-F10	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	22.3		
0.5-F11	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	22.5		
0.5-F11	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	22.5		
0.5-U1	RECTIFIER 24VDC / 20A	DRP-48024	180-264V		MEAN WELL	1	21.0		
0.7-H1	BUZZER ESZ-11N-A	37-786-02	24VDC		ELFA	1	34.7		
0.7-H2	INDICATION LAMP RED KL1-100R	1SFA616400R1001			ABB	1	34.7		
0.7-H2	INDICATION LAMP HOLDER KCBH-001	1SFA616600R1200			ABB	1	34.7		
0.7-H2	LED LAMP RED	1SFA616921R2021	24VDC		ABB	1	34.7		
0.7-K1	SAFETY RELAY C571	1SAR501020R0001	24V AC/DC		ABB	1	32.5		
0.7-K2	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	32.6		
0.7-K2	AUX. CONTACT BLOCK 1nc	1SBN010010R1001			ABB	1	32.6		
0.7-K3	CONTACTOR AE9-30-00	1SBL149001R8100	24VDC		ABB	1	85.5		
0.7-K4	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	34.6		
0.7-K4	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	34.6		
0.7-K5	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	34.7		
0.7-K7	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	34.6		
0.7-K7	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	34.6		
0.7-K8	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	34.6		
0.7-K8	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	34.6		
0.7-S1	PUSHBUTTON BLUE KP1-10L	1SFA616100R1004			ABB	1	34.3		
0.7-S1	CONTACT BLOCK KCBH-10	1SFA616600R1101			ABB	1	34.3		
0.7-S2	PUSHBUTTON BLACK KP1-10B	1SFA616100R1006			ABB	1	32.8		
0.7-S2	CONTACT BLOCK KCBH-10	1SFA616600R1101			ABB	1	32.8		
0.7-S3	EMERGENCY STOP	704.064.2			EAO	1	32.0		
0.7-S3	CONTACT BLOCK 1no 1nc	704.910.5			EAO	1	32.0		
0.7-S3	PROTECTIVE SHROUD	704.927.0			EAO	1	32.0		
0.7-S3	SIGN "EMERGENCY STOP" 90mm	704.963.1			EAO	1	32.0		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



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SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB4

DESIGN
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SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
32

CONT. SHEET
33

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.10-K1	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	77B.6		
0.10-K1	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	77B.6		
0.10-S2	SELECTOR SWITCH 2-POS	XB4BD21		3732624	TELEMECANIQUE	1	77B.3		
0.11-K1	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	44.5		
0.11-K1	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	44.5		
0.11-U1	TENSION SENSOR TRANSMITTER	AST-3P	24VDC		THERMO NOBEL	1	44.4		
0.11-U2	TENSION SENSOR TRANSMITTER	AST-3P	24VDC		THERMO NOBEL	1	44.4		
0.11-U3	ISOLATION AMPLIFIER MCR-C-U/U-DC	2814469	24VDC		PHOENIX	1	44.6		
0.11-U4	ISOLATION AMPLIFIER MCR-C-U/U-DC	2814469	24VDC		PHOENIX	1	44.6		
2.0-S6	KEY SELECTOR SWIYCH K2SSK2-101	1SFA616281R1001			ABB	1	95.4		
2.0-S6	CONTACT BLOCK KCBH-20	1SFA616600R1102			ABB	1	95.4		
3.2-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	90.3		
3.2-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	90.3		
3.2-K4	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	90.7		
3.2-K4	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	90.7		
3.2-S6	SELECTOR SWIYCH K3SS2-10B	1SFA616211R1006			ABB	1	90.3		
3.2-S6	CONTACT BLOCK KCBH-11	1SFA616600R1111			ABB	1	90.3		
4.0-K1	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	78.3		
4.0-K1	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	78.3		
4.0-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	78.3		
4.0-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	78.3		
4.0-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	78.3		
4.0-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	78.3		
4.0-K4	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	78.3		
4.0-K4	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	78.3		
4.0-K5	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	83.7		
4.0-K5	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	83.7		
4.0-K6	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	83.7		
4.0-K6	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	83.7		
4.0-K7	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	83.7		
4.0-K7	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	83.7		
4.0-K8	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	84.7		
4.0-K8	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	84.7		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB4

DESIGN
M.D.

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SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
33

CONT. SHEET
34

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
4.0-K9	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	84.7		
4.0-K9	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	84.7		
4.0-K10	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	84.7		
4.0-K10	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	84.7		
4.0-U1	CONVERTER SSI->ANALOG	IV251	24VDC		MKS	1	83.3		
4.0-U2	CONVERTER SSI->ANALOG	IV251	24VDC		MKS	1	84.3		
4.3-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.3-K4	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K4	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.3-K5	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K5	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.3-K6	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K6	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.3-K7	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K7	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.3-K8	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K8	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.3-K9	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K9	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.3-K10	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K10	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.3-K11	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K11	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.3-K12	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	81.3		
4.3-K12	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	81.3		
4.4-K1	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	82.4		
4.4-K1	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	82.4		
4.4-K2	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	82.4		
4.4-K2	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	82.4		
4.5-K10	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	85.5		
4.5-K10	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	85.5		
4.5-K11	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	86.6		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB4

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
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CONT. SHEET
35

reproduced, or exploited in any other unauthorized way.

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
4.5-K11	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	86.6		
4.5-K12	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	86.4		
4.5-K12	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	86.4		
4.5-K13	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	86.6		
4.5-K13	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	86.6		
4.5-K14	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	86.6		
4.5-K14	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	86.6		
4.5-K21	INTERFACE RELAY 1P	G2R-1SN(110VAC)	110VAC		OMRON	1	86.4		
4.5-K21	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	86.4		
4.5-K22	INTERFACE RELAY 1P	G2R-1SN(110VAC)	110VAC		OMRON	1	86.4		
4.5-K22	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	86.4		
4.5-K23	INTERFACE RELAY 1P	G2R-1SN(110VAC)	110VAC		OMRON	1	86.4		
4.5-K23	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	86.4		
4.5-K24	INTERFACE RELAY 1P	G2R-1SN(110VAC)	110VAC		OMRON	1	86.4		
4.5-K24	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	86.4		
5.0-K9	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	51.2		
5.0-K9	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	51.2		
5.0-K10	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	52.2		
5.0-K10	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	52.2		
5.0-K11	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	53.6		
5.0-K11	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	53.6		
5.0-S12	KEY SELECTOR SWIYCH K2SSK2-101	1SFA616281R1001			ABB	1	53.3		
5.0-S12	CONTACT BLOCK KCBH-20	1SFA616600R1102			ABB	1	53.3		
5.2-K5	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	59.3		
5.2-K5	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	59.3		
6.0-S6	KEY SELECTOR SWIYCH K2SSK2-101	1SFA616281R1001			ABB	1	66.3		
6.0-S6	CONTACT BLOCK KCBH-20	1SFA616600R1102			ABB	1	66.3		
TP1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	8			
TP1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
TP1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
U100	QJ71PB92D PROFIBUS DP-MASTER	313270020			mitsubishi	1	23.3		
U101	QJ71E71-100 TCP/IP COMMUNICATION MODULE	313242040			mitsubishi	1	23.4		
U102	QJ71MT91 TCP/IP COMMUNICATION MODULE	100556			mitsubishi	1	23.5		



AUTOMATION

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SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB4

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SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
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DRAWING NO.
A524 - C

SHEET
35

CONT. SHEET
36

REV.	MODIFICATION	DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
U103	QX80 DIGITAL INPUT MODULE 16in	313231010	24VDC		mitsubishi	1	23.6		
U104	QX80 DIGITAL INPUT MODULE 16in	313231010	24VDC		mitsubishi	1	23.7		
U105	QX80 DIGITAL INPUT MODULE 16in	313231010	24VDC		mitsubishi	1	23.7		
U106	QX80 DIGITAL INPUT MODULE 16in	313231010	24VDC		mitsubishi	1	24.2		
U107	QX80 DIGITAL INPUT MODULE 16in	313231010	24VDC		mitsubishi	1	24.2		
U108	QY80 DIGITAL OUTPUT MODULE 16out	313232020	24VDC		mitsubishi	1	24.3		
U109	QY80 DIGITAL OUTPUT MODULE 16out	313232020	24VDC		mitsubishi	1	24.4		
U110	QY80 DIGITAL OUTPUT MODULE 16out	313232020	24VDC		mitsubishi	1	24.5		
U120	Q68ADV ANALOG INPUT MODULE 8in +/-10V	313244005			mitsubishi	1	25.2		
U121	Q68DAV ANALOG OUTPUT MODULE 8out +/-10V	313244210			mitsubishi	1	25.3		
UX100	Q61P-A2 POWER SUPPLY MODULE	313213000	200-240VAC		mitsubishi	1	23.1		
UX100	Q312B-E MAIN BASE UNIT 12 SLOT	313221015			mitsubishi	1	23.1		
UX100	Q06H-CPU	313221120			mitsubishi	1	23.1		
UX101	Q65B EXTENSION BASE UNIT 5 SLOT	313211055			mitsubishi	1	25.1		
UX101	QC06B CABLE TO EXTENSION BASE UNIT 0,6M	313211100			mitsubishi	1	25.1		
UX101	Q61P-A2 POWER SUPPLY MODULE	313213000	200-240VAC		mitsubishi	1	25.1		
X4	TERMINAL WDU 4 SL	9537440000			weidmuller	35			
X4	END SUPPORT ZEW 35	9540000000			weidmuller	2			
X4	ENDPLATE WAP 2,5-10	1050000000			weidmuller	1			
X20	TERMINAL WDU 4 SL	9537440000			weidmuller	17			
X20	END SUPPORT ZEW 35	9540000000			weidmuller	2			
X20	ENDPLATE WAP 2,5-10	1050000000			weidmuller	1			
X27	TERMINAL WDU 4 SL	9537440000			weidmuller	230			
X27	END SUPPORT ZEW 35	9540000000			weidmuller	2			
X27	ENDPLATE WAP 2,5-10	1050000000			weidmuller	1			
X28	TERMINAL WDU 4 SL	9537440000			weidmuller	50			
X28	END SUPPORT ZEW 35	9540000000			weidmuller	2			
X28	ENDPLATE WAP 2,5-10	1050000000			weidmuller	1			
XU2	TERMINAL WDU 4 SL	9537440000			weidmuller	10			
XU2	END SUPPORT ZEW 35	9540000000			weidmuller	2			
XU2	ENDPLATE WAP 2,5-10	1050000000			weidmuller	1			



AUTOMATION

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BMH

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SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

SWITCHBOARD SB4

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SHEET
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CONT. SHEET
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0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.12-K14	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	46.6		
0.12-K14	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	46.6		
0.12-K15	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	47.6		
0.12-K15	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	47.6		
0.12-K16	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	47.6		
0.12-K16	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	47.6		
0.12-K17	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	47.6		
0.12-K17	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	47.6		
0.12-K18	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	47.7		
0.12-K18	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	47.7		
1.1-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	105.5		
1.1-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	105.5		
1.1-U1	ISOLATION AMPLIFIER MCR-C-U/U-DC	2814469	24VDC		PHOENIX	1	104.6		
1.1-U2	THERMISTOR RELAY CM-MSS	1SVR430800R9100	24V AC/DC		ABB	1	105.5		
1.1-U3	Pt-100 AMPLIFIER MCR-PT100/U-DC/3/7, 0-200C	2810311	24VDC		PHOENIX	1	105.5		
2.0-U1	Pt-100 AMPLIFIER MCR-PT100/U/DC/3/5, 0-100C	2810311	24VDC		PHOENIX	1	95.4		
2.0-U2	ISOLATION AMPLIFIER MCR-C-I/U-4-DC	2814511	24VDC		PHOENIX	1	95.4		
2.0-U3	VIBRATION MONITOR	VM42-2/A	220-240V		SEMA-TEC	1	98.4		
2.1-K5	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	99.5		
2.1-K5	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	99.5		
2.1-U2	THERMISTOR RELAY CM-MSS	1SVR430800R9100	24V AC/DC		ABB	1	99.5		
2.1-U3	Pt-100 AMPLIFIER MCR-PT100/U-DC/3/7, 0-200C	2810311	24VDC		PHOENIX	1	99.5		
2.2-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	102.5		
2.2-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	102.5		
2.2-U2	THERMISTOR RELAY CM-MSS	1SVR430800R9100	24V AC/DC		ABB	1	102.5		
2.2-U3	Pt-100 AMPLIFIER MCR-PT100/U-DC/3/7, 0-200C	2810311	24VDC		PHOENIX	1	102.5		
3.1-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	93.5		
3.1-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	93.5		
3.1-U2	THERMISTOR RELAY CM-MSS	1SVR430800R9100	24V AC/DC		ABB	1	93.5		
3.1-U3	Pt-100 AMPLIFIER MCR-PT100/U-DC/3/7, 0-200C	2810311	24VDC		PHOENIX	1	93.5		
5.0-U1	ISOLATION AMPLIFIER MCR-C-I/U-4-DC	2814511	24VDC		PHOENIX	1	41.6		
6.0-U1	Pt-100 AMPLIFIER MCR-PT100/U/DC/3/5, 0-100C	2810311	24VDC		PHOENIX	1	66.3		
6.1-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	68.5		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

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BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

CABIN

DESIGN
M.D.

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SCALE
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PROJECT NO.
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reproduced, or exploited in any other unauthorized way.

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BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
6.1-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	68.5		
6.1-U2	THERMISTOR RELAY CM-MSS	1SVR430800R9100	24V AC/DC		ABB	1	68.5		
6.1-U3	Pt-100 AMPLIFIER MCR-PT100/U-DC/3/7, 0-200C	2810311	24VDC		PHOENIX	1	68.5		
6.5-A1	COMPONENT MODULE EMG 25-B4	2948335			PHOENIX	1	72.5		
6.5-K1	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	72.6		
6.5-K1	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	72.6		
6.5-U2	ISOLATION AMPLIFIER MCR-C-I/U-4-DC	2814511	24VDC		PHOENIX	1	72.3		
6.5-U3	ISOLATION AMPLIFIER MCR-C-U/U-DC	2814469	24VDC		PHOENIX	1	72.6		
6.6-A1	COMPONENT MODULE EMG 25-B4	2948335			PHOENIX	1	73.5		
6.6-K1	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	73.6		
6.6-K1	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	73.6		
6.6-U2	ISOLATION AMPLIFIER MCR-C-I/U-4-DC	2814511	24VDC		PHOENIX	1	73.3		
6.6-U3	ISOLATION AMPLIFIER MCR-C-U/U-DC	2814469	24VDC		PHOENIX	1	73.6		
7.1-K7	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	74.5		
7.1-K7	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	74.5		
11.1-F1	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	14.1		
11.1-F1	AUTOMATIC FUSE S202 C10 2P	2CDS252001R0104			ABB	1	14.1		
11.1-F2	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	14.3		
11.1-F2	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	14.3		
11.1-F3	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	14.5		
11.1-F3	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	14.5		
11.1-F4	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	14.9		
11.1-F4	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	14.9		
11.1-F5	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	15.0		
11.1-F5	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	15.0		
11.1-F6	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	15.5		
11.1-F6	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	15.5		
11.1-F7	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	15.7		
11.1-F7	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	15.7		
11.1-F8	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	16.5		
11.1-F8	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	16.5		
11.1-F9	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	16.7		
11.1-F9	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	16.7		



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SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

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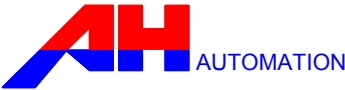
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BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
11.1-F10	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	16.8		
11.1-F10	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	16.8		
11.1-F11	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	17.4		
11.1-F11	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	17.4		
11.1-F12	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	17.6		
11.1-F12	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	17.6		
11.1-F13	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	18.1		
11.1-F13	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	18.1		
11.1-F14	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	18.3		
11.1-F14	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	18.3		
11.1-F15	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	18.4		
11.1-F15	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	18.4		
11.1-F16	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	18.5		
11.1-F16	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	18.5		
11.1-F17	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	15.2		
11.1-F17	AUTOMATIC FUSE S202 C10 2P	2CDS252001R0104			ABB	1	15.2		
11.1-F31	EARTH-FAULT PROTECTION F202 A-25/0,03	2CSF202101R1250			ABB	1	14.5		
11.1-F31	AUX. CONTACT S9-X	16056100			ABB	1	14.5		
11.1-F51	AUTOMATIC FUSE S201 C20 1P	2CDS251001R0204			ABB	1	15.2		
11.1-F52	AUTOMATIC FUSE S201 C4 1P	2CDS251001R0044			ABB	1	15.4		
11.1-F71	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	15.7		
11.1-F71	AUTOMATIC FUSE S201 C32 1P	2CDS251001R0324			ABB	1	15.7		
11.1-F72	AUX. CONTACT S2C-H6R	2CDS200912R0001			ABB	1	15.8		
11.1-F72	AUTOMATIC FUSE S201 C10 1P	2CDS251001R0104			ABB	1	15.8		
11.1-F81	EARTH-FAULT PROTECTION F202 A-25/0,03	2CSF202101R1250			ABB	1	16.5		
11.1-F81	AUX. CONTACT S9-X	16056100			ABB	1	16.5		
11.1-F91	EARTH-FAULT PROTECTION F202 A-25/0,03	2CSF202101R1250			ABB	1	16.7		
11.1-F91	AUX. CONTACT S9-X	16056100			ABB	1	16.7		
11.1-F101	EARTH-FAULT PROTECTION F202 A-25/0,03	2CSF202101R1250			ABB	1	16.8		
11.1-F101	AUX. CONTACT S9-X	16056100			ABB	1	16.8		
11.1-H3	PORTABLE EMERGENCY LIGHT 2110-HPL3	7322030	230V / 50Hz	7322030	KAMIC	1	14.7		
11.1-H3	SHELF FOR 2110-HPL3	7322036		7322036	KAMIC	1	14.7		
11.1-H20	INDICATION LAMP RED LED	XB4BVB4	24V AC/DC	3732638	TELEMECANIQUE	1	77.1		



SMÖGEN/UDDEVALLA SWEDEN

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SWEDEN

PROJECT 15050007
SHIPUNLOADER MERSEY DOCKS
ELECTRICAL EQUIPMENT
PARTS LIST
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SCALE
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BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
11.1-H30	INDICATION LAMP GREEN LED	XB4BVB3	24V AC/DC	3732637	TELEMECANIQUE	1	77.8		
11.1-H31	BUZZER ESZ-11N-A	37-786-02	24VDC		ELFA	1	34.8		
11.1-K1	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	65.3		
11.1-K1	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	65.3		
11.1-K2	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	16.4		
11.1-K3	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	16.4		
11.1-K3	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	16.4		
11.1-K4	CONTACTOR A9-30-10	1SBL141001R8810	230-260V / 50-60Hz		ABB	1	17.3		
11.1-K5	INTERFACE RELAY 1P	G2R-1SN(24VDC)	24VDC	4014429	OMRON	1	17.2		
11.1-K5	RELAY SOCKET	P2RF-05E		4014490	OMRON	1	17.2		
11.1-P1	INSTRUMENT 0-10V / 0-130%	PQ144			CELSA	1	104.9		
11.1-Q1	CIRCUIT BREAKER OT 40 M3	1SCA022497R490			ABB	1	14.0		
11.1-S20	JOYSTICK 2x2kOHM	V10B101ZP+01ZP			GESSMANN	1	53.0		
11.1-S21	SELECTOR SWITCH 2-POS	XB4BD21		3732624	TELEMECANIQUE	1	77.1		
11.1-S22	SELECTOR SWITCH 2-POS	XB4BD21		3732624	TELEMECANIQUE	1	77.1		
11.1-S23	DOUBLE PUSH-BUTTON WHITE/BLACK	ZB4BA8112		3732802	TELEMECANIQUE	1	74.1		
11.1-S23	CONTACT BLOCK 2no	ZB4BZ103		3732711	TELEMECANIQUE	1	74.1		
11.1-S24	SELECTOR SWITCH 3-POS	ZB4BD3		3732858	TELEMECANIQUE	1	77.1		
11.1-S24	CONTACT BLOCK 2nc	ZB4BZ104		3732712	TELEMECANIQUE	1	77.1		
11.1-S25	PUSH BUTTON BLACK 1no	XB4BA21		3732600	TELEMECANIQUE	1	16.0		
11.1-S26	PUSH BUTTON BLACK 1no	XB4BA21		3732600	TELEMECANIQUE	1	77.1		
11.1-S27	SELECTOR SWITCH 3-POS	XB4BD53		3732627	TELEMECANIQUE	1	77.1		
11.1-S30	JOYSTICK 2x2kOHM	V10B101ZP+01ZP			GESSMANN	1	60.0		
11.1-S31	SELECTOR SWITCH 2-POS	XB4BD21		3732624	TELEMECANIQUE	1	104.1		
11.1-S32	PUSH BUTTON BLACK 1no	XB4BA21		3732600	TELEMECANIQUE	1	104.1		
11.1-S33	ILLUM. PUSHBUTTON RED LED	XB4BW34B5	24V AC/DC	3732670	TELEMECANIQUE	1	34.8		
11.1-S34	PUSH BUTTON BLACK 1no	XB4BA21		3732600	TELEMECANIQUE	1	32.8		
11.1-S35	DOUBLE PUSHBUTTON WITH YELLOW LAMP	XB4BW84B5	24V		TELEMECANIQUE	1	65.1		
11.1-S36	DOUBLE PUSHBUTTON 1no + 1nc	XB4BL845			TELEMECANIQUE	1	77.8		
11.1-S37	DOUBLE PUSHBUTTON 1no + 1nc	XB4BL845			TELEMECANIQUE	1	77.8		
11.1-S38	SELECTOR SWITCH 2-POS	XB4BD21		3732624	TELEMECANIQUE	1	77.8		
11.1-T2	RECTIFIER MINI-PS 10-15VDC 2A	2938756	100-240VAC		PHOENIX	1	15.5		
11.1-T3	RECTIFIER 24VDC 40A	6EP1337-3BA00	110-240V		SIEMENS	1	15.7		



AUTOMATION

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REV.	MODIFICATION	DATE

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BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
11.1-T4	RECTIFIER 24VDC 10A	ABL7RE2410	100-240VAC	5213114	TELEMECANIQUE	1	15.8		
11.1-X1	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	46			
11.1-X1	TERMINAL WDU 6 SL	9537460000			WEIDMULLER	4			
11.1-X1	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
11.1-X1	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
11.1-X2	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	138			
11.1-X2	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
11.1-X2	TERMINAL WDU 35	1020500000			WEIDMULLER	2			
11.1-X2	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
11.1-X2	ENDPLATE WAP 16-35	1050100000			WEIDMULLER	1			
11.1-X3	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	67			
11.1-X3	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
11.1-X3	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
11.1-X4	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	22			
11.1-X4	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
11.1-X4	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
11.1-X5	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	32			
11.1-X5	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
11.1-X5	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
OP1	E900 OPERATOR PANEL	321001304	100-240VAC		BEIJERS	1	20.3		
OP1	IFC ETTP EXTENSION CARD FOR TCP/IP	321010145			BEIJERS	1	20.3		
TP2	TERMINAL WDU 4 SL	9537440000			WEIDMULLER	8			
TP2	END SUPPORT ZEW 35	9540000000			WEIDMULLER	2			
TP2	ENDPLATE WAP 2,5-10	1050000000			WEIDMULLER	1			
U401	POWER SUPPLY ST1PSD	313178005	24VDC		mitsubishi	1	29.2		
U401	SCREW CONNECTION ST1B-E4P2-H-SET	313179026			mitsubishi	1	29.2		
U402	INPUT MODULE 16DI ST1X16-DE1	313178013	24VDC		mitsubishi	1	29.3		
U402	SCREW CONNECTION ST1B-E4X16	313179032			mitsubishi	1	29.3		
U403	INPUT MODULE 16DI ST1X16-DE1	313178013	24VDC		mitsubishi	1	29.4		
U403	SCREW CONNECTION ST1B-E4X16	313179032			mitsubishi	1	29.4		
U404	INPUT MODULE 16DI ST1X16-DE1	313178013	24VDC		mitsubishi	1	29.5		
U404	SCREW CONNECTION ST1B-E4X16	313179032			mitsubishi	1	29.5		
U405	OUTPUT MODULE 16DO ST1Y16-TPE3	313178026	24VDC		mitsubishi	1	29.5		



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BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
U405	SCREW CONNECTION ST1B-E3Y16	313179038			mitsubishi	1	29.5		
U406	INPUT MODULE 2AI ST1AD2-V	313178037			mitsubishi	1	29.6		
U406	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	29.6		
U407	INPUT MODULE 2AI ST1AD2-V	313178037			mitsubishi	1	29.7		
U407	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	29.7		
U408	INPUT MODULE 2AI ST1AD2-V	313178037			mitsubishi	1	29.8		
U408	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	29.8		
U409	OUTPUT MODULE 2AO ST1DA2-V	313178048			mitsubishi	1	29.9		
U409	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	29.9		
U411	POWER SUPPLY ST1PSD	313178005	24VDC		mitsubishi	1	30.2		
U411	SCREW CONNECTION ST1B-E4P2-H-SET	313179026			mitsubishi	1	30.2		
U412	INPUT MODULE 16DI ST1X16-DE1	313178013	24VDC		mitsubishi	1	30.3		
U412	SCREW CONNECTION ST1B-E4X16	313179032			mitsubishi	1	30.3		
U413	INPUT MODULE 16DI ST1X16-DE1	313178013	24VDC		mitsubishi	1	30.4		
U413	SCREW CONNECTION ST1B-E4X16	313179032			mitsubishi	1	30.4		
U414	INPUT MODULE 16DI ST1X16-DE1	313178013	24VDC		mitsubishi	1	30.5		
U414	SCREW CONNECTION ST1B-E4X16	313179032			mitsubishi	1	30.5		
U415	OUTPUT MODULE 16DO ST1Y16-TPE3	313178026	24VDC		mitsubishi	1	30.6		
U415	SCREW CONNECTION ST1B-E3Y16	313179038			mitsubishi	1	30.6		
U416	POWER SUPPLY ST1PSD	313178005	24VDC		mitsubishi	1	30.7		
U416	SCREW CONNECTION ST1B-E4P2-R-SET	313179025			mitsubishi	1	30.7		
U417	INPUT MODULE 2AI ST1AD2-I	313178038			mitsubishi	1	31.1		
U417	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	31.1		
U418	INPUT MODULE 2AI ST1AD2-V	313178037			mitsubishi	1	31.2		
U418	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	31.2		
U419	INPUT MODULE 2AI ST1AD2-V	313178037			mitsubishi	1	31.2		
U419	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	31.2		
U420	INPUT MODULE 2AI ST1AD2-V	313178037			mitsubishi	1	31.3		
U420	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	31.3		
U421	INPUT MODULE 2AI ST1AD2-V	313178037			mitsubishi	1	31.4		
U421	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	31.4		
U422	INPUT MODULE 2AI ST1AD2-V	313178037			mitsubishi	1	31.5		
U422	SCREW CONNECTION ST1B-E4IR2	313179042			mitsubishi	1	31.5		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

CABIN

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - C

SHEET
47

CONT. SHEET
48

REV.	MODIFICATION	DATE

0	1	2	3	4	5	6	7	8	9
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[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PARTS LIST CABIN		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-24	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - C		APPR.	FILE NAME A524.DB	SHEET 48	CONT. SHEET 51				
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.3-E1	HEATING FAN 3kW	8711103		8711103	FRICO	1	6.5		
0.3-H11	ILLUMINATOR 2x36W	7272924	230V/50Hz		EXAKTOR	1	8.2		
0.3-H12	ILLUMINATOR 2x36W	7272924	230V/50Hz		EXAKTOR	1	8.2		
0.3-H13	ILLUMINATOR 2x36W	7272924	230V/50Hz		EXAKTOR	1	8.2		
0.3-H14	ILLUMINATOR 2x36W	7272924	230V/50Hz		EXAKTOR	1	8.4		
0.3-H15	ILLUMINATOR 2x36W	7272924	230V/50Hz		EXAKTOR	1	8.4		
0.3-H16	ILLUMINATOR 2x36W	7272924	230V/50Hz		EXAKTOR	1	8.2		
0.3-H46	PORTABLE EMERGENCY LIGHT 2110-HPL3	7322030	230V / 50Hz	7322030	KAMIC	1	12.0		
0.3-H46	SHELF FOR 2110-HPL3	7322036		7322036	KAMIC	1	12.0		
0.3-U1	SMART-UPS XL 1000VA USB	SUA1000XLI	240V / 50Hz		APC	1	9.0		
0.4-U1	UPS 1500VA	SUA1500	240V / 50Hz		APC	1	19.6		
0.8-A1	T1008 8-ZONE FIRE ALARM PANEL	508.022.035			TYCO	1	11.0		
0.8-S1	601CH CARBON MONOXIDE FIRE DETECTOR	516.600.004.Y			TYCO	1	40.6		
0.8-S2	601CH CARBON MONOXIDE FIRE DETECTOR	516.600.004.Y			TYCO	1	40.7		
1.1-A1	FREQUENCY CONVERTER 132kW	ACS800-02-0170-3			ABB	1	107.4		
1.1-A1	PROFIBUS MODULE	RPBA-01			ABB	1	107.4		
1.1-A1	DDCS COMMUNICATION MODULE	RDCO-03			ABB	1	107.4		
5.0-A1	PROFIBUS MODULE	RPBA-01			ABB	1	55.4		
5.0-A1	PULSE ENCODER MODULE	RTAC-01			ABB	1	55.4		
5.0-A1	DDCS COMMINCATION MODULE	RDCO-03			ABB	1	55.4		
5.0-A1	FREQUENCY CONVERTER 90kW	ACS800-01-0100-3			ABB	1	55.4		
5.2-A1	DDCS COMMINCATION MODULE	RDCO-03			ABB	1	62.4		
5.2-A1	FREQUENCY CONVERTER 22kW	ACS800-01-0025-3			ABB	1	62.4		
5.2-A1	PROFIBUS-DP MODULE	RPBA-01			ABB	1	62.4		
5.2-A1	PULSE ENCODER MODULE	RTAC-01			ABB	1	62.4		
5.2-R1	BREAKE RESISTOR	SACE15RE13			ABB	1	62.9		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

ELHOUSE

DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-24	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - C	
APPR.	FILE NAME A524.DB	SHEET 51	CONT. SHEET 1

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.0-M1	MOTOR (GENERAL)	MOTOR			BMH	1	39.8		
0.3-H1	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	7.4		
0.3-H2	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	7.4		
0.3-H3	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	7.5		
0.3-H4	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	7.5		
0.3-H5	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	7.6		
0.3-H6	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	7.6		
0.3-H7	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	7.7		
0.3-H8	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	7.7		
0.3-H9	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	8.1		
0.3-H10	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	8.1		
0.3-H21	AVIATION OBSTRUCTION LIGHT	OBI-S-R-CM	230V		ERRI	1	9.7		
0.3-H32	FLOODLIGHT 400W	1076015-044	240VAC		AQUA SIGNAL	1	16.5		
0.3-H33	FLOODLIGHT 400W	1076015-044	240VAC		AQUA SIGNAL	1	16.7		
0.3-H34	FLOODLIGHT 400W	1076015-044	240VAC		AQUA SIGNAL	1	16.8		
0.3-H40	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	9.3		
0.3-H41	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	9.3		
0.3-H42	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	9.3		
0.3-H43	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	9.4		
0.3-H44	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	9.6		
0.3-H45	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	9.6		
0.3-H47	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	17.4		
0.3-H48	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	17.4		
0.3-H49	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	17.5		
0.3-H50	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	17.5		
0.3-H51	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	17.6		
0.3-H52	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	17.6		
0.3-H53	ILLUMINATOR 2x18W	9795907	230V / 50Hz		SOLAR	1	17.6		
0.3-H54	ILLUMINATOR 2x36W	9795912	230V / 50Hz		SOLAR	1	17.8		
0.3-S10	CONTROL-BOX 1BUTTON "I" no	XALD102		3734700	TELEMECANIQUE	1	7.0		
0.3-S11	CONTROL-BOX 1BUTTON "I" no	XALD102		3734700	TELEMECANIQUE	1	7.0		
0.3-S12	CONTROL-BOX 1BUTTON "I" no	XALD102		3734700	TELEMECANIQUE	1	7.0		
0.3-S13	CONTROL-BOX 1BUTTON "I" no	XALD102		3734700	TELEMECANIQUE	1	7.0		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

EXTERNAL

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - D

SHEET
1

CONT. SHEET
2

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
0.7-S4	CONTACT BLOCK 1no 1nc	704.910.5			EAO	1	32.0		
0.7-S4	PROTECTIVE SHROUD	704.927.0			EAO	1	32.0		
0.7-S4	SIGN "EMERGENCY STOP" 90mm	704.963.1			EAO	1	32.0		
0.7-S4	ENCLOSURE	704.944.33			EAO	1	32.0		
0.7-S4	EMERGENCY STOP	704.064.2			EAO	1	32.0		
0.7-S5	PROTECTIVE SHROUD	704.927.0			EAO	1	32.0		
0.7-S5	SIGN "EMERGENCY STOP" 90mm	704.963.1			EAO	1	32.0		
0.7-S5	ENCLOSURE	704.944.33			EAO	1	32.0		
0.7-S5	EMERGENCY STOP	704.064.2			EAO	1	32.0		
0.7-S5	CONTACT BLOCK 1no 1nc	704.910.5			EAO	1	32.0		
0.7-S6	PROTECTIVE SHROUD	704.927.0			EAO	1	32.0		
0.7-S6	SIGN "EMERGENCY STOP" 90mm	704.963.1			EAO	1	32.0		
0.7-S6	ENCLOSURE	704.944.33			EAO	1	32.0		
0.7-S6	EMERGENCY STOP	704.064.2			EAO	1	32.0		
0.7-S6	CONTACT BLOCK 1no 1nc	704.910.5			EAO	1	32.0		
0.7-S7	SIGN "EMERGENCY STOP" 90mm	704.963.1			EAO	1	33.0		
0.7-S7	ENCLOSURE	704.944.33			EAO	1	33.0		
0.7-S7	EMERGENCY STOP	704.064.2			EAO	1	33.0		
0.7-S7	CONTACT BLOCK 1no 1nc	704.910.5			EAO	1	33.0		
0.7-S7	PROTECTIVE SHROUD	704.927.0			EAO	1	33.0		
0.8-S4	601CH CARBON MONOXIDE FIRE DETECTOR	516.600.004.Y			TYCO	1	40.9		
0.10-H1	YODALIGHT, YL5 (WR-IP66)	1YL050318	24VDC		CLIFFORD & SNELL	1	77B.8		
0.10-H2	YODALIGHT, YL5 (WR-IP66)	1YL050318	24VDC		CLIFFORD & SNELL	1	77B.8		
0.10-H3	YODALIGHT, YL5 (WR-IP66)	1YL050318	24VDC		CLIFFORD & SNELL	1	77B.8		
0.10-S1	INDUCTIVE PROXIMITY SENSOR	XS7C40PC440	24VDC	3847800	TELEMECANIQUE	1	77B.1		
0.11-S1	TENSION SENSOR	STT-1			VISHAY NOBEL	1	44.0		
0.11-S2	TENSION SENSOR	STT-1			VISHAY NOBEL	1	44.0		
1.1-M1	MOTOR, INLET FEEDER	M3BP315SMB-4	415VAC		ABB	1	107.8		
1.1-R1	THERMISTOR, MOTOR INLET FEEDER	C011-160			ABB	1	105.0		
1.1-R2	Pt-100 SENSOR, MOTOR INLET FEEDER	PTC			ABB	1	105.0		
1.1-R3	HEATER, MOTOR INLET FEEDER	-		INCLUDED IN MOTOR	ABB	1	105.0		
1.1-S1	SPEED-GUARD	XSAV12373	24VDC	3878020	TELEMECANIQUE	1	105.0		
1.1-S2	SPEED-GUARD	XSAV11373	24VDC	3878010	TELEMECANIQUE	1	105.0		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

EXTERNAL

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - D

SHEET
2

CONT. SHEET
3

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
2.0-E1	HEATER, GEARBOX VERT. CONV.	LACM07-4	415VAC		OILTECH	1	96.7		
2.0-E2	HEATER, GEARBOX VERT. CONV.	LACM07-4	415VAC		OILTECH	1	96.8		
2.0-R1	Pt-100 SENSOR, GEARBOX VERT. CONV.	66RCS06513			INOR	1	95.0		
2.0-S1	SPEED-GUARD	XSAV11373	24VDC	3878010	TELEMECANIQUE	1	98.0		
2.0-S4	PRESSURE GUARD, IN GEARBOX	01DIV00013			INOR	1	95.0		
2.1-M1	MOTOR, VERTICAL CONVEYOR	M3BP355M6A-4	415VAC		ABB	1	101.8		
2.1-R1	THERMISTOR, MOTOR VERT. CONV.	C011-160			ABB	1	99.0		
2.1-R2	Pt-100 SENSOR, MOTOR VERT. CONV.	PTC			ABB	1	99.0		
2.1-R3	HEATER, MOTOR VERT. CONV.	-		INCLUDED IN MOTOR	ABB	1	99.0		
2.2-M1	MOTOR, VERTICAL CONVEYOR	M3BP355MA-4	415VAC		ABB	1	103.8		
2.2-R1	THERMISTOR, MOTOR VERT. CONV.	C011-160			ABB	1	102.0		
2.2-R2	Pt-100 SENSOR, MOTOR VERT. CONV.	PTC			ABB	1	102.0		
2.2-R3	HEATER, MOTOR VERT. CONV.	-		INCLUDED IN MOTOR	ABB	1	102.0		
2.3-M1	MOTOR, COOLING FAN	-			OILTECH	1	97.8		
3.0-E1	HEATER, GEARBOX HOR. CONV.	-			ABB	1	92.9		
3.1-M1	MOTOR, HORIZONTAL CONVEYOR	-			ABB	1	94.8		
3.1-R1	THERMISTOR, MOTOR HOR. CONV.	C011-160			ABB	1	93.0		
3.1-R2	Pt-100 SENSOR, MOTOR HOR. CONV.	PTC			ABB	1	93.0		
3.1-R3	HEATER, MOTOR HOR. CONV.	-		INCLUDED IN MOTOR	ABB	1	93.0		
3.1-S1	SPEED-GUARD	XSAV11373	24VDC	3878010	TELEMECANIQUE	1	93.0		
3.2-M1	MOTOR, GREASE PUMP	T-DA63MB	415VAC		BROOKE CROMPTON	1	91.8		
3.2-S1	LEVEL GUARD, GREASE PUMP	LEVEL GUARD		901143	ASSALUB	1	90.0		
3.2-S2	LEVEL GUARD, GREASE PUMP	LEVEL GUARD		901143	ASSALUB	1	90.0		
3.2-S3	LEVEL GUARD, GREASE PUMP	LEVEL GUARD		901143	ASSALUB	1	90.0		
3.2-S4	LEVEL GUARD, GREASE BARREL	LEVEL GUARD		1117069	ASSALUB	1	90.0		
3.2-Y1	VALVE, REPLENISHMENT GREASE	101847			ASSALUB	1	90.0		
3.2-Y2	VALVE, BARREL PUMP	101310			ASSALUB	1	90.9		
4.0-M1	MOTOR, GATE A	M2VAD80B			ABB	1	83B.8		
4.0-M2	MOTOR, GATE B	M2VAD80B			ABB	1	84B.8		
4.0-S5	LIMIT SWITCH	XCM3915H29EX	24VDC		TELEMECANIQUE	1	83.0		
4.0-S6	LIMIT SWITCH	XCM3915H29EX	24VDC		TELEMECANIQUE	1	83.0		
4.0-S7	ABSOLUTE ENCODER	CE58M	24VDC		TR-ELECTRONIC	1	83.0		
4.0-S8	LIMIT SWITCH	XCM3915H29EX	24VDC		TELEMECANIQUE	1	84.0		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

EXTERNAL

DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-24	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - D	
APPR.	FILE NAME A524.DB	SHEET 3	CONT. SHEET 4

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
4.0-S9	LIMIT SWITCH	XCM3915H29EX	24VDC		TELEMECANIQUE	1	84.0		
4.0-S10	ABSOLUTE ENCODER	CE58M	24VDC		TR-ELECTRONIC	1	84.0		
4.3-M1	MOTOR, FILTER	AM132SZA	415VAC		AEG	1	80.8		
4.3-Y1	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.3-Y2	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.3-Y3	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.3-Y4	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.3-Y5	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.3-Y6	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.3-Y7	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.3-Y8	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.3-Y9	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.3-Y10	VALVE, FILTER CLEANING	RCA	24VDC		ASCO	1	81.8		
4.4-M1	MOTOR, COMPRESSOR	-		SEE SEPARATE MANUAL	ATLAS COPCO	1	82.7		
4.4-Y1	DRAIN VALVE	-		SEE SEPARATE MANUAL	KAESER	1	82.8		
4.5-M1	MOTOR, FILTER	-		CUSTOMER SUPPLY		1	87.7		
4.6-M1	MOTOR, FILTER	-		CUSTOMER SUPPLY		1	88.7		
5.0-B1	WIND SENSOR "COMPACT"	4.3519.00.141	24V / 4-20mA		THIES CLIMA	1	41.1		
5.0-H1	YODALIGHT, YL5 (WR-IP66)	1YL050318	24VDC		CLIFFORD & SNELL	1	53.9		
5.0-H2	YODALIGHT, YL5 (WR-IP66)	1YL050318	24VDC		CLIFFORD & SNELL	1	53.8		
5.0-H3	YODALIGHT, YL5 (WR-IP66)	1YL050318	24VDC		CLIFFORD & SNELL	1	53.8		
5.0-H4	YODALIGHT, YL5 (WR-IP66)	1YL050318	24VDC		CLIFFORD & SNELL	1	53.8		
5.0-M1	MOTOR, STORMLOCK	-			HILLMAR	1	51.9		
5.0-M2	MOTOR, STORMLOCK	-			HILLMAR	1	52.9		
5.0-S1	ABSOLUTE ENCODER PROFIBUS, TRAVELLING	CE65M	24VDC		TR-ELECTRONIC	1	50.8		
5.0-S2	MAGNETIC SENSOR	637.3252.068	24VDC		BERNSTEIN	1	50.0		
5.0-S2	MAGNET	630.1262.039			BERNSTEIN	1	50.0		
5.0-S3.1	INDUCTIVE PROXIMITY SENSOR	XS7C40PC440	24VDC	3847800	TELEMECANIQUE	1	50.0		
5.0-S3.2	INDUCTIVE PROXIMITY SENSOR	XS7C40PC440	24VDC	3847800	TELEMECANIQUE	1	50.0		
5.0-S4.1	INDUCTIVE PROXIMITY SENSOR	XS7C40PC440	24VDC	3847800	TELEMECANIQUE	1	50.0		
5.0-S4.2	INDUCTIVE PROXIMITY SENSOR	XS7C40PC440	24VDC	3847800	TELEMECANIQUE	1	50.0		
5.0-S5	LIMIT SWITCH	XCRE181			TELEMECANIQUE	1	50.8		
5.0-S6	LIMIT SWITCH	XCRE181			TELEMECANIQUE	1	50.8		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

EXTERNAL

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - D

SHEET
4

CONT. SHEET
5

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
BMH_ITEM_LIST									
POSITION	DENOMINATION	TYPE	VOLT/Hz	ARTICLE No.	MANUFACTURE	QTY.	SHEET		
5.1-M1	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	56.3		
5.1-M2	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	56.4		
5.1-M3	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	56.5		
5.1-M4	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	56.7		
5.1-M5	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	56.8		
5.1-M6	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	57.3		
5.1-M7	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	57.4		
5.1-M8	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	57.5		
5.1-M9	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	57.7		
5.1-M10	MOTOR, TRAVELLING	BN132S4	415VAC		BONFIGLIOLI	1	57.8		
5.1-S1	PULSE ENCODER, TRAVELLING	DGS65GAK02048			SICK	1	55.4		
5.2-S1	ABSOLUTE ENCODER PROFIBUS, SLEWING	CE65M			TR-ELECTRONIC	1	59.8		
5.2-S2	MAGNETIC SENSOR	637.3252.068	24VDC		BERNSTEIN	1	59.0		
5.2-S2	MAGNET	630.1262.039			BERNSTEIN	1	59.0		
5.2-S3	INDUCTIVE PROXIMITY SENSOR	XS7C40PC440	24VDC	3847800	TELEMECANIQUE	1	59.1		
5.2-X3	SEE SEPARATE DRAWING	.				1	59.0		
5.2-Y1	VALVE, SLEWING LUBR.	101847	24VDC		ASSALUB	1	59.0		
5.3-M1	MOTOR, SLEWING	BN160L6	415VAC		BONFIGLIOLI	1	63.3		
5.3-M2	MOTOR, SLEWING	BN160L6	415VAC		BONFIGLIOLI	1	63.5		
5.3-S1	PULSE ENCODER, SLEWING	DGS65GAK02048	24VDC		SICK	1	62.4		
6.0-E1	HEATER, HYDRAUL OIL TANK	OE361			BACKER	1	67.8		
6.0-R1	Pt-100 SENSOR, HYDRAUL	MBT5250/200			DANFOSS	1	66.0		
6.0-S1	LEVEL GUARD, HYDRAUL	HFMB-V900			HEMOMATIK	1	66.0		
6.0-S2	FILTER GUARD, HYDRAUL	RFMBN/851G10C1.X			HYDAC	1	66.0		
6.0-S3	FILTER GUARD, MOVEMENTS	OFBNHC500G10C1.11B			HYDAC	1	66.0		
6.1-M1	MOTOR, HYDRAUL PUMP	M3AA250SMB-4			ABB	1	69.8		
6.1-R1	THERMISTOR, HYDRAUL PUMP	C011-160			ABB	1	68.0		
6.1-R2	Pt-100 SENSOR, HYDRAUL PUMP	PTC			ABB	1	68.0		
6.1-R3	HEATER, HYDRAUL PUMP	-		INCLUDED IN MOTOR	ABB	1	68.0		
6.2-M1	MOTOR, OIL COOLER	LAC044-4-A-00			OILTECH	1	70.8		
6.3-A1	SEE SEPARATE DRAWING	.				1	71.0		
6.3-M1	MOTOR, CIRC. PUMP	M3AA112M			ABB	1	71.9		
6.3-S1	PRESSURE GUARD, CIRC. PUMP	PS055-PODL-32			DENISON	1	71.0		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PARTS LIST

EXTERNAL

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - D

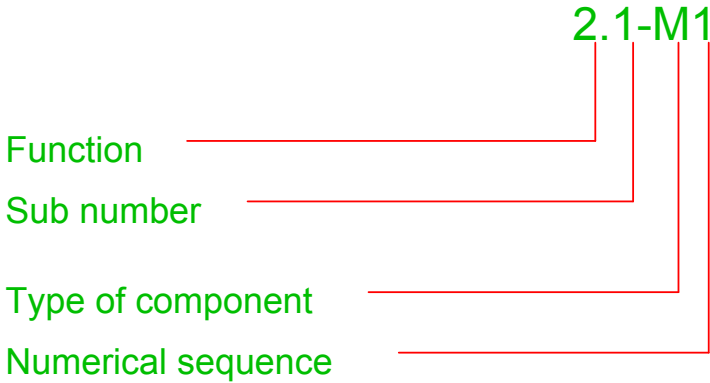
SHEET
5

CONT. SHEET
6

REV.	MODIFICATION	DATE

CONTENTS CIRCUIT DIAGRAM

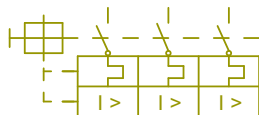
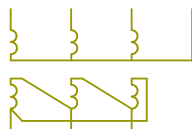
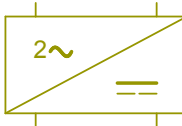
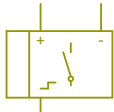
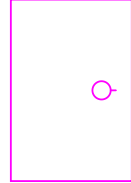
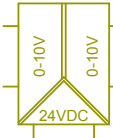
SIDE	REV.	CONTENTS	SIDE	REV.	CONTENTS	SIDE	REV.	CONTENTS
001		CONTENTS CIRCUIT DIAGRAM	35B		FUSE SUPERVISION	74		CONTROL CIRCUIT AUXILIARY HOIST
002		SYSTEM CONFIGURATION	36		FUSE SUPERVISION	75		MOTORSTARTER AUXILIARY HOIST
003		SYMBOL LIST	37		FUSE SUPERVISION	76		MOTORSTARTER AUXILIARY HOIST
004		COLOUR CODE	38		TRANSFORMER GUARDS	77		START / STOP CONVEYORS
1		MAIN POWER SUPPLY	39		COOLING FAN TRANSFORMER ROOM	77B		START / STOP CONVEYORS
2		STAND-BY POWER SUPPLY	40		FIRE DETECTION	78		OUTLOADING CONTROL
3		MAIN POWER DISTRIBUTION	41		ANEMOMETER	79		OUTLOADING CONTROL
4		DISTRIBUTION 240/415V	42		LIMITSWITCHES POWER CABLE REEL	80		DUST FILTER FAN ON UNLOADER
5		DISTRIBUTION 240/415V	43		POWER CABLE REEL	81		DUST-CLEANING EQUIPMENT
6		DISTRIBUTION 240/415V	44		FORCE REDUCING SYSTEM	82		SUPPLY AIR-COMPRESSOR
7		DISTRIBUTION 240/415V	45		RADAR SYSTEM	83		SLIDE GATE, BELT RB1
8		DISTRIBUTION 240/415V	46		RADAR SYSTEM	83B		SLIDE GATE, BELT RB1
9		DISTRIBUTION 240/415V	47		RADAR SYSTEM	84		SLIDE GATE, BELT RB2
10		DISTRIBUTION 240/415V	48		RADAR SYSTEM	84B		SLIDE GATE, BELT RB2
11		DISTRIBUTION 240/415V	49		SPARE SIGNALS BETWEEN J-BOX A500 6 CABIN	85		DUST FILTER FAN & OUTLOADING
12		DISTRIBUTION 240/415V	50		TRAVELLING POSITION SENSORS	86		DUST FILTER FAN & OUTLOADING
13		DISTRIBUTION 110V	51		TRAVELLING AUTOMATIC STORMLOCK 1, LEFT	87		DUST FILTER FAN OUTLOADING BELT RB1
14		DISTRIBUTION 240/415V IN CABIN	52		TRAVELLING AUTOMATIC STORMLOCK 2, RIGHT	88		DUST FILTER FAN OUTLOADING BELT RB2
15		DISTRIBUTION 240/415V IN CABIN	53		TRAVELLING JOYSTICK AND ALARM	90		LUBRICATION PUMP HORIZONTAL CONVEYOR
16		DISTRIBUTION 240/415V IN CABIN	54		SUPPLY FREQUENCY CONVERTER TRAVELLING	91		LUBRICATION PUMP HORIZONTAL CONVEYOR
17		DISTRIBUTION 240/415V IN CABIN	55		FREQUENCY CONVERTER TRAVELLING	92		HEATER IN GEARBOX HORIZONTAL CONVEYOR
18		DISTRIBUTION 240/415V IN CABIN	56		TRAVELLING MOTORS	93		CONTROL CIRCUIT HORIZONTAL CONVEYOR
19		DISTRIBUTION 240V CONTROL VOLTAGE	57		TRAVELLING MOTORS	94		MOTORSTARTER HORIZONTAL CONVEYOR
20		DISTRIBUTION 240V UPS CONTROL VOLTAGE	58		TRAVELLING PARKING BRAKES	95		CONTROL CIRCUIT VERTICAL GEARBOX
21		DISTRIBUTION 24VDC CONTROL VOLTAGE	59		SLEWING POSITION SENSOR	96		HEATER VERTICAL GEARBOX
22		DISTRIBUTION 24VDC CONTROL VOLTAGE	60		SLEWING JOYSTICK	97		COOLING FAN VERTICAL GEARBOX
23		POWER SUPPLY PLC, SB4	61		SUPPLY FREQUENCY CONVERTER SLEWING	98		CONTROL CIRCUIT VERTICAL CONVEYORS
24		POWER SUPPLY PLC, SB4	62		FREQUENCY CONVERTER SLEWING	99		CONTROL CIRCUIT VERTICAL CONVEYOR 1
25		POWER SUPPLY PLC, SB4	63		SLEWING MOTORS	100		MOTORSTARTER VERTICAL CONVEYOR 1
26		POWER SUPPLY PLC, SB1	64		SLEWING PARKING BRAKES	101		MOTORSTARTER VERTICAL CONVEYOR 1
27		POWER SUPPLY PLC, SB1	65		START / STOP MOVEMENTS	102		CONTROL CIRCUIT VERTICAL CONVEYOR 2
28		POWER SUPPLY PLC, SB3	66		CONTROL CIRCUIT HYDRAULIC	103		MOTORSTARTER VERTICAL CONVEYOR 2
29		POWER SUPPLY PLC, CABIN	67		HYDRAULIC HEATER	104		INLET FEEDER CONTROL CIRCUIT
30		POWER SUPPLY PLC, CABIN	68		CONTROL CIRCUIT HYDRAULIC MAIN PUMP	105		INLET FEEDER CONTROL CIRCUIT
31		POWER SUPPLY PLC, CABIN	69		MOTORSTARTER HYDRAULIC MAIN PUMP	106		SUPPLY FREQUENCY CONVERTER INLET FEEDER
32		EMERGENCY STOP CIRCUITS	70		MOTORSTARTER HYDRAULIC COOLING FAN	107		FREQUENCY CONVERTER INLET FEEDER
33		EMERGENCY STOP CIRCUITS	71		MOTORSTARTER HYDRAULIC CIRCULATION PUMP	108		MEASUREMENT I/O
34		FAULT SUPERVISION AND RADIO	72		LUFFING MOVEMENTS			
35		FUSE SUPERVISION	73		PENDULUM MOVEMENTS			



EX; 2.1-M1: Vertical conveyor, Drive 1, Motor No: 1

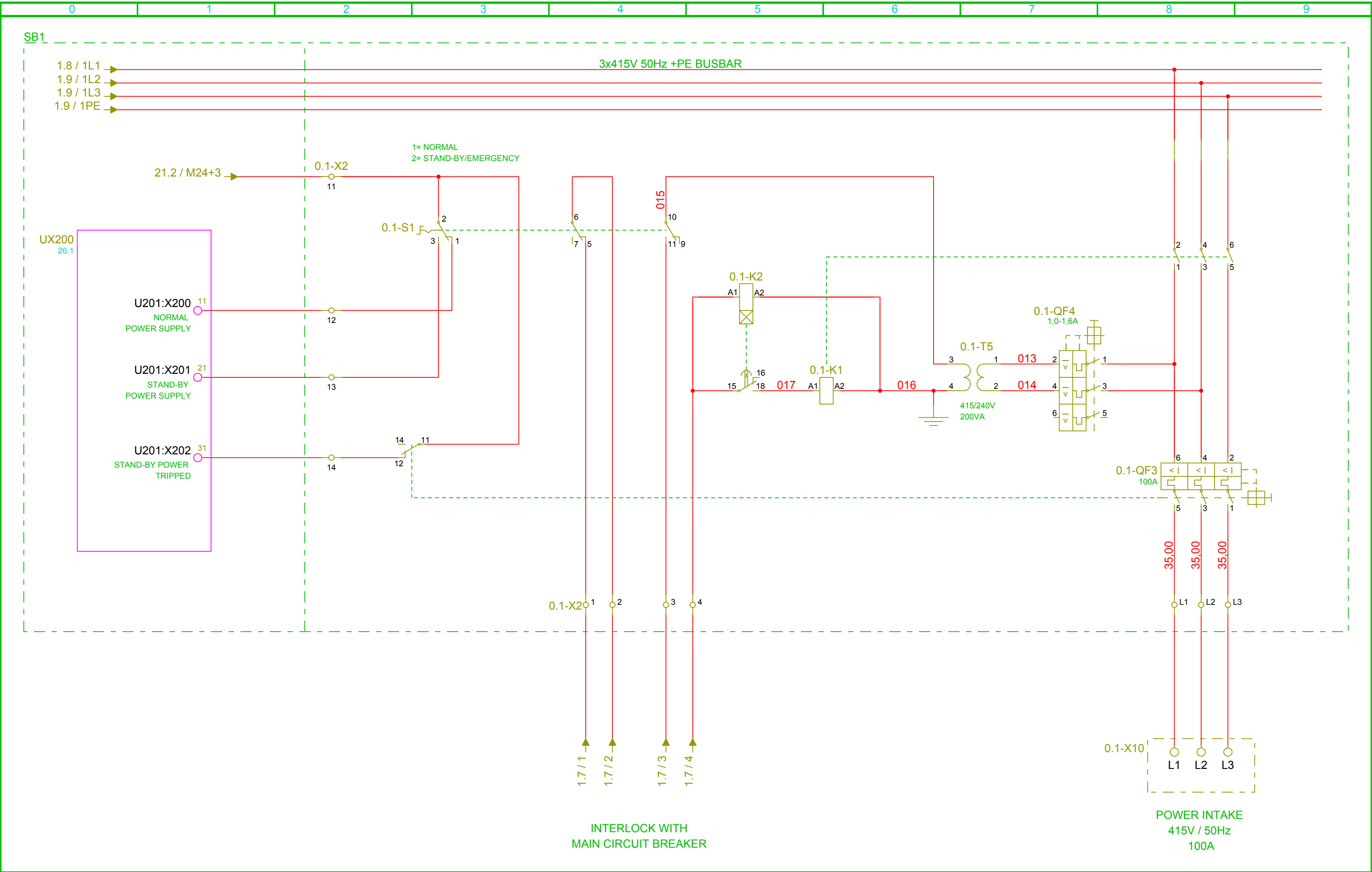
Type:	Denomination:
A	Unit, sub-unit
B	Inductive proximity sensor
C	Capacitor
F	Fuse, Overload protection
H	Optic & acustic signal, Illumination
K	Contactor, Relay, Time-relay
L	Inductor
M	Motor
P	Instrument
QF	Breaker with protection, MCCB, Motorprotection
R	Resistor, Heater
S	Electric switch (manual / limit etc.)
T	Transformer
U	Converter
X	Outlet, Terminal, Junction box
Y	Solenoid valve, Brake etc.

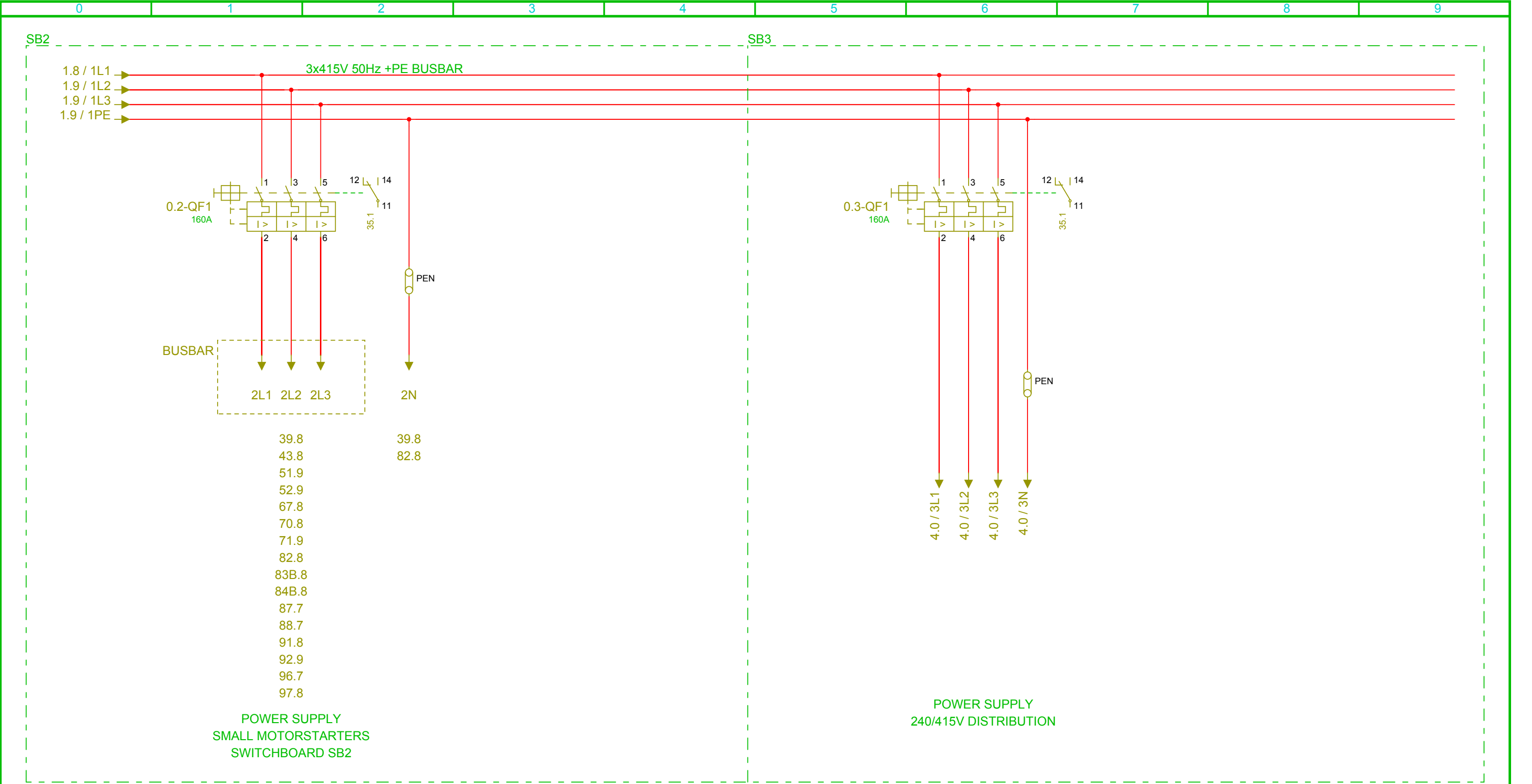
Function:	0 = Power distribution / auxiliaries	1 = Inlet feeder	2 = Vertical conveyor	3 = Horizontal conveyor	4 = Outloading, dustfilter eq.
Sub number:	.0 = High voltage transformer .1 = Main 415V distribution .2 = Distribution 415V, Small motorstarters .3 = Distribution 415/240V .31 = Distribution 110V .4 = Distribution / Control voltage 240V .5 = Distribution / Control voltage 24VDC .6 = .7 = Emergency stop / Alarm handling .8 = Fire alarm .9 = .10 = .11 = Force reducing system .12 = Radar system .13 =	.1 = Drive	.0 = Common control & Heater .1 = Conveyor drive Motor 1 .2 = Conveyor drive Motor 2 .3 = Cooling fan	.0 = Common control & Heater .1 = Conveyor drive .2 = Lubrication pump	.0 = Outloading control .1 = .2 = .3 = Dustfilter on unloader .4 = Compressor .5 = Dustfilter outloading belt A .6 = Dustfilter outloading belt B
Function:	5 = Travelling & Slewing	6 = Hydraulic equipment	7 = Auxiliary hoist	8 = Cable reel	11 = Cabin
Sub number:	.0 = Travelling feeder & common aux. .1 = Travelling motors .2 = Slewing feeder & common aux. .3 = Slewing motors	.0 = Control circuit & Heater .1 = Main pump .2 = Cooling fan .3 = Circulation pump .4 = .5 = Luffing movements .6 = Pendulum movements	.1 = Auxiliary hoist	.1 = Power cable reel	.1 = Cabin equipment

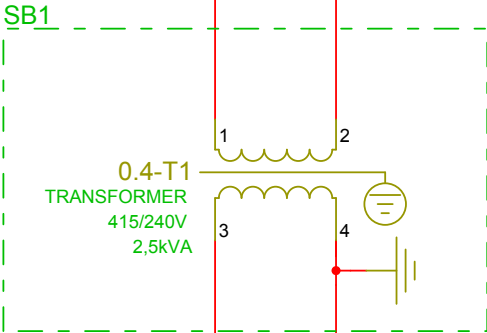
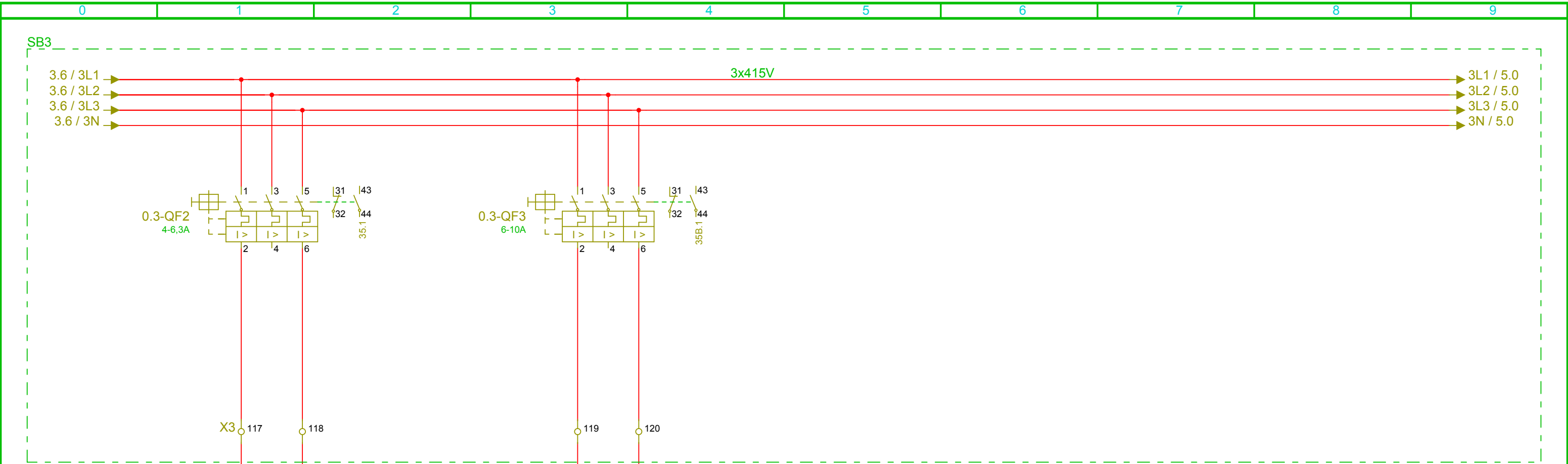
0	1	2	3	4	5	6	7	8	9
 TERM. AND MAGN. OVERLOAD PROT.	 CONTACTOR & RELAY COIL	 SWITCH PUSH- BUTTON SWITCH	 CURRENT TRANSFORMER	 OUTLET	 THERMOSTAT				
 3-PHASE TRANSFORMER DYN	 IMPULSE RELAY COIL	 EMERGENCY STOP	 CABLE SHIELDING	 CABLE LINE	 LEVEL-GUARD				
 1 & 2 PHASE TRANSFORMER	 N.C CHANGE-OVER CONTACT CONTACT	 TIME-DELAY RELAY	 TERMINAL PLUG	 HEATER	 PRESSURE-GUARD				
 RECTIFIER	 TIME-DELAY THERMAL CONTACT CONTACT	 WORK-TIME METER	 3-PHASE MOTOR	 CAPACITORS (DELTA CONNECTION)	 GUARD, GENERAL				
 MINIATYRE CIRCUIT BREAKER 3-POL	 CONTACTOR CONTACTS	 OVERLOAD PROTECTION	 POTENTIOMETER	 LAMP/ LIGHT BUZZER	 INDUCTIVE PROXIMITY SWITCH				
 MINIATYRE CIRCUIT BREAKER 1-POL + AUX. CONTACT	 SOLENOID VALVE	 PLC INPUT/OUTPUT	 RESISTOR	 FLUORESCENT LAMP	 ISOLATION AMPLIFIER				
 FUSE 3-POL	 BRAKE	 VOLT- METER AMPERE- METER	 PHOTORESISTOR	 INTERRUPTION POINT	 EARTH				

COLOUR CODE FOR INTERNAL CABLES

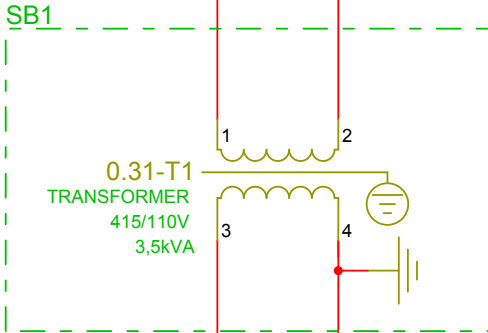
Colour	Denomination
Black	Power/Control circuits >50V AC
Blue	Neutral conductor
Yellow/Green	Protective conductor PE (Earth)
Red	Control circuits <50V DC+
White	Control circuits <50V DC- / 0V
Brown	Power/Control circuits <50V AC
Violet or screened cable	Analog signals
Orange	External supplied (foreign) control circuits AC/DC



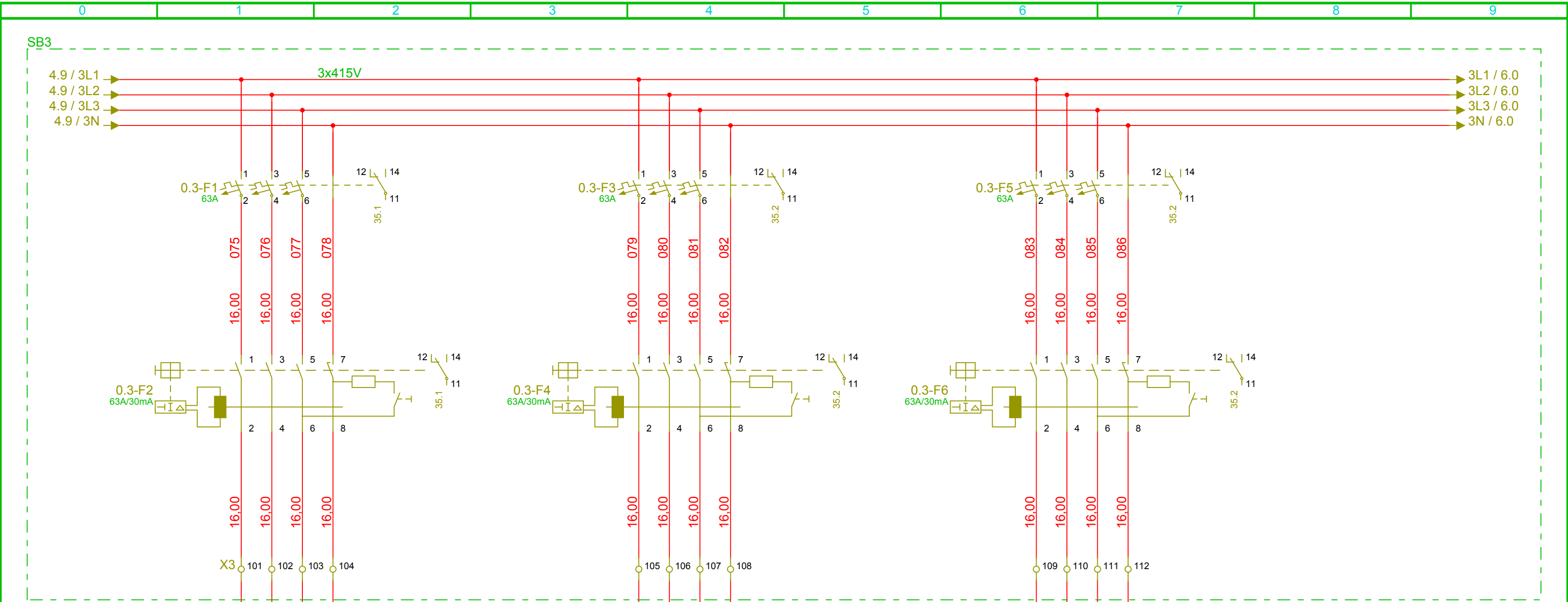


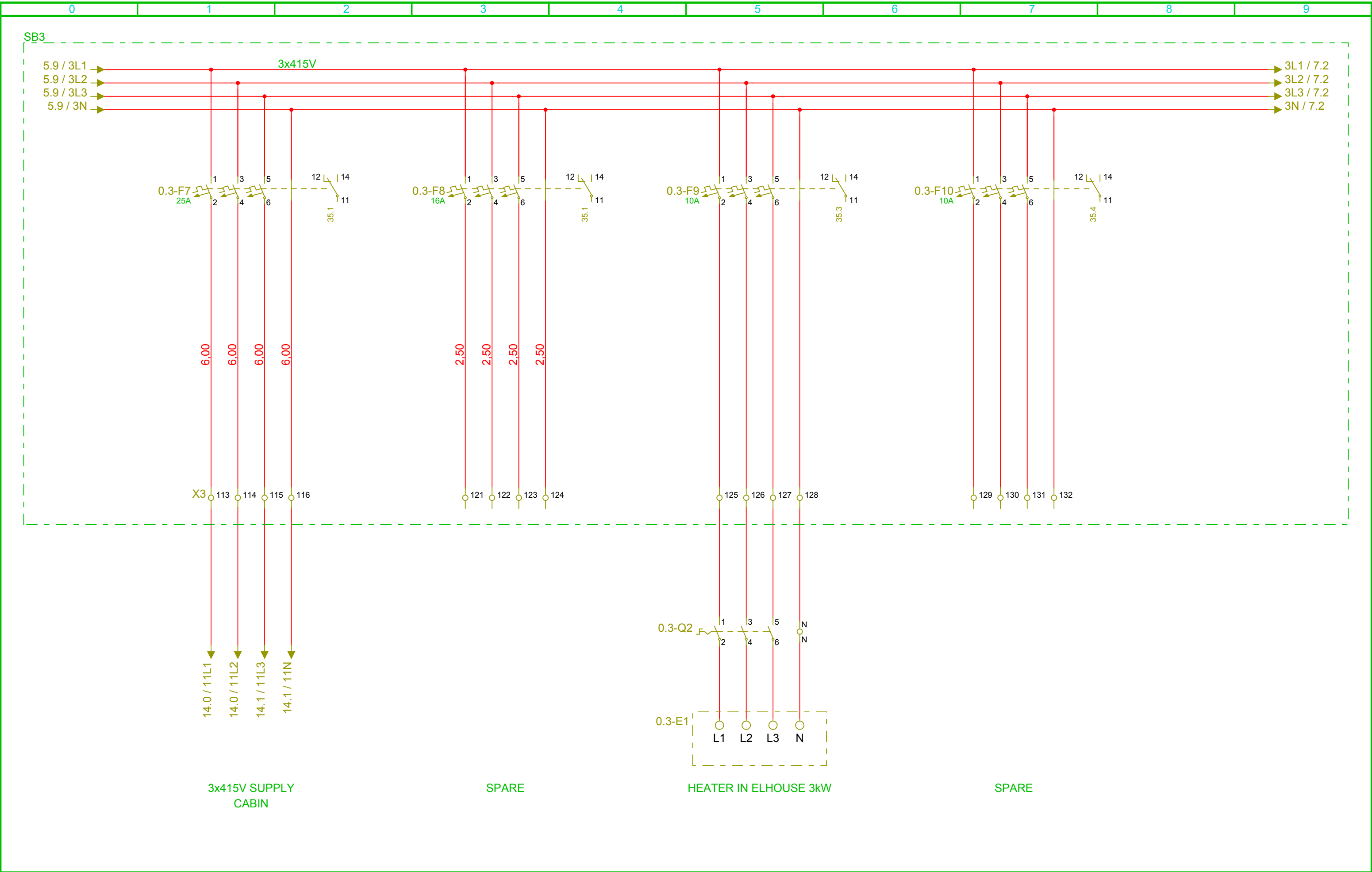


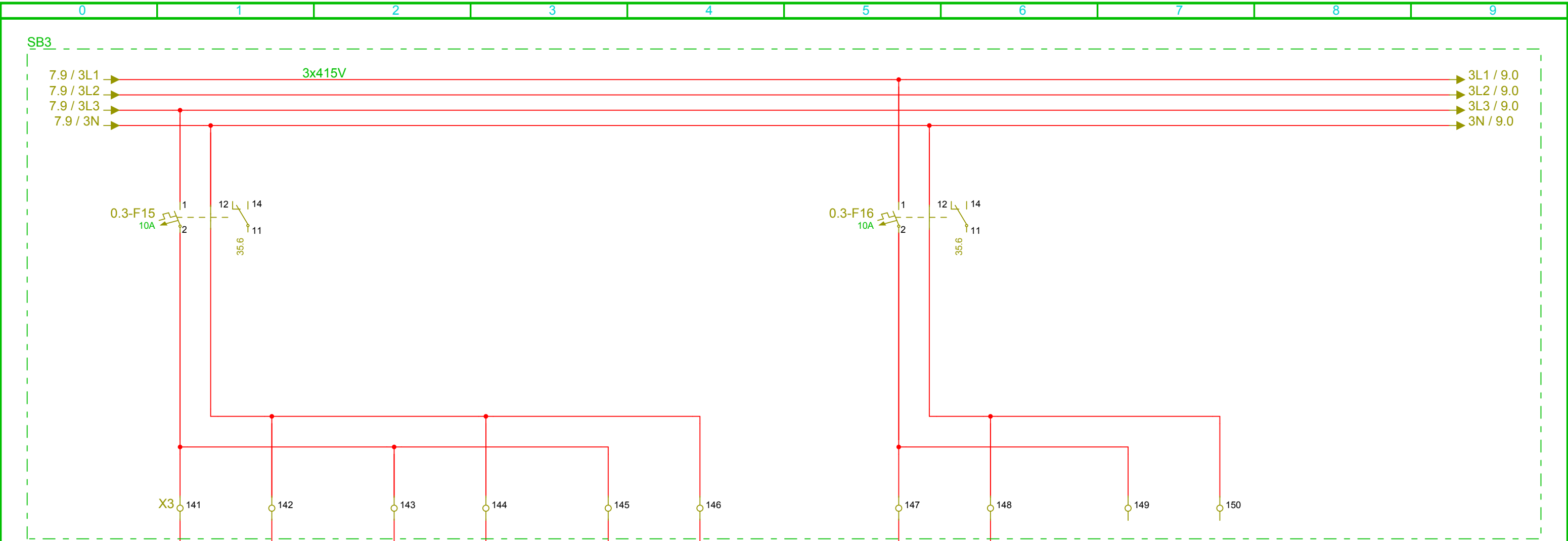
240V SUPPLY
SWITCHBOARD SB4

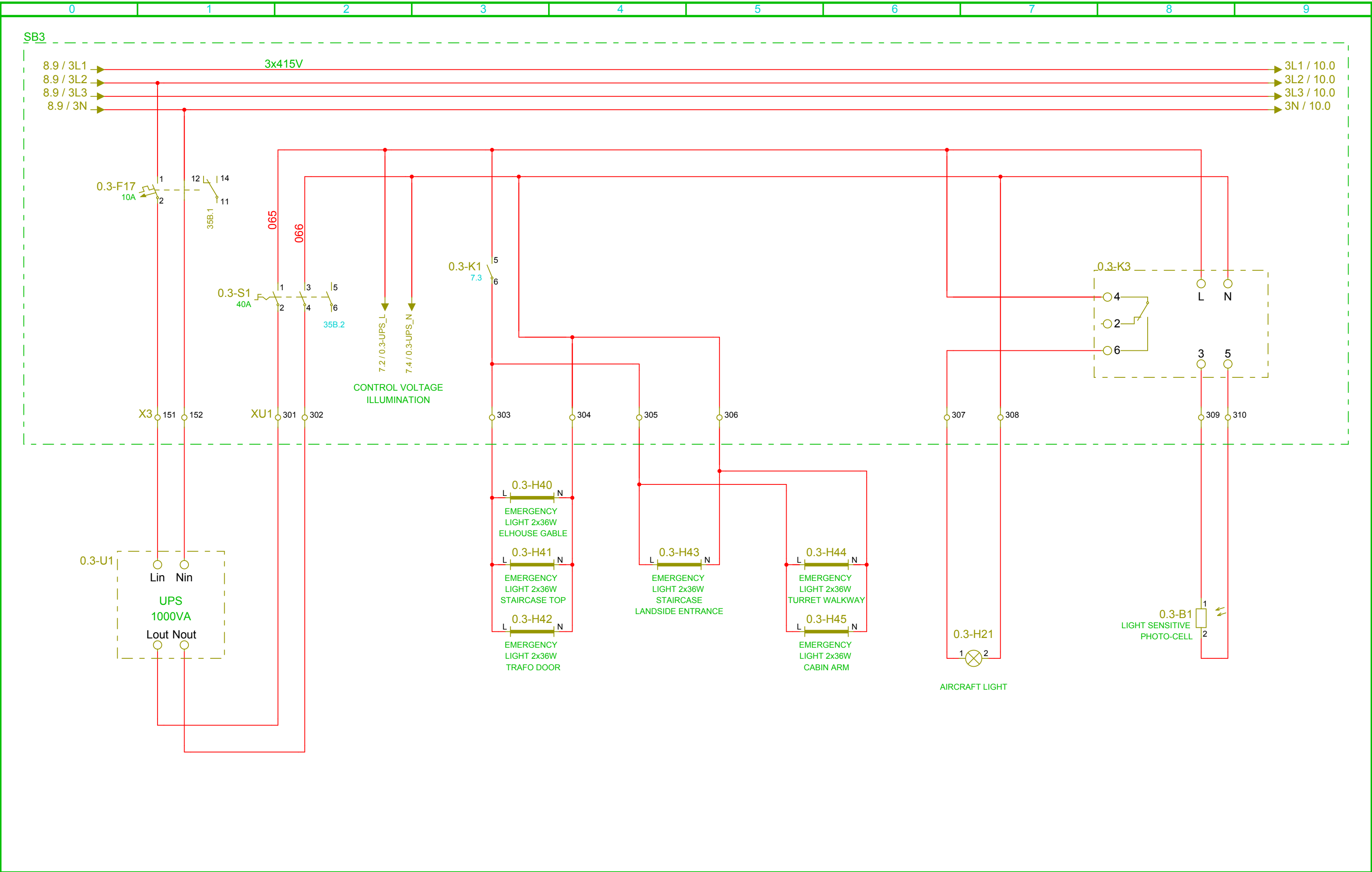


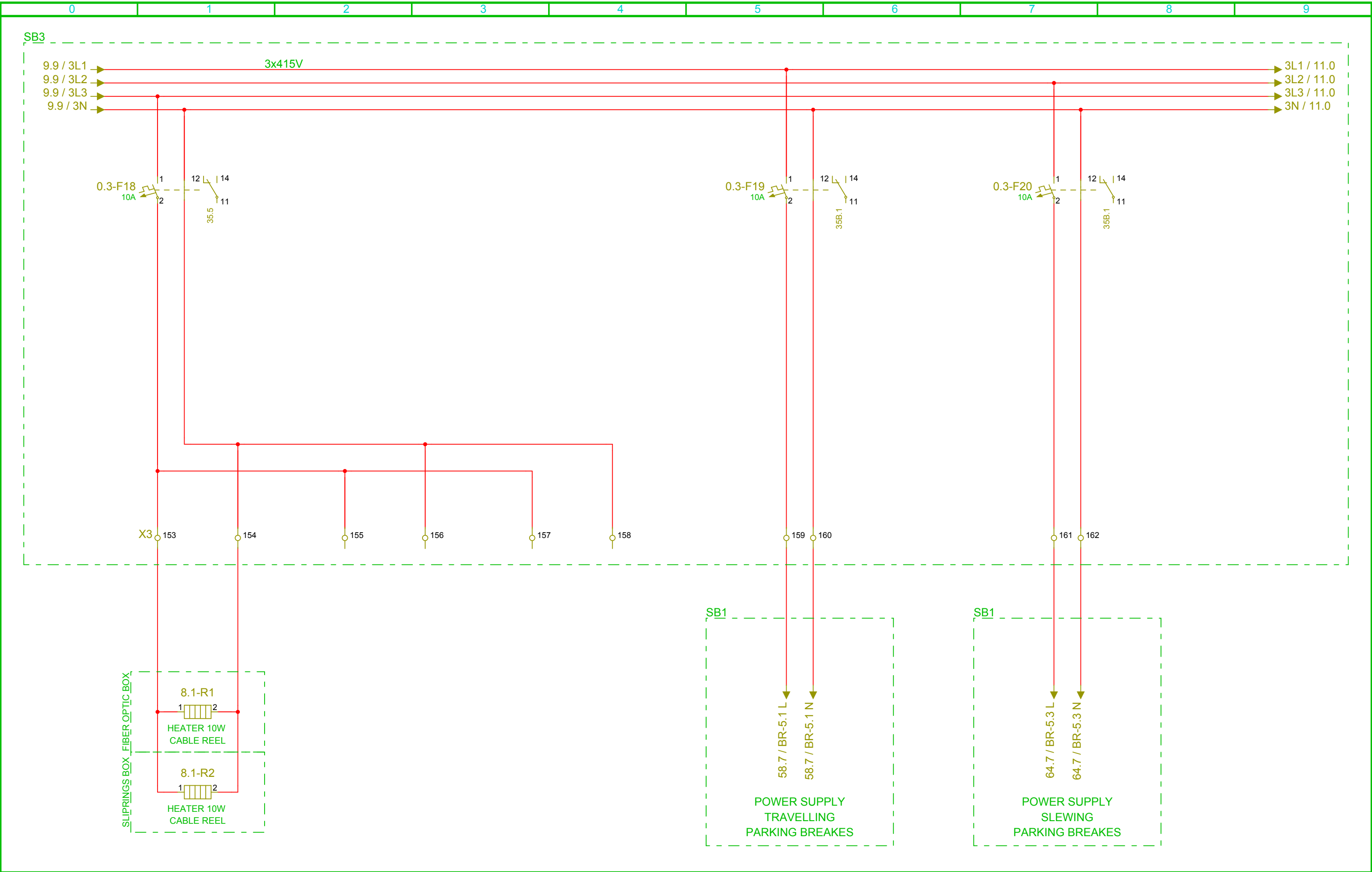
110V
DISTRIBUTION

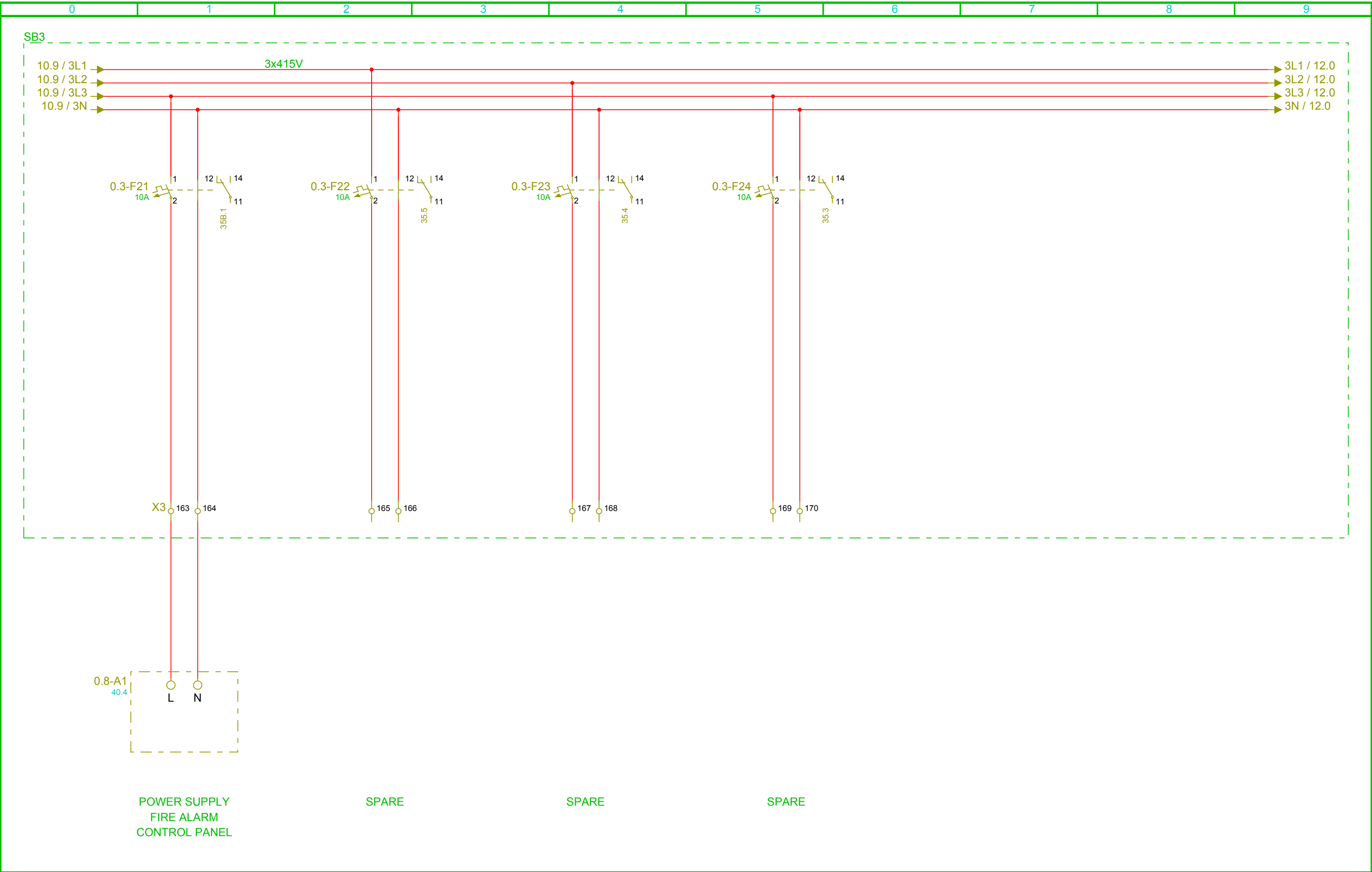


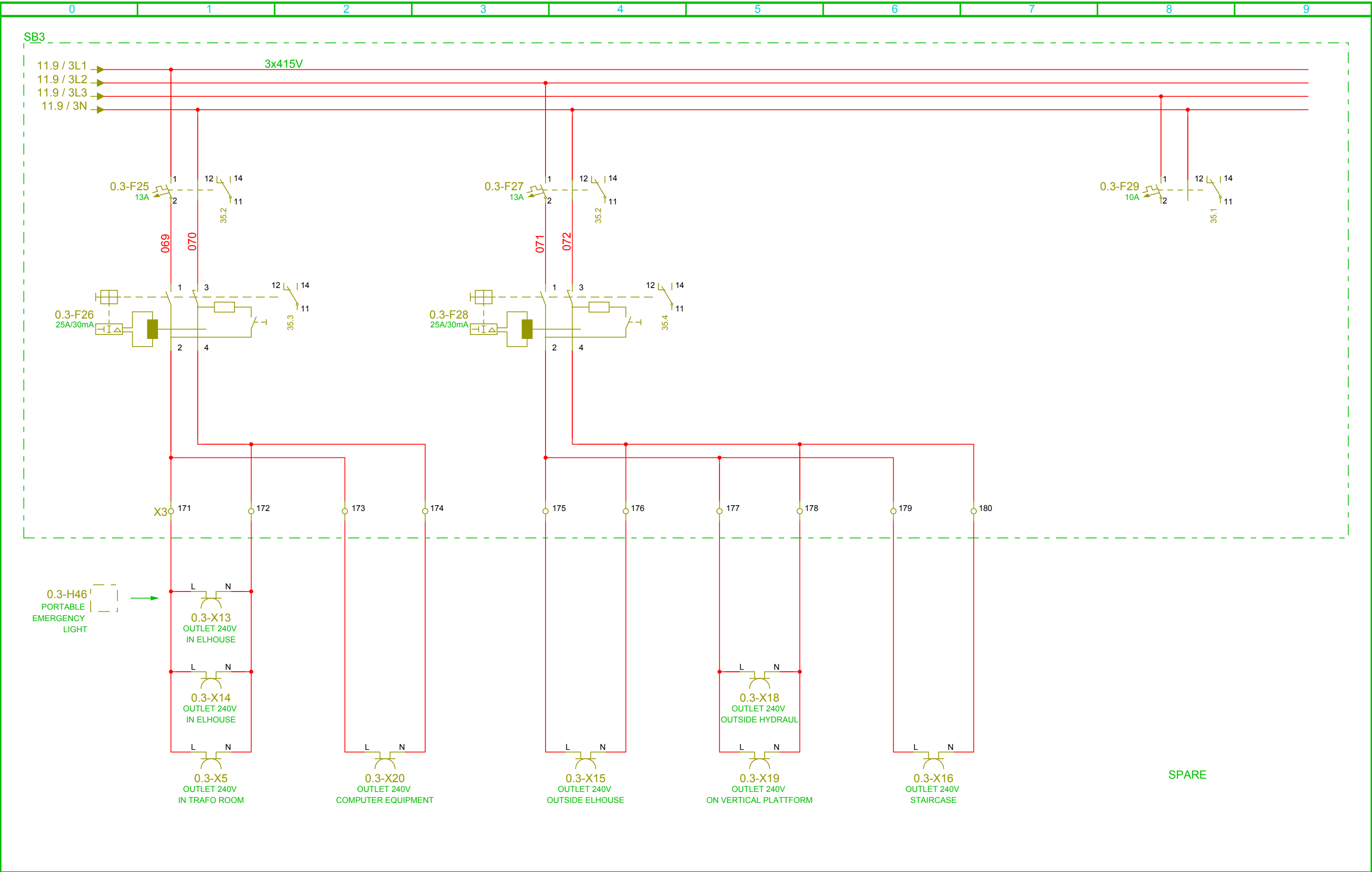


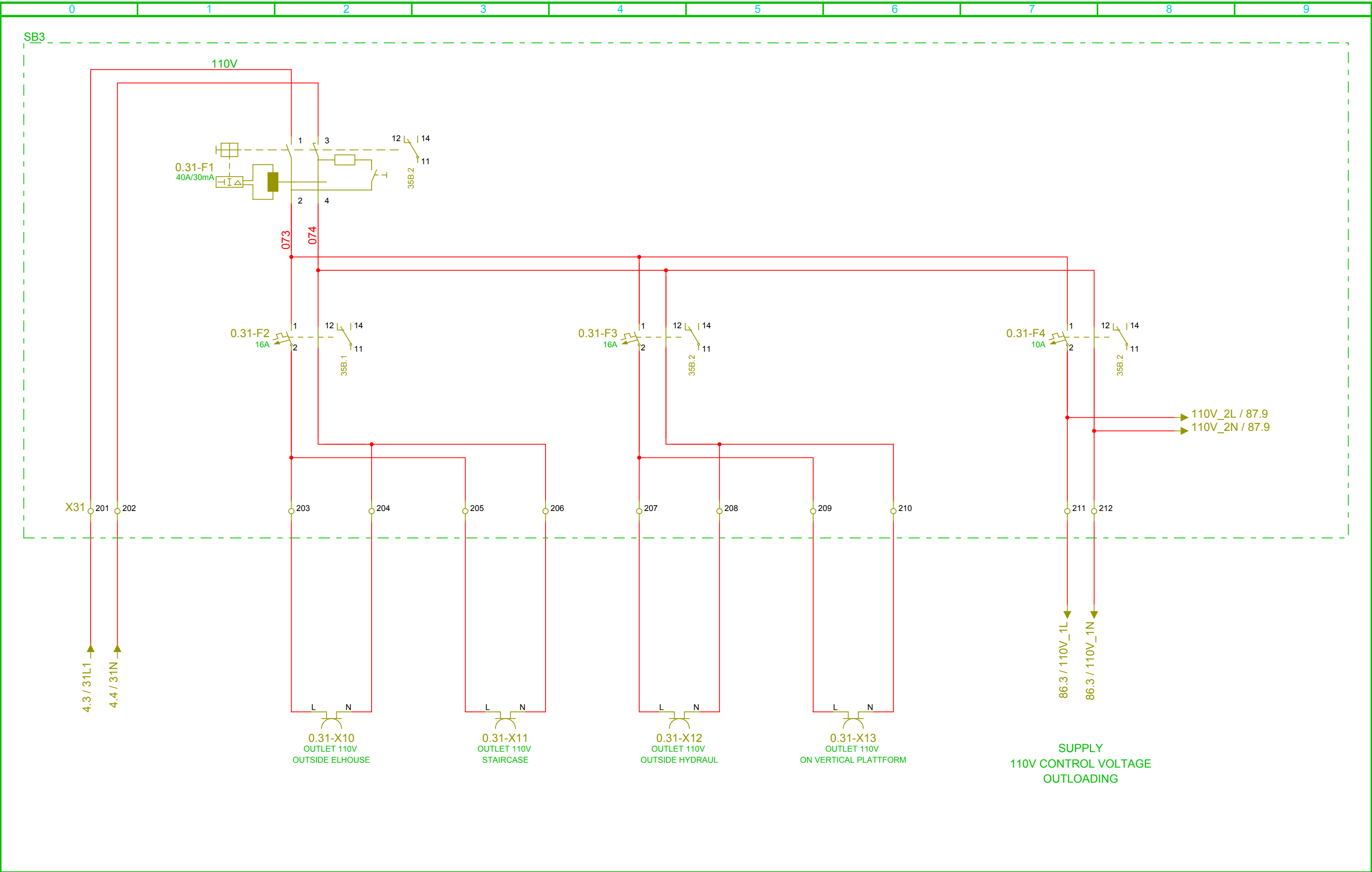


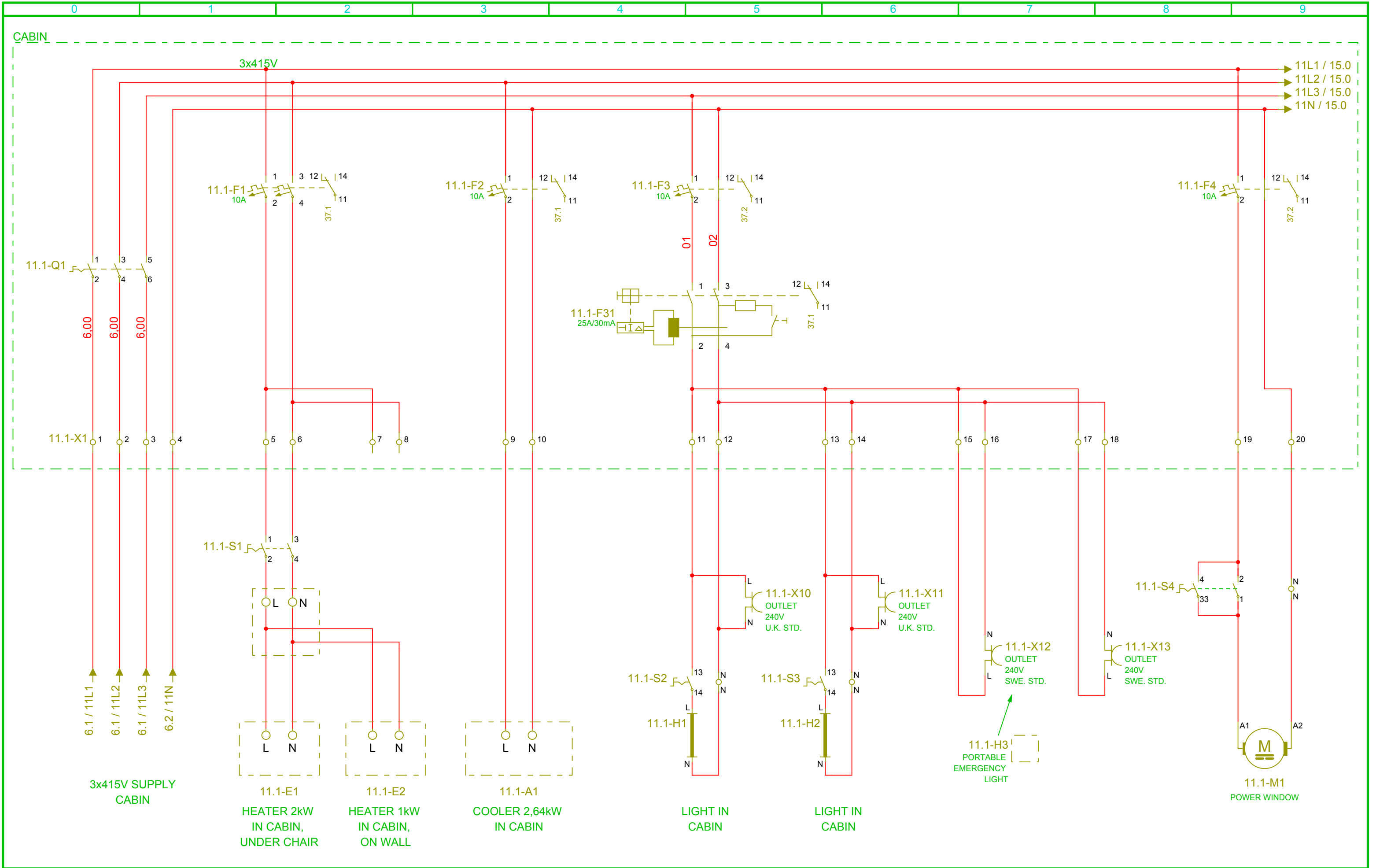


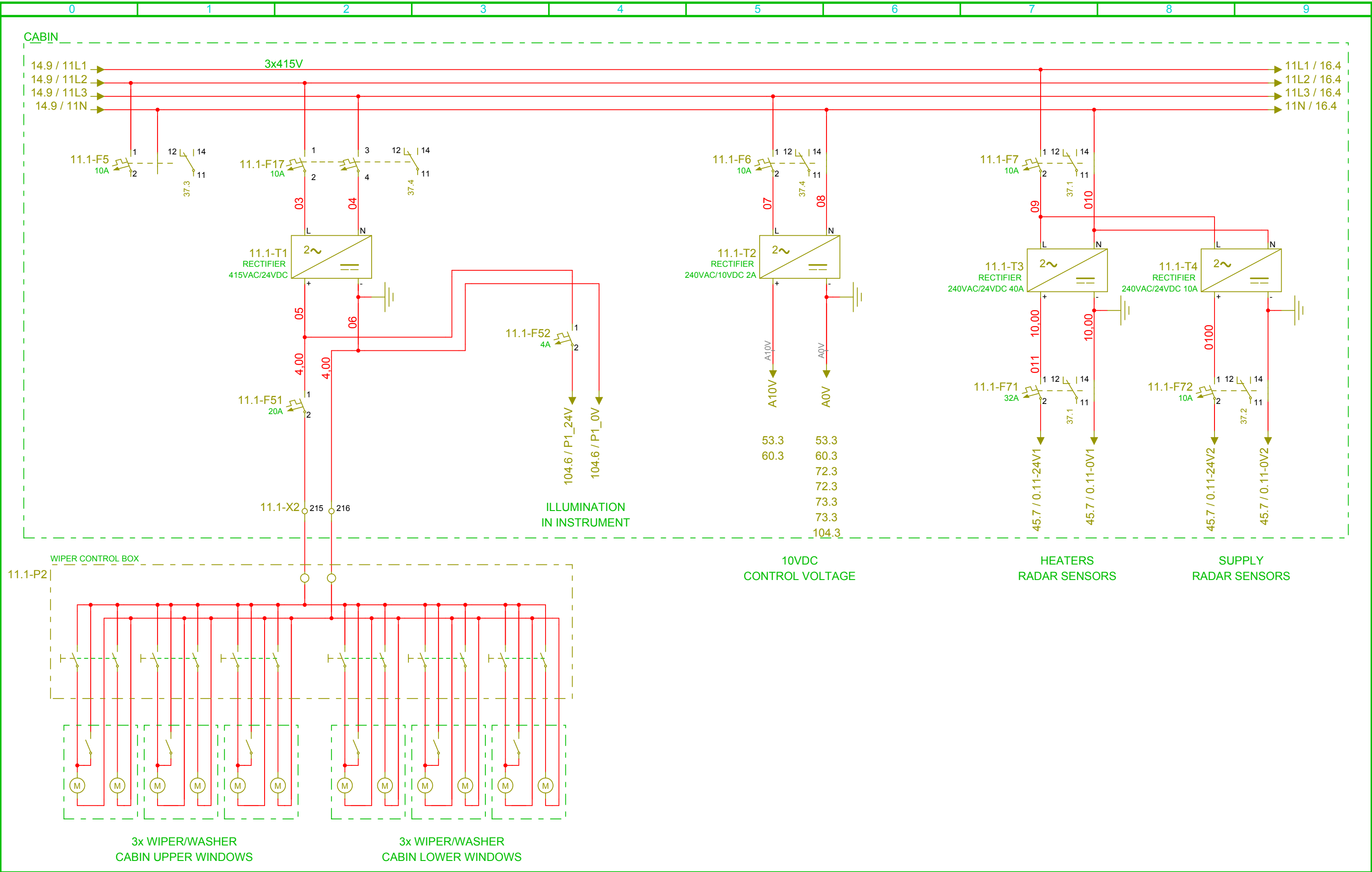


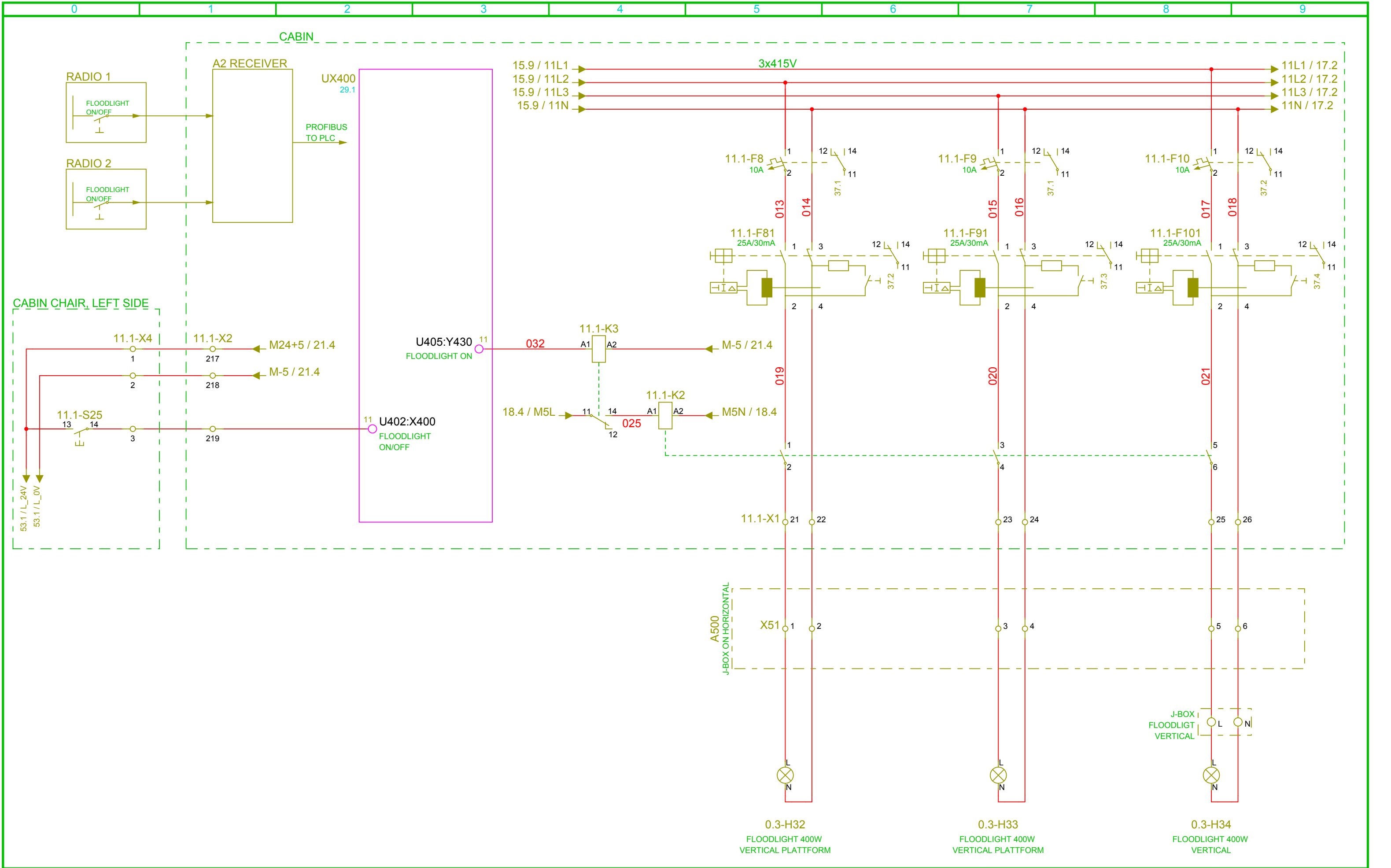


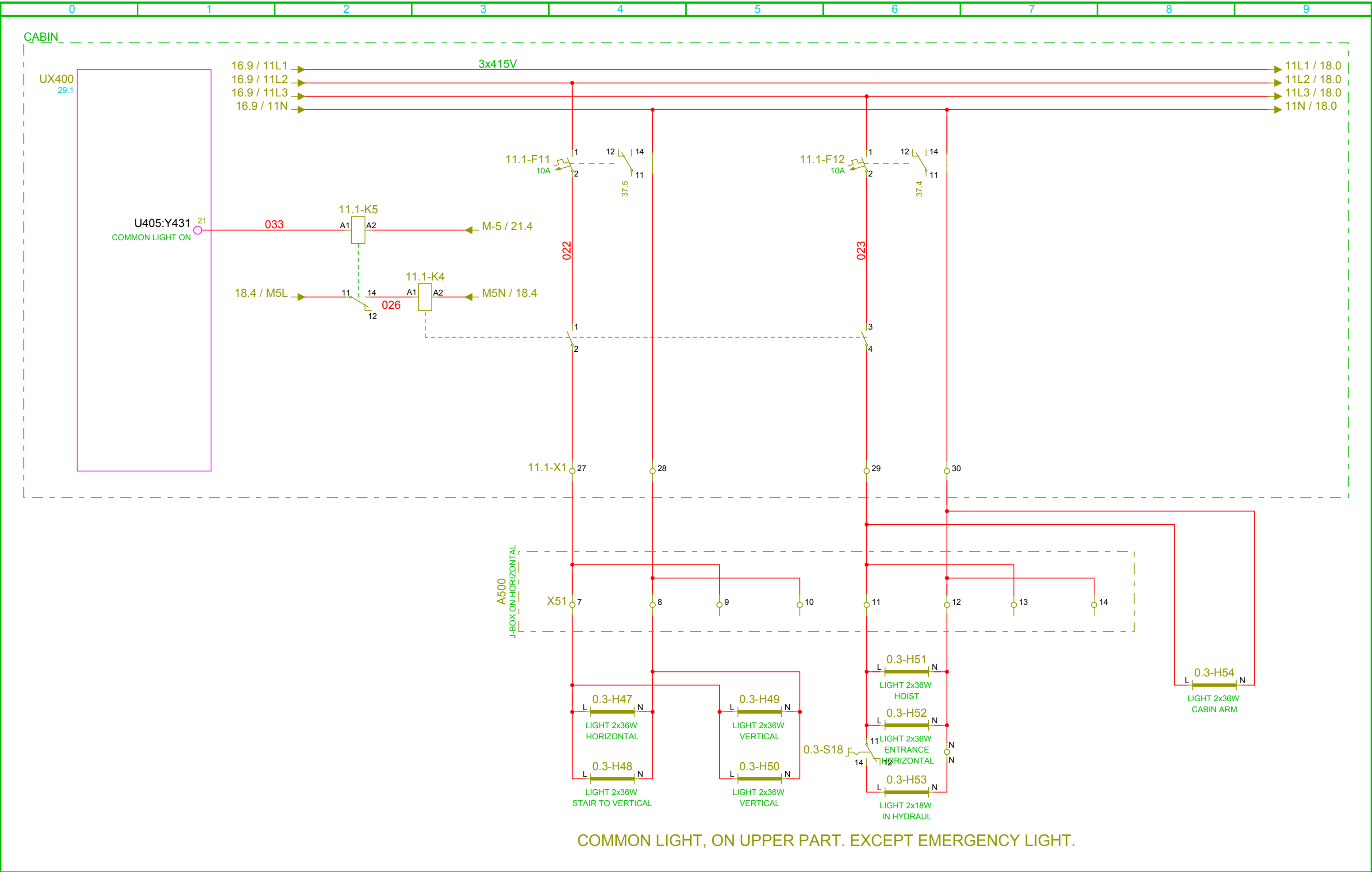


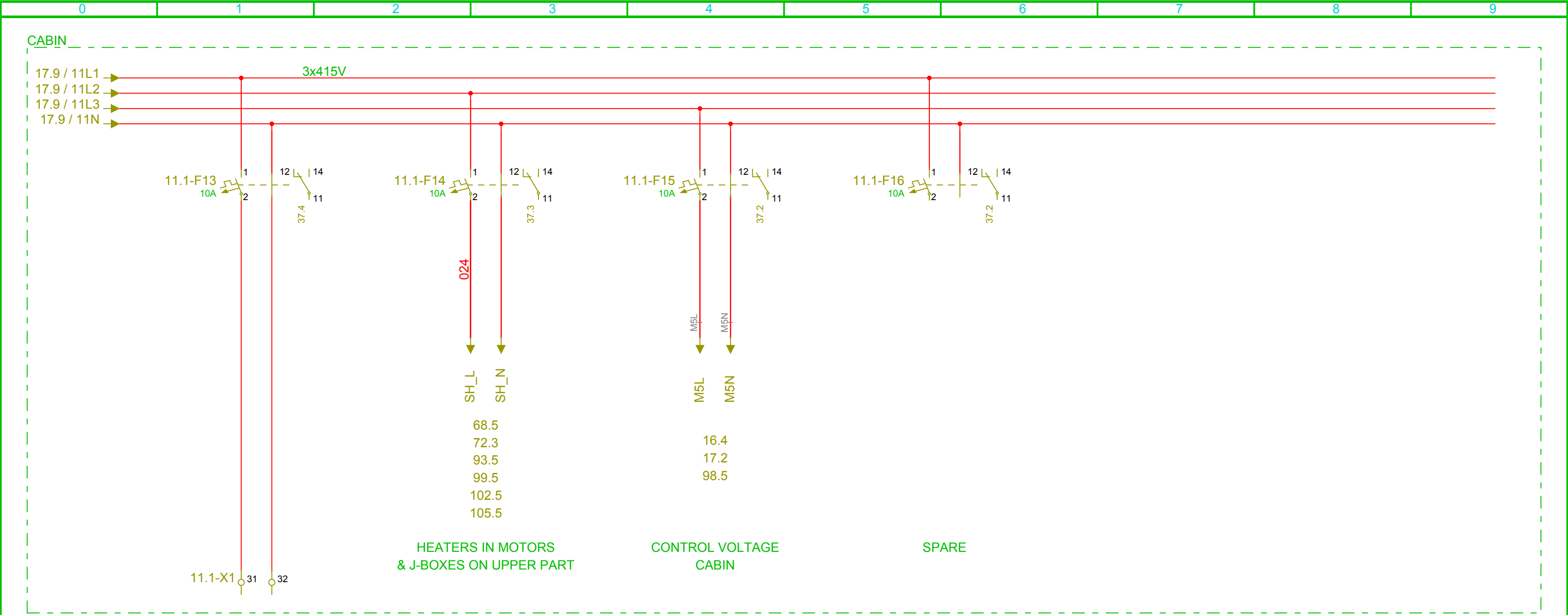


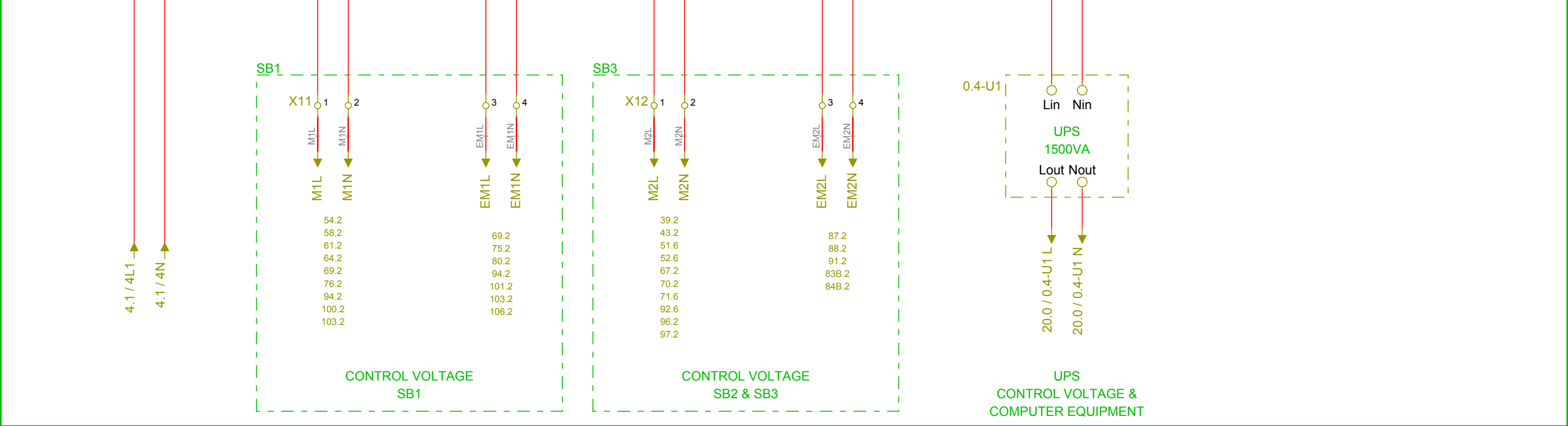
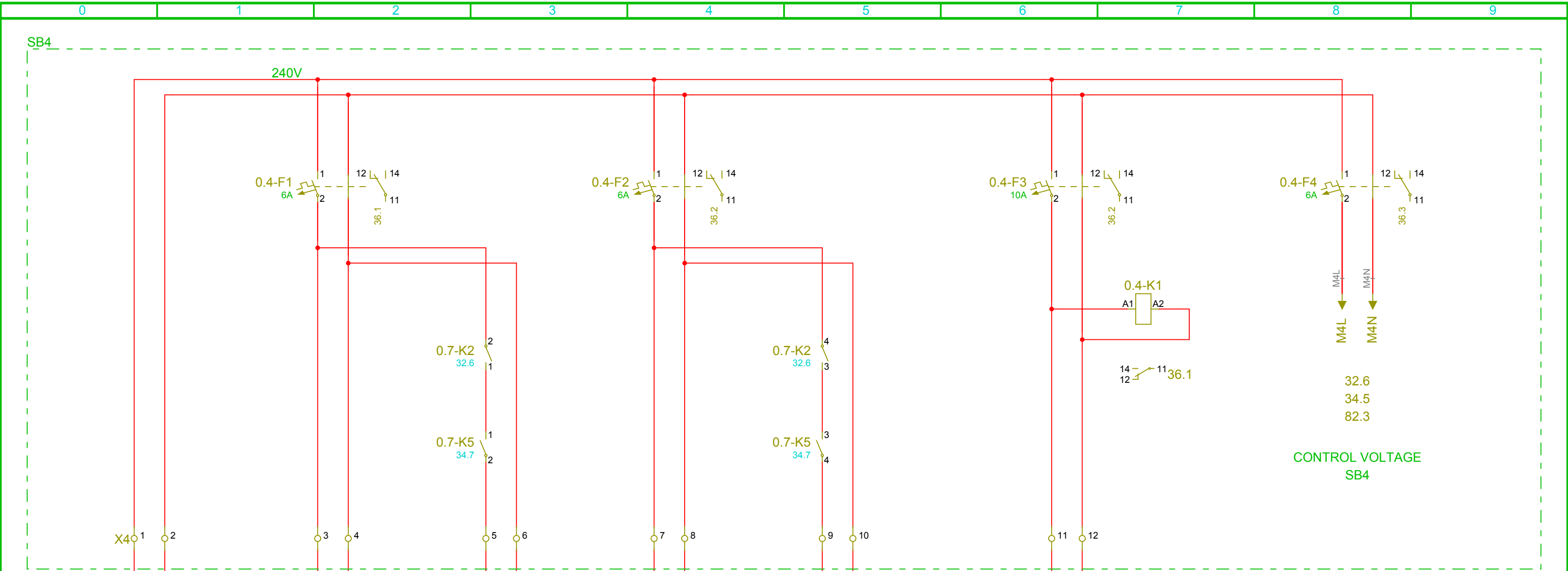


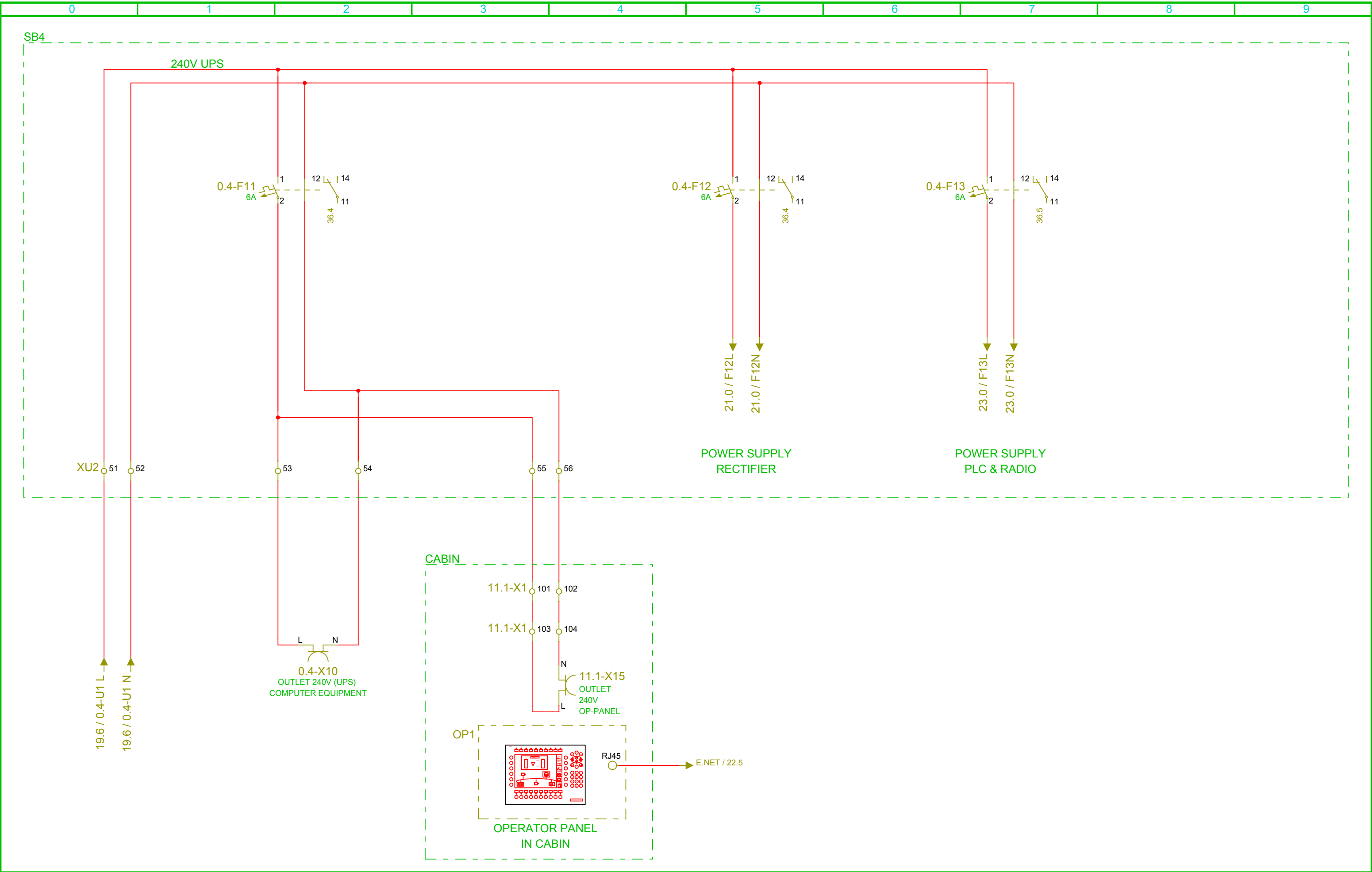


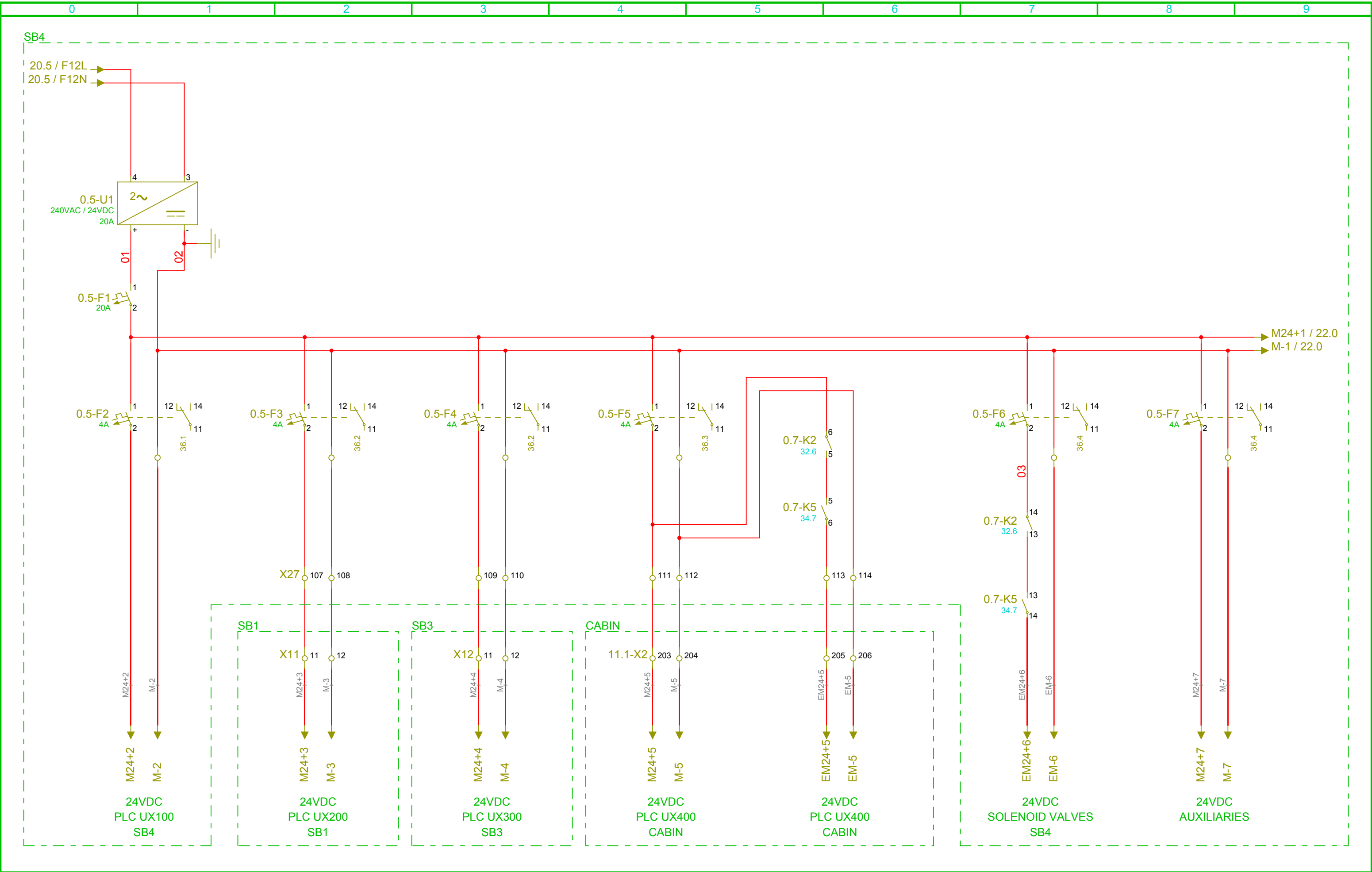


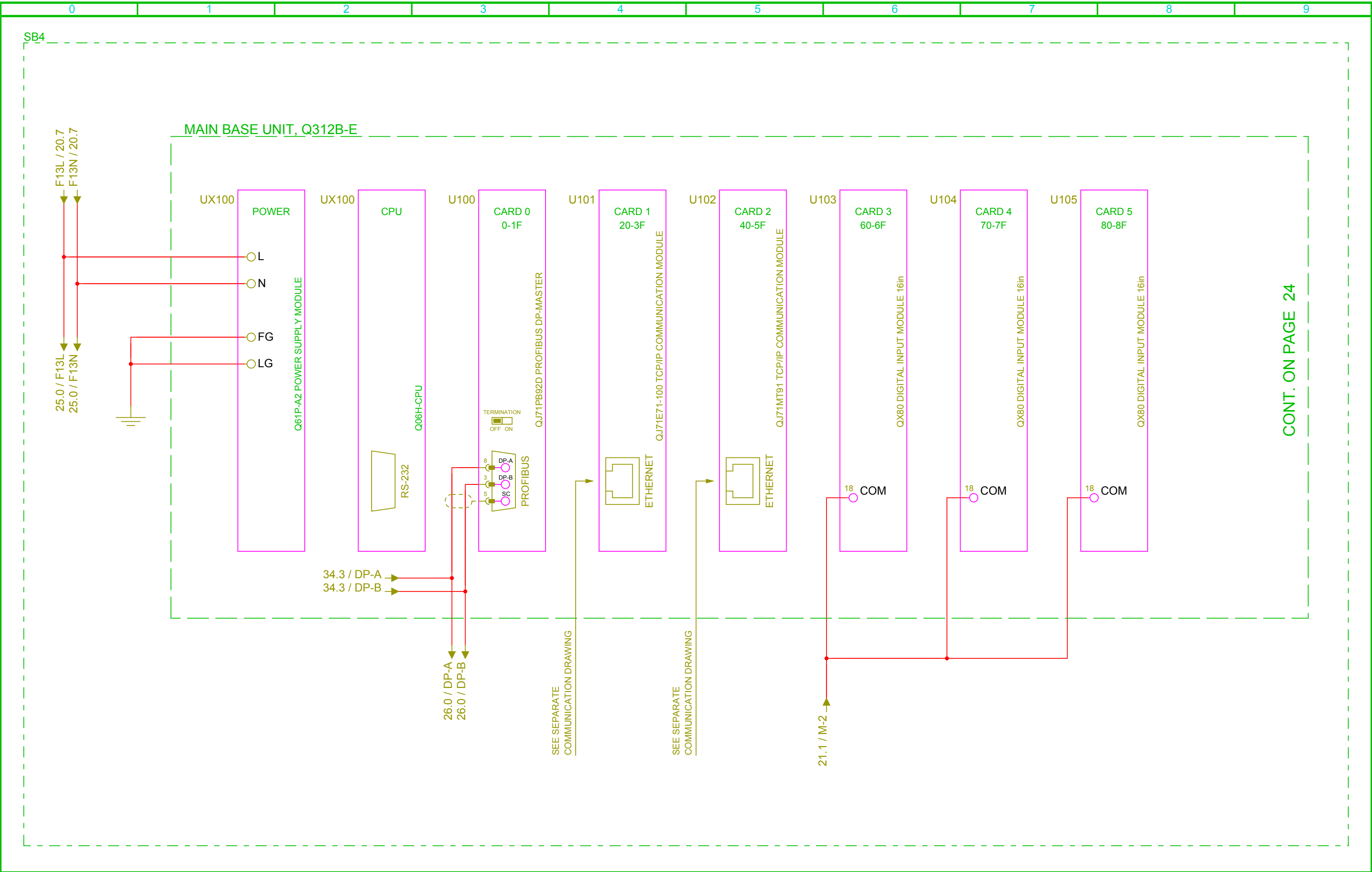


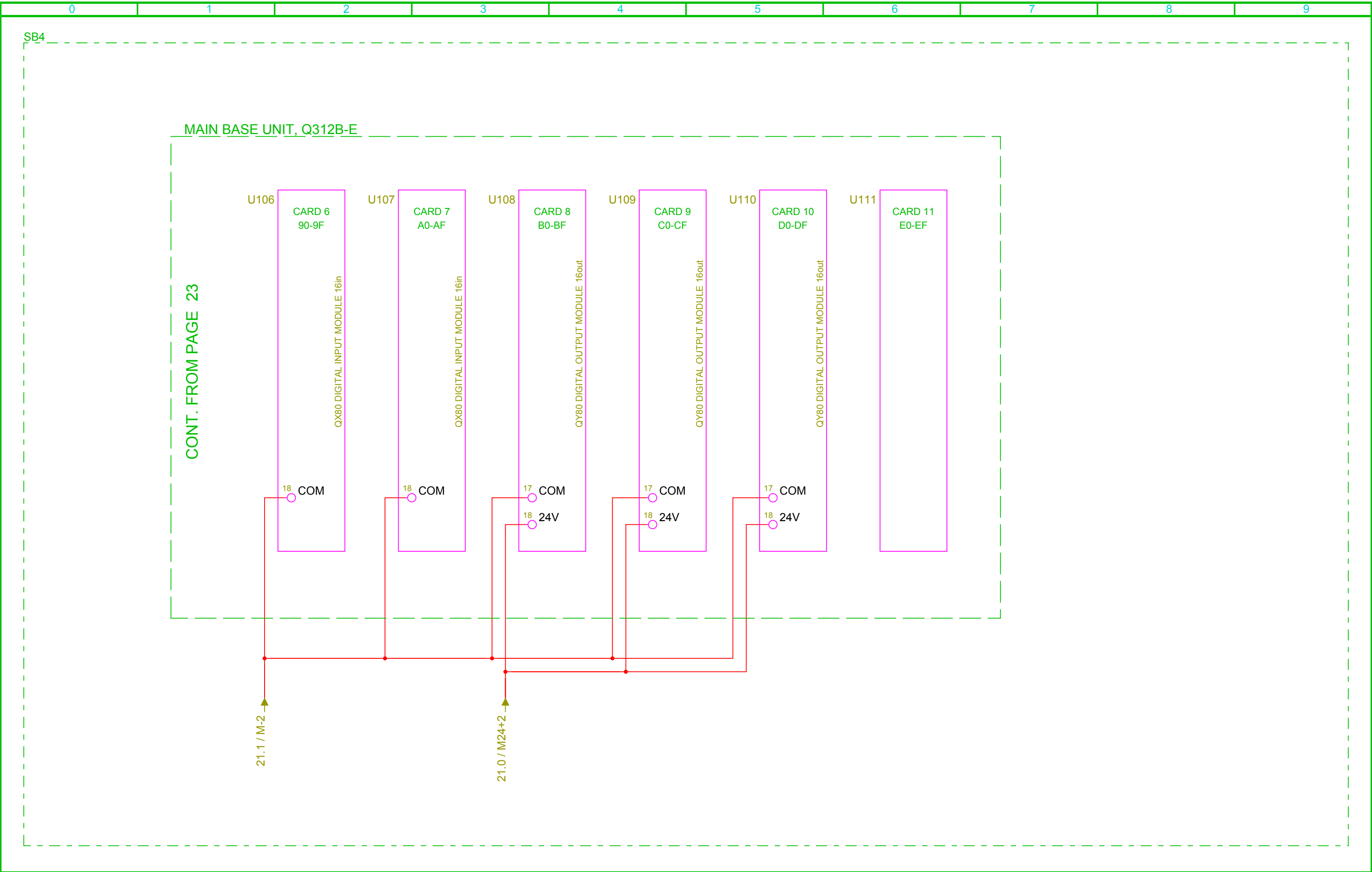


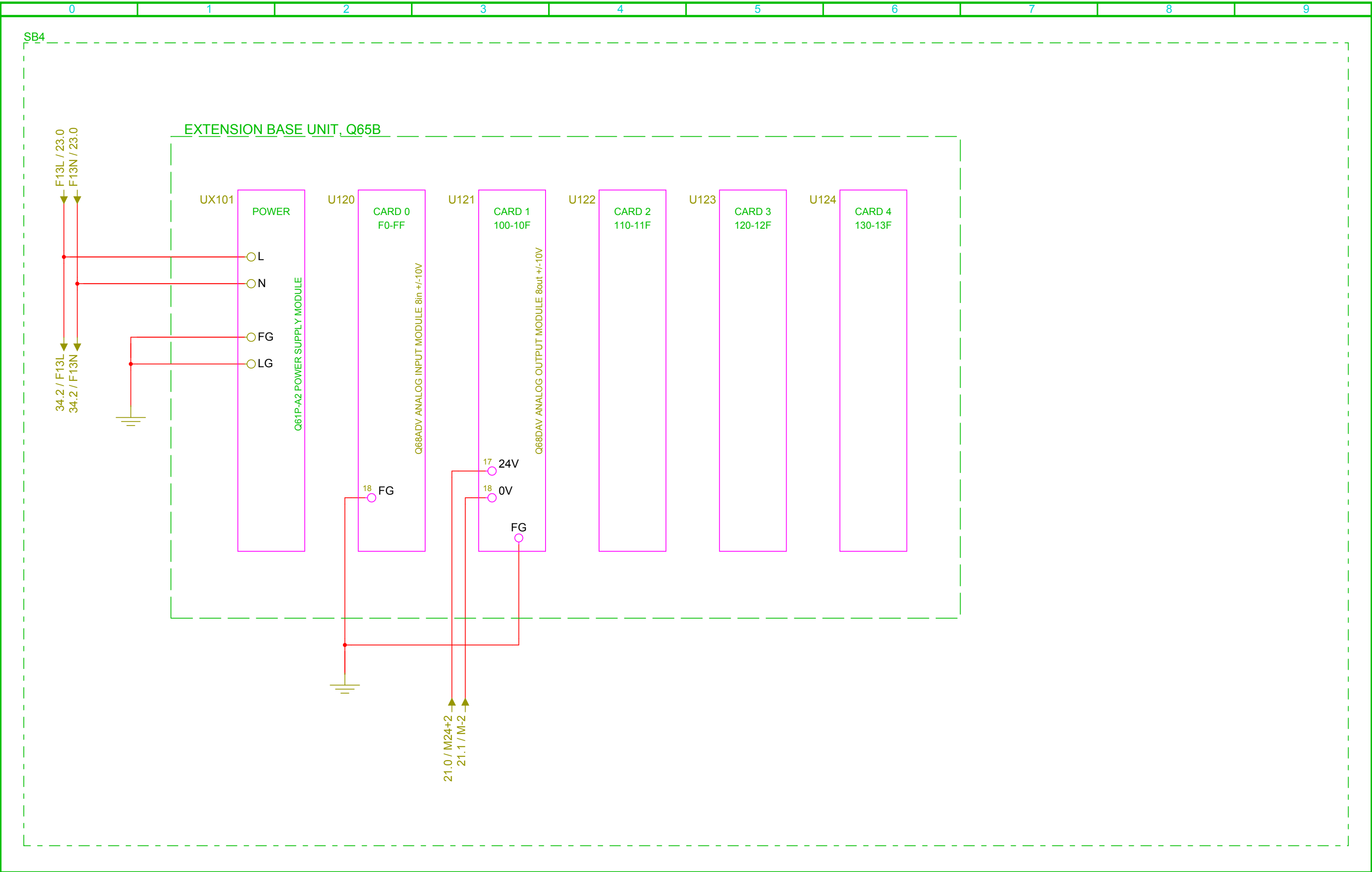


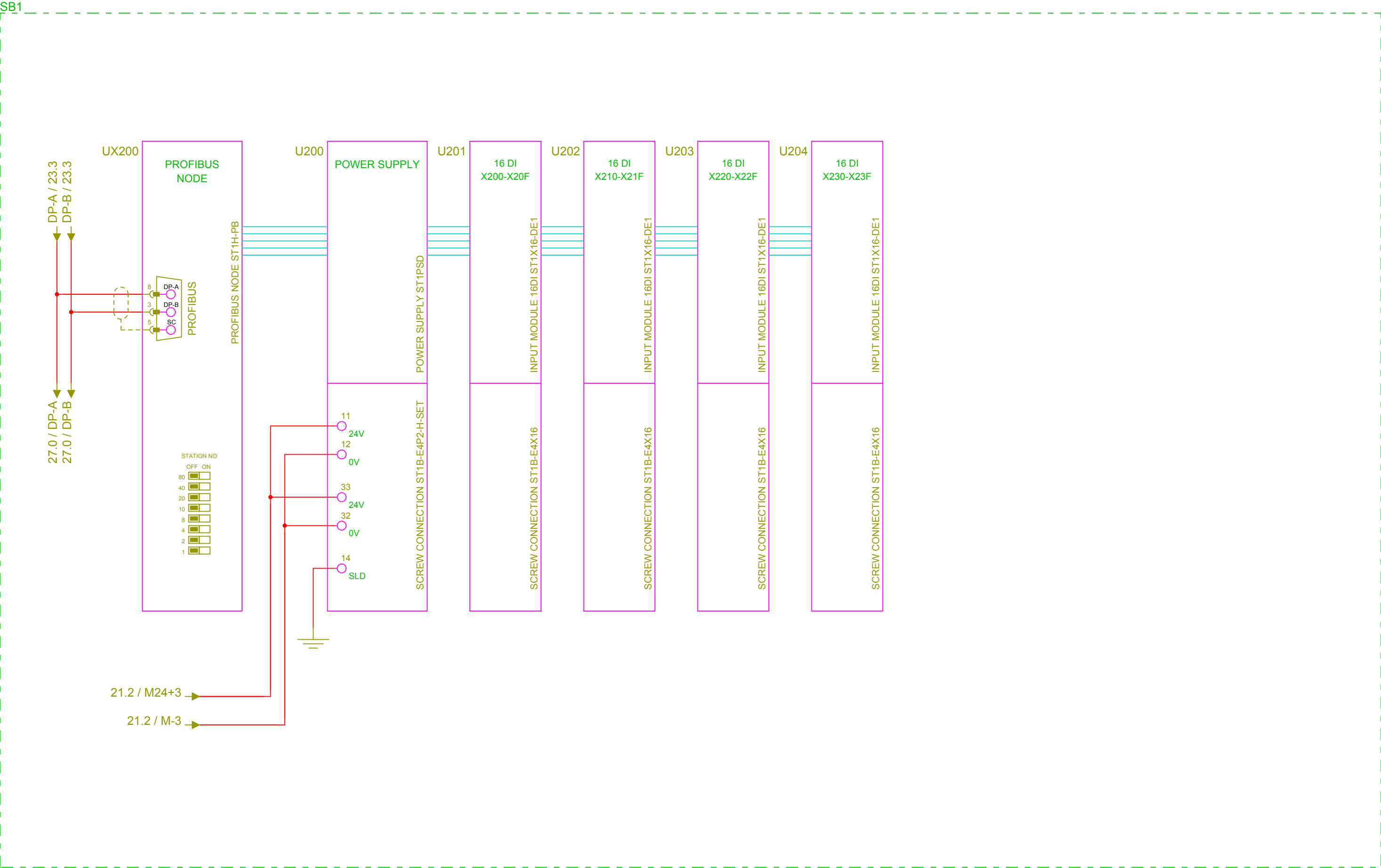


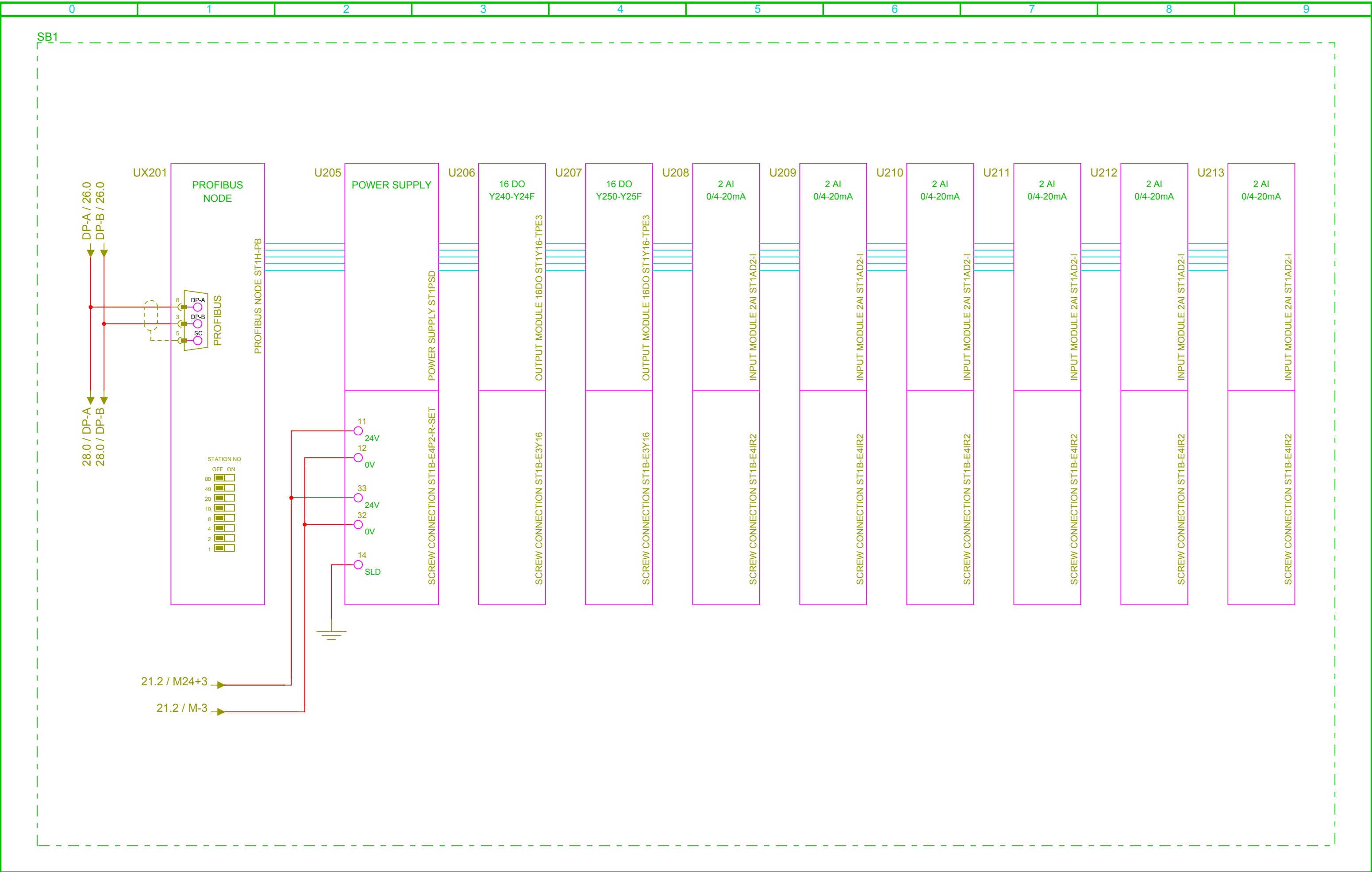


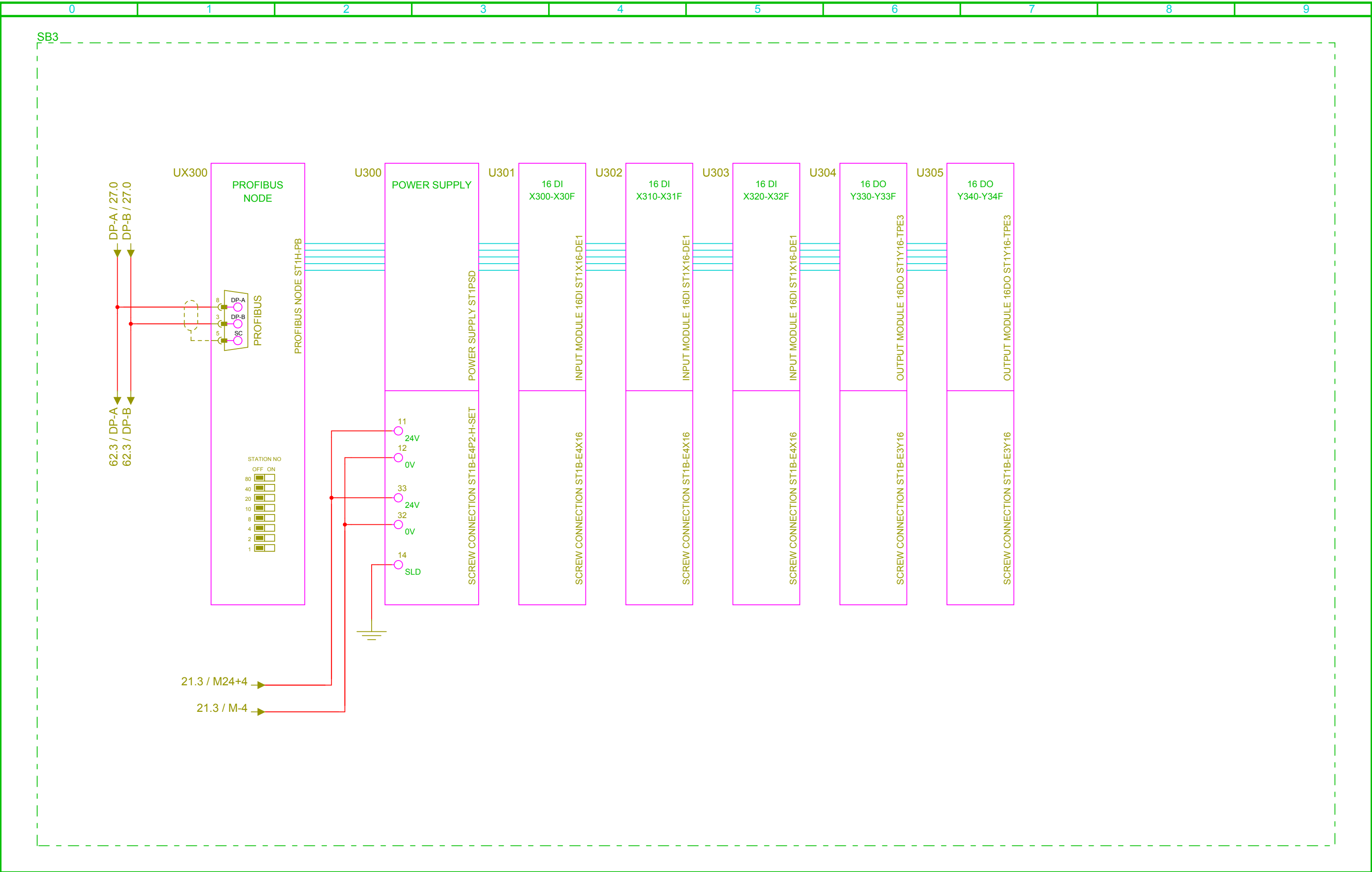


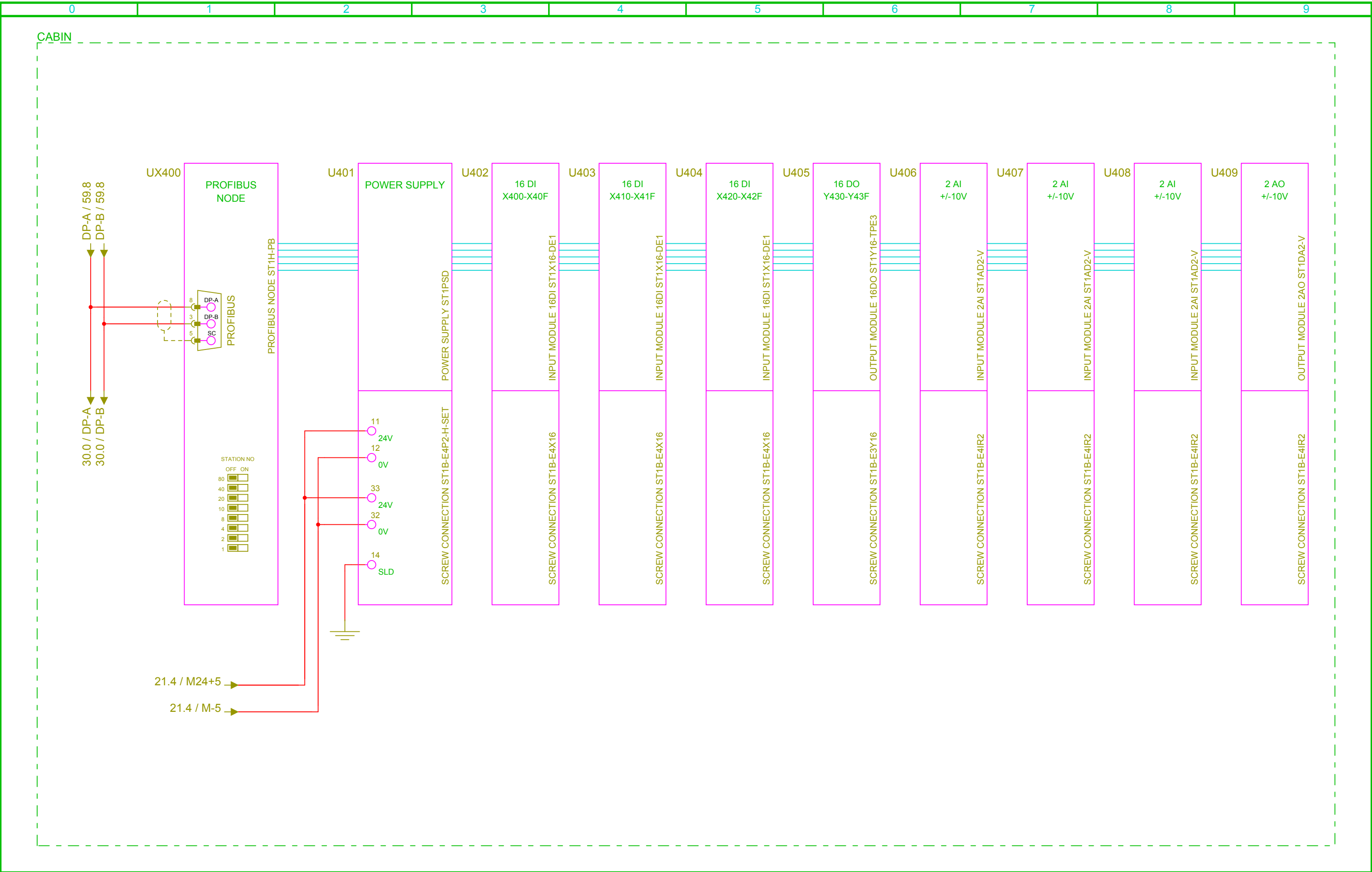


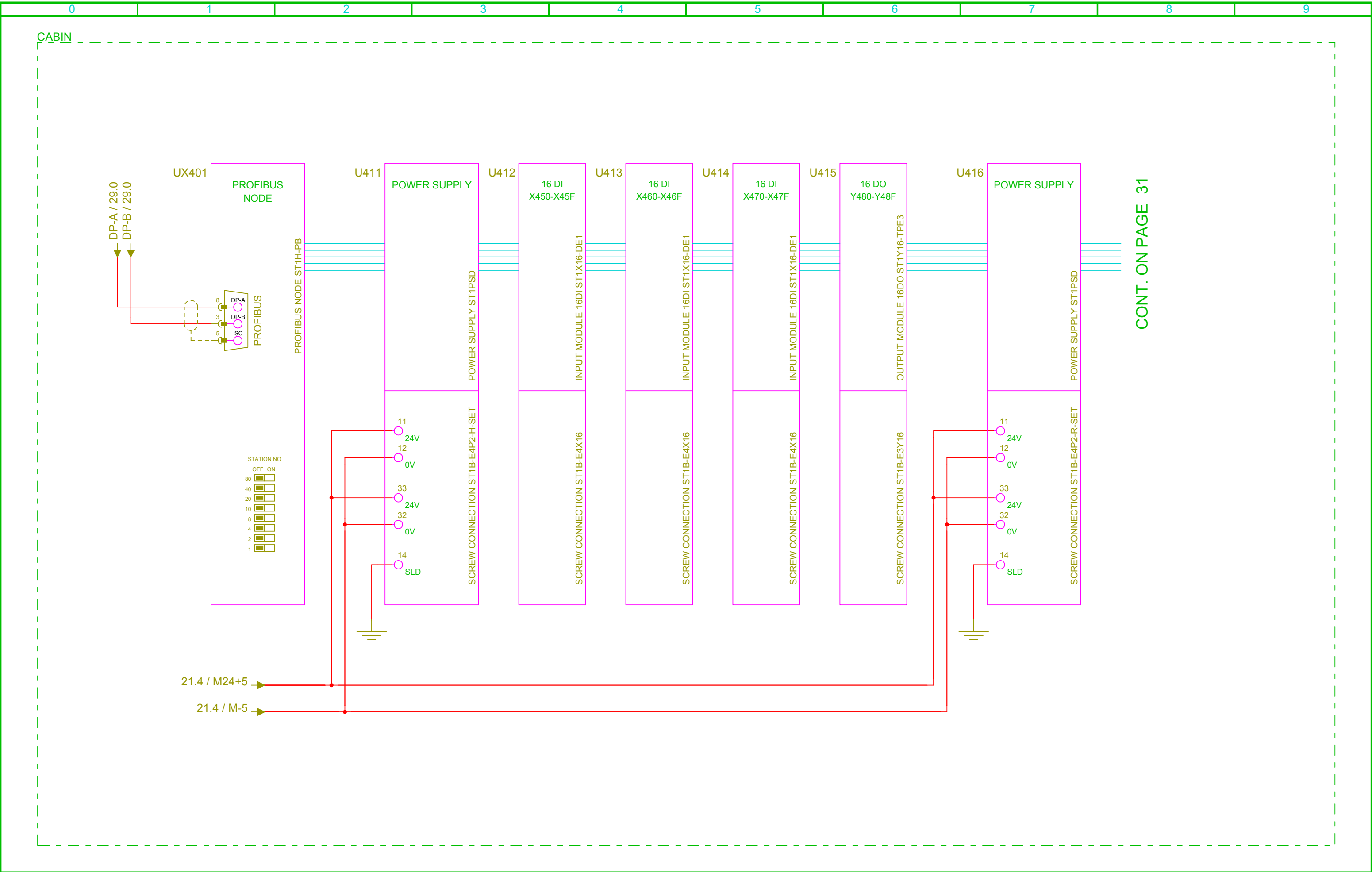


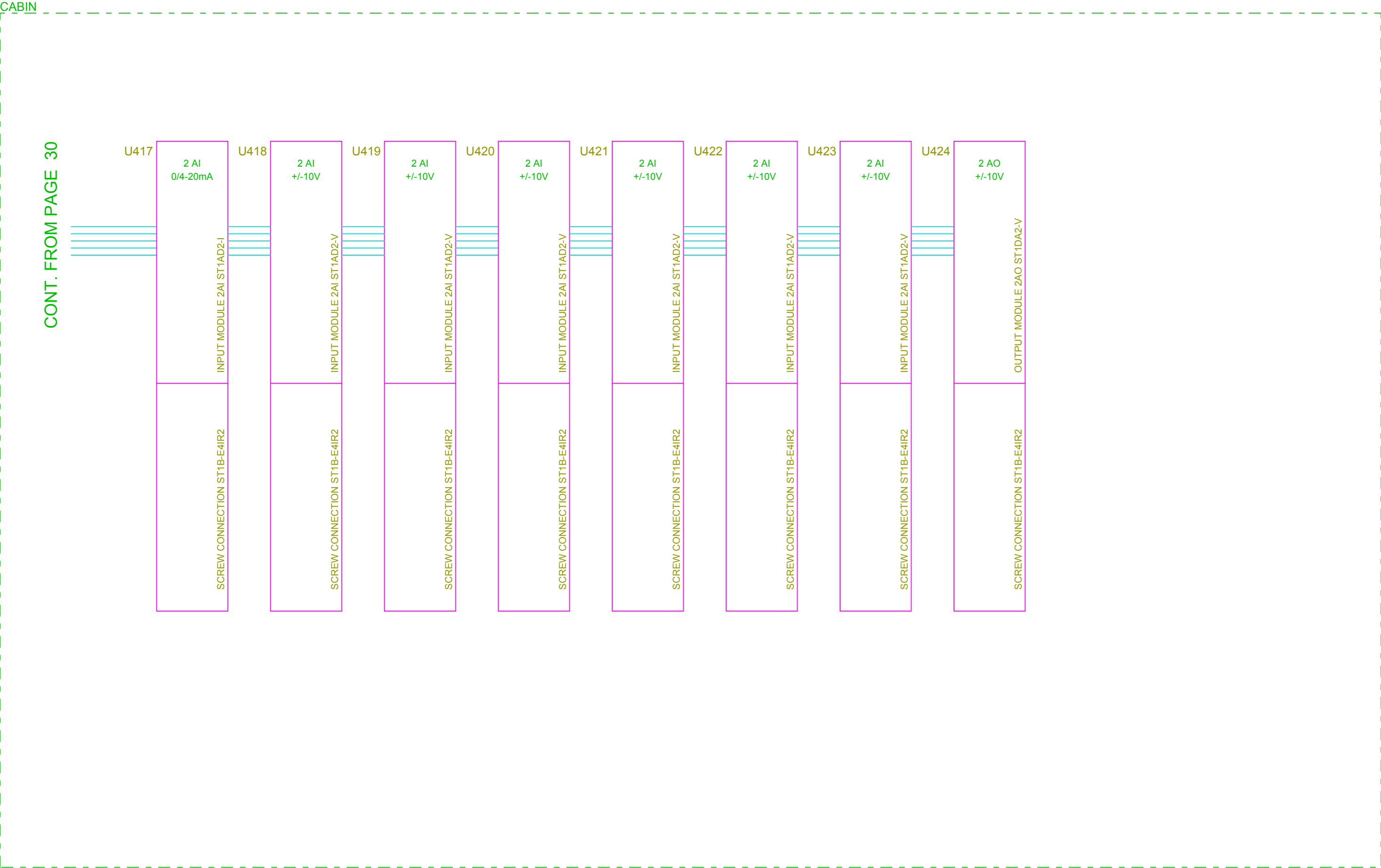




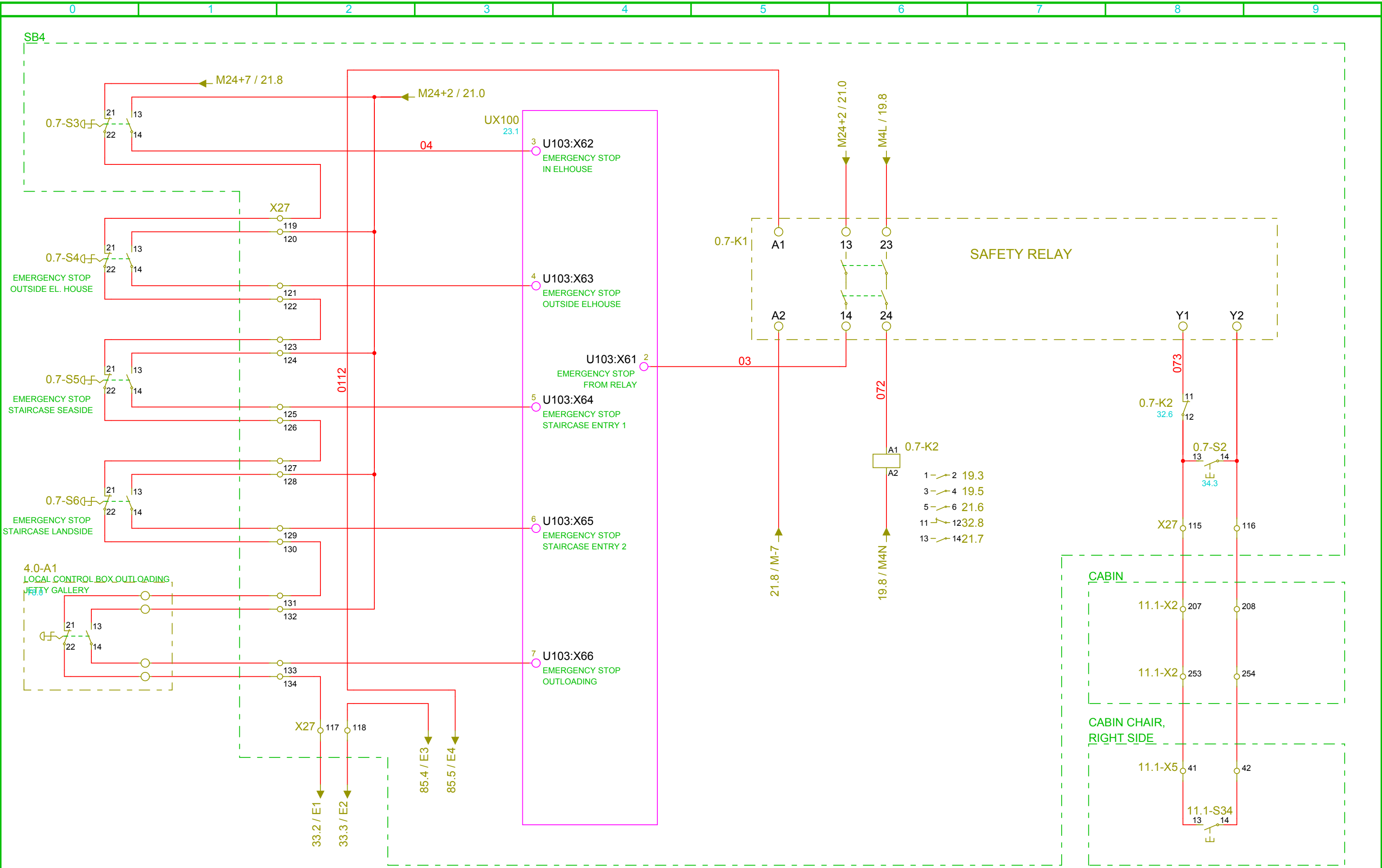


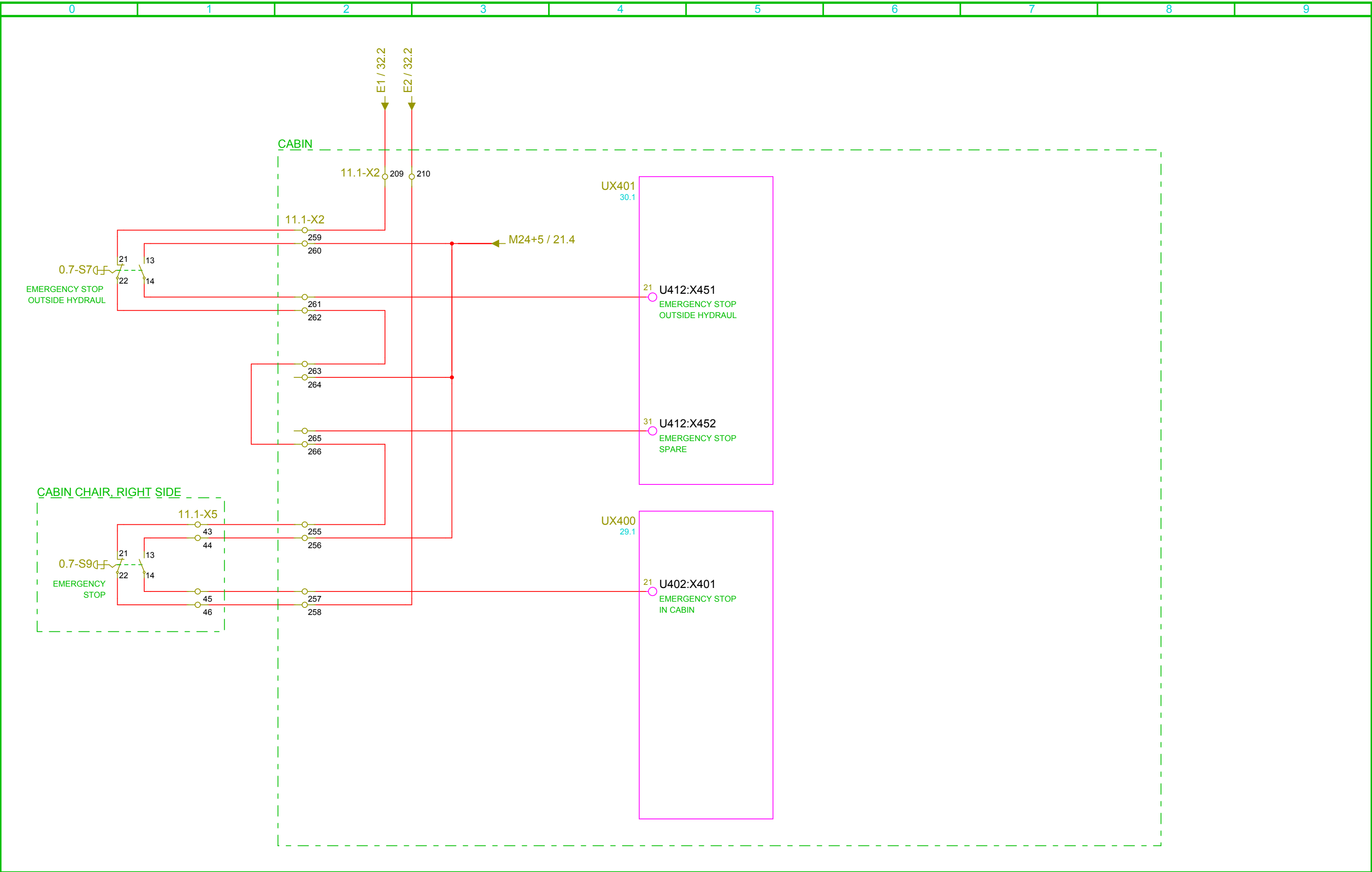


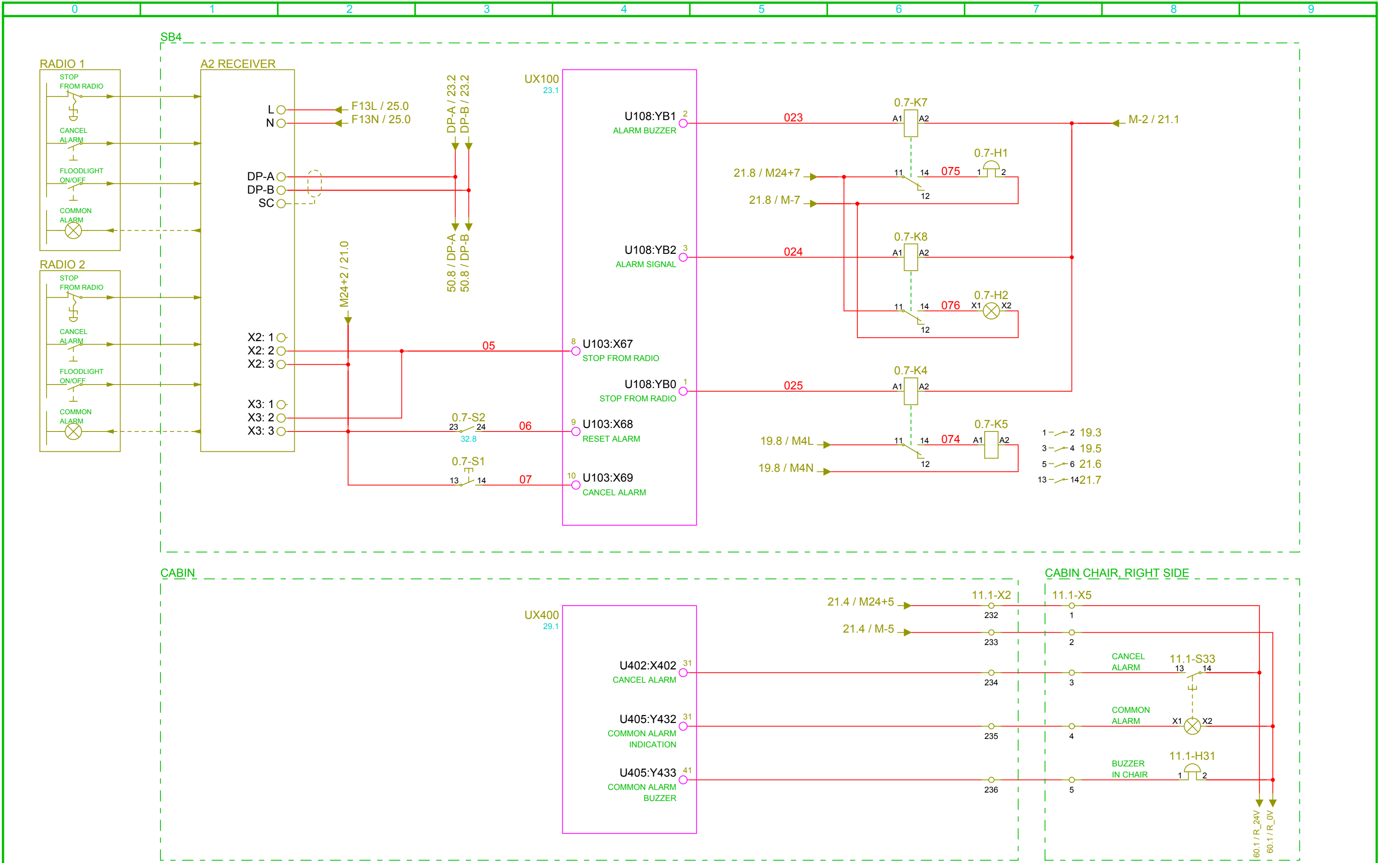


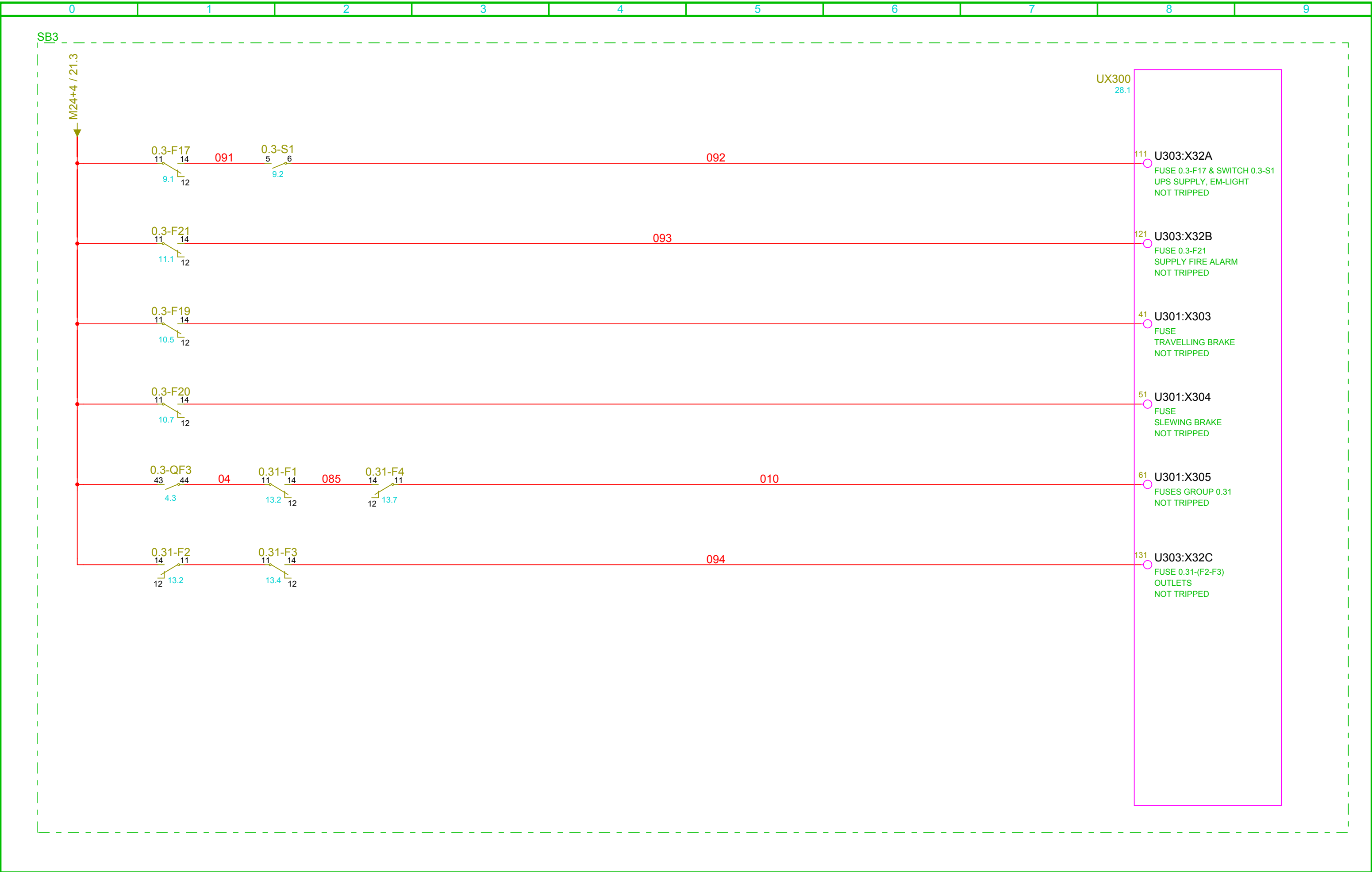


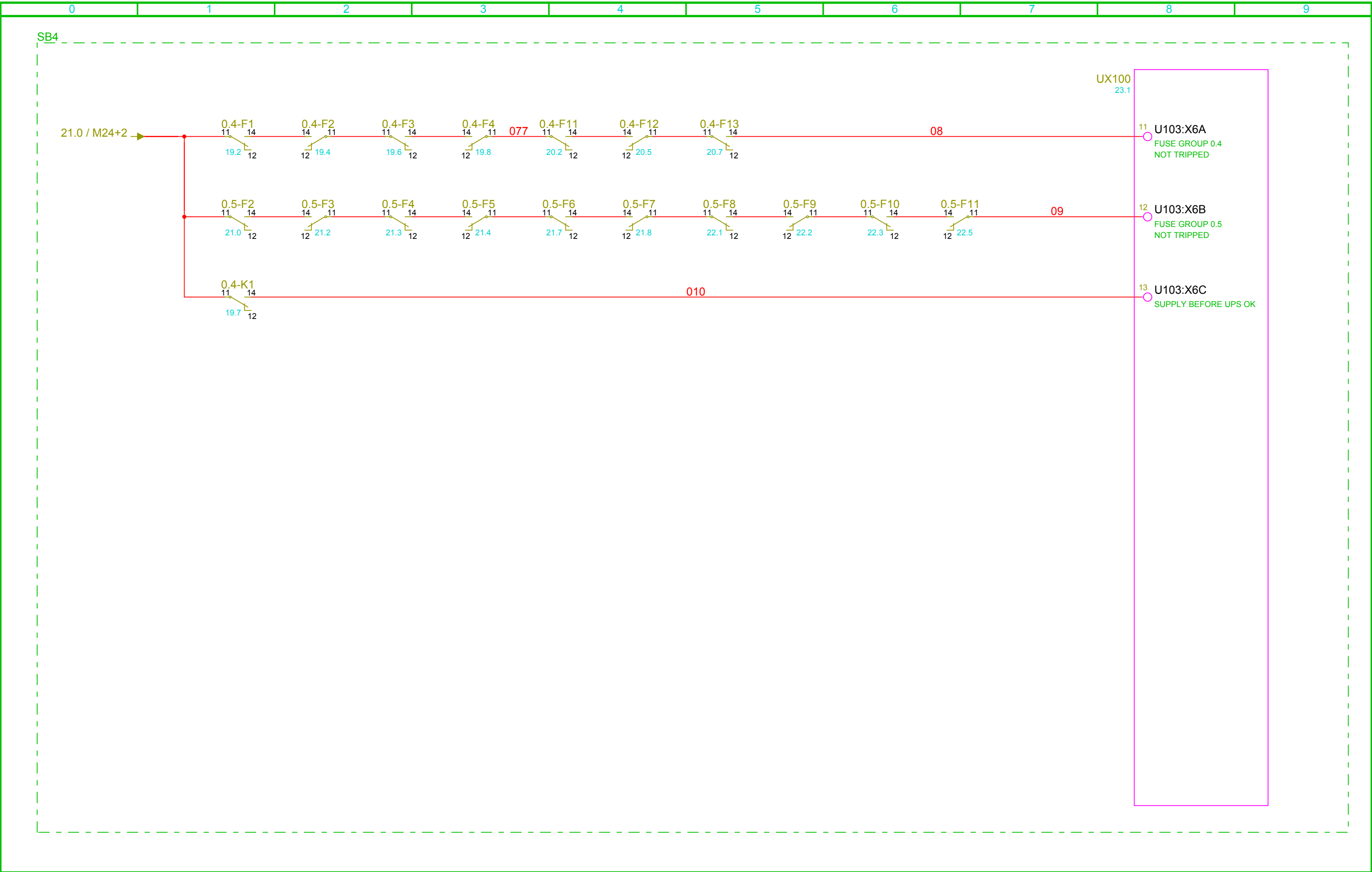
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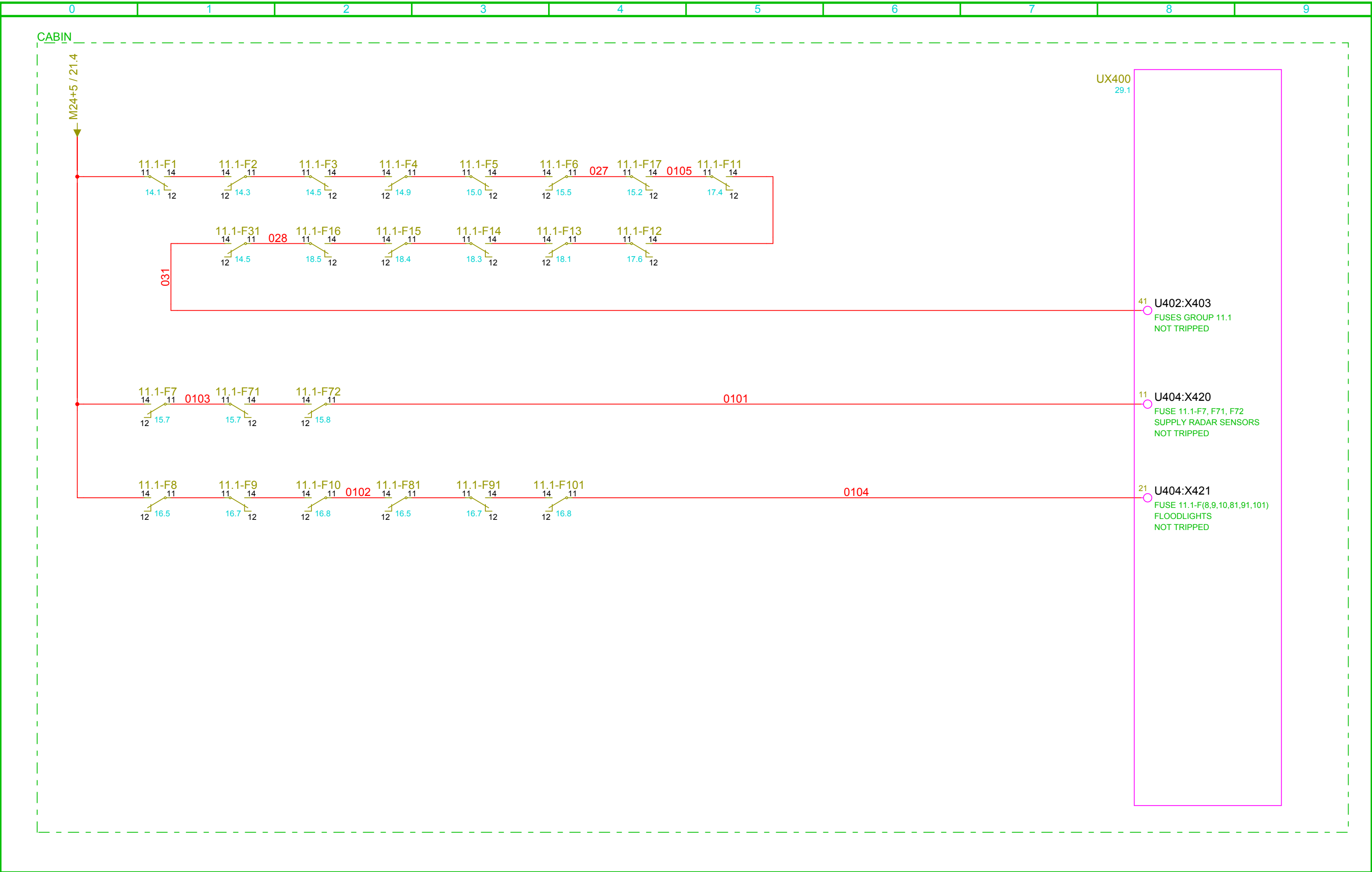






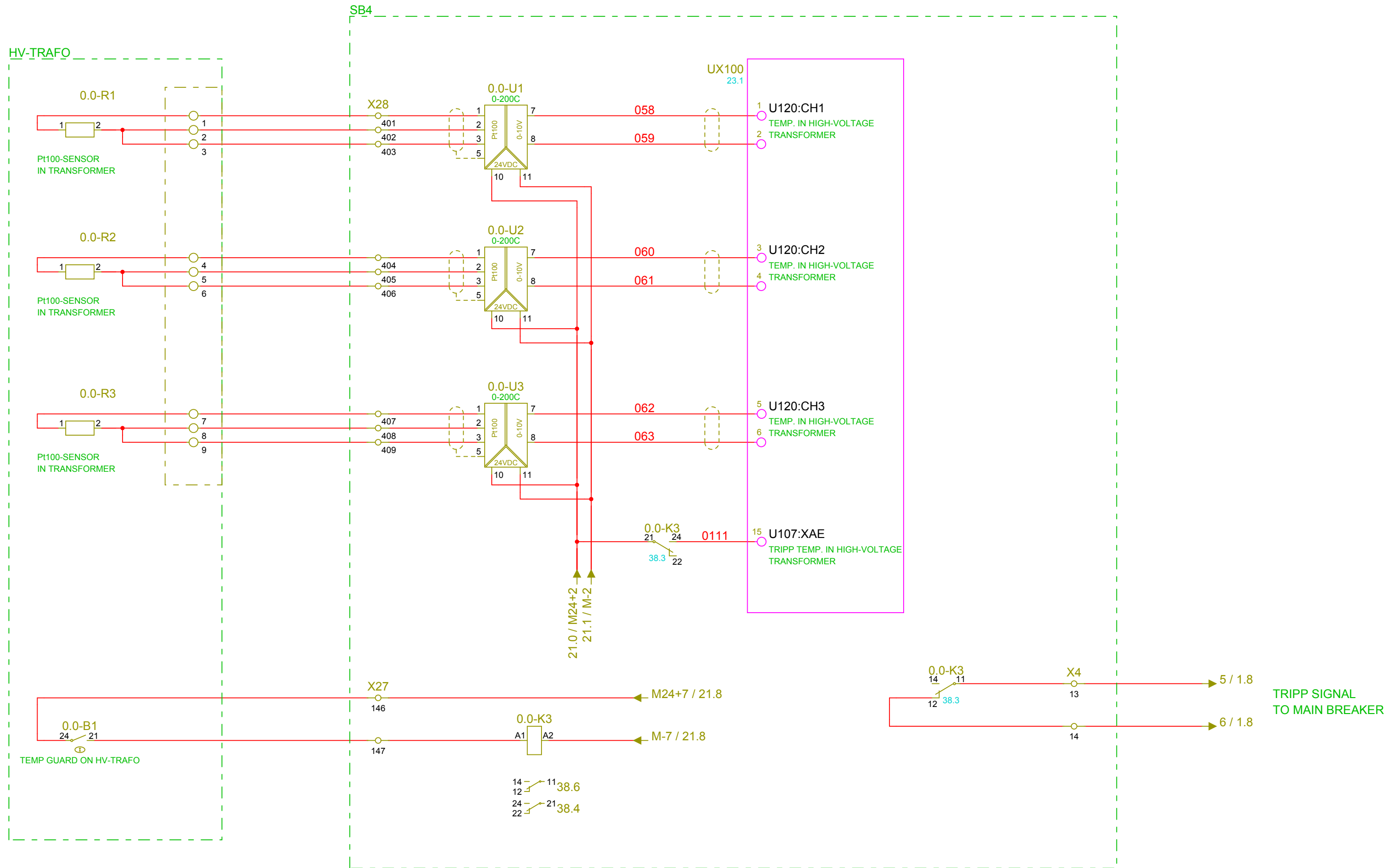


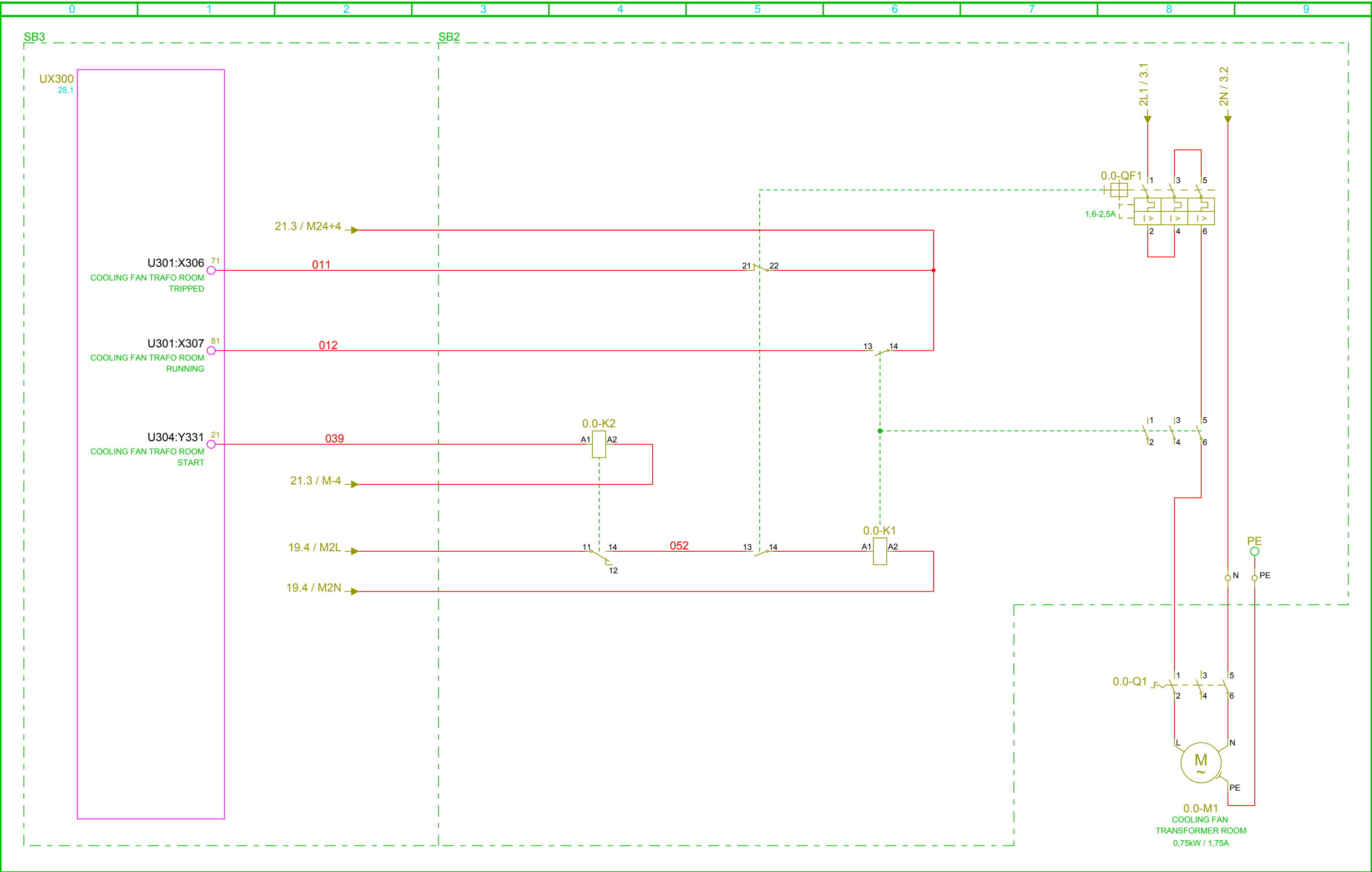


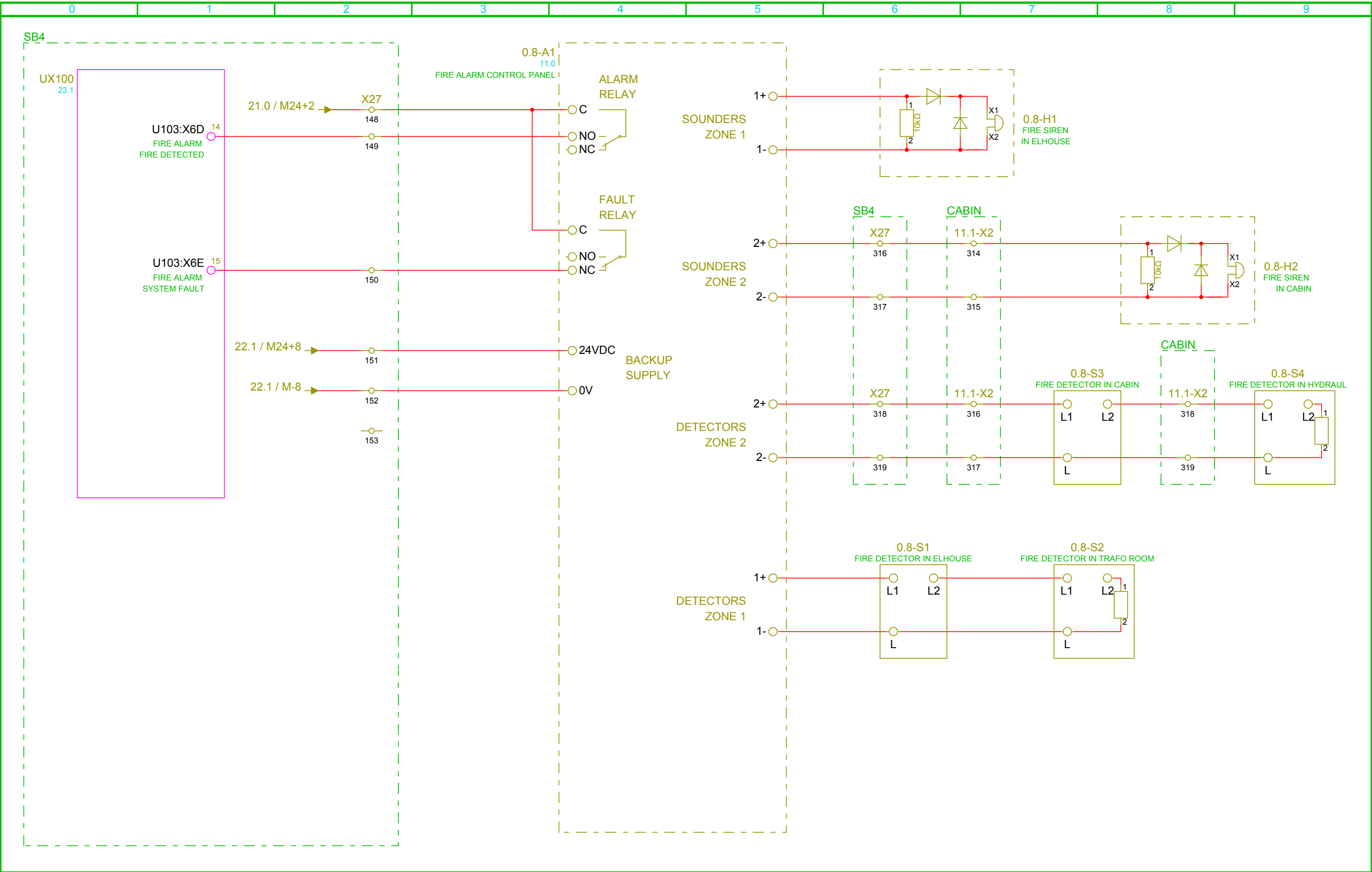


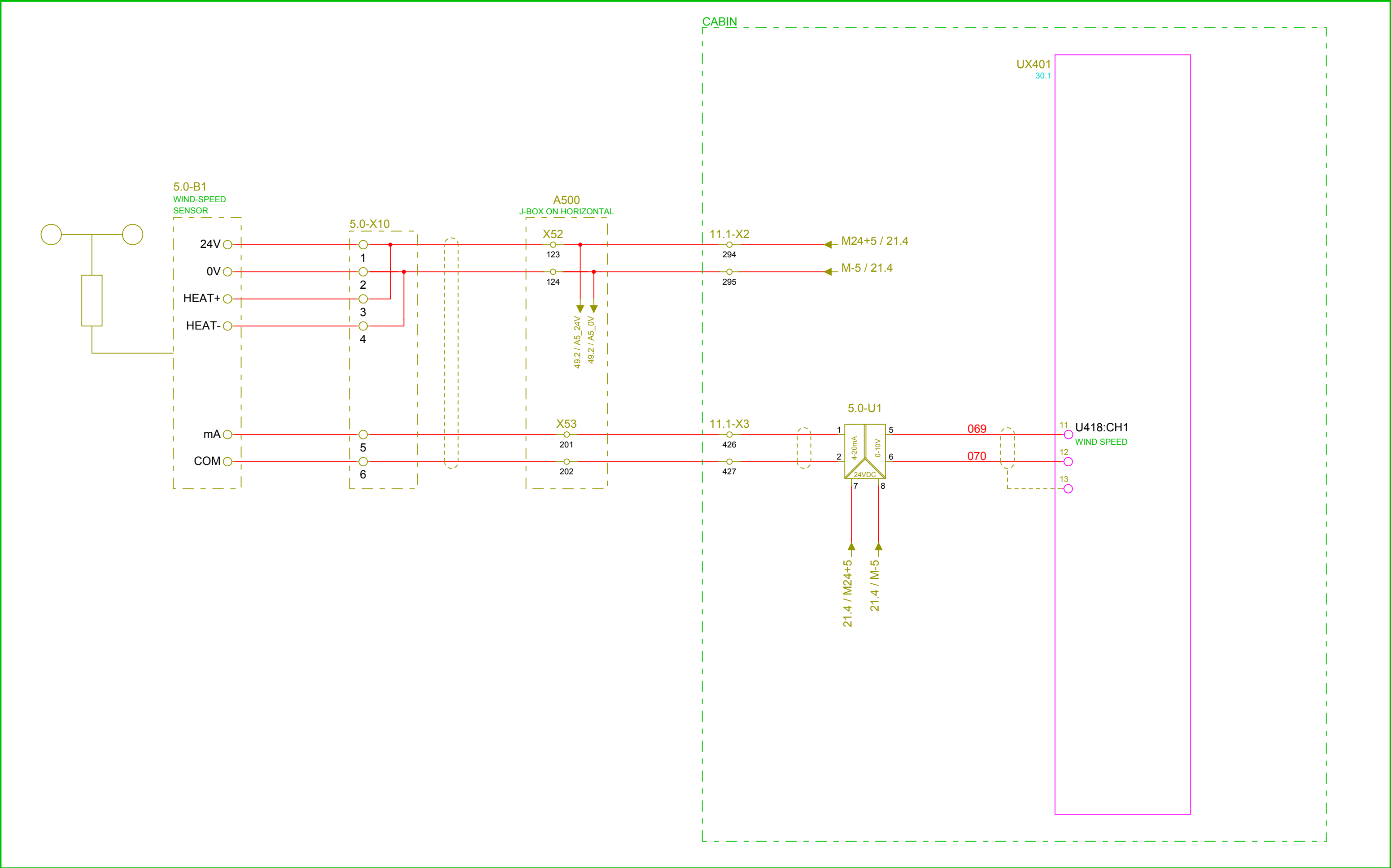
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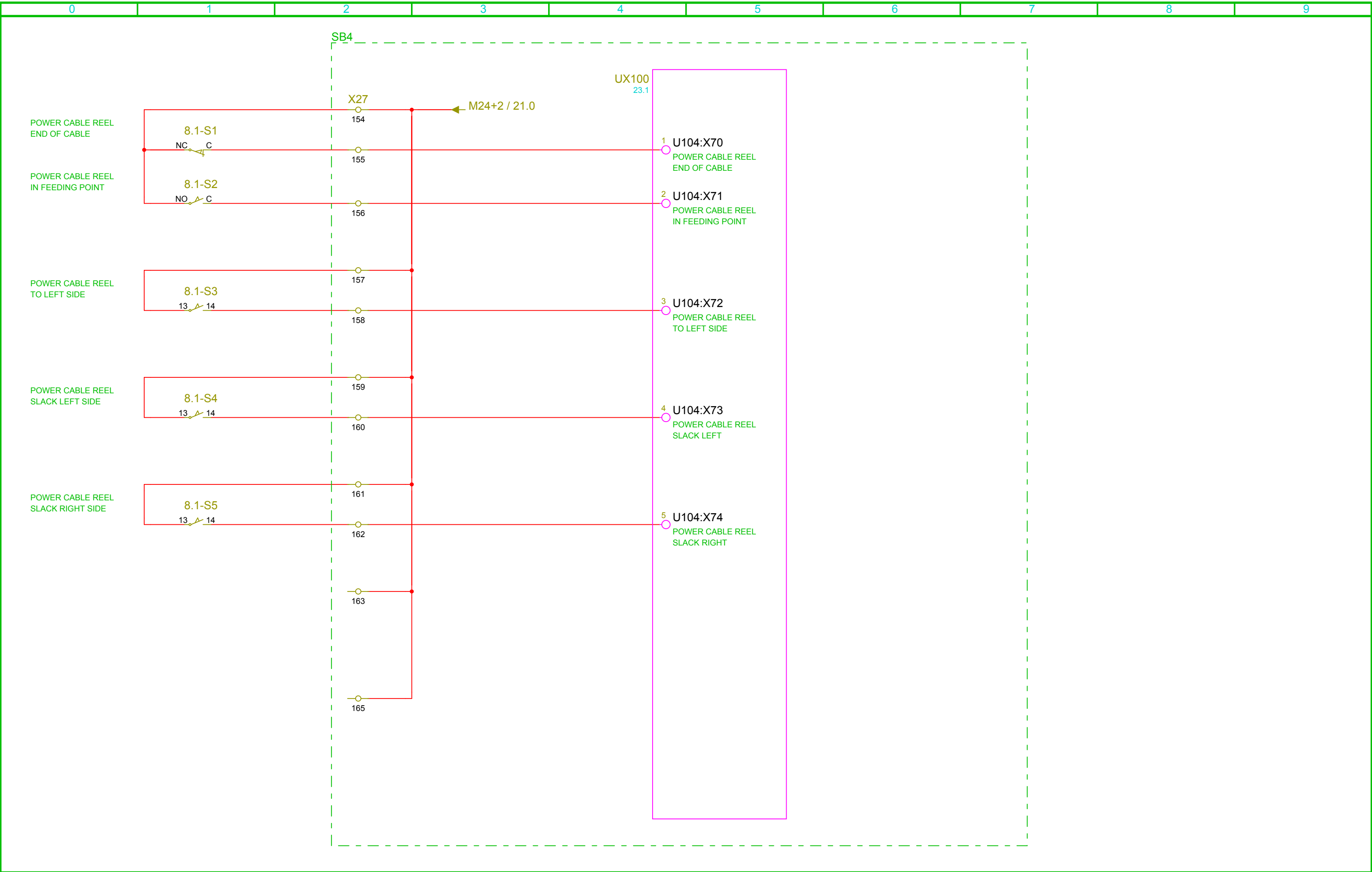
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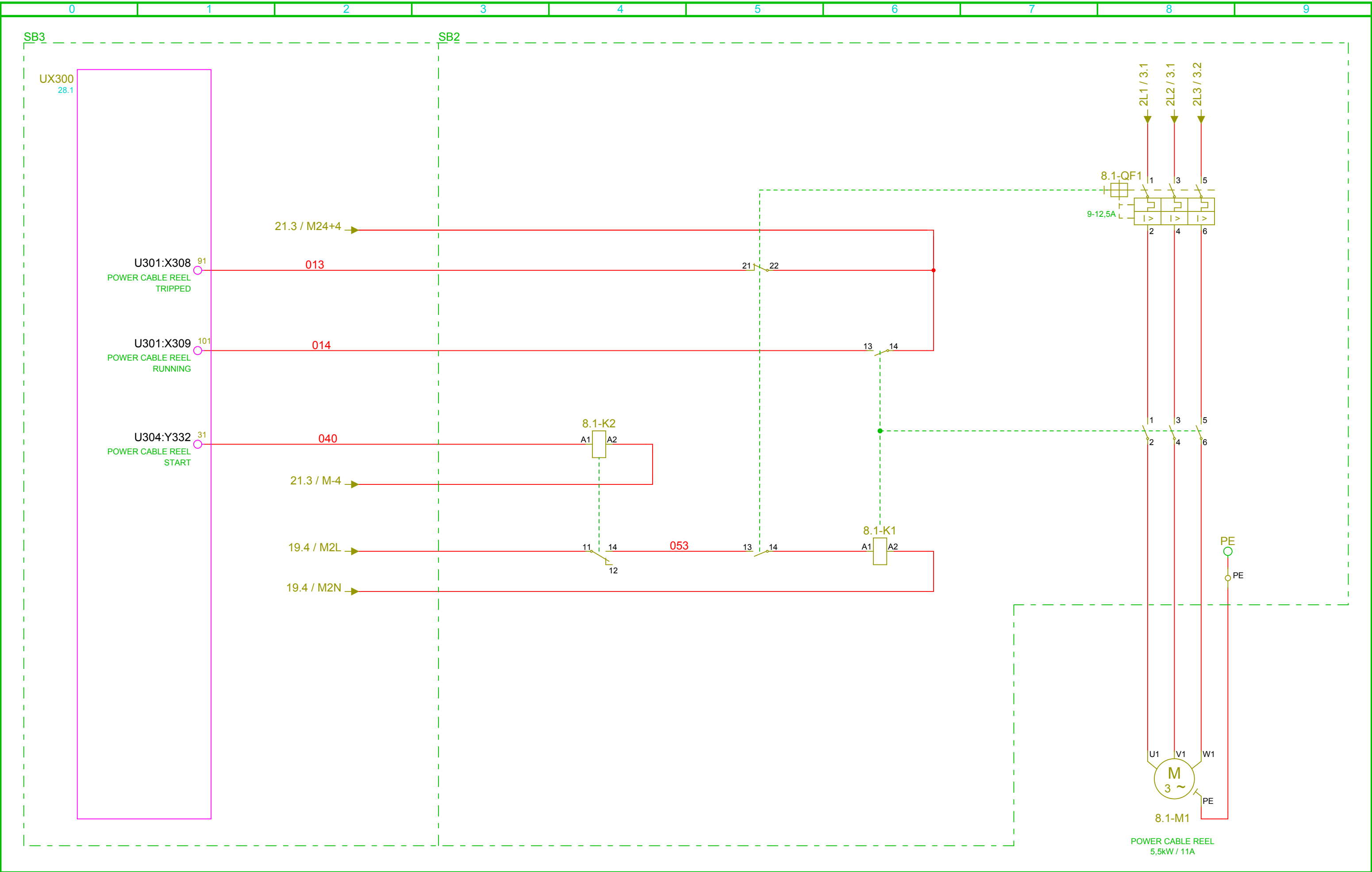


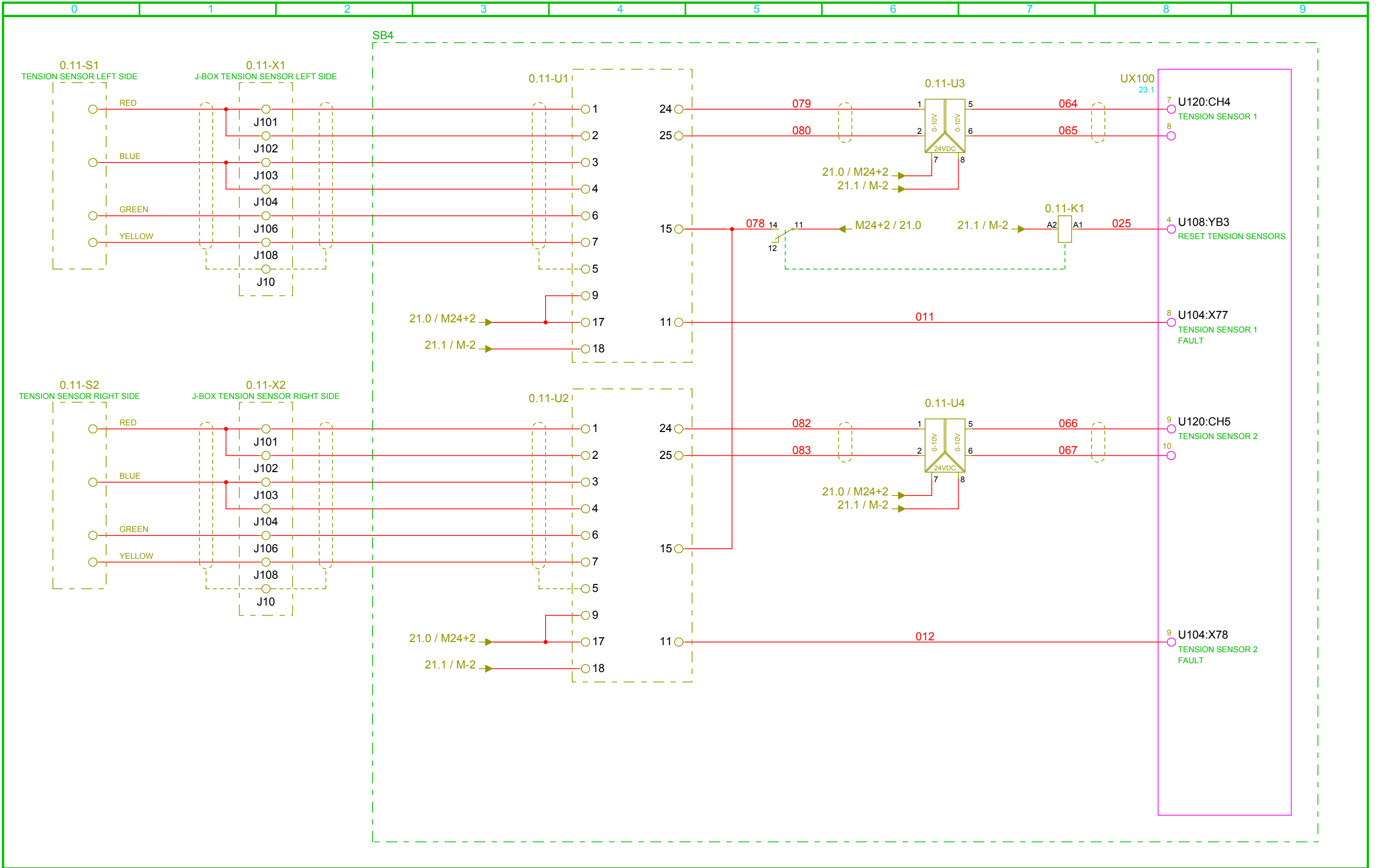


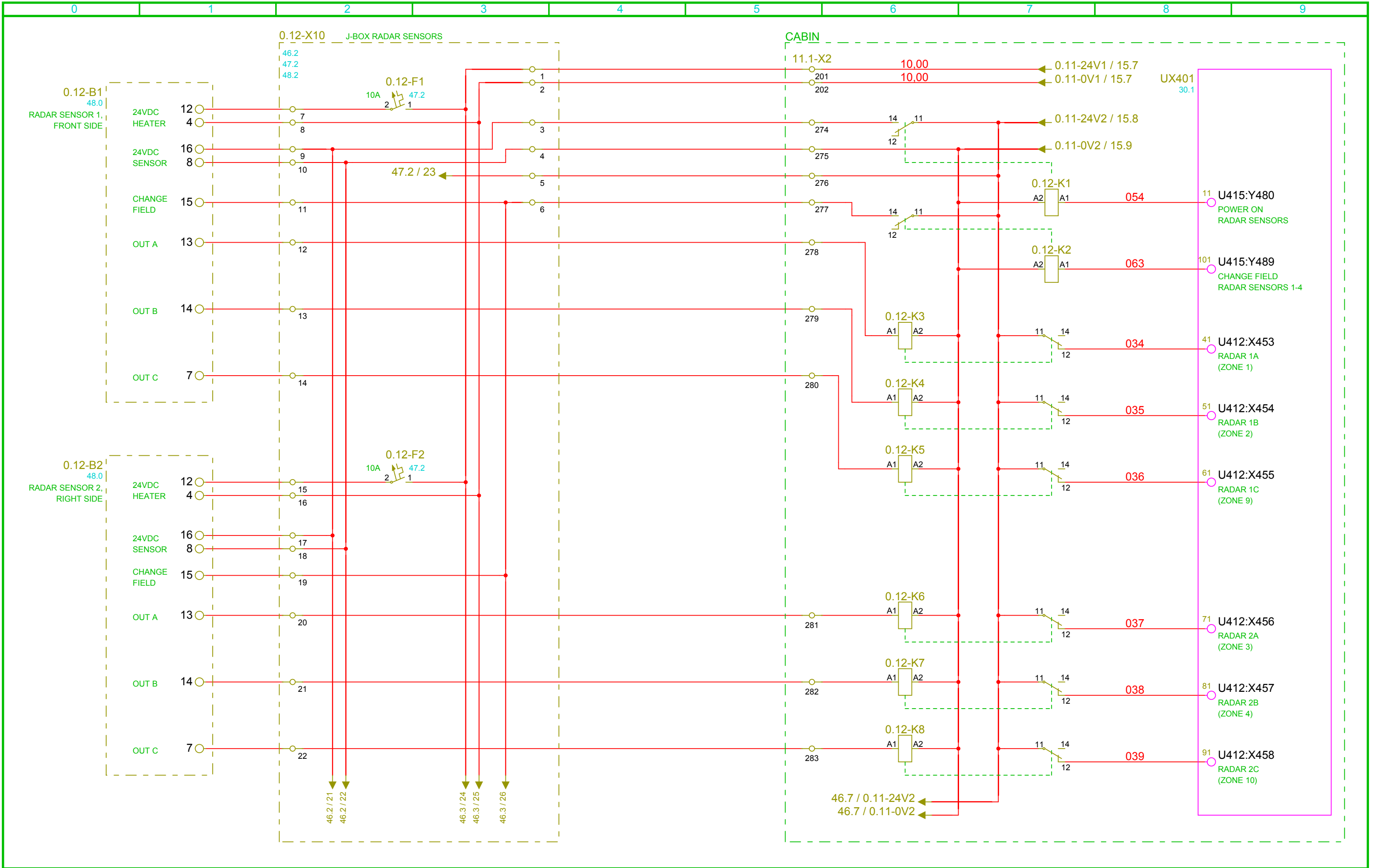


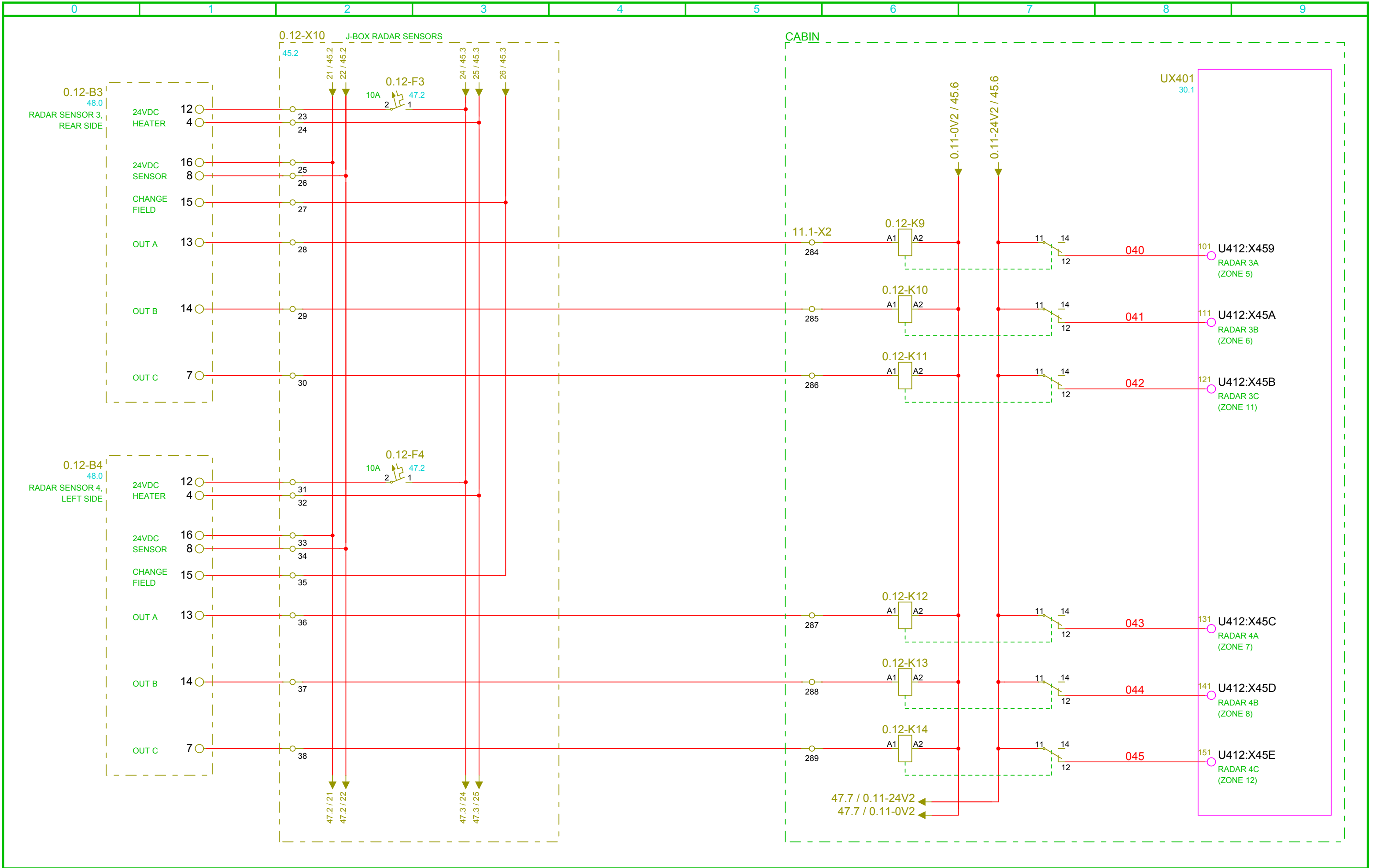


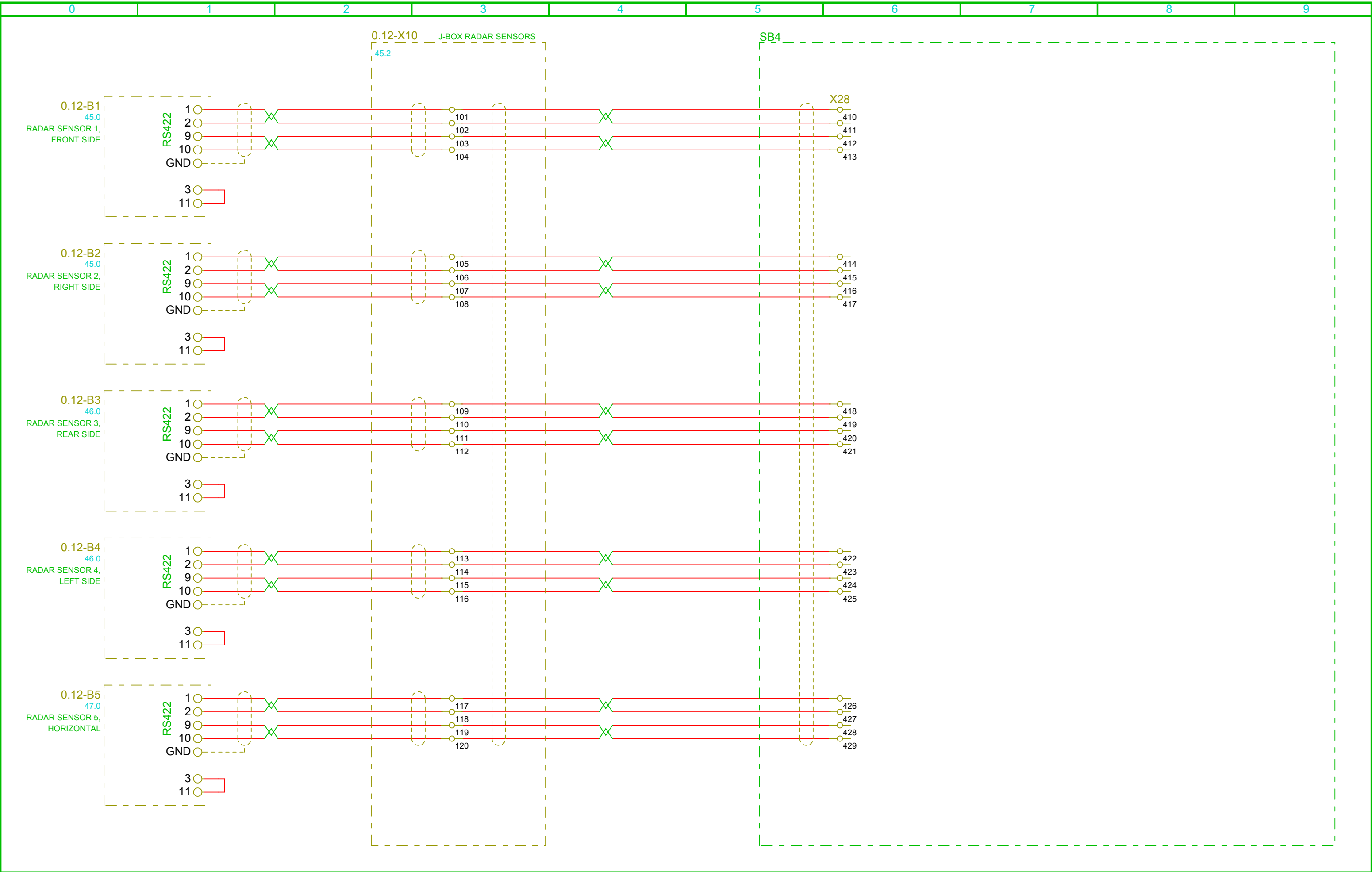


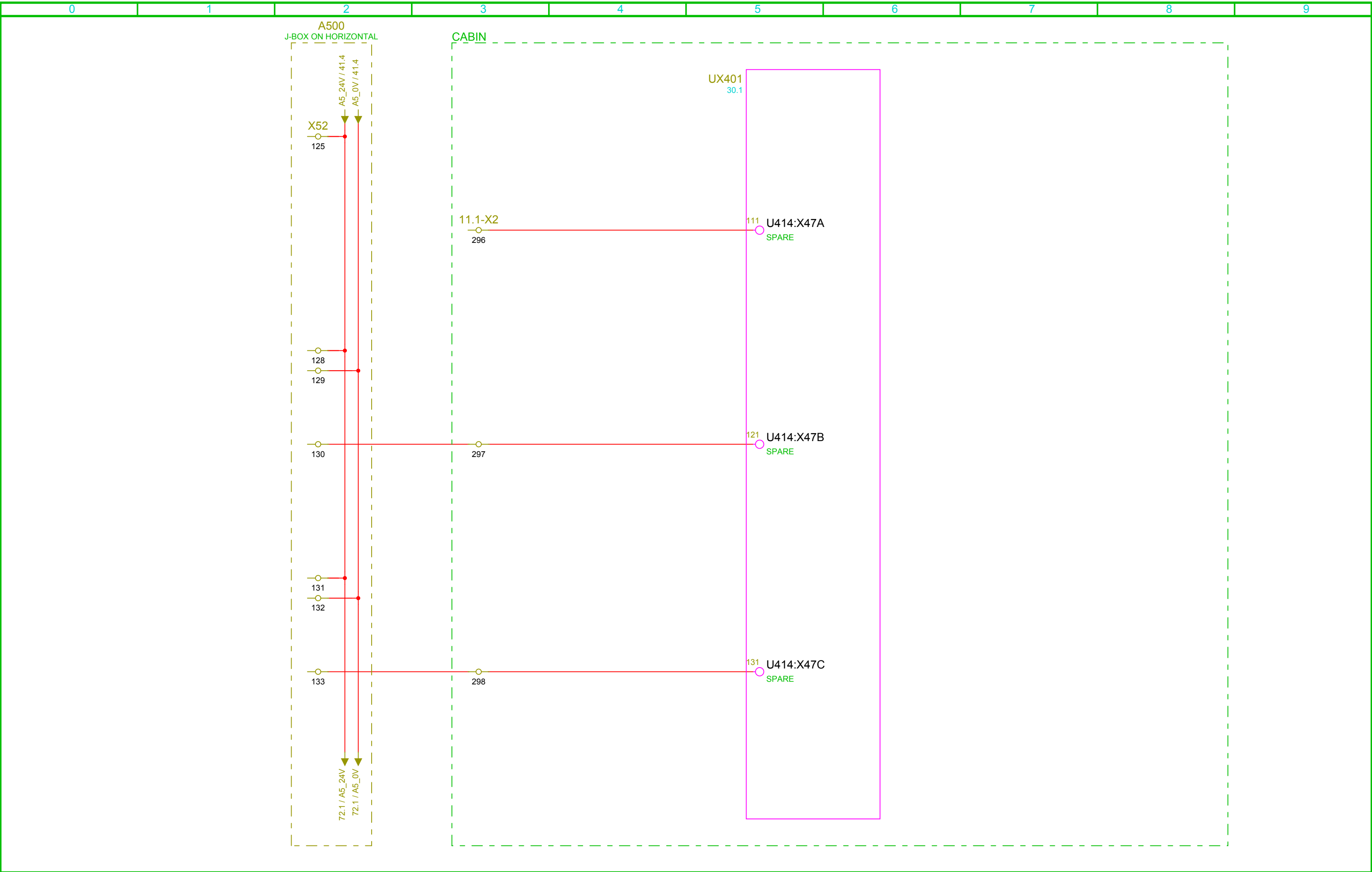


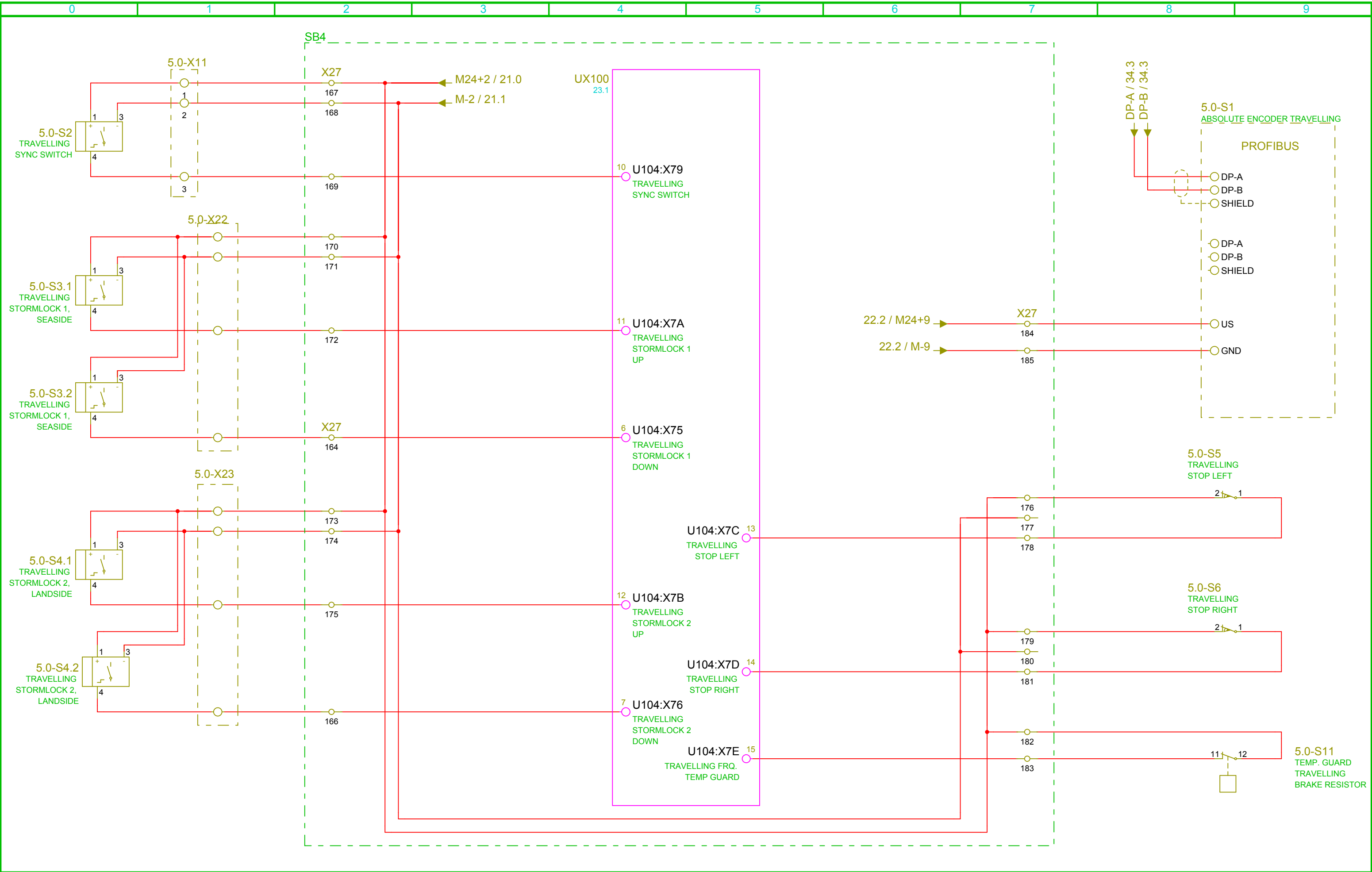


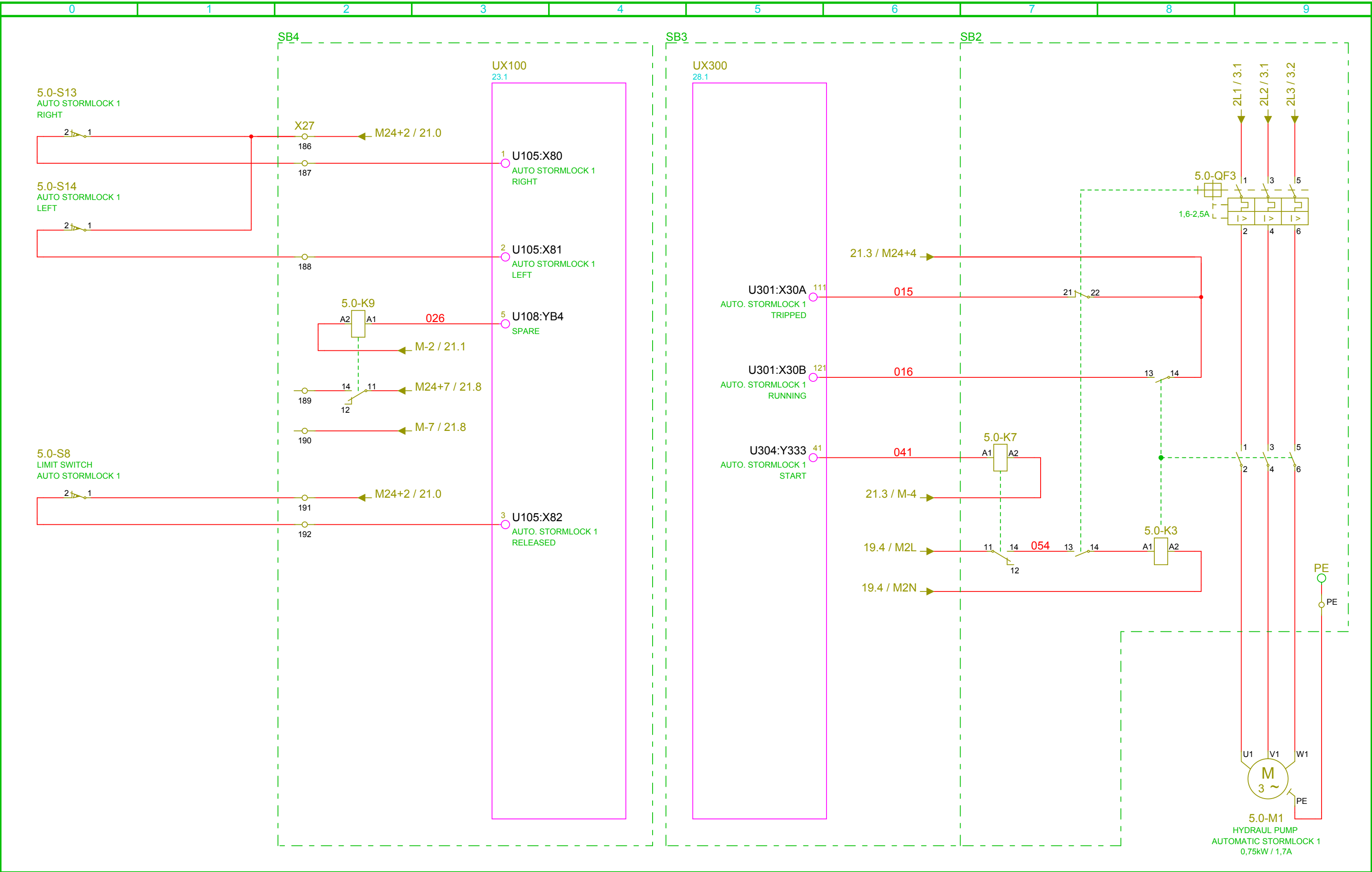


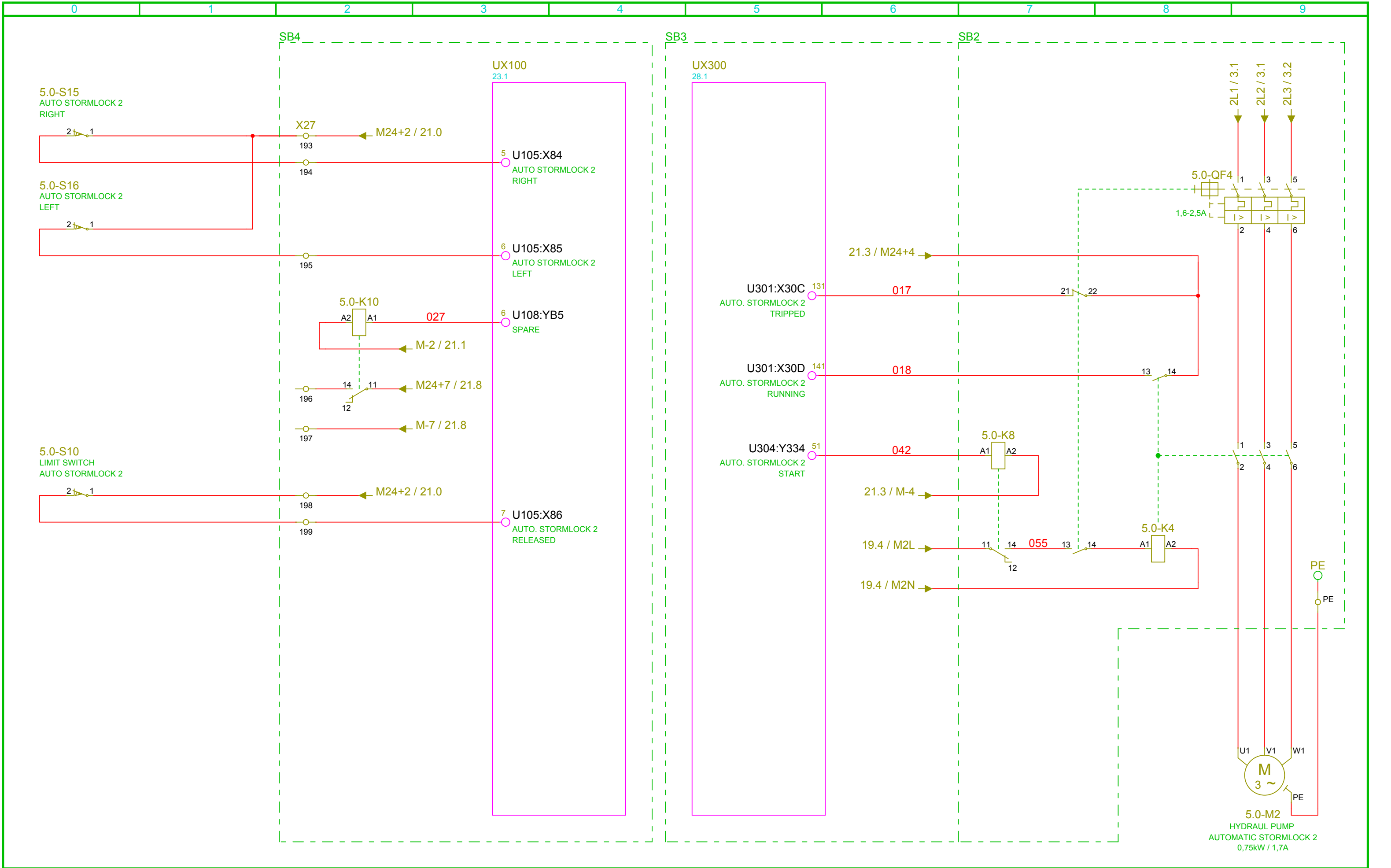


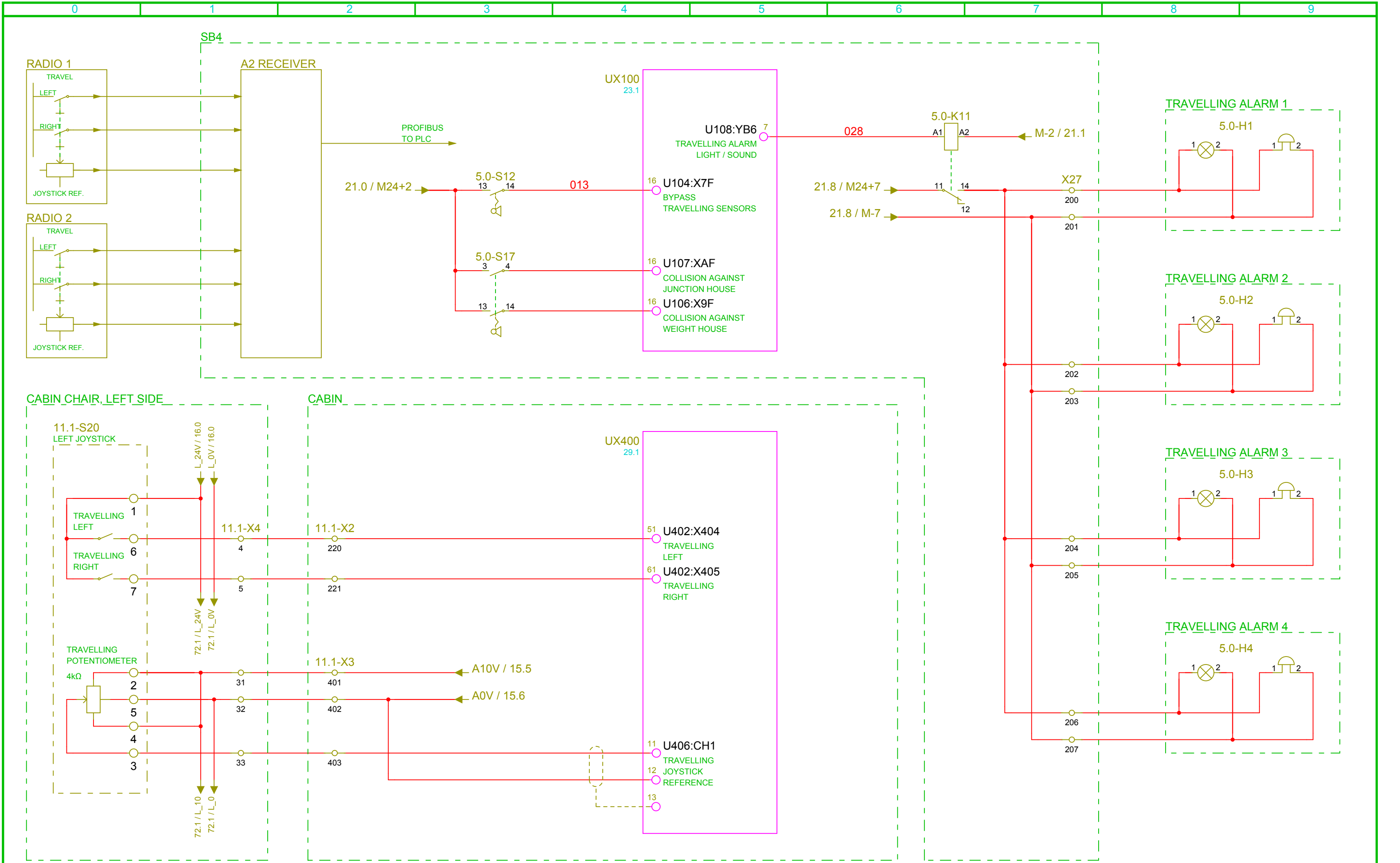












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SB1

UX200
26.1

U201:X203
TRAVELLING SUPPLY
TRIPPED

U201:X204
TRAVELLING SUPPLY
CONNECTED

U201:X205
TRAVELLING HEATING
TRIPPED

U201:X206
TRAVELLING HEATING
ON

U206:Y240
TRAVELLING SUPPLY
START

U206:Y241
TRAVELLING HEATING
START

21.2 / M24+3

5.0-X1

11

13

14

15

16

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17

18

12

19.2 / M1L

19.2 / M1N

21.2 / M-3

24

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08

09

10.00

10.00

5.0-F1

40A

5.0-X1

45

46

43

44

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5.0-QF1
125A

5.0-QF2
2,5-4A

5.0-K1

5.0-K2

5.0-K5

5.0-K6

5.0-T1
TRANSFORMER
415/30V
1,2kVA

415V SUPPLY
FREQUENCY CONVERTER

30V SUPPLY
MOTOR HEATING



AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CIRCUIT DIAGRAM

SUPPLY FREQUENCY CONVERTER TRAVELLING

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.

FILE NAME

A524.DB

DATE
2006-10-16

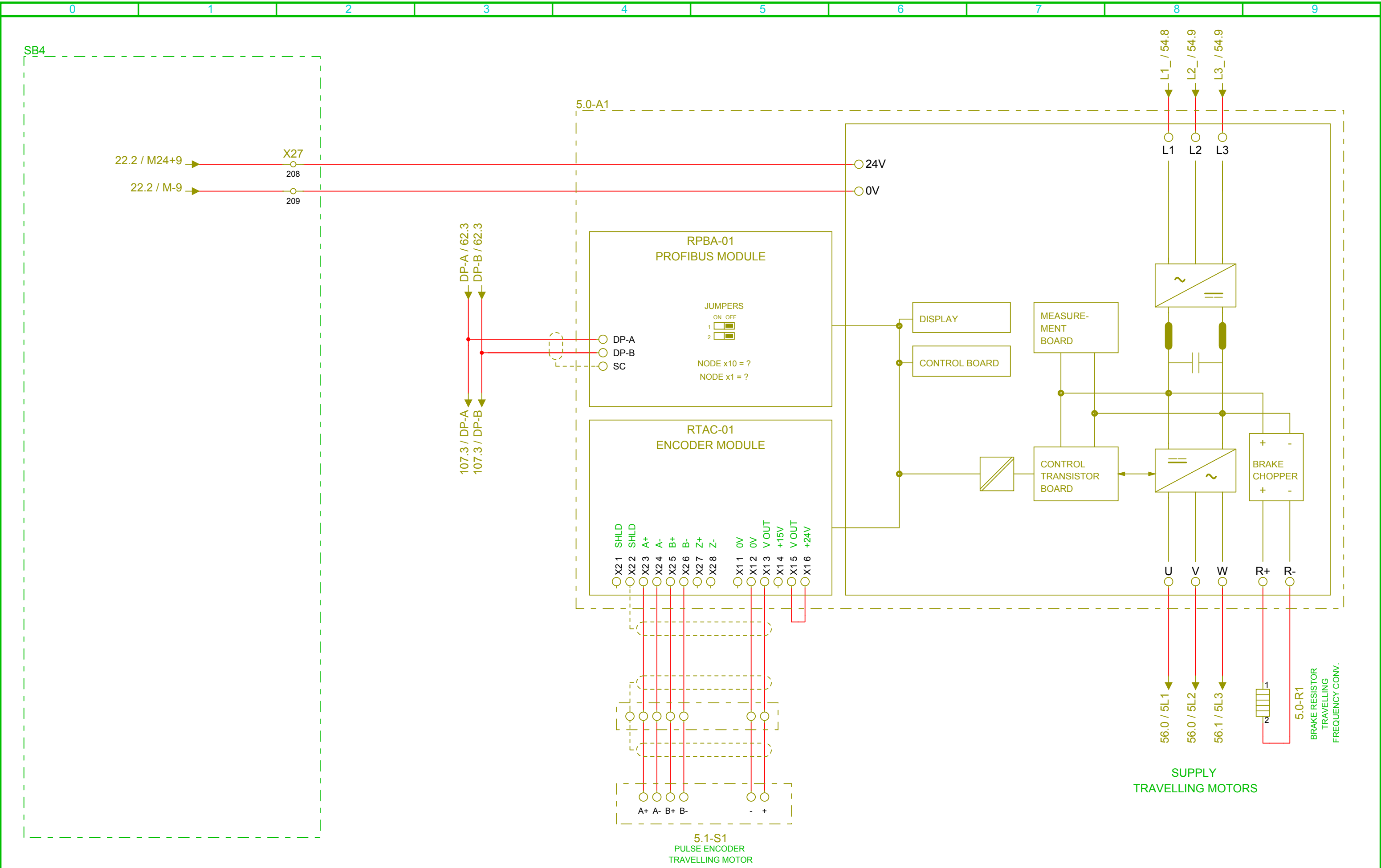
DRAWING NO.

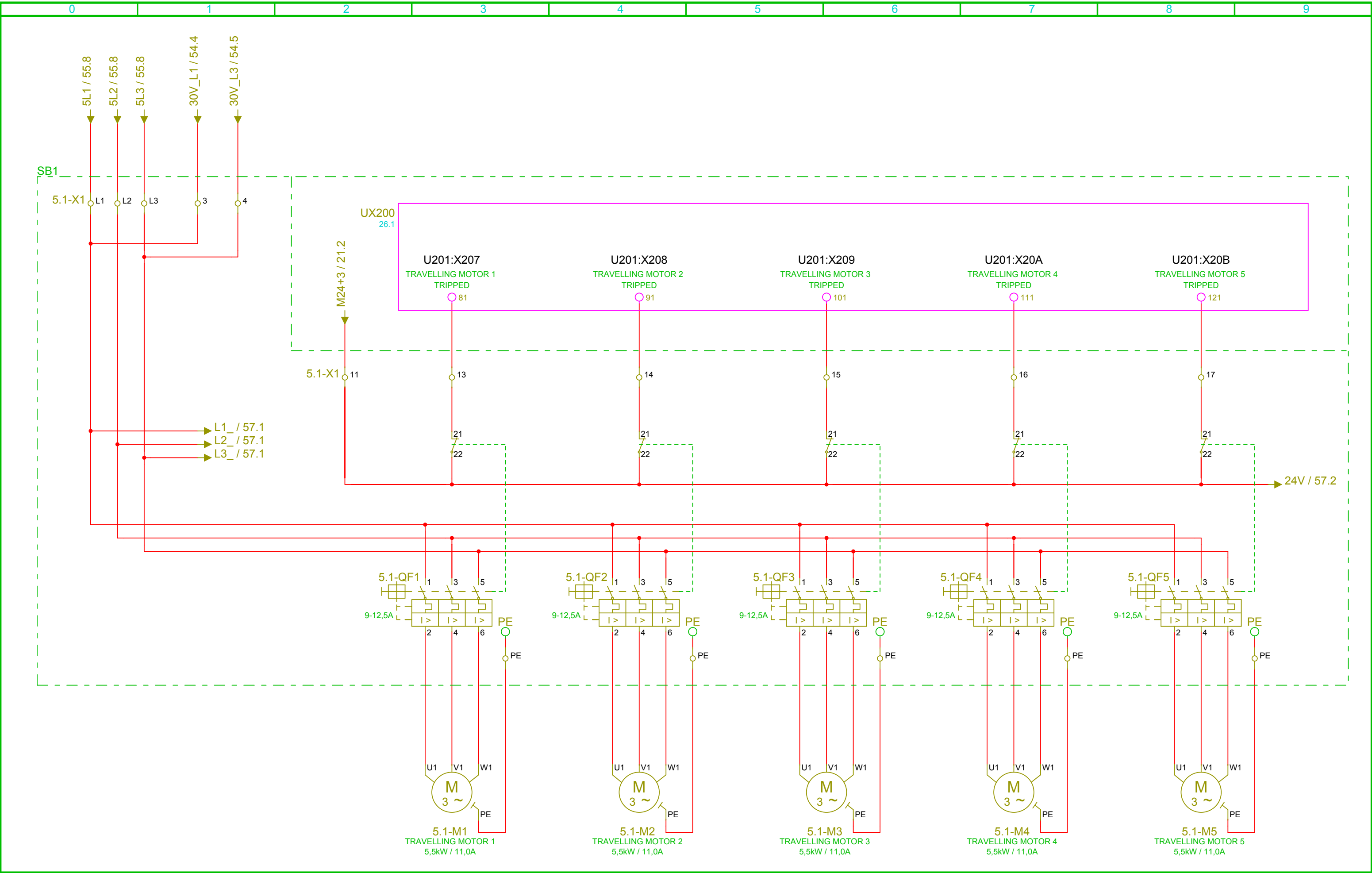
SHEET

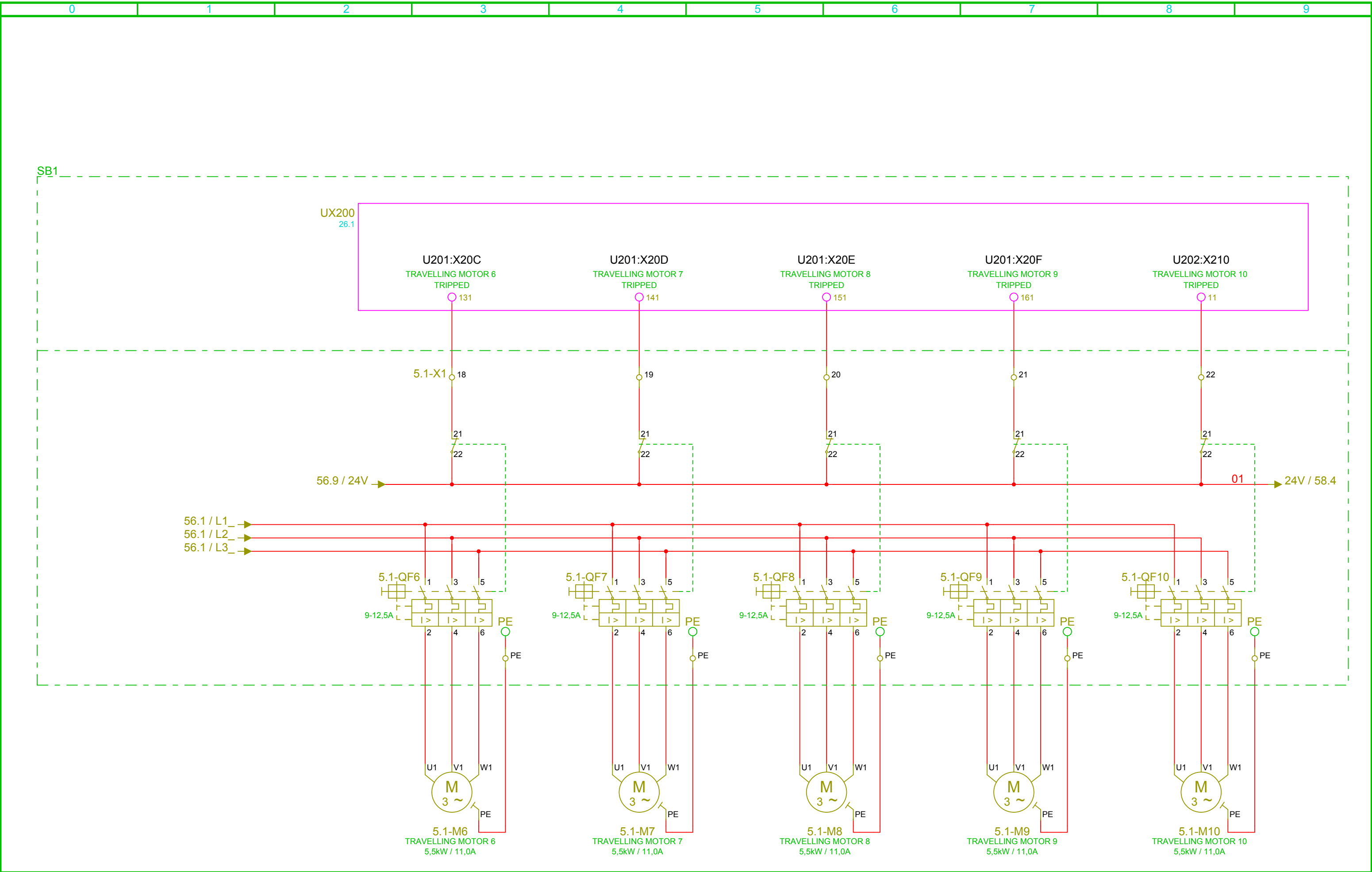
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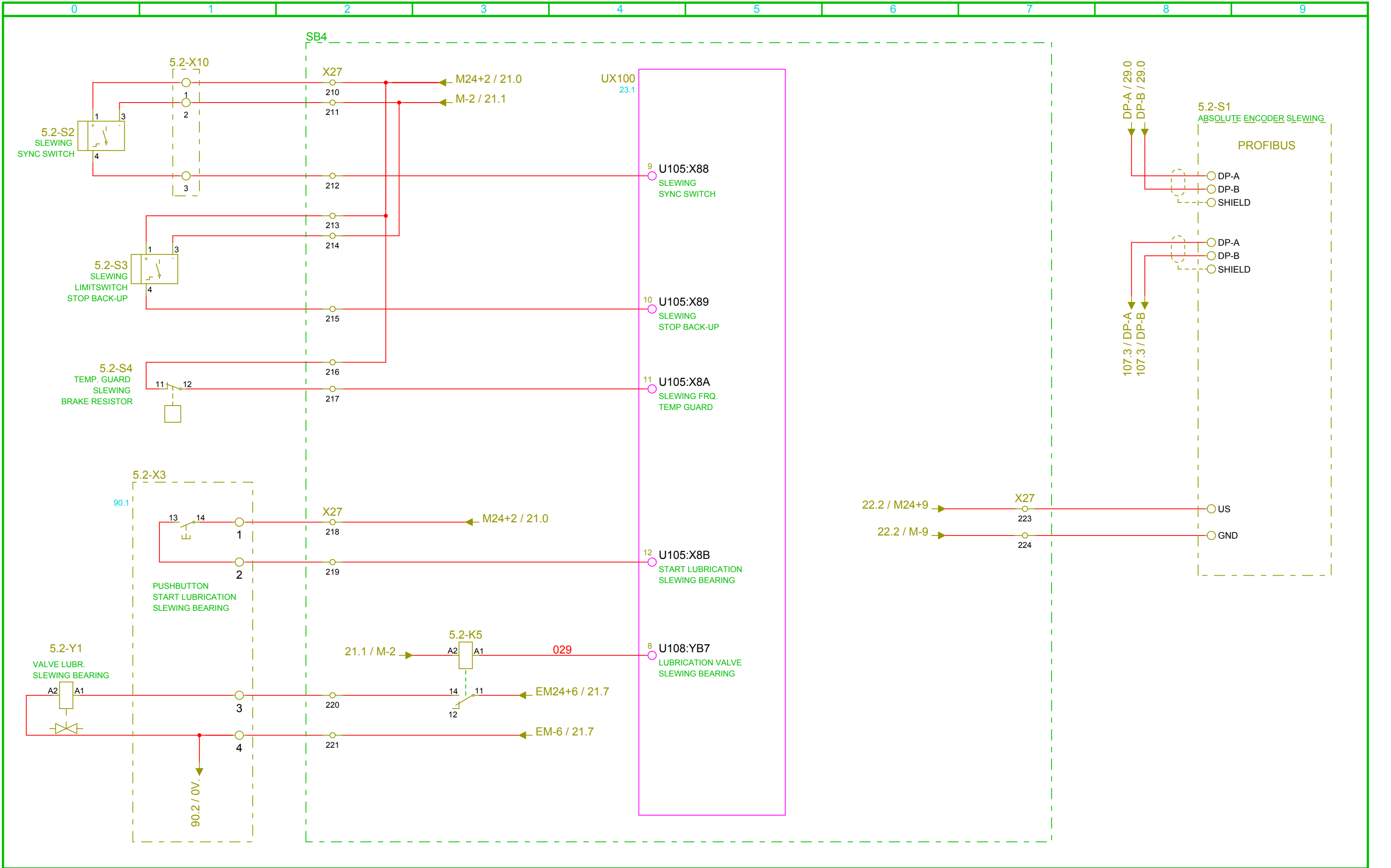
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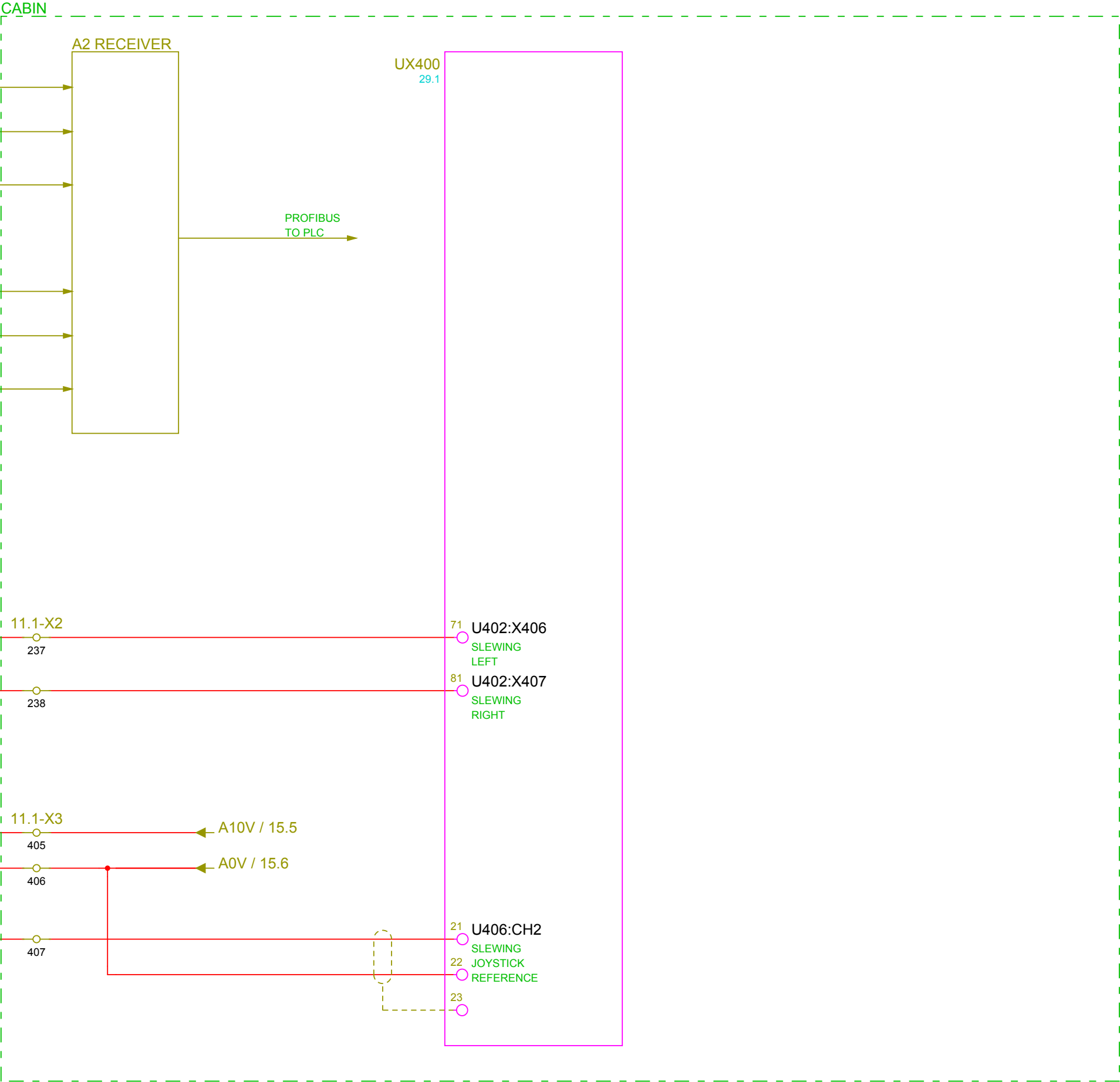
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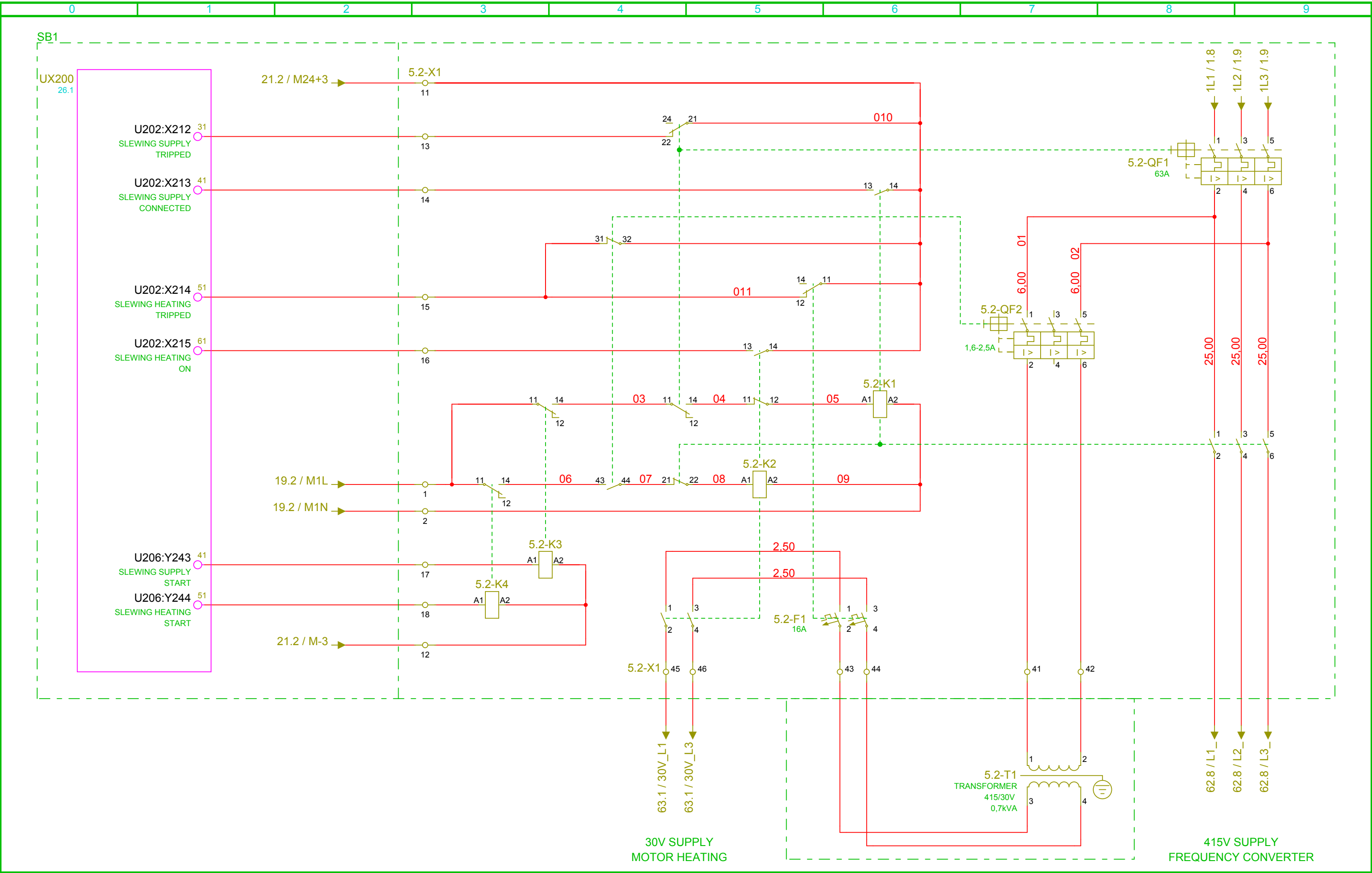


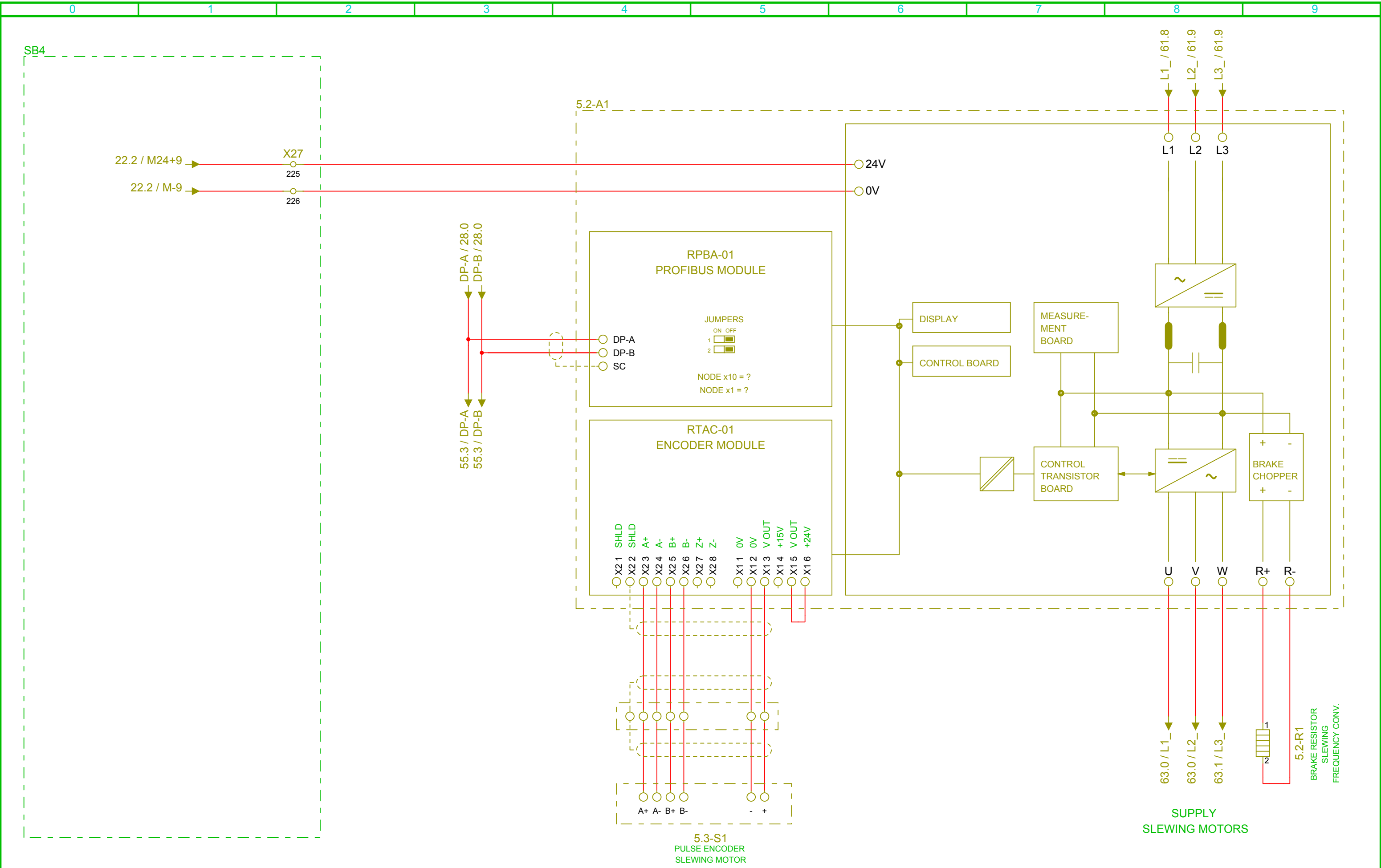


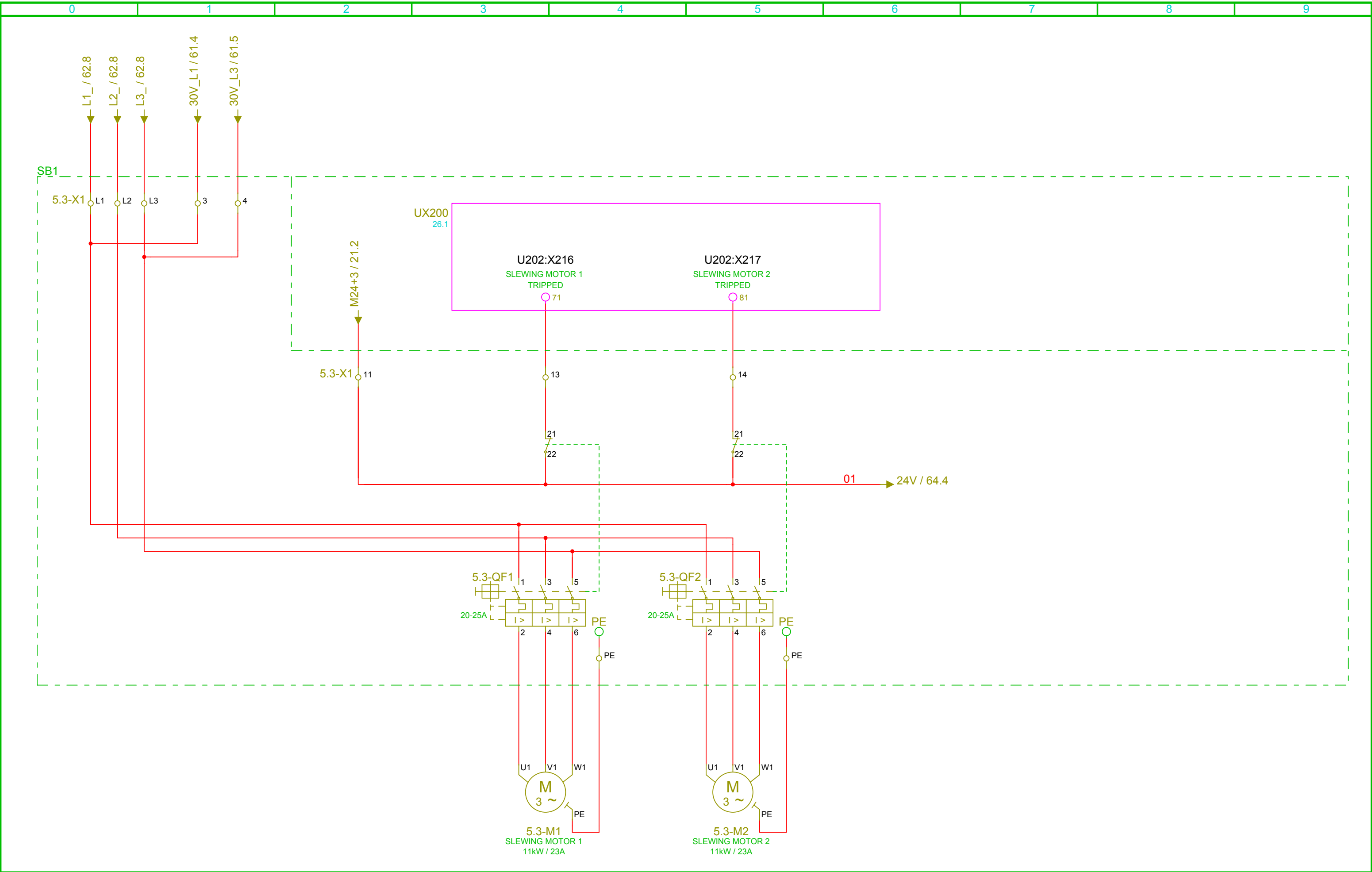


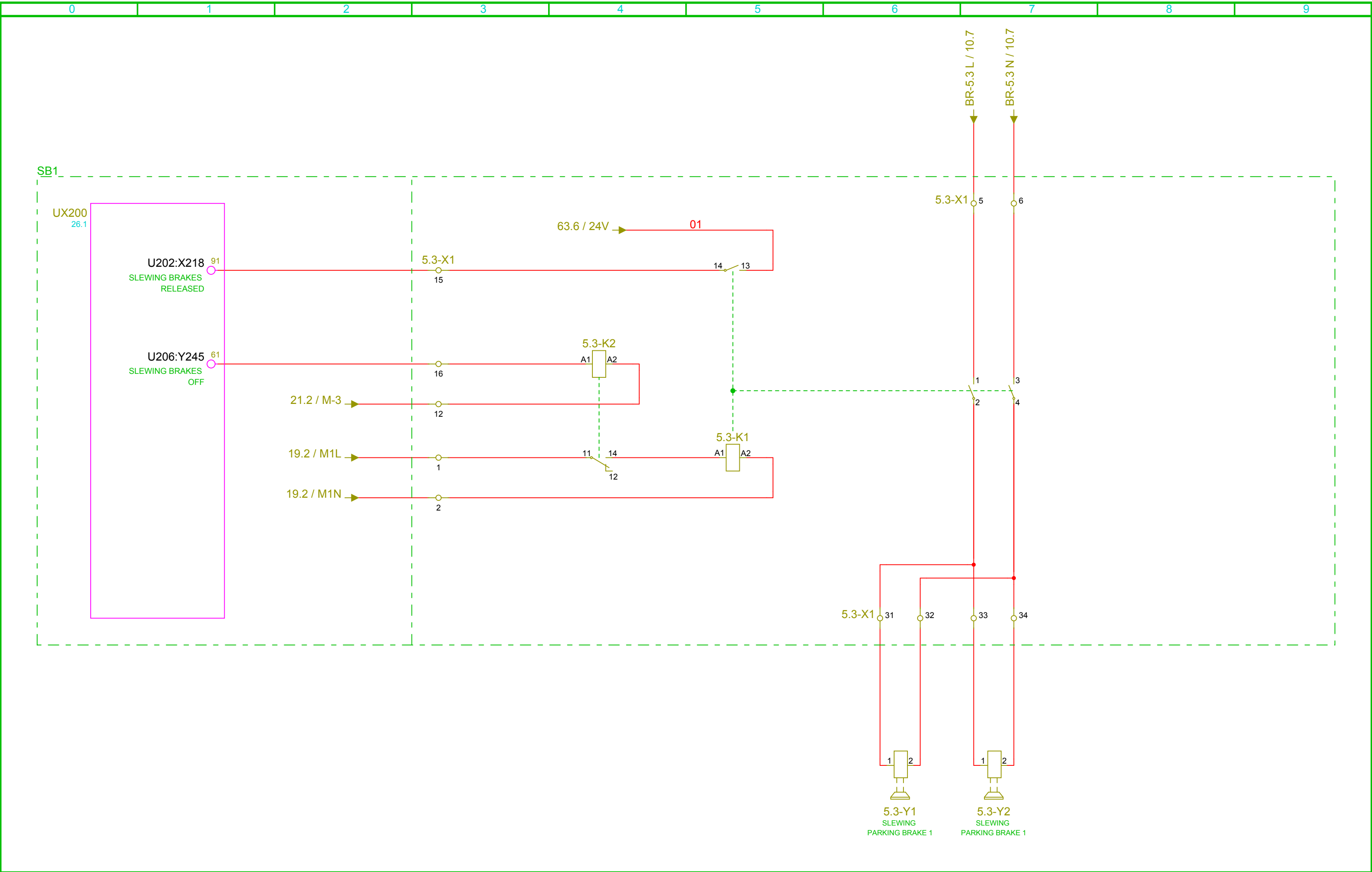










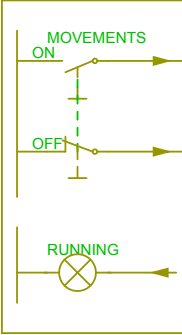


AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN			AS BUILT			BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN			PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT CIRCUIT DIAGRAM SLEWING PARKING BRAKES			DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
CHECKED			PROJECT NO. 202A6086			DRAWING NO. A524 - E			FILE NAME A524.DB			SHEET 64			
APPR.			SHEET 64			CONT. SHEET 65									

REV.	MODIFICATION	DATE

CABIN

RADIO 1



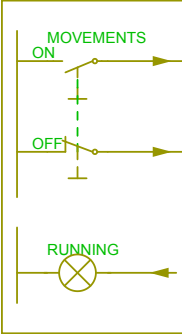
A2 RECEIVER



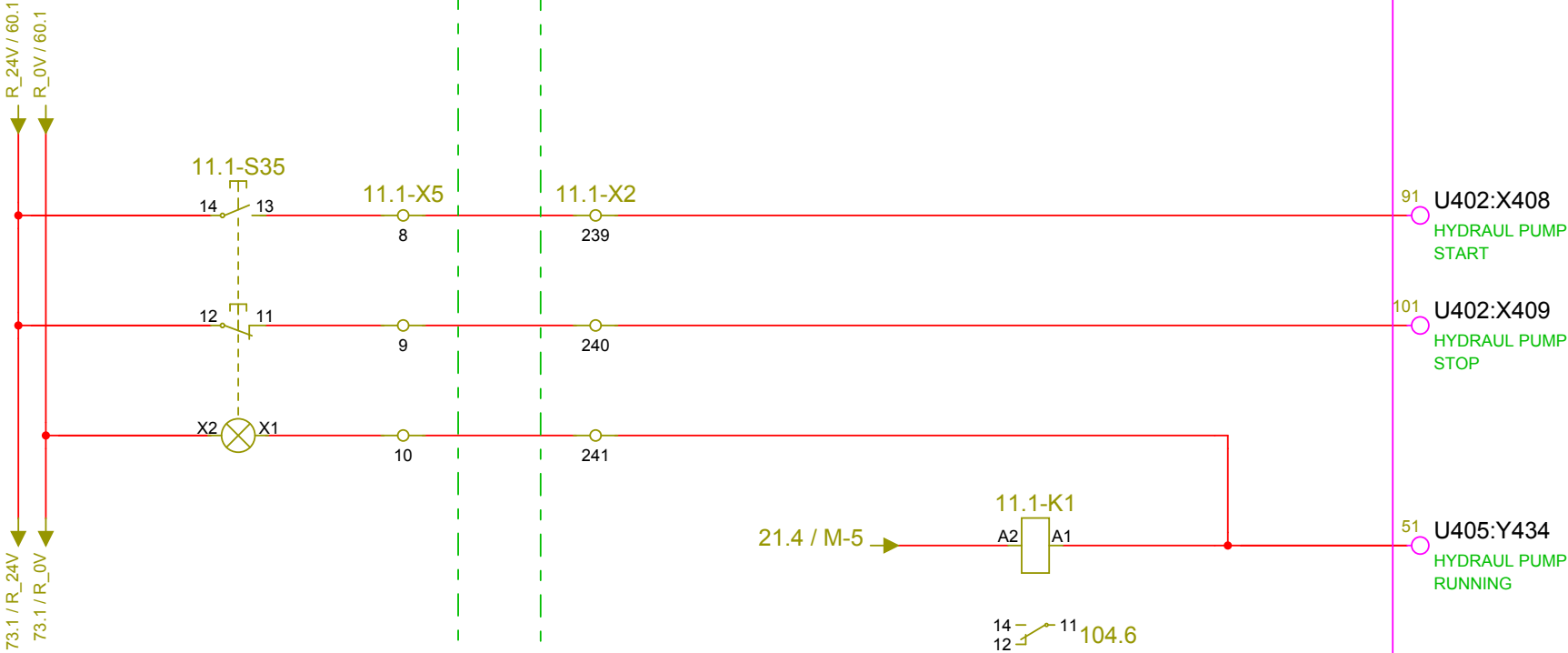
UX400
29.1

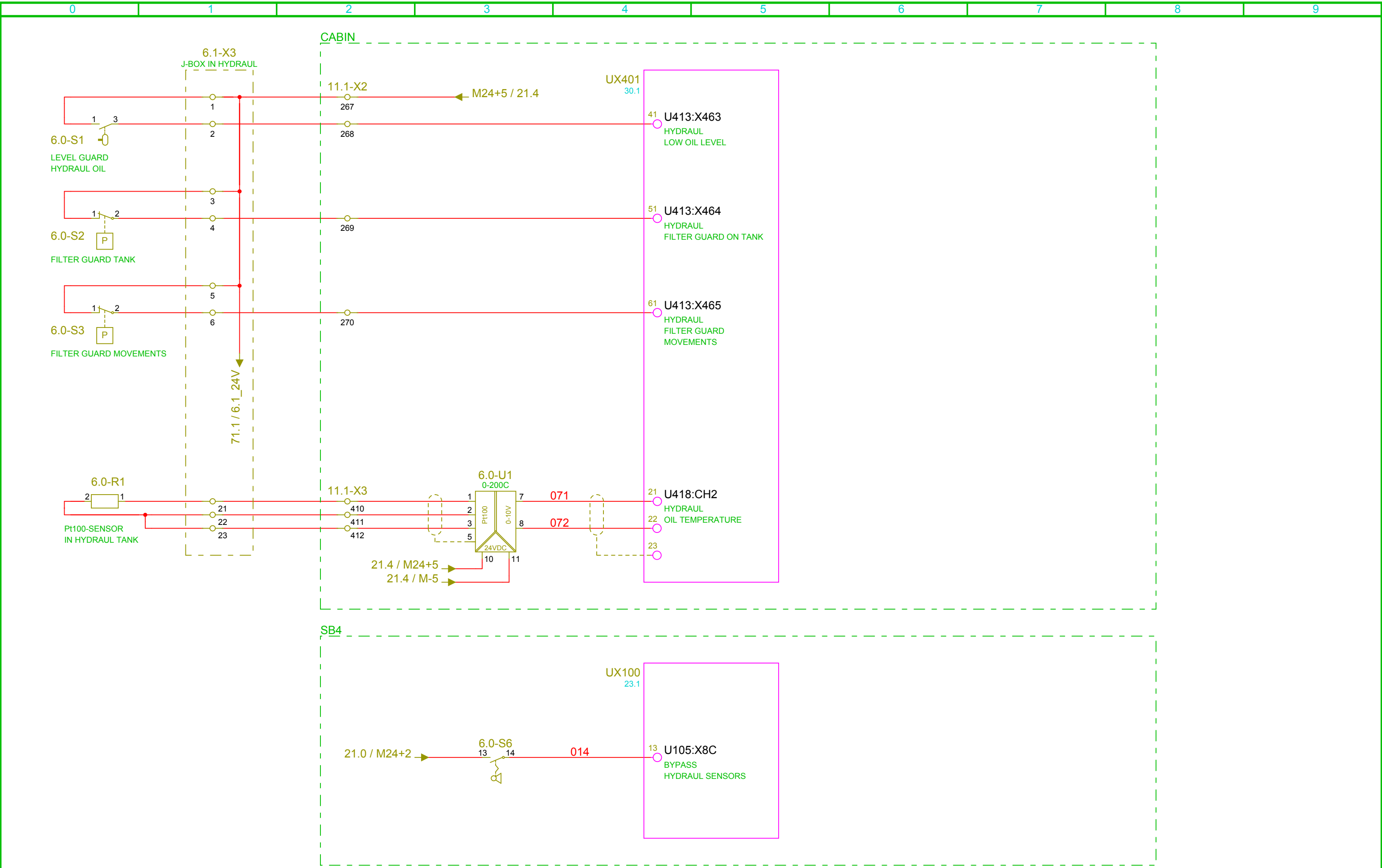
PROFIBUS
TO PLC

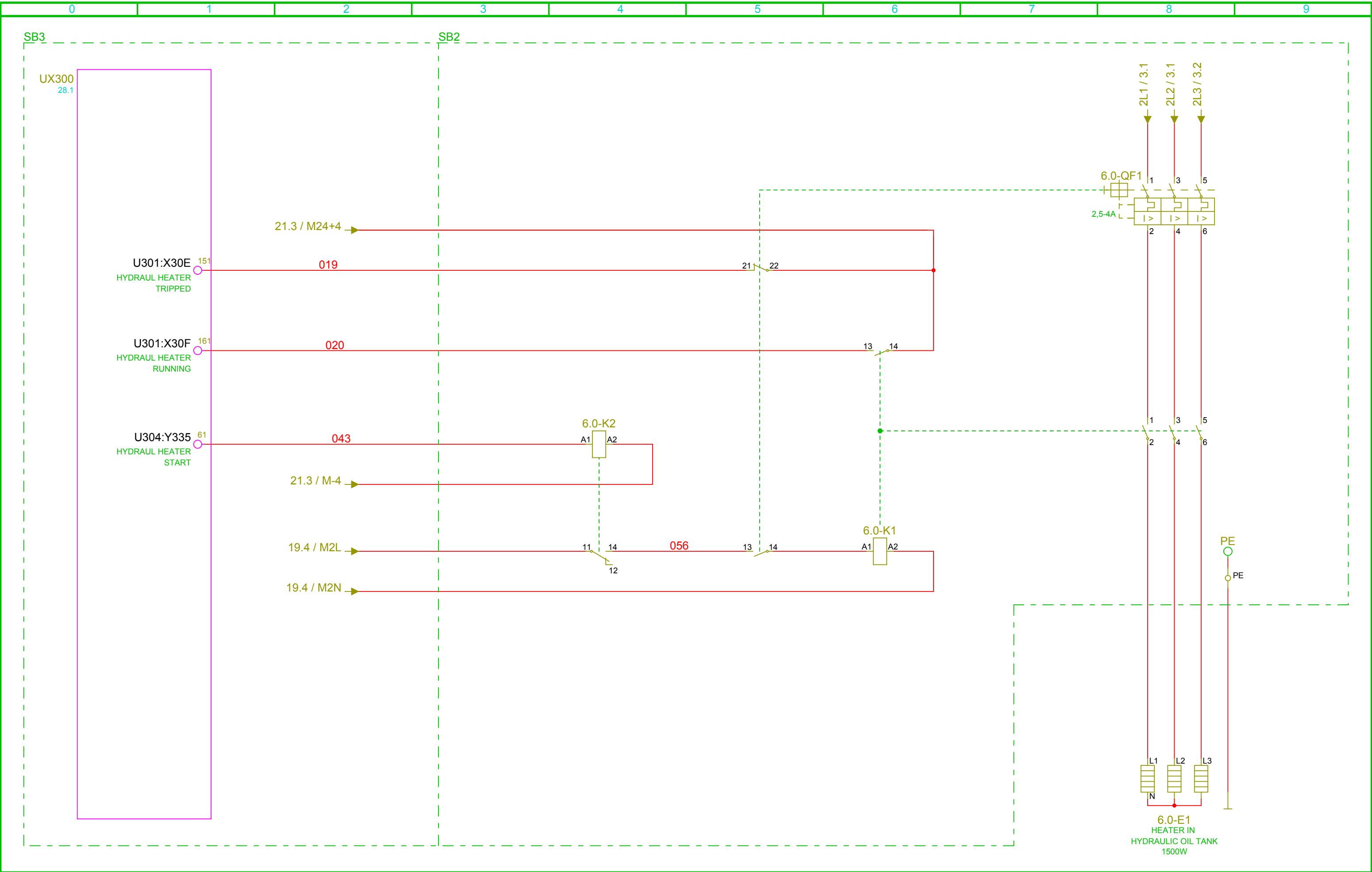
RADIO 2



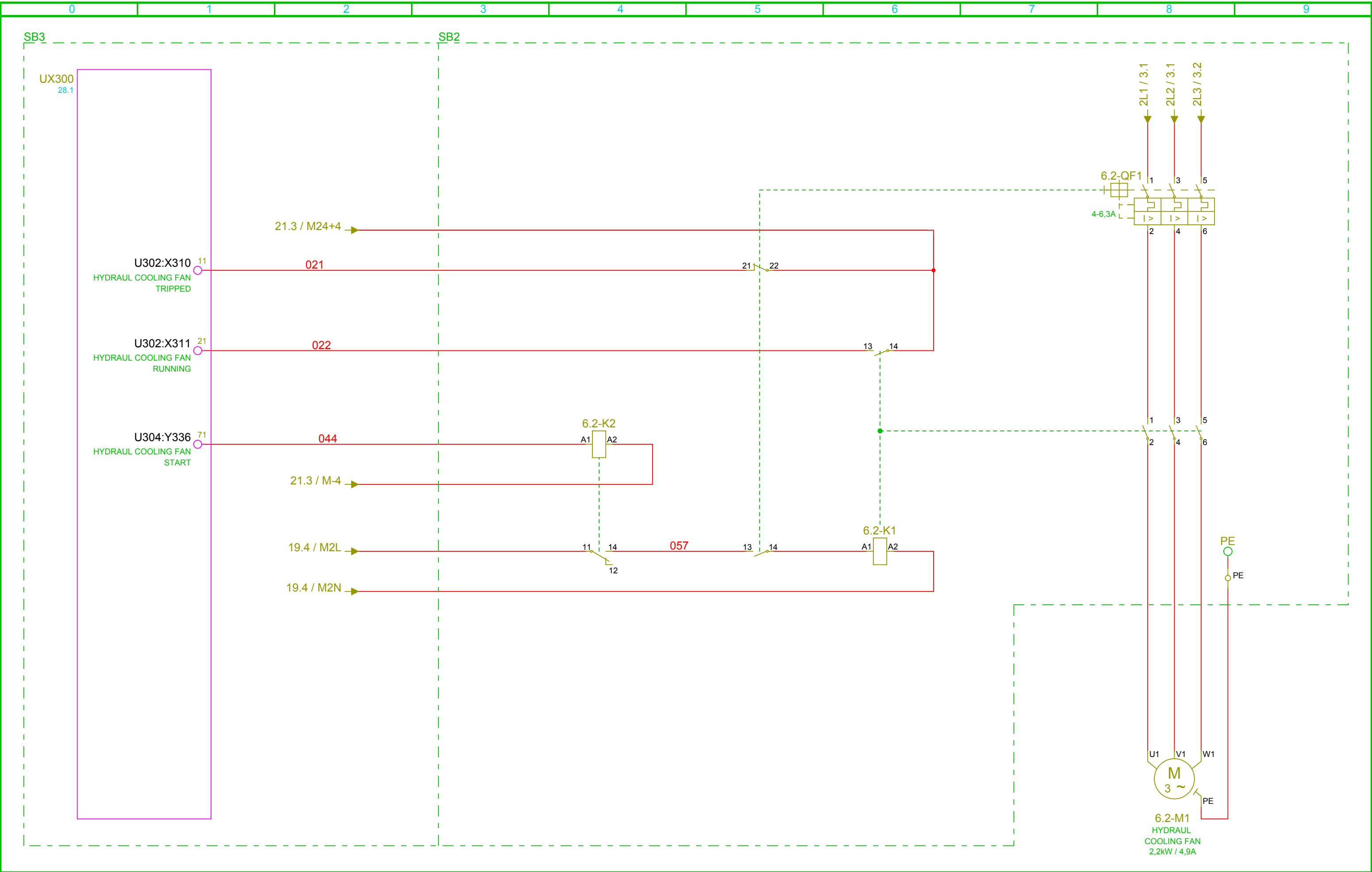
CABIN CHAIR_RIGHT SIDE

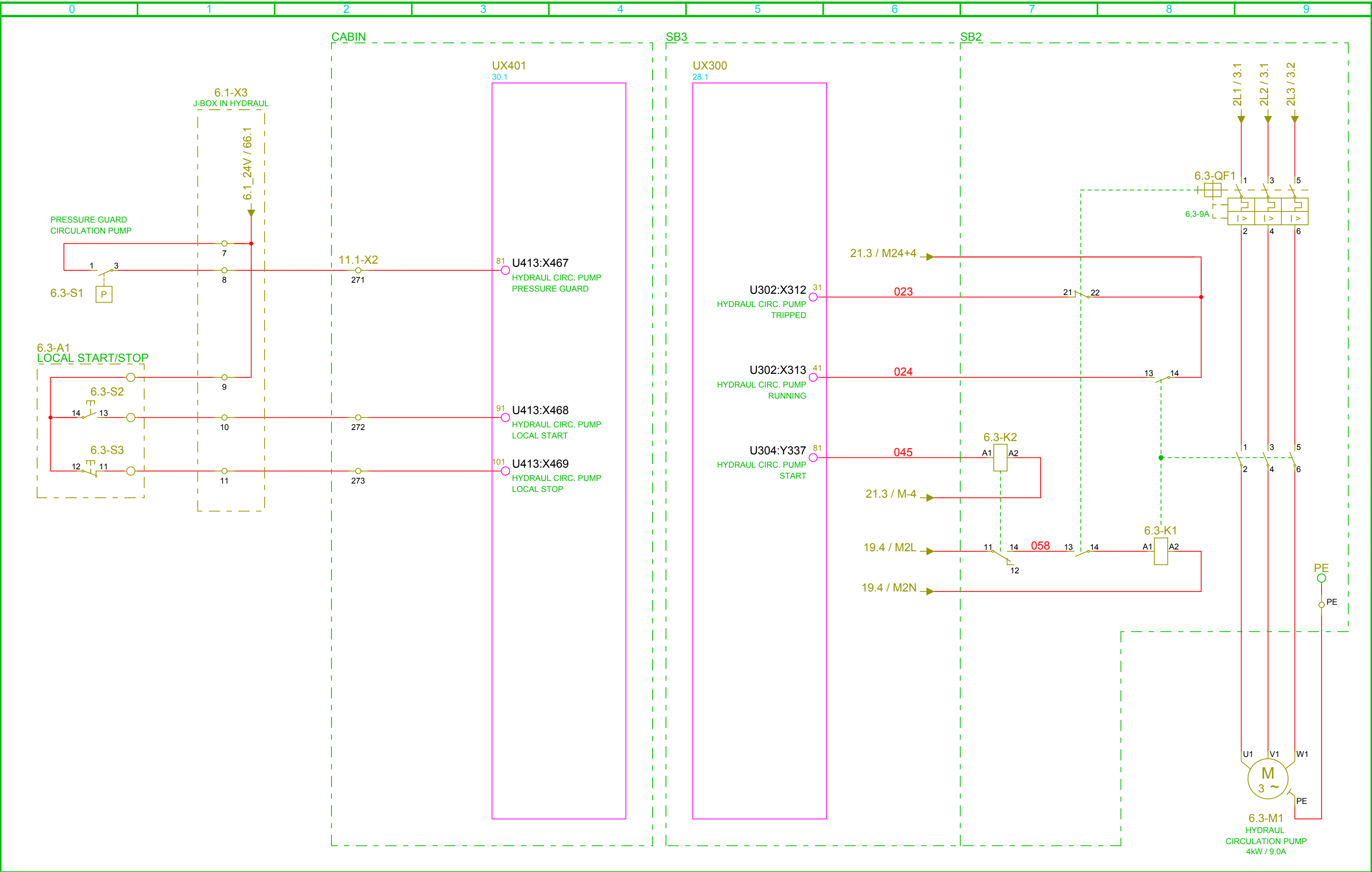


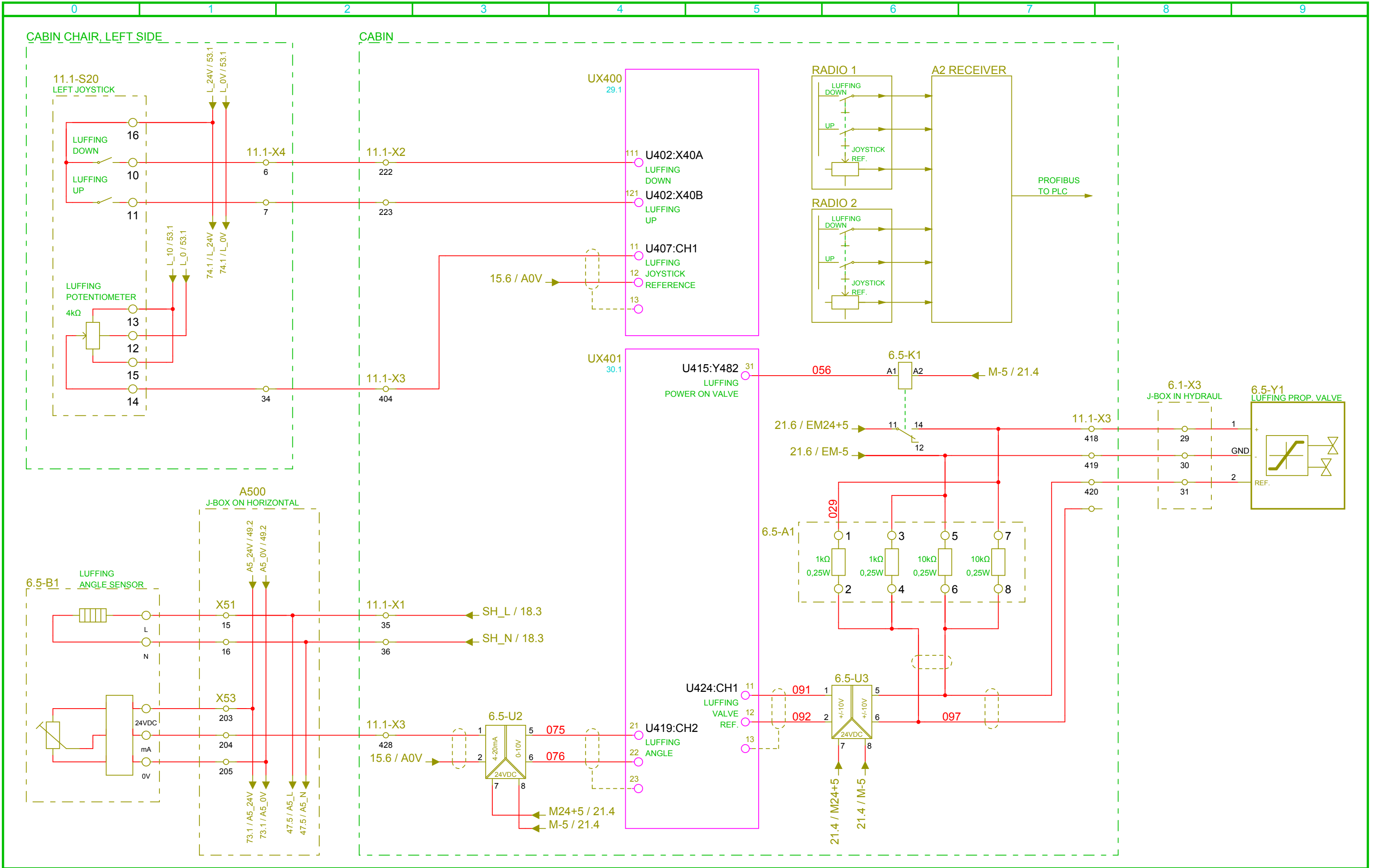






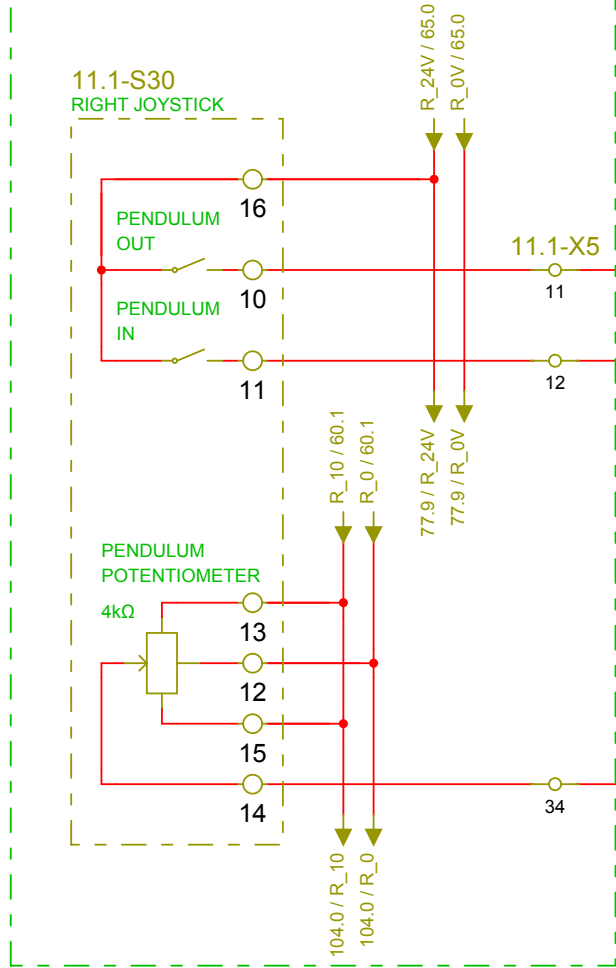




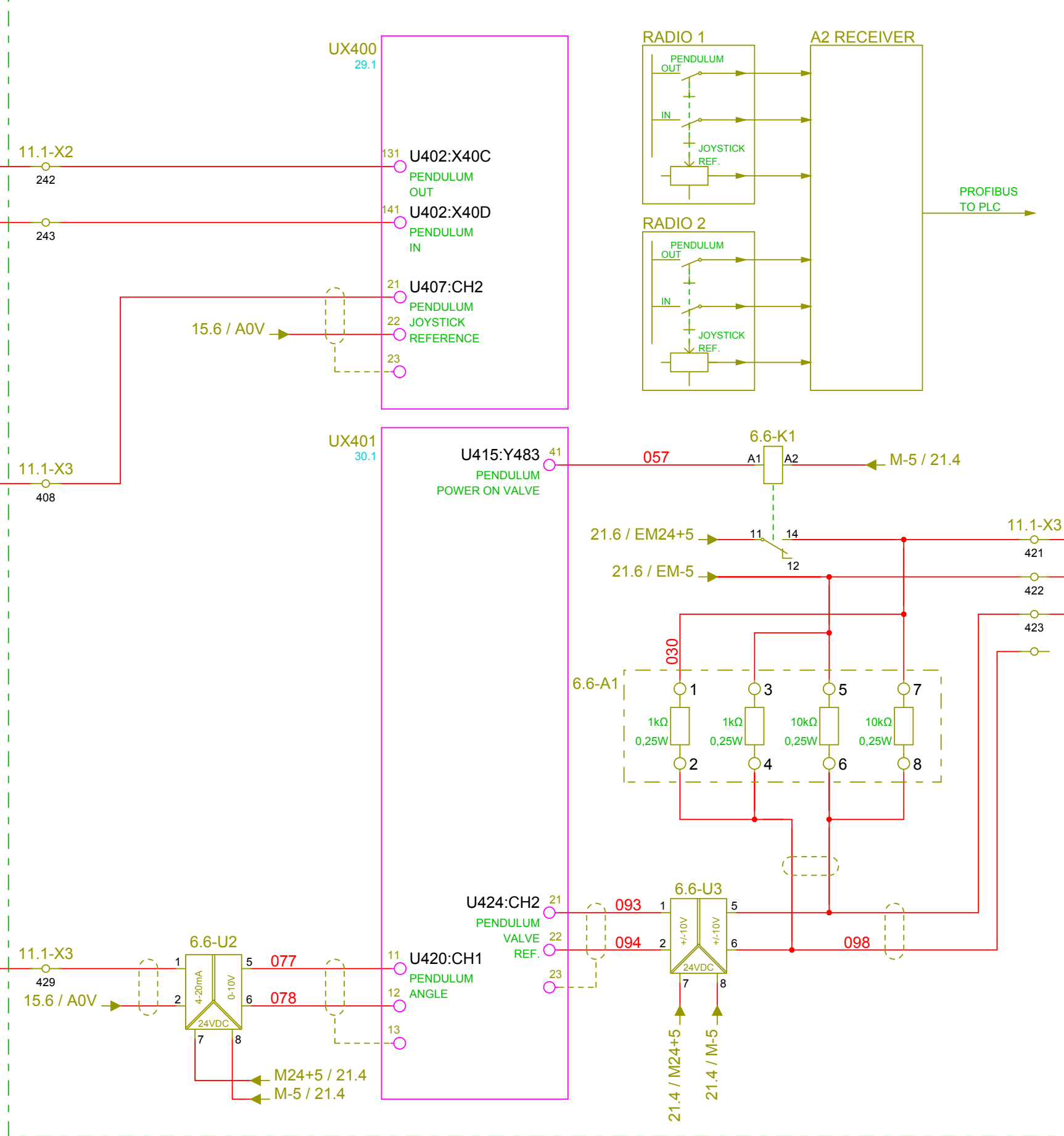


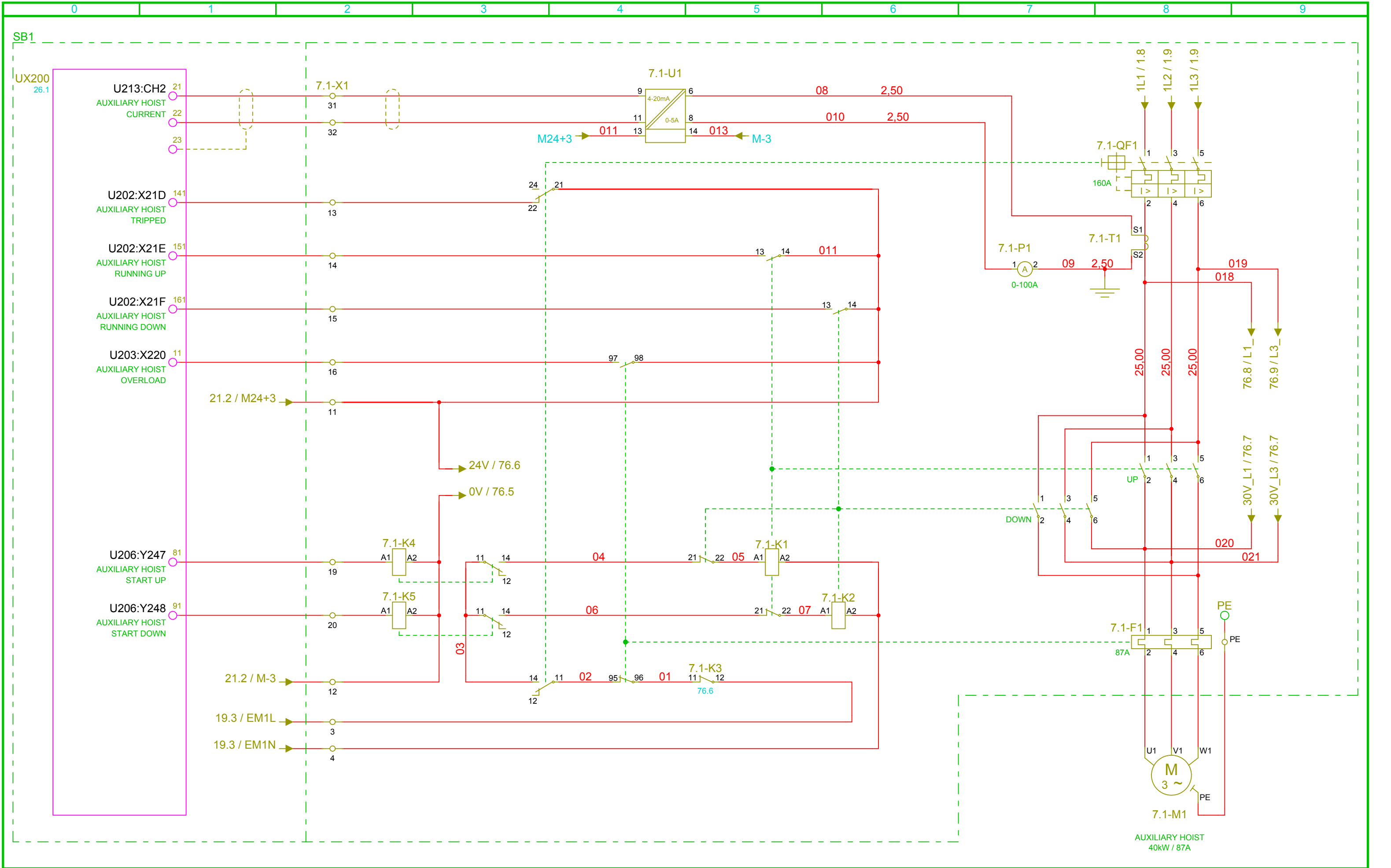
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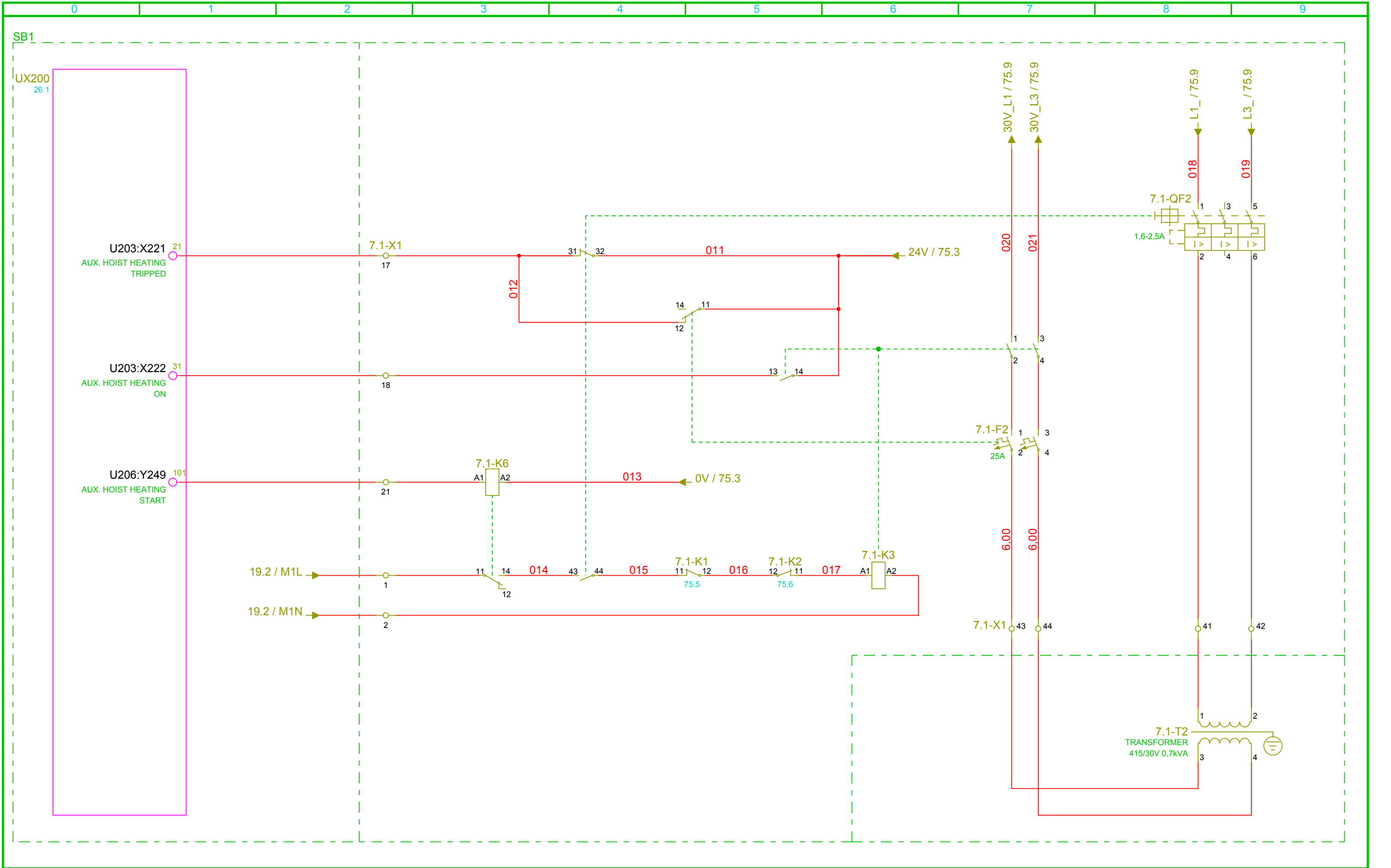
CABIN CHAIR, RIGHT SIDE

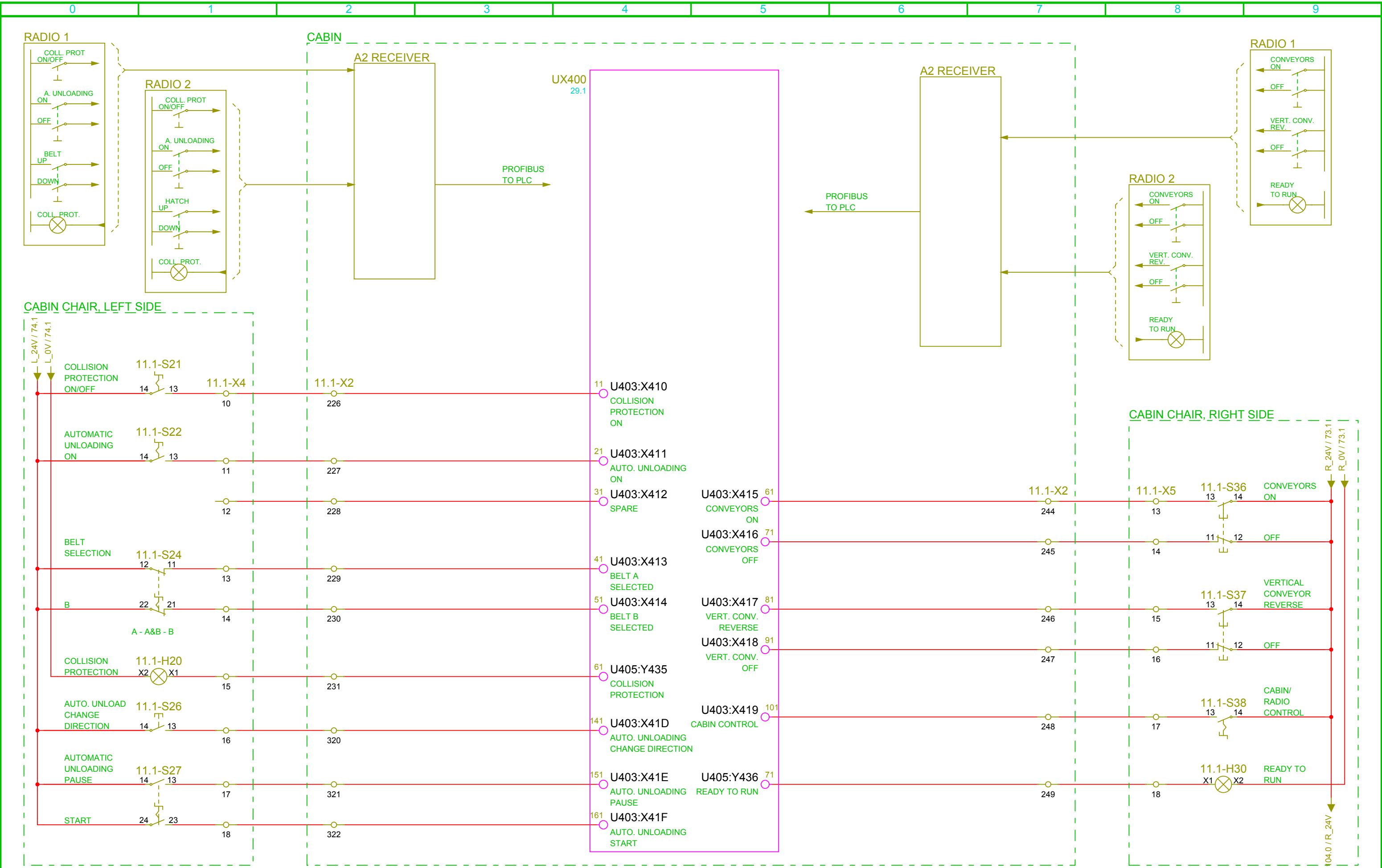


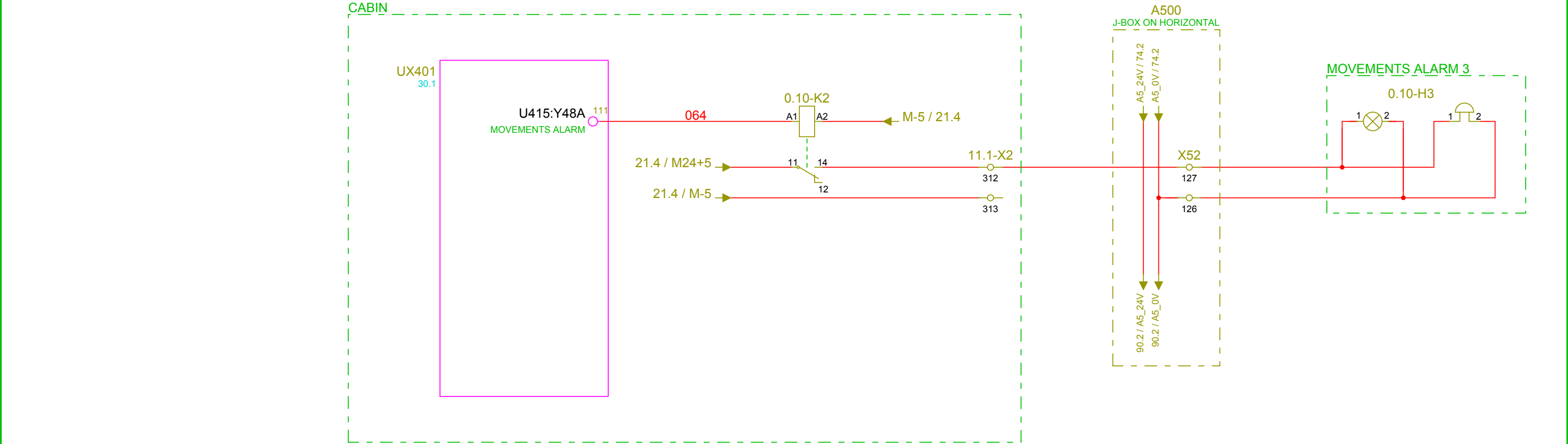
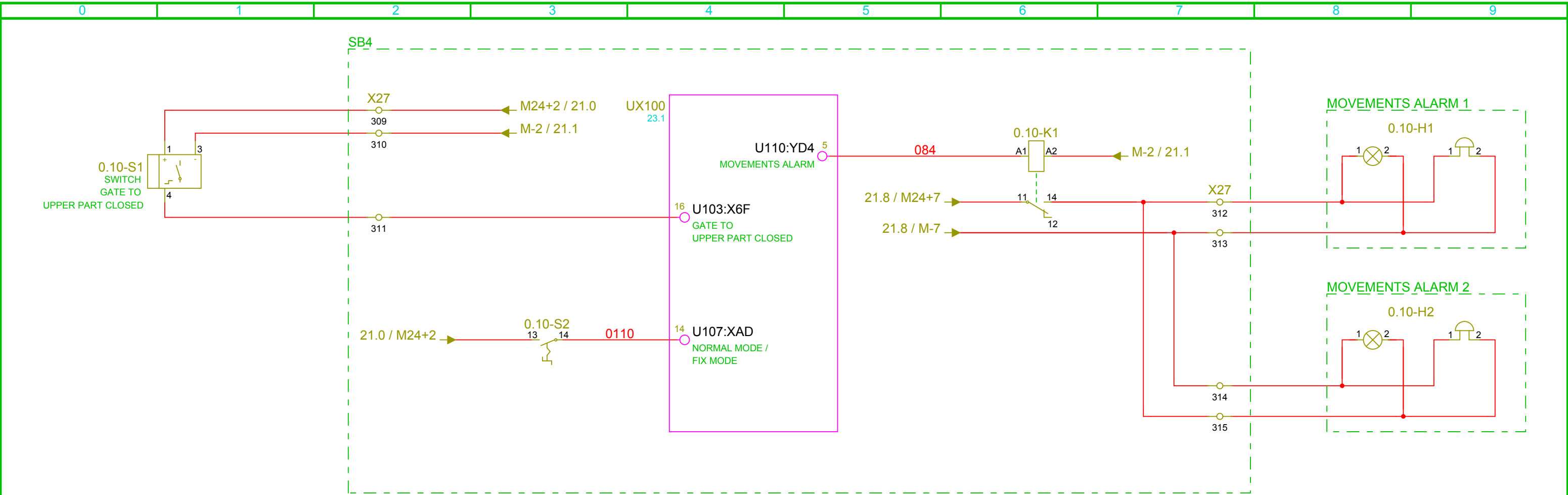
CABIN



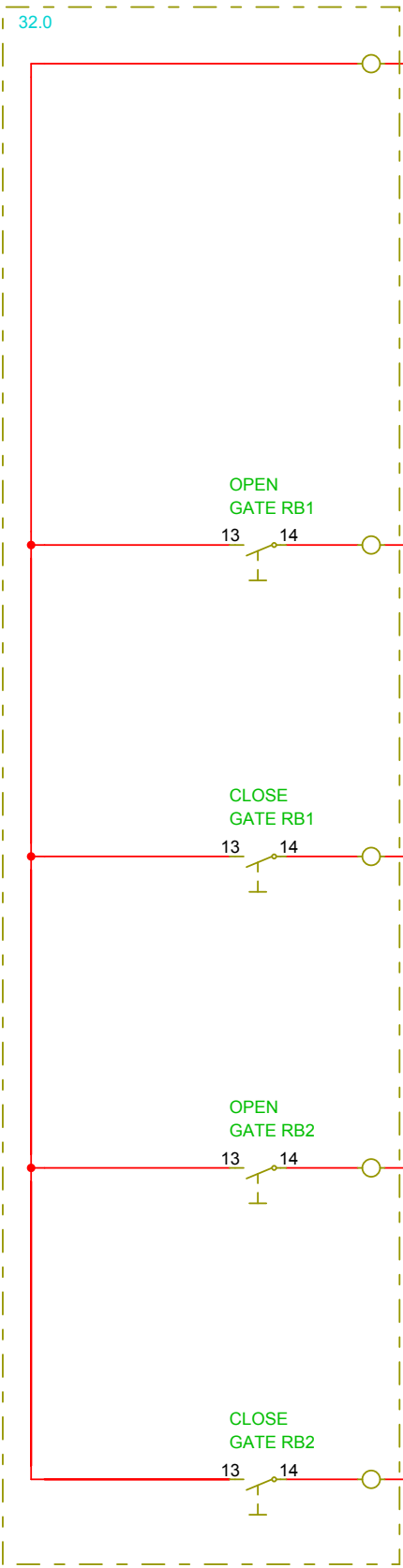








4.0-A1
LOCAL CONTROL BOX OUTLOADING
JETTY GALLERY



SB4

LOCAL/
REMOTE



X27

135

136

137

138

139

140

141

142

143

144

145

4.0-K1

A2 A1

14 11

12

4.0-K2

A2 A1

14 11

12

4.0-K3

A2 A1

14 11

12

4.0-K4

A2 A1

14 11

12

M24+2 / 21.0

M-2 / 21.1

UX100

23.1

030

031

032

033

14 U105:X8D

OUTLOADING LOCAL CONTROL
IN LOCAL MODE

9 U108:YB8

SPARE

15 U105:X8E

OUTLOADING LOCAL CONTROL
OPEN GATE RB1

10 U108:YB9

SPARE

16 U105:X8F

OUTLOADING LOCAL CONTROL
CLOSE GATE RB1

11 U108:YBA

SPARE

1 U106:X90

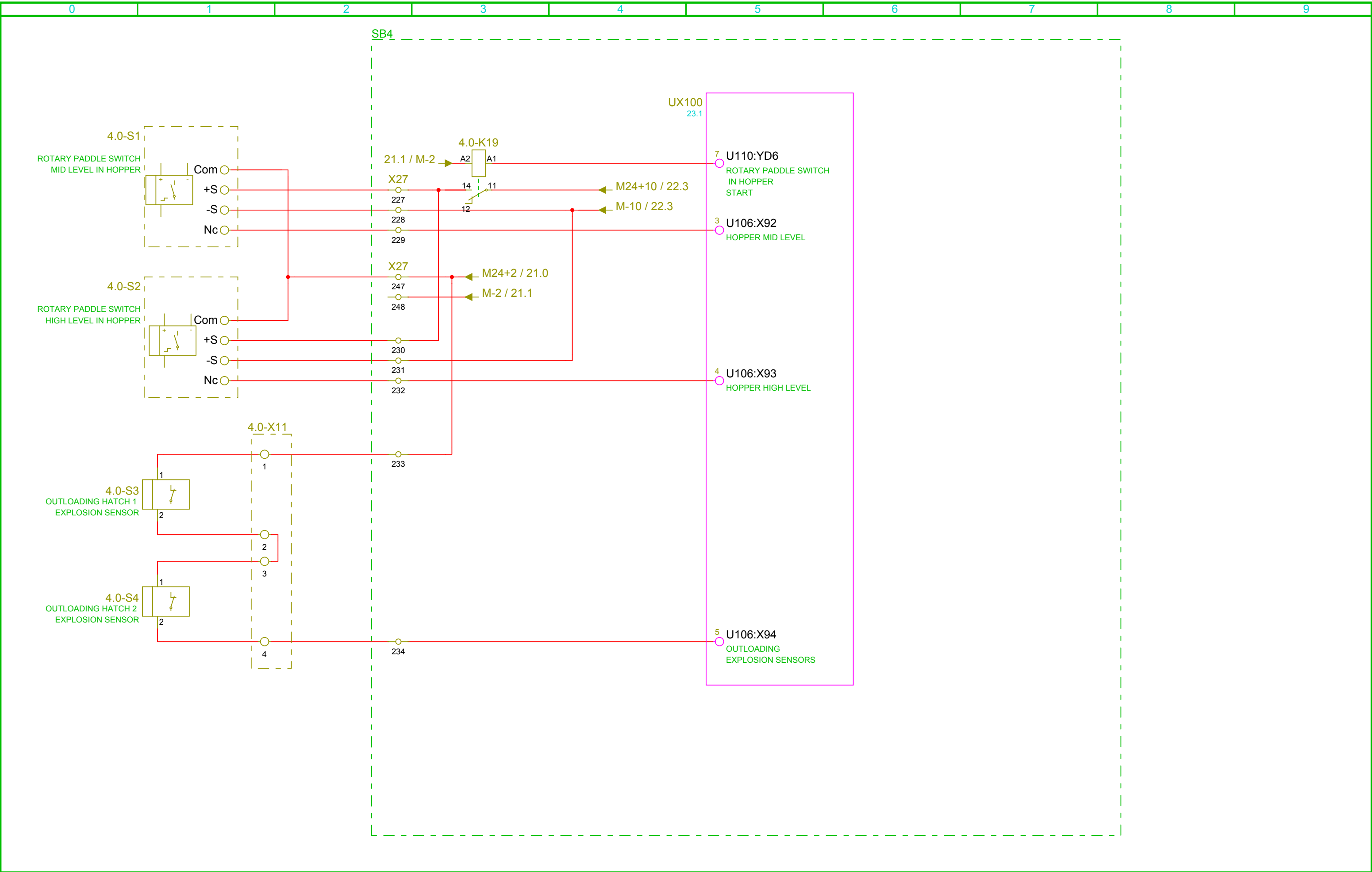
OUTLOADING LOCAL CONTROL
OPEN GATE RB2

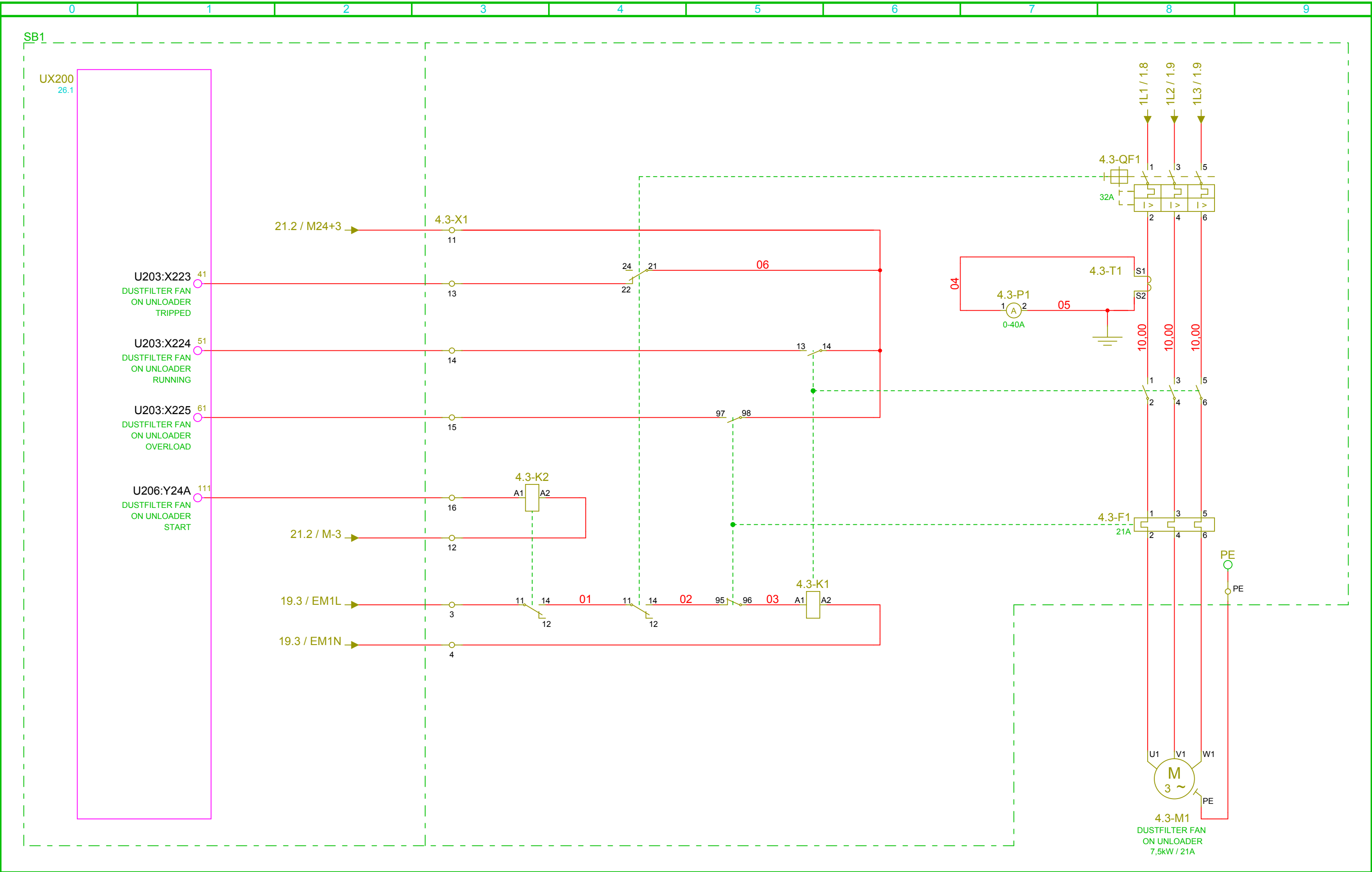
12 U108:YBB

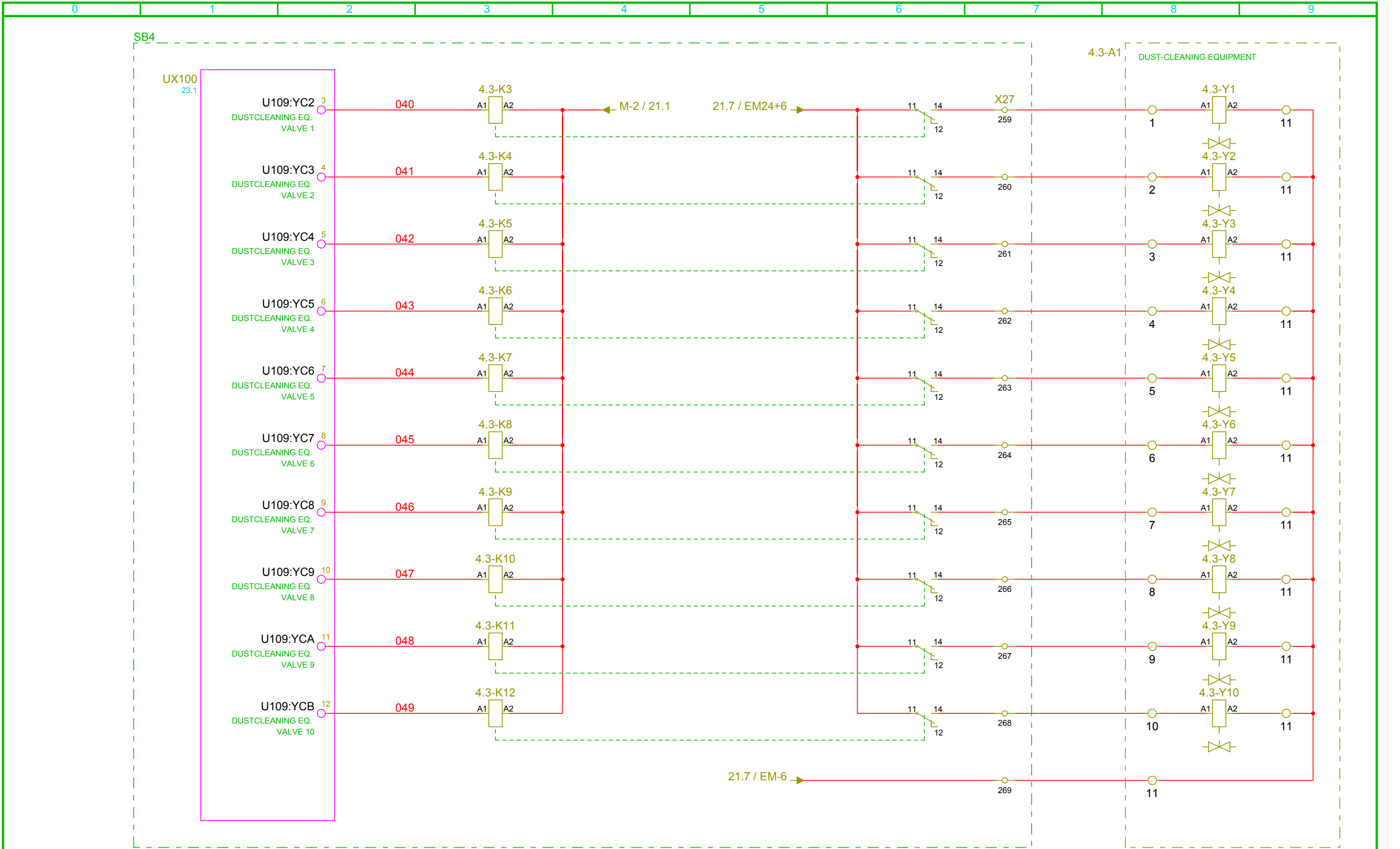
SPARE

2 U106:X91

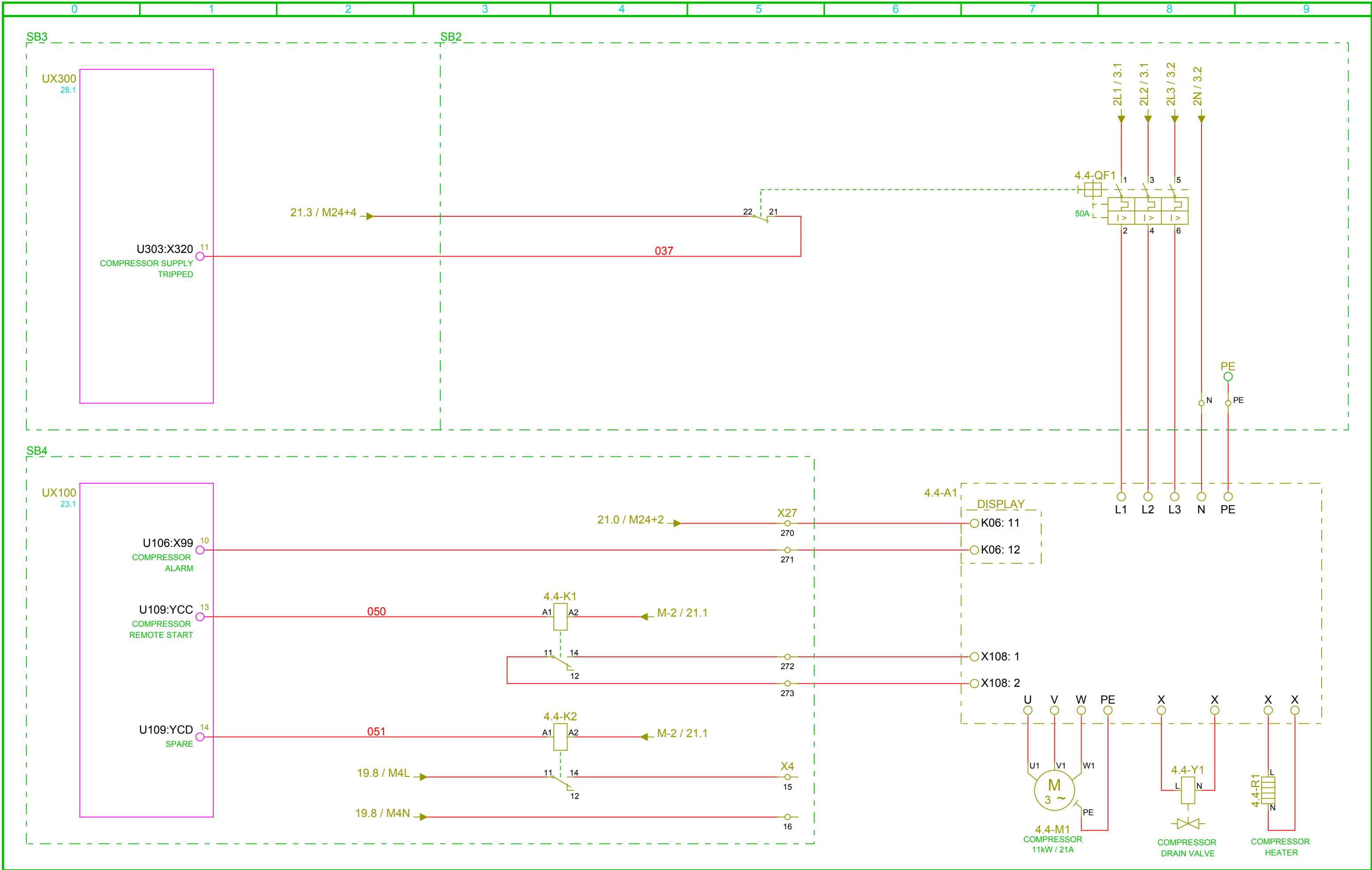
OUTLOADING LOCAL CONTROL
CLOSE GATE RB2





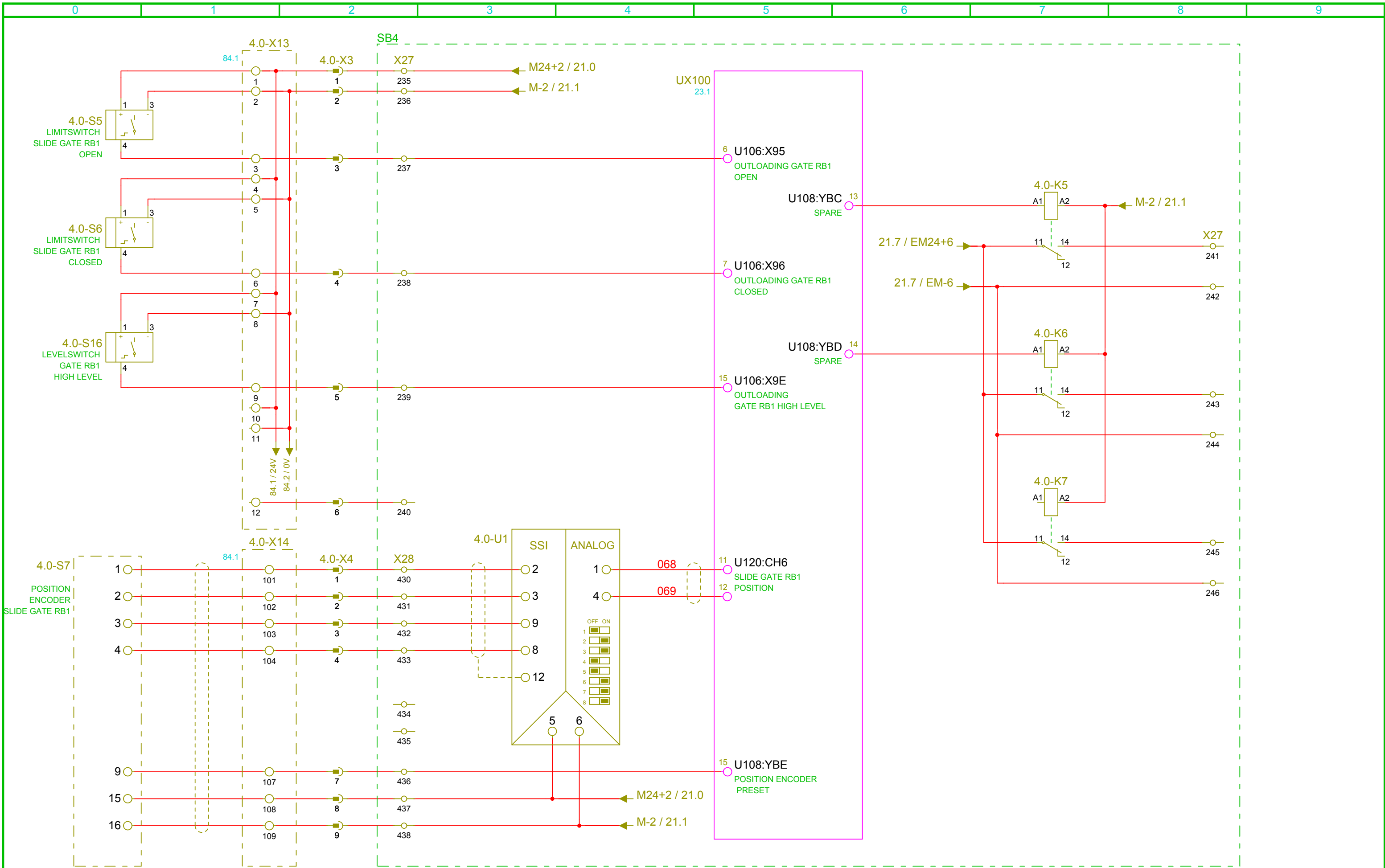


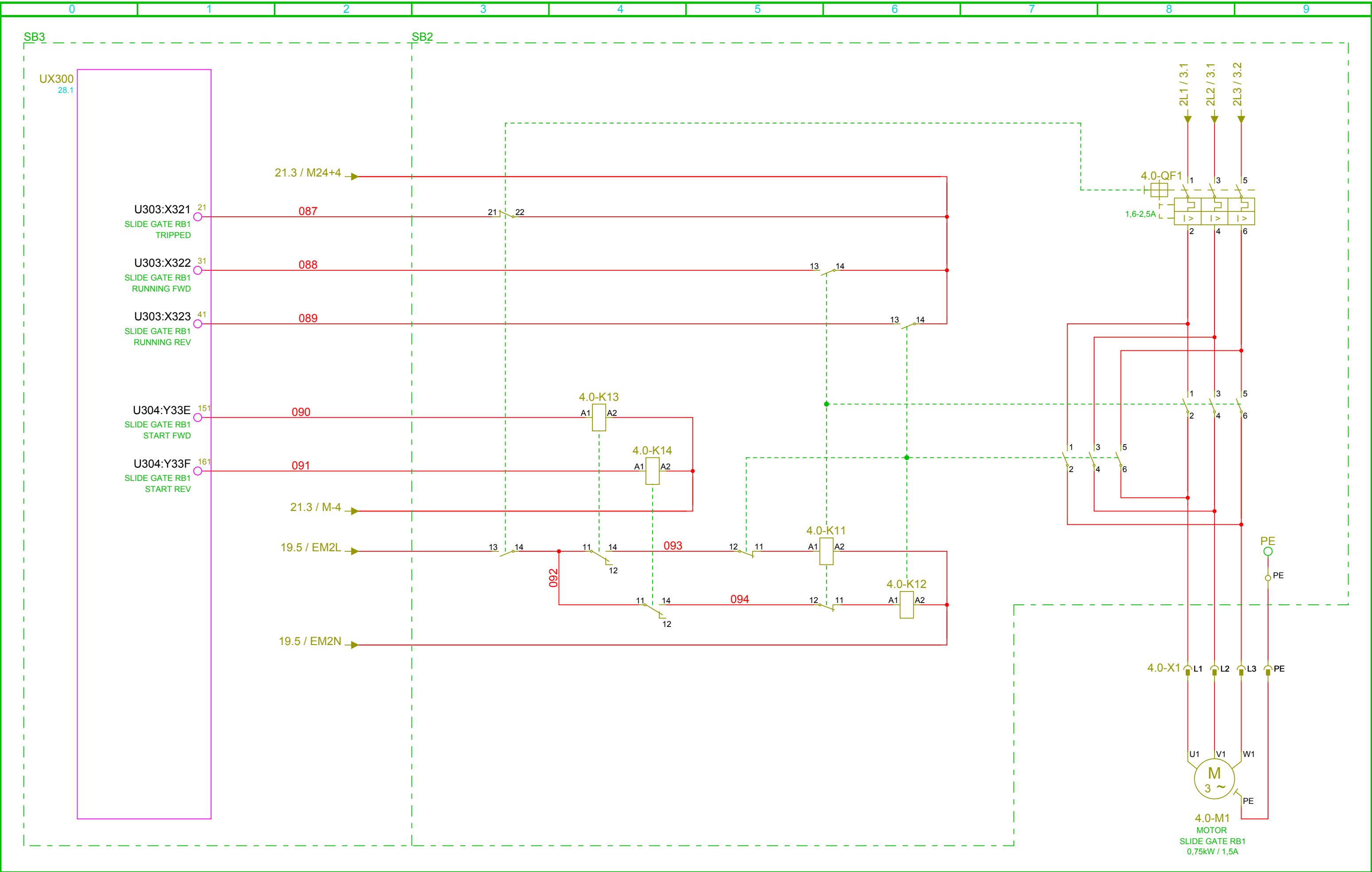
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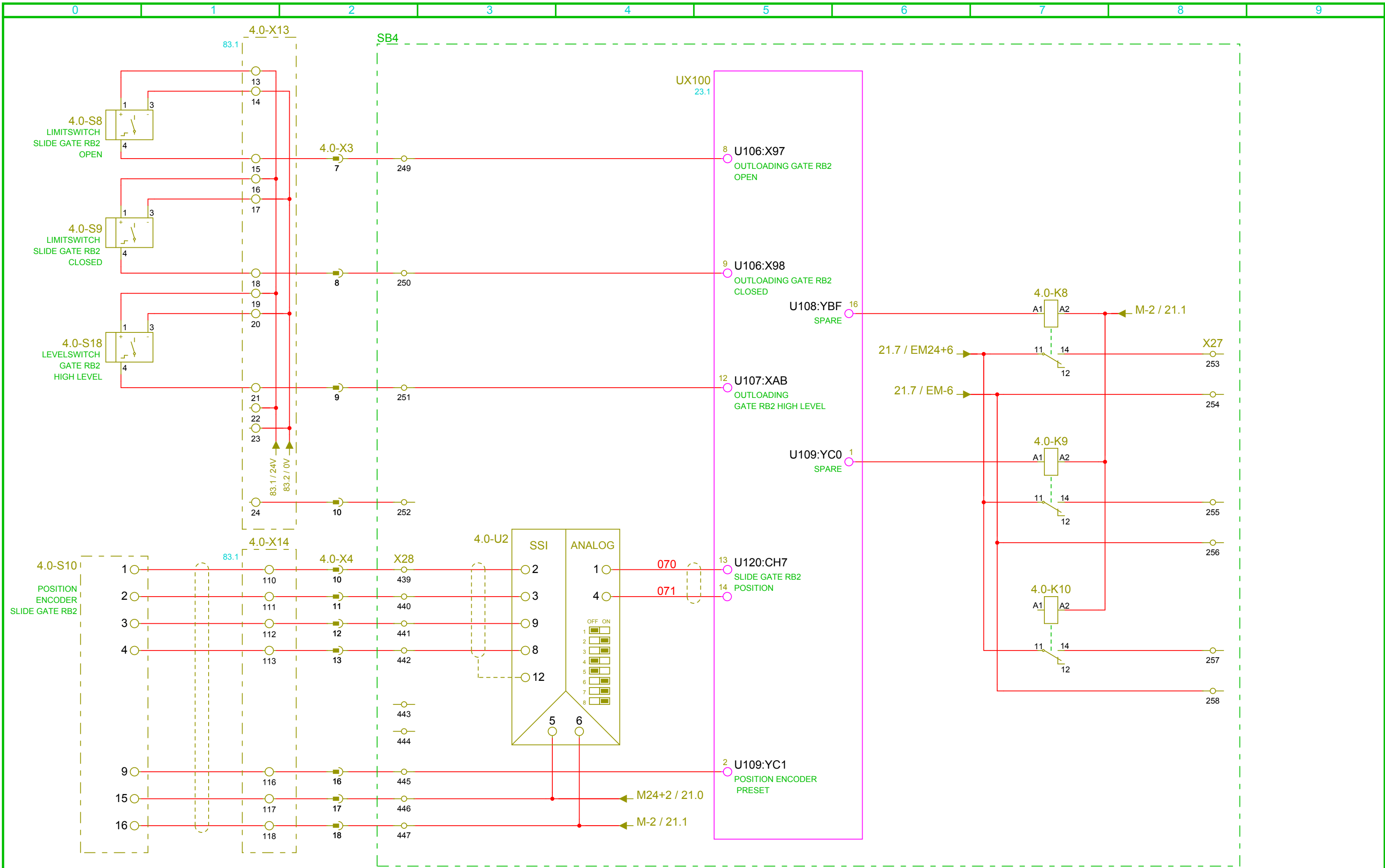
AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN			AS BUILT			BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN			PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT CIRCUIT DIAGRAM SUPPLY AIR-COMPRESSOR			DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-17	
CHECKED			PROJECT NO. 202A6086			DRAWING NO. A524 - E			FILE NAME A524.DB			SHEET 82	CONT. SHEET 83		
APPR.															

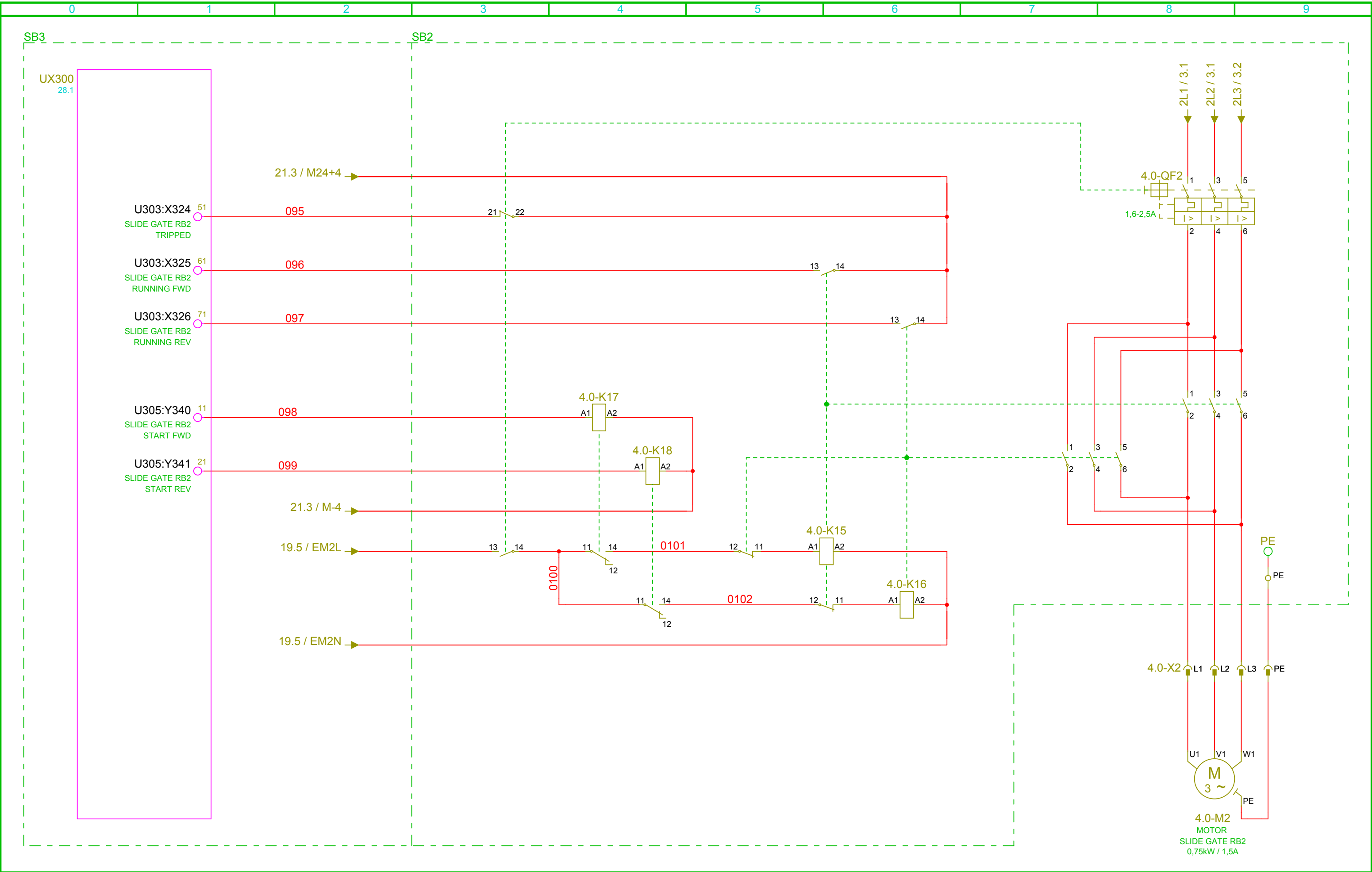
reproduced, or exploited in any other unauthorized way.

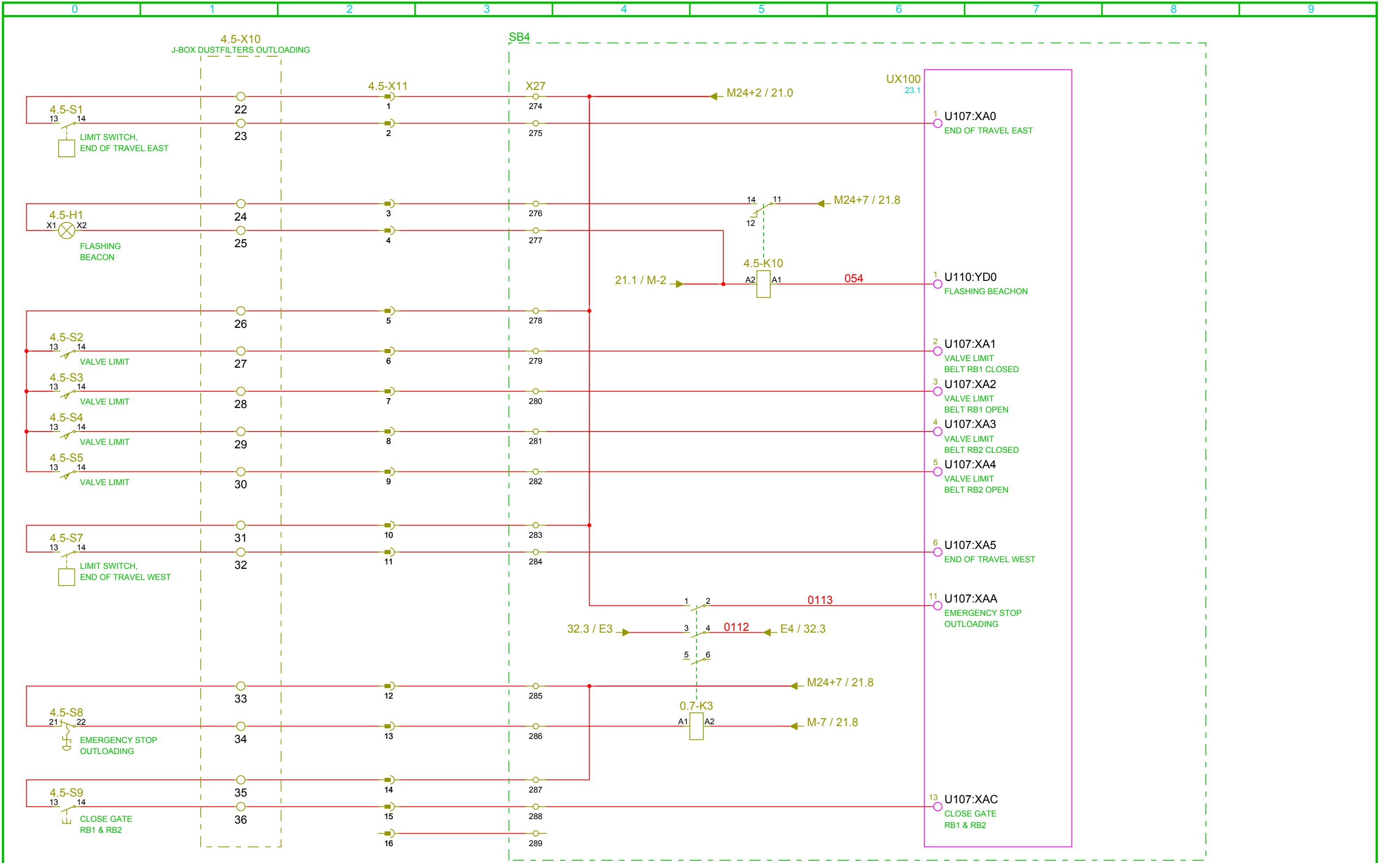


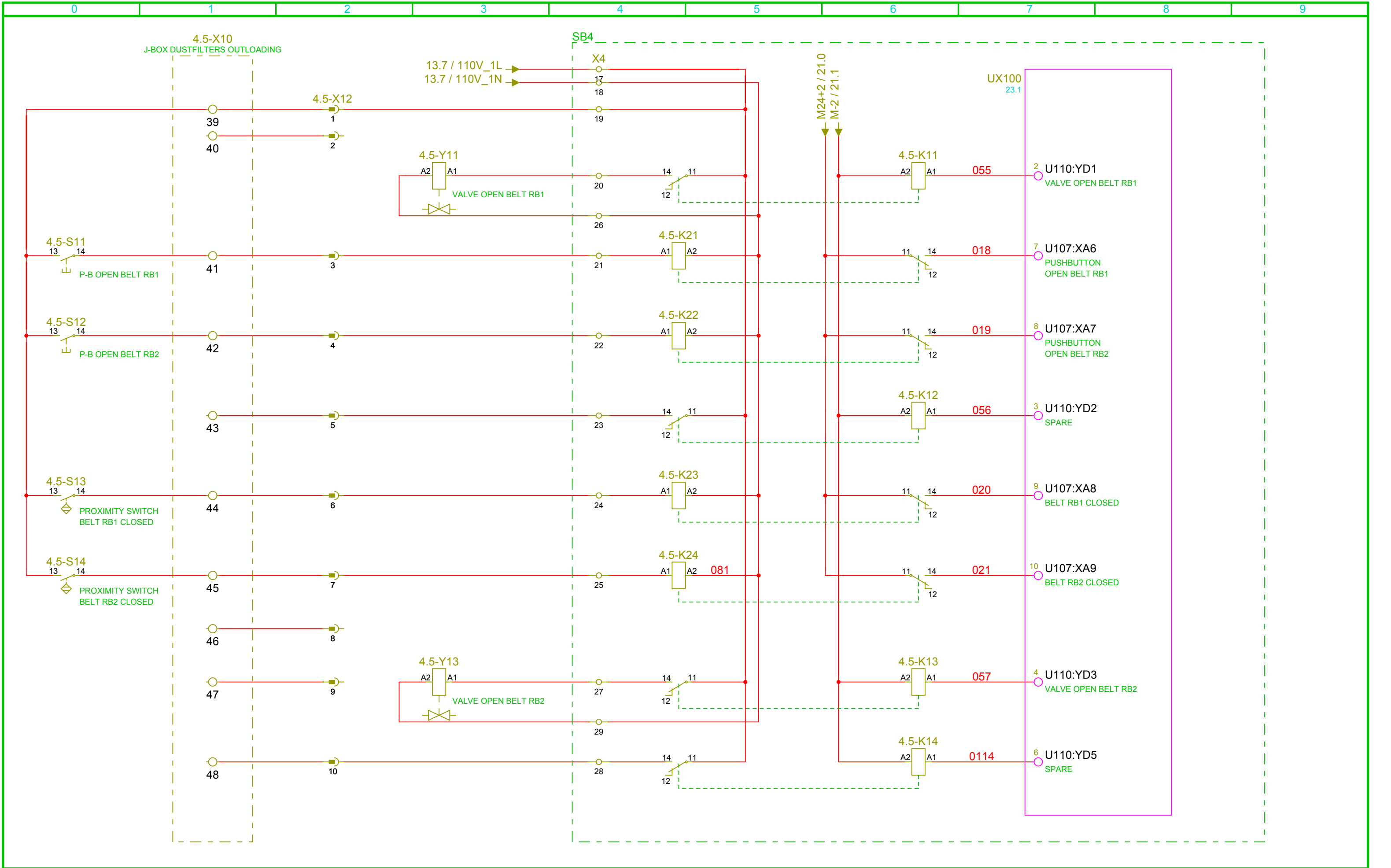


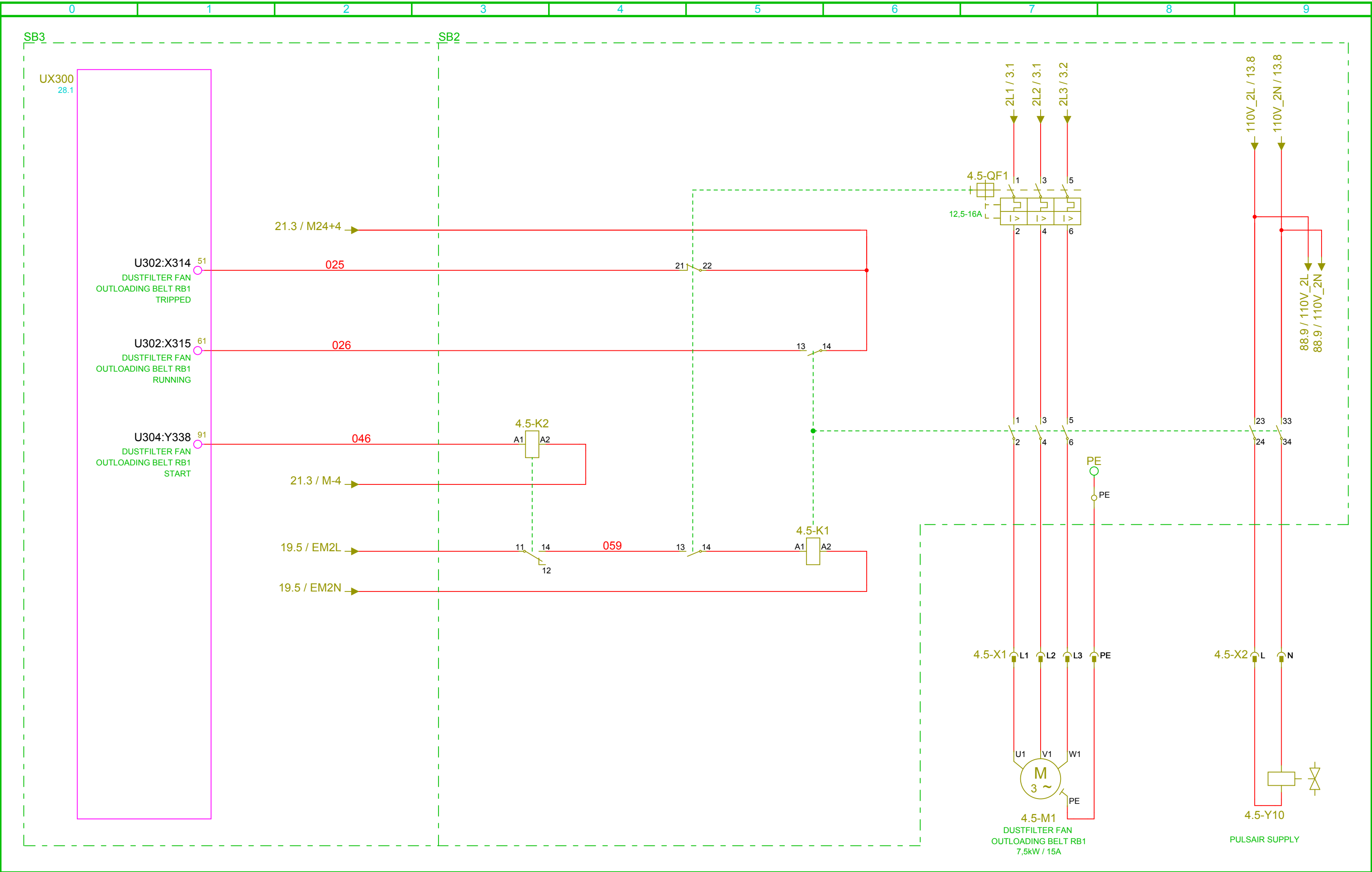
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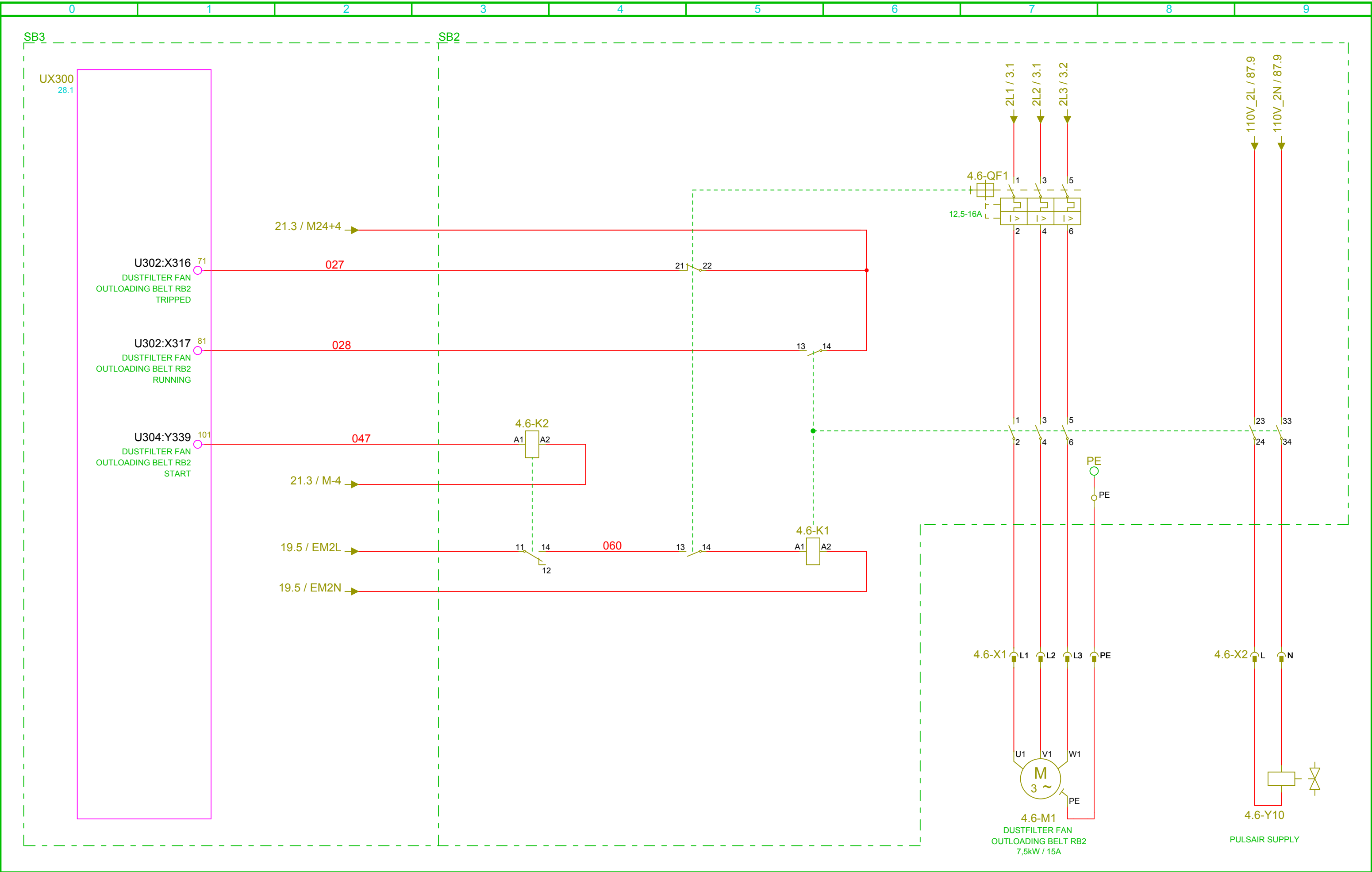


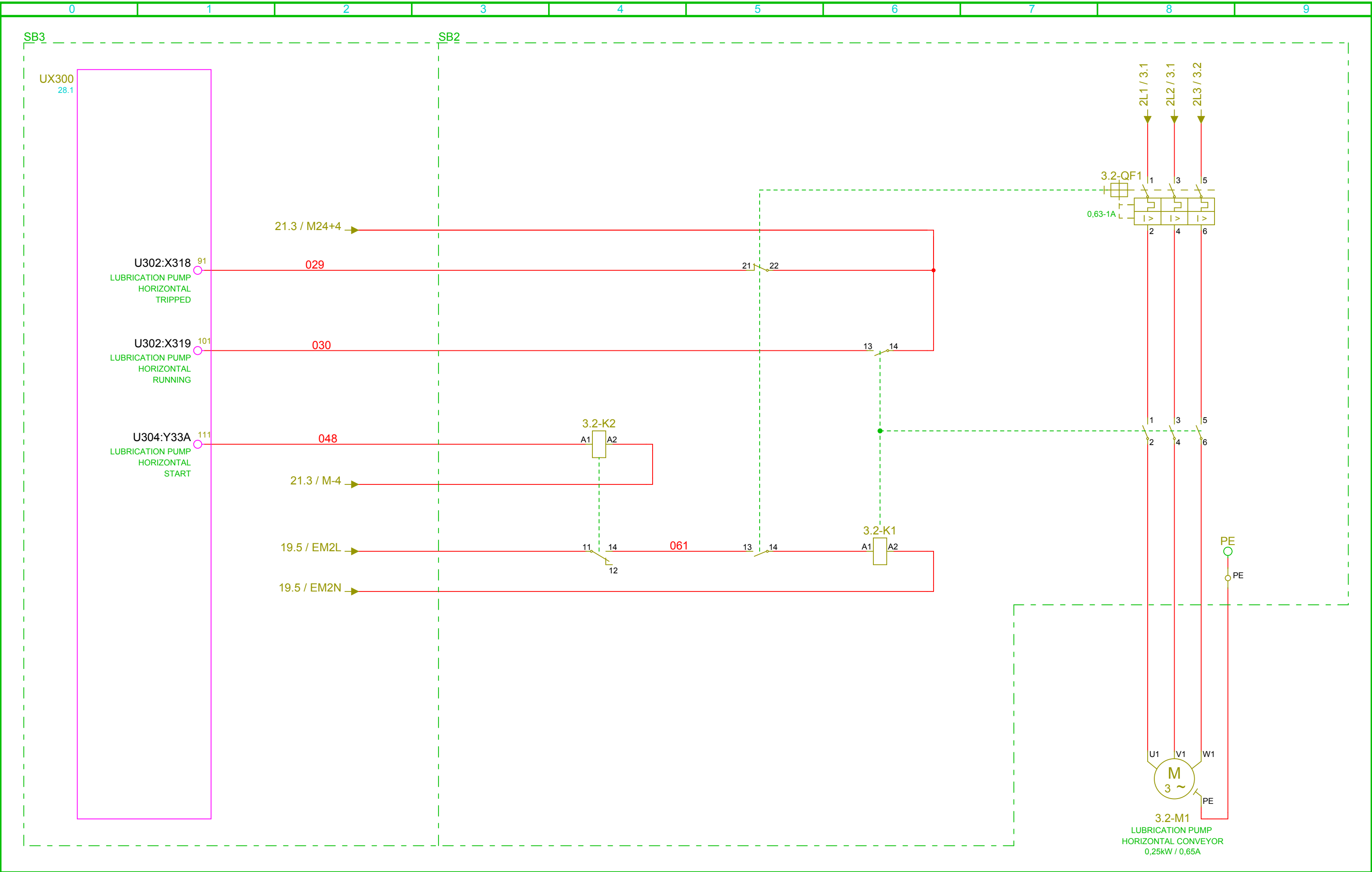


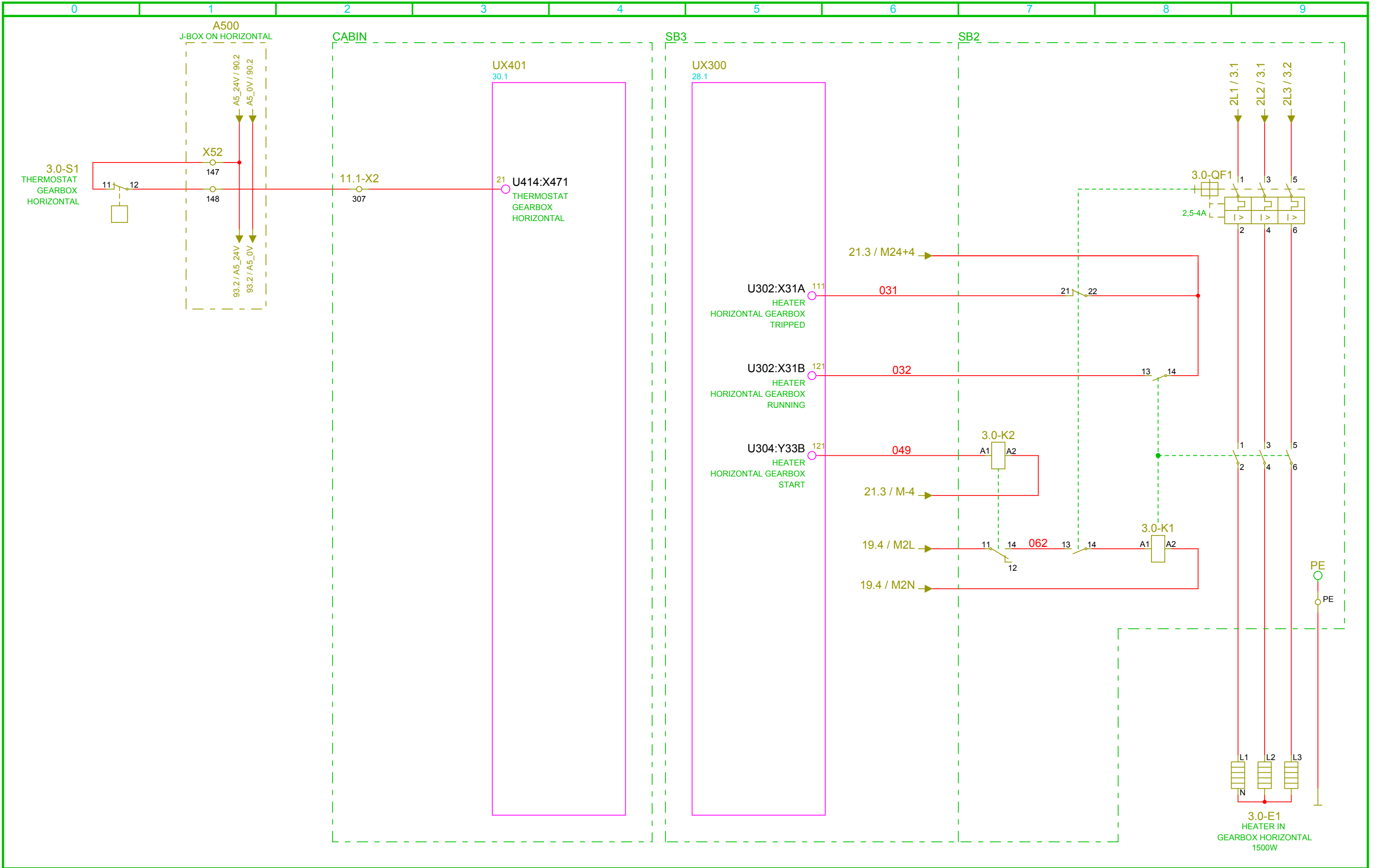




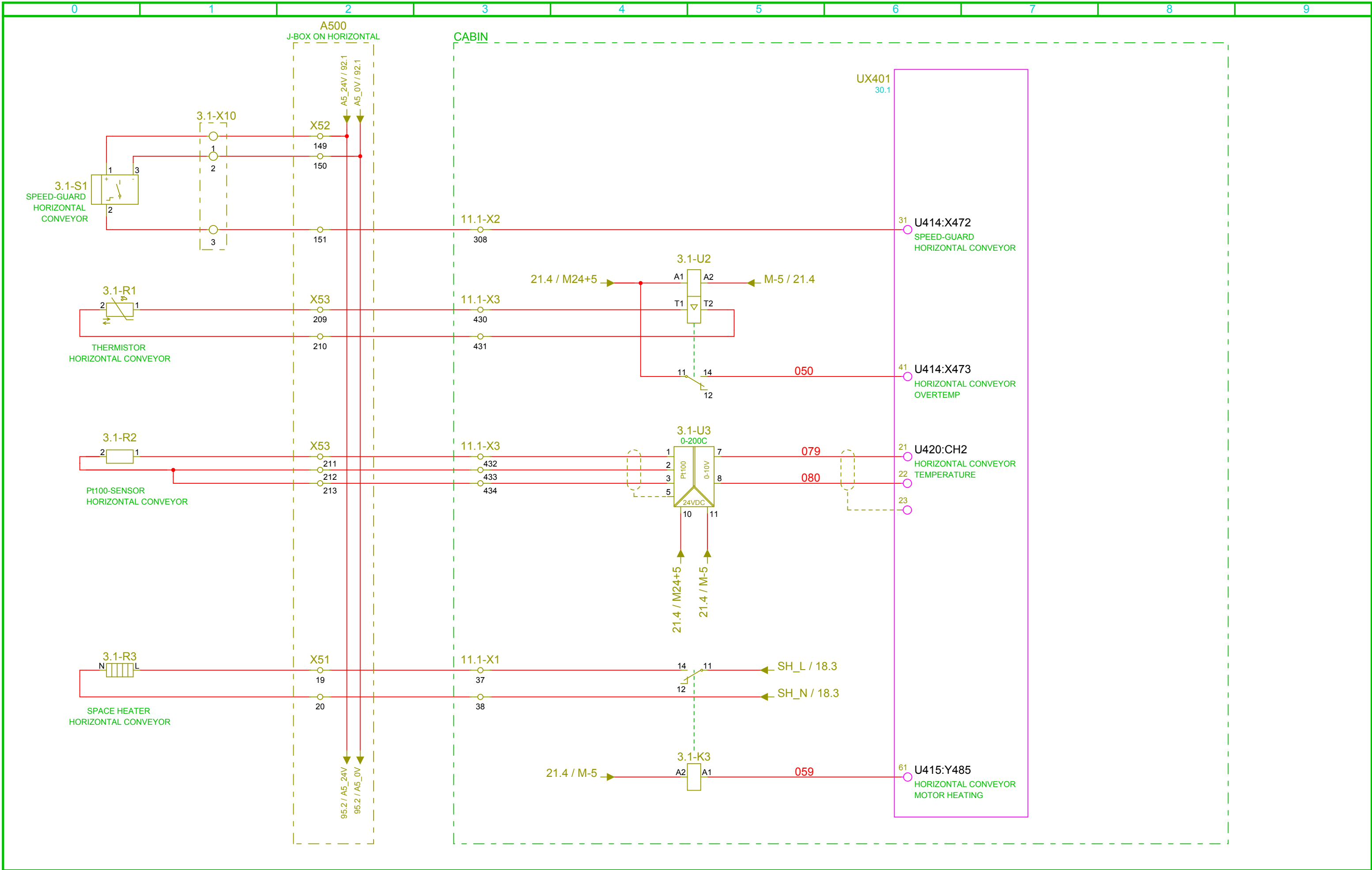






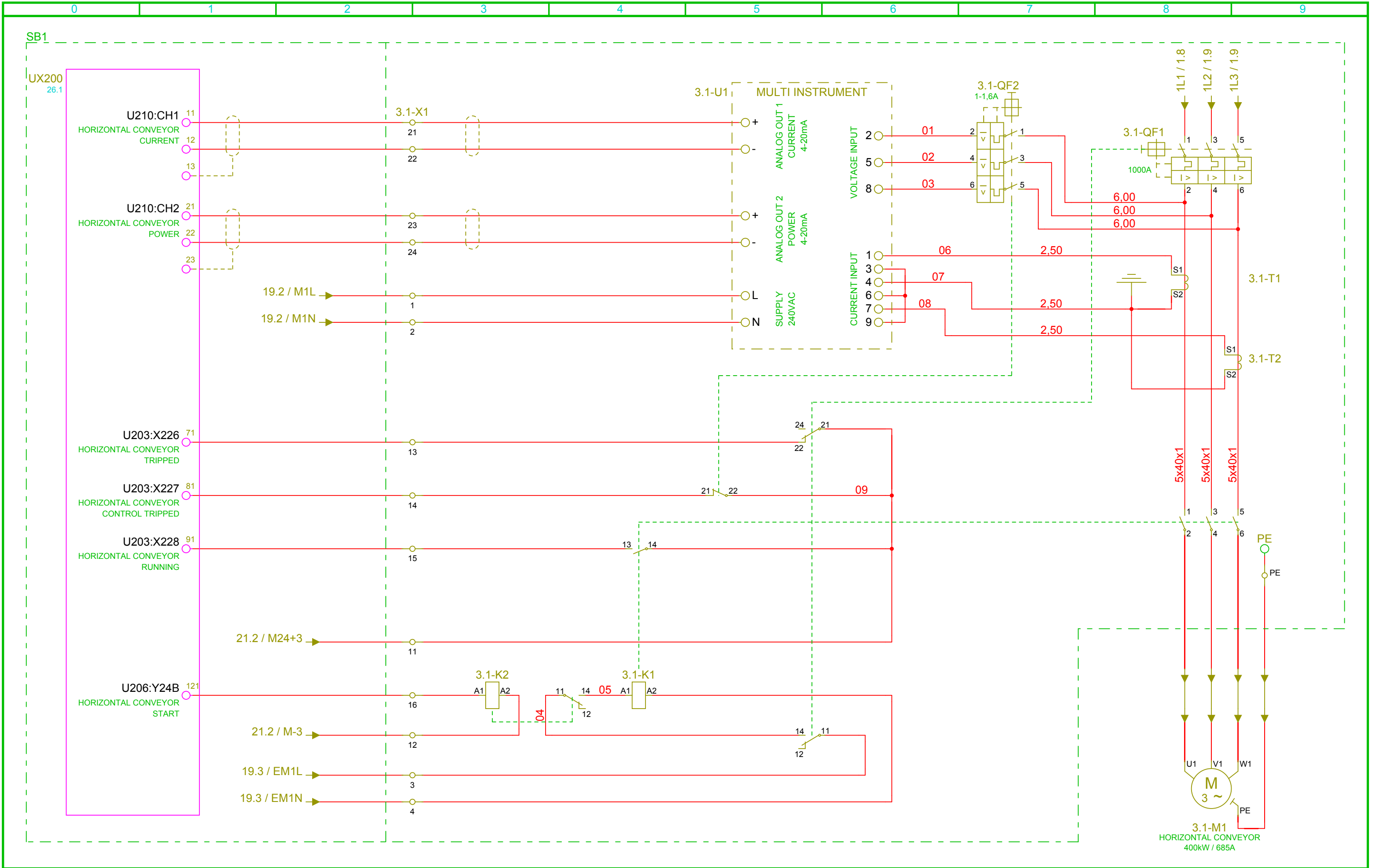


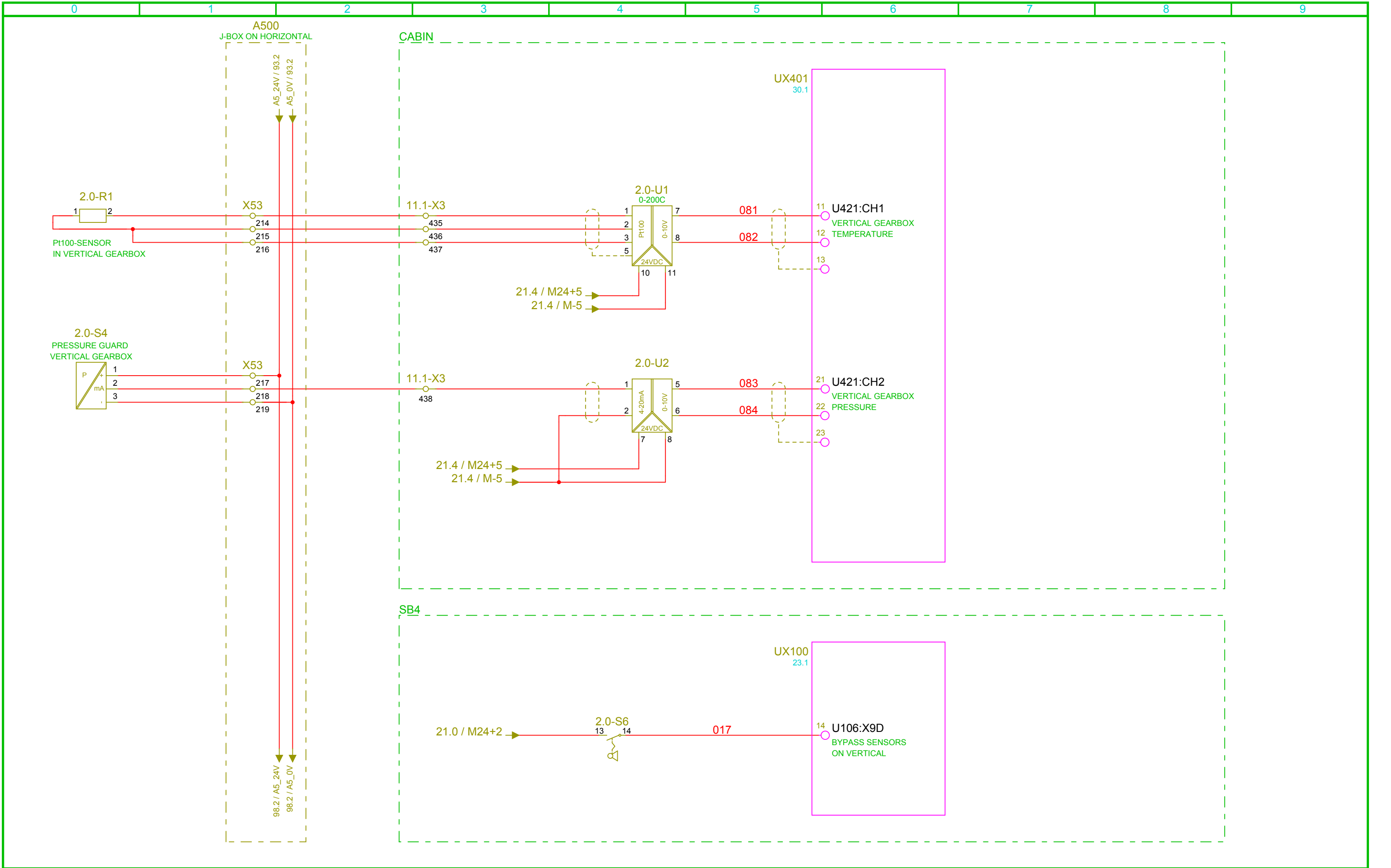
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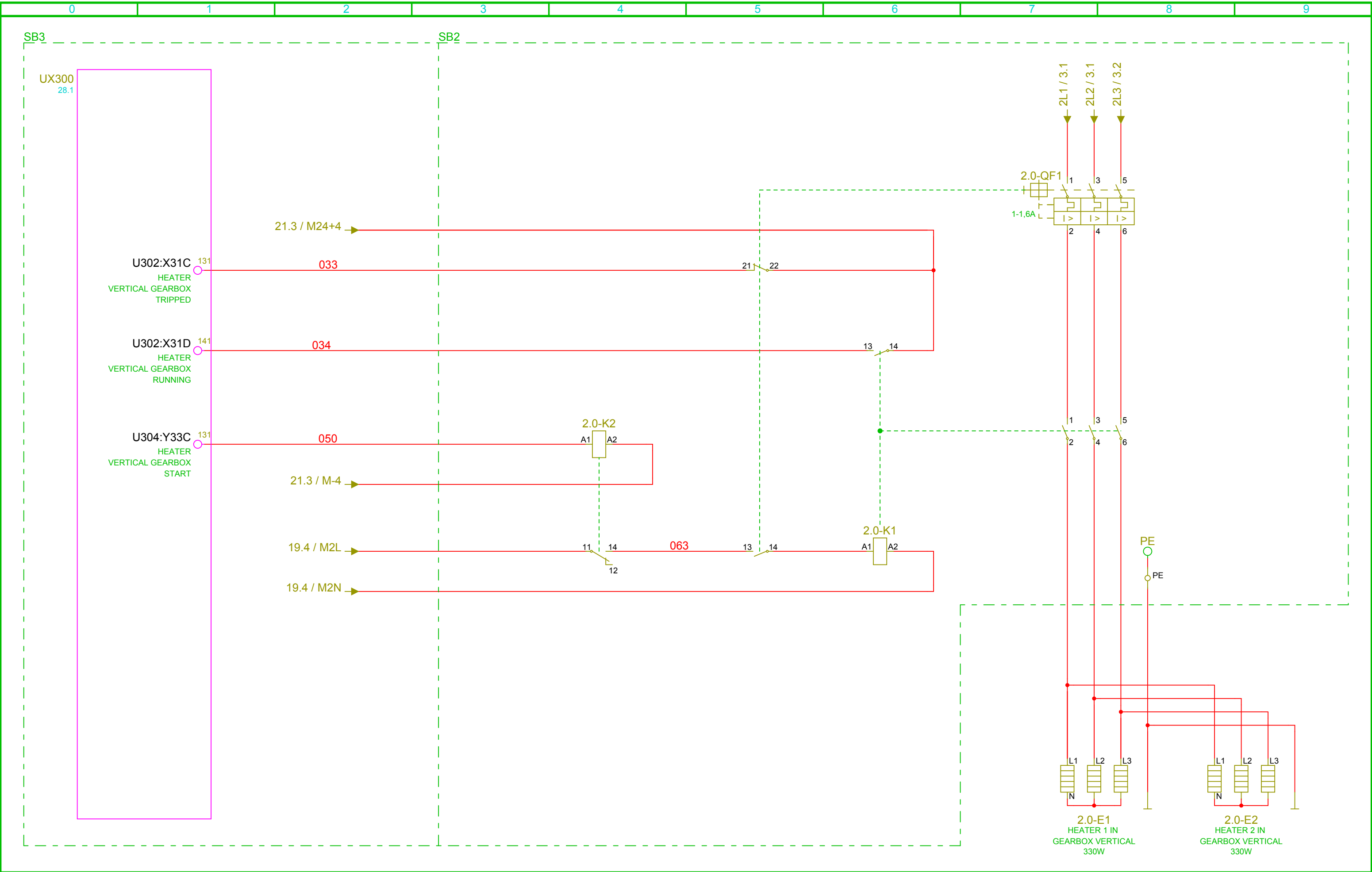


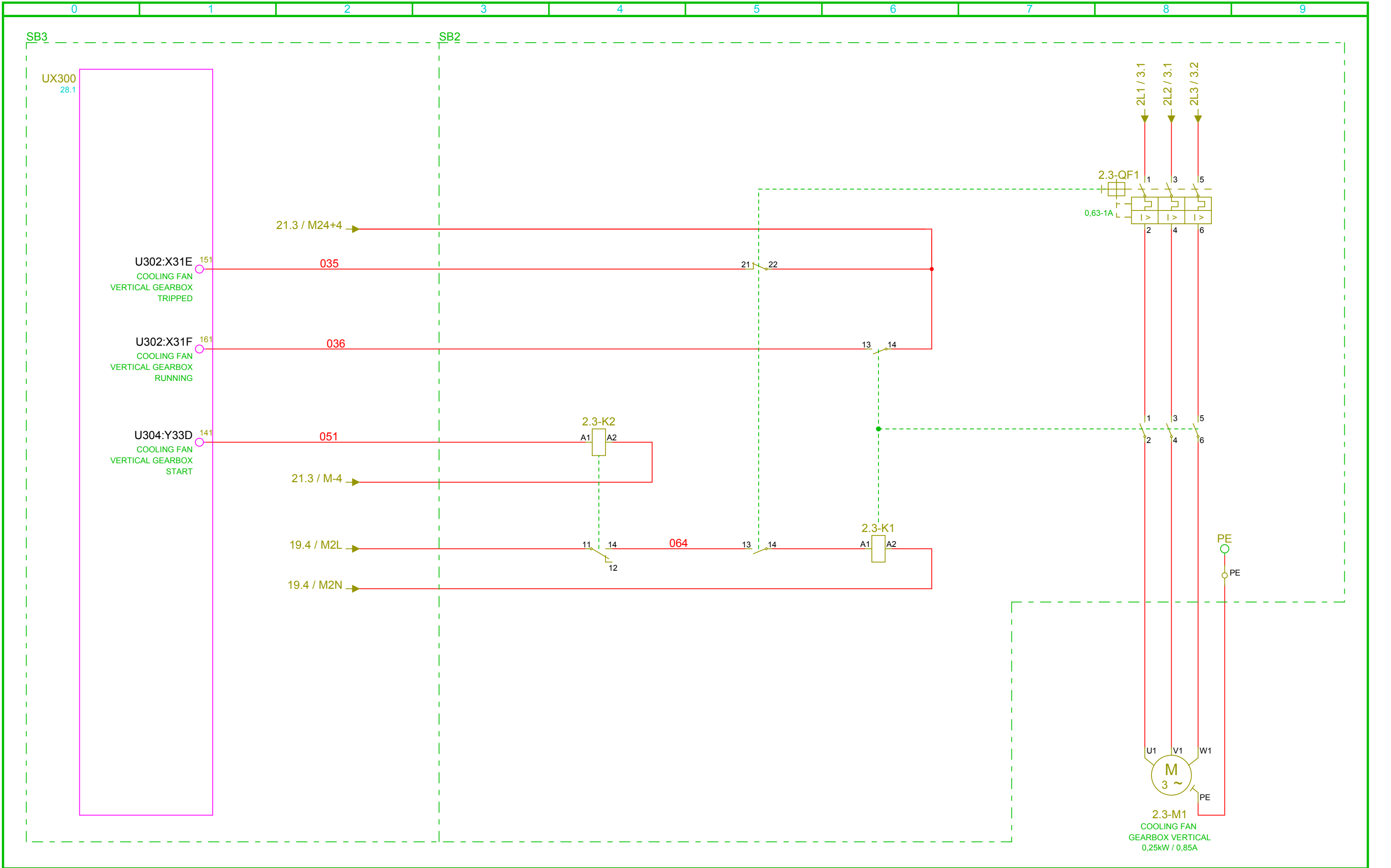
AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN			AS BUILT			BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN			PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT CIRCUIT DIAGRAM CONTROL CIRCUIT HORIZONTAL CONVEYOR			DESIGN M.D.	SCALE 1 : 1		DATE 2006-10-16	
CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - E													
APPR.	FILE NAME A524.DB		SHEET 93	CONT. SHEET 94												
REV.	MODIFICATION				DATE											

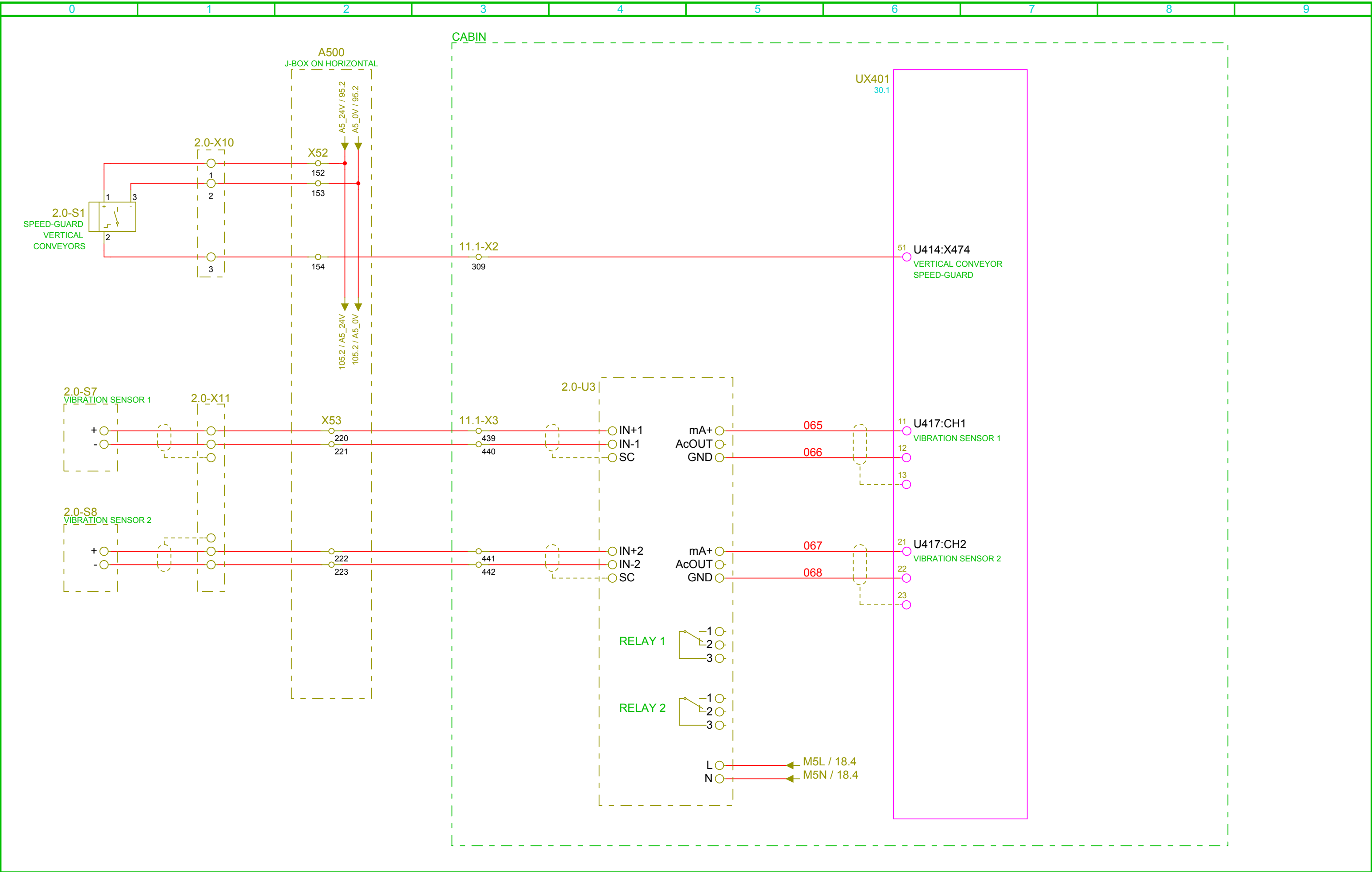
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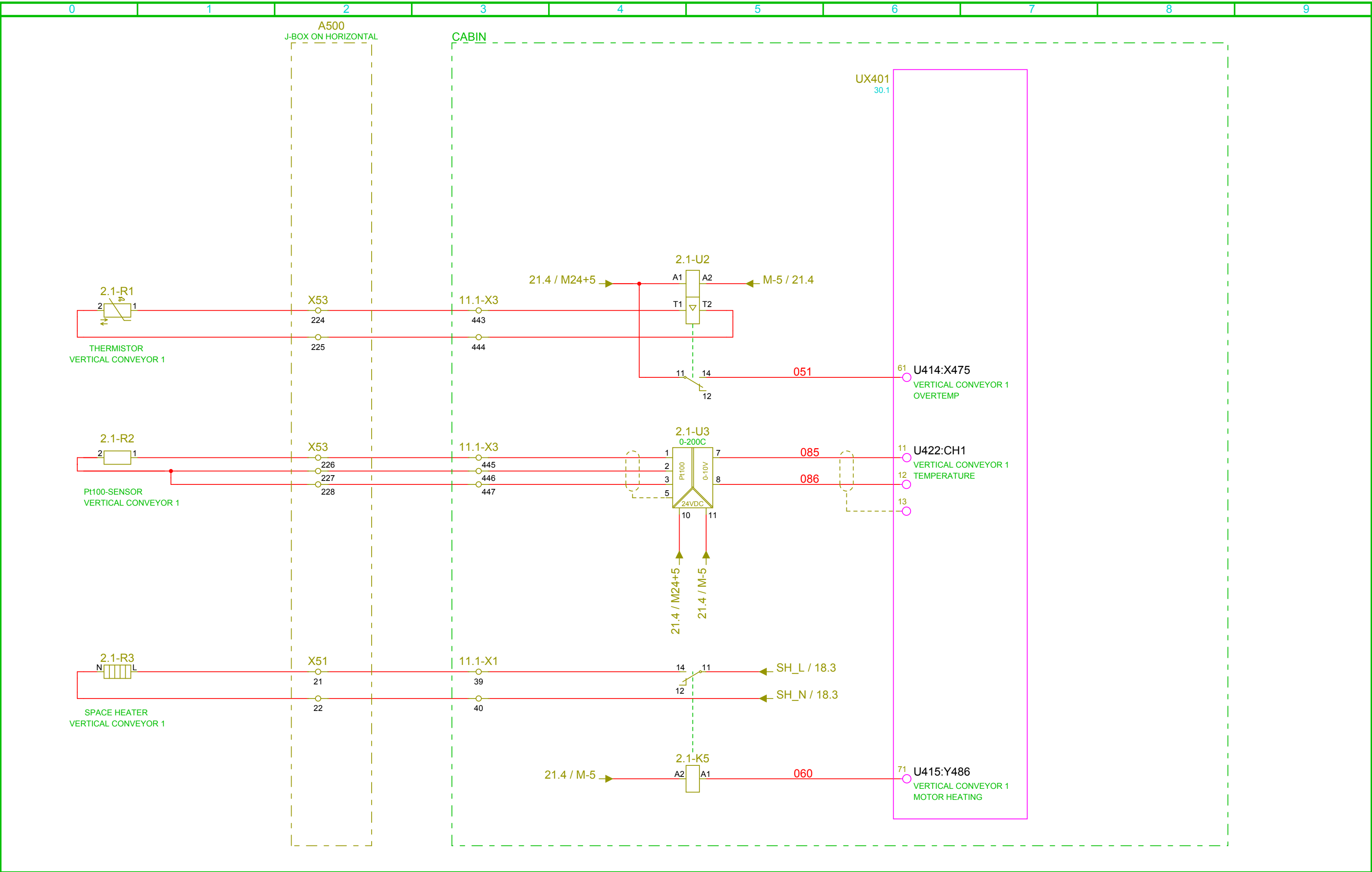


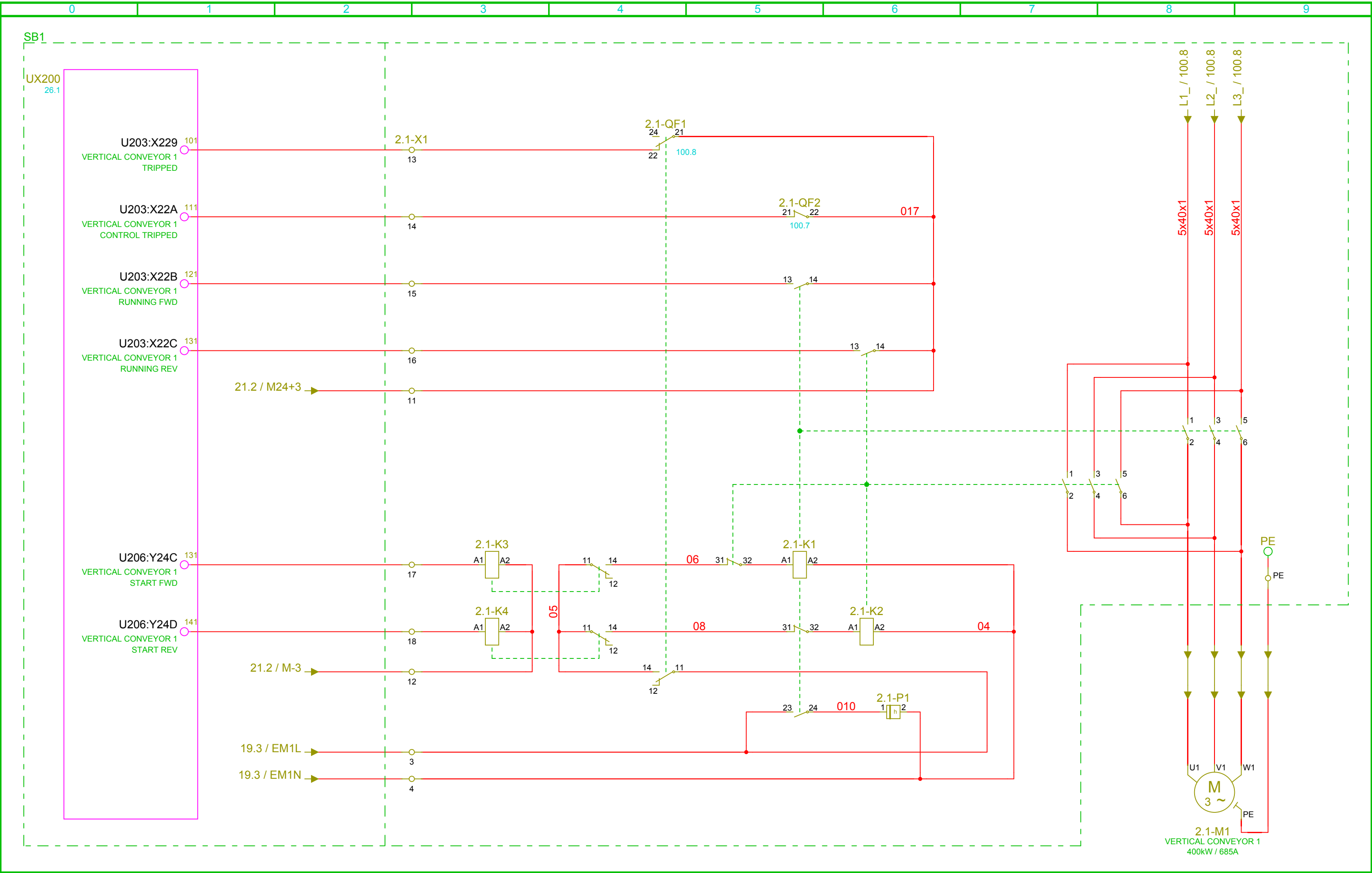


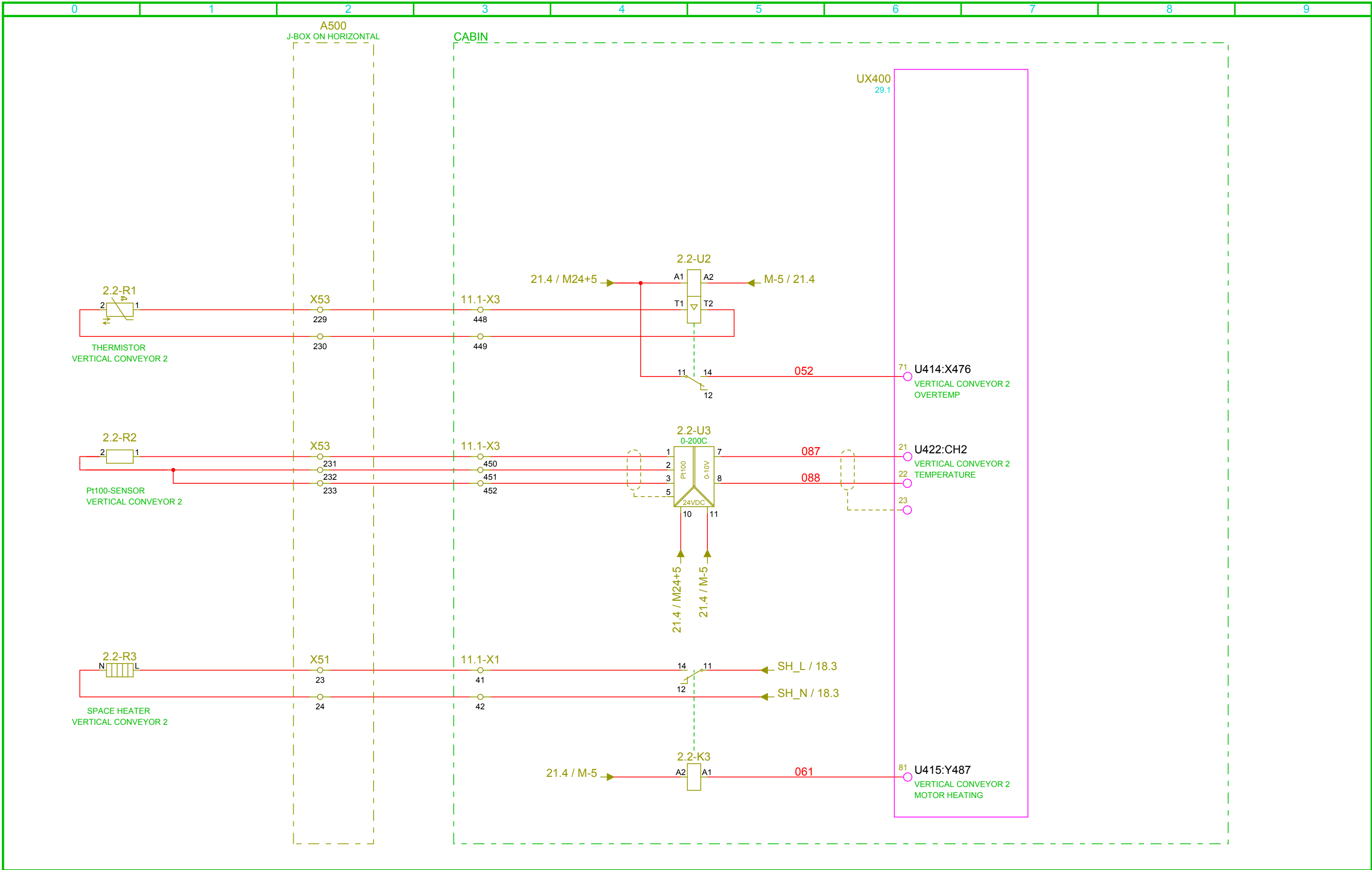




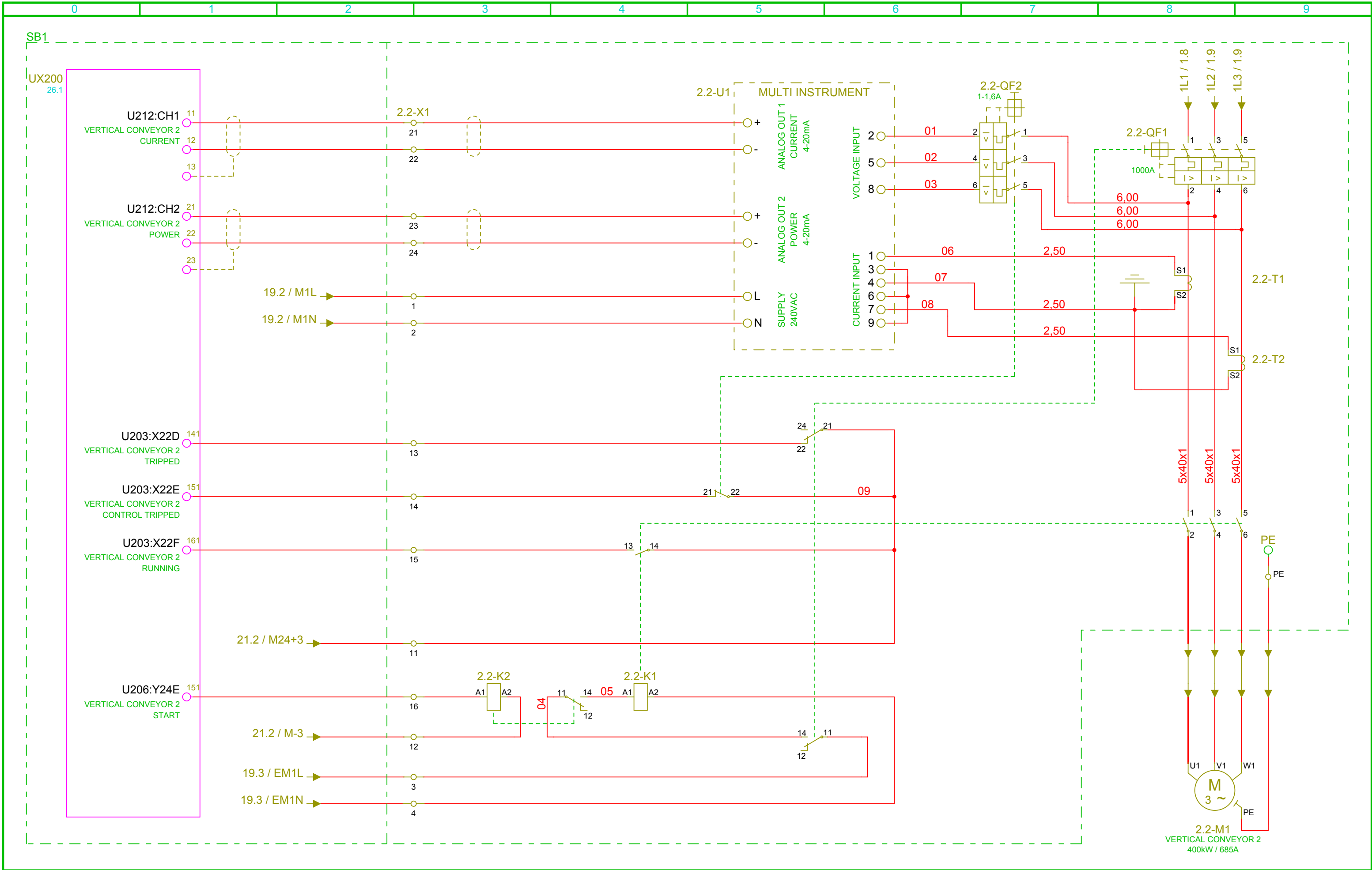


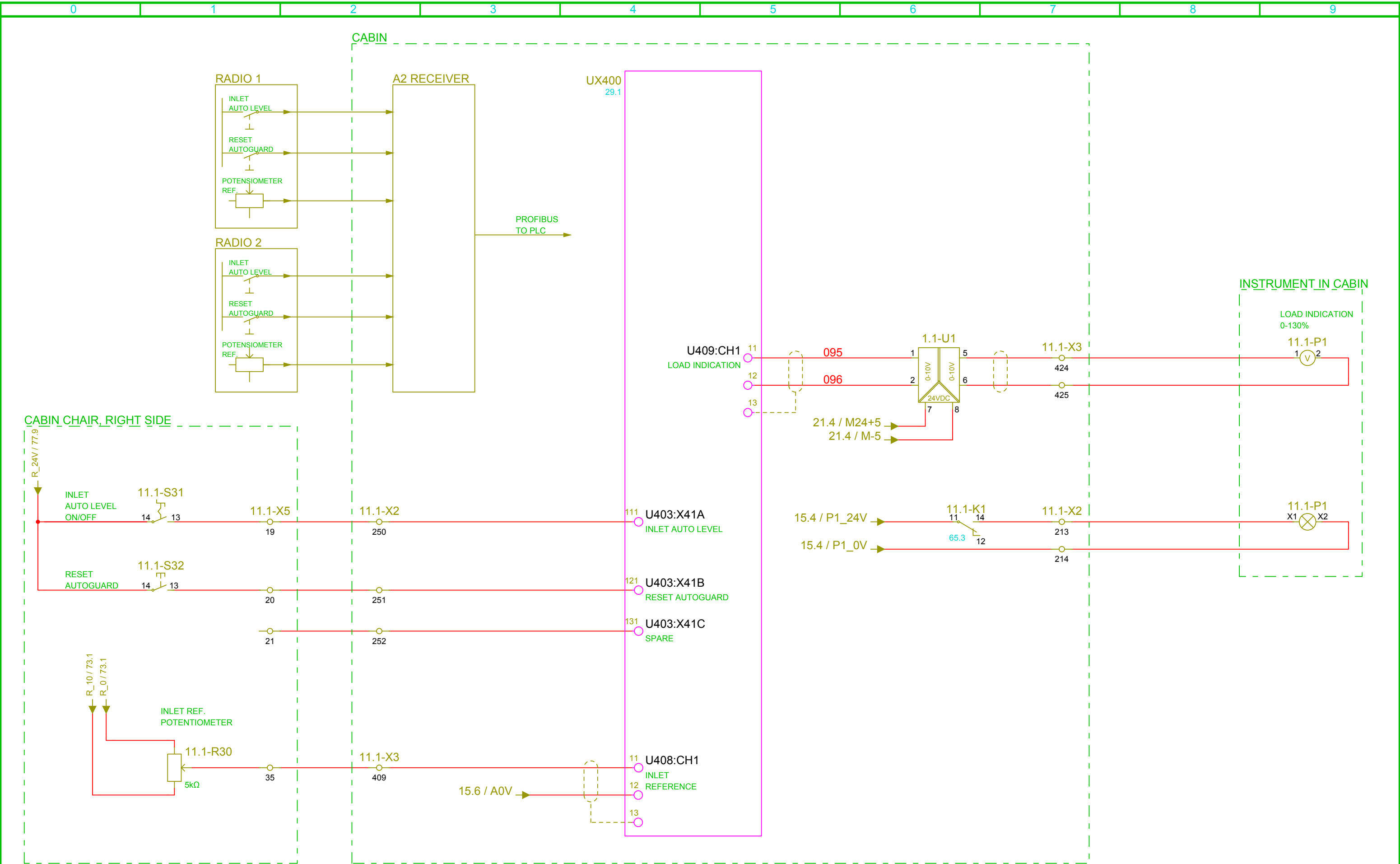


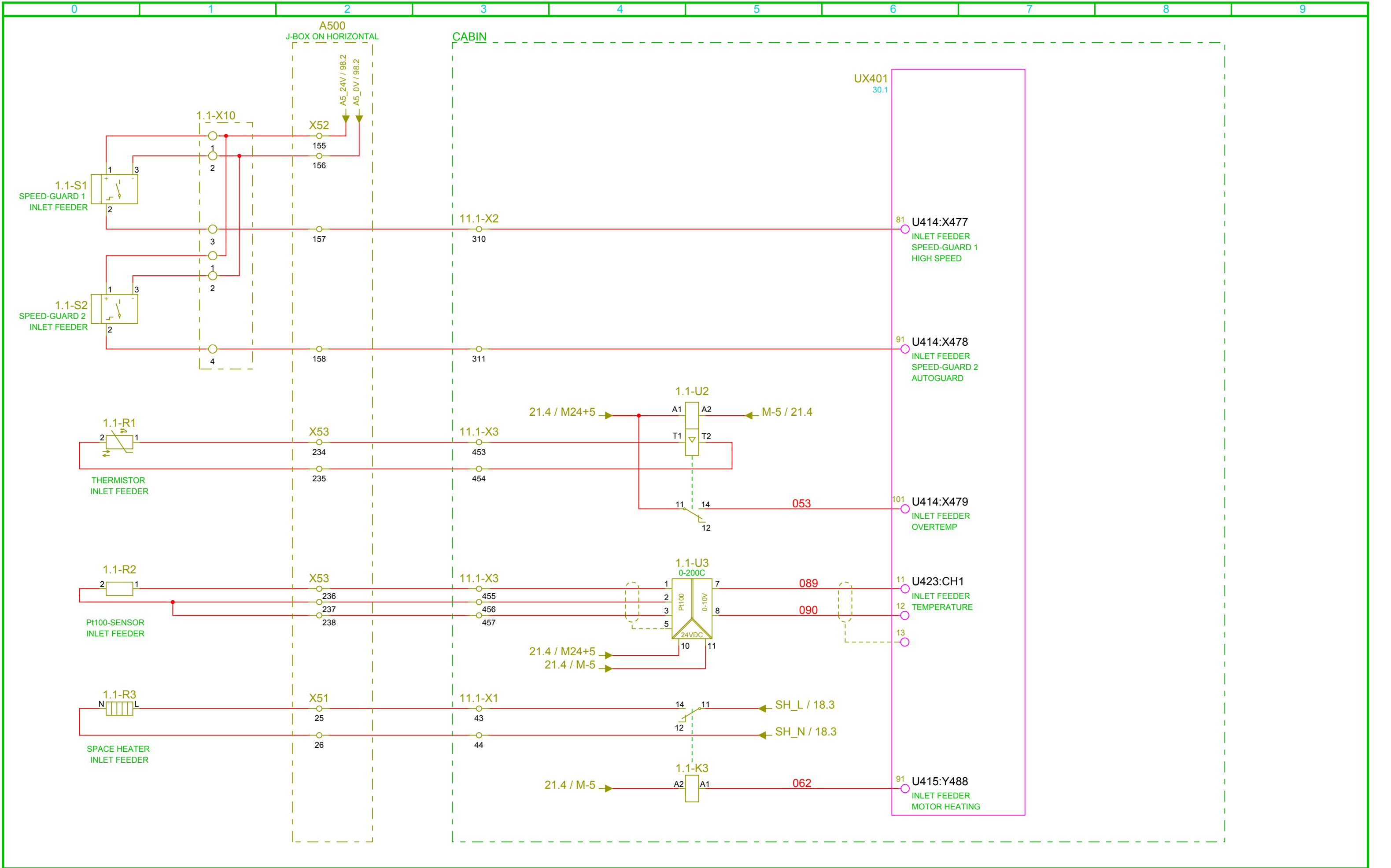


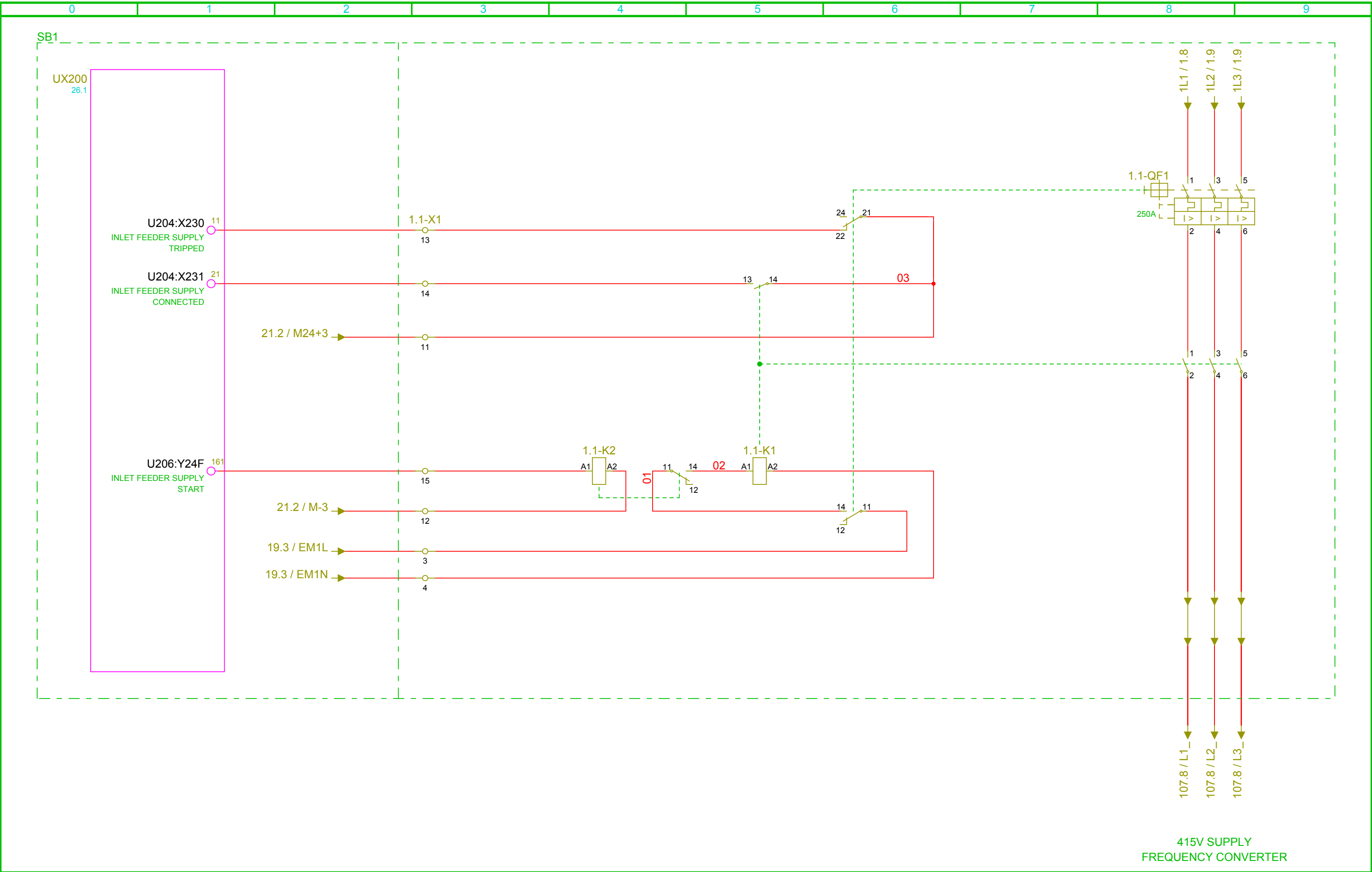


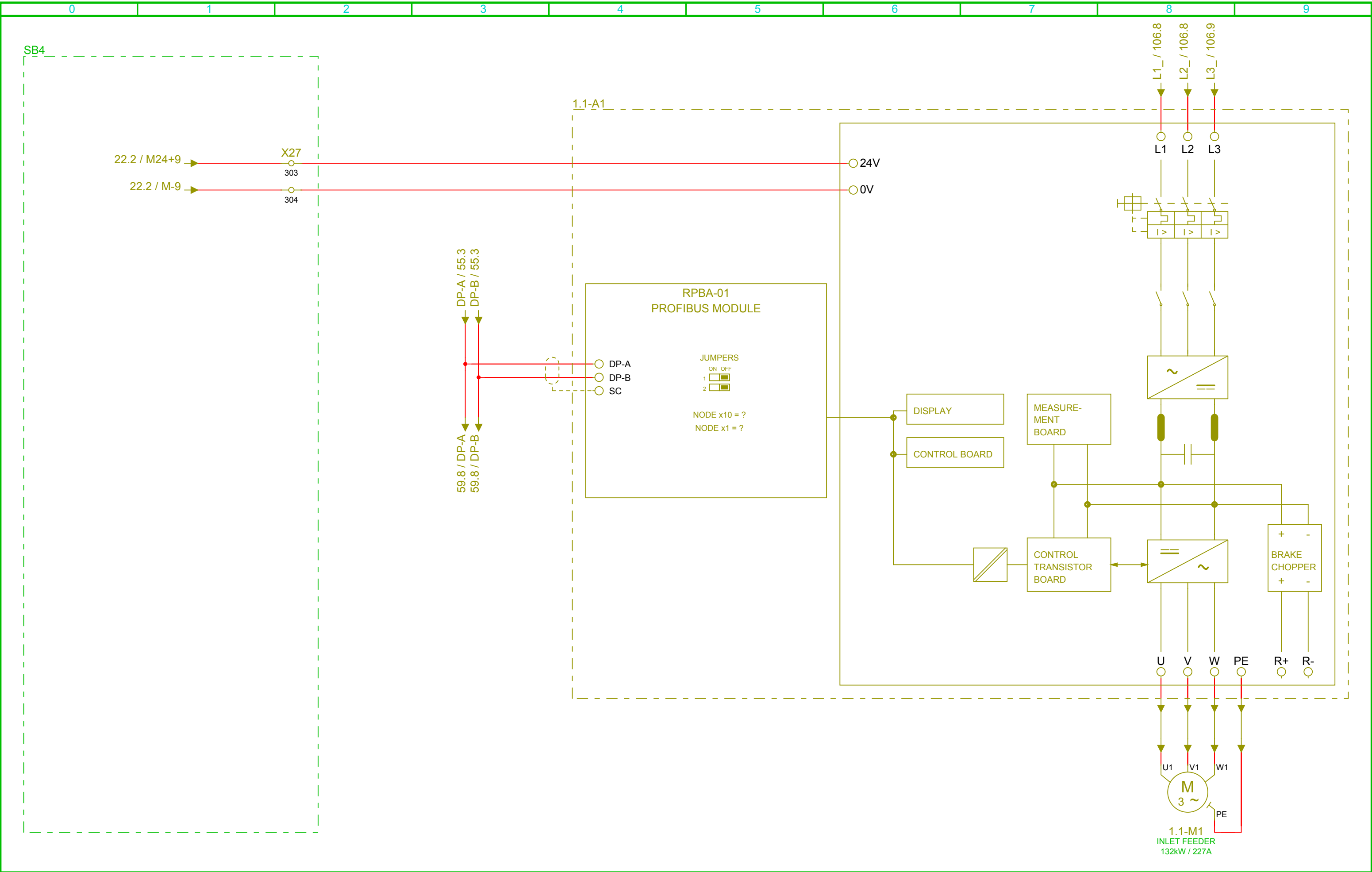
reproduced, or exploited in any other unauthorized way.

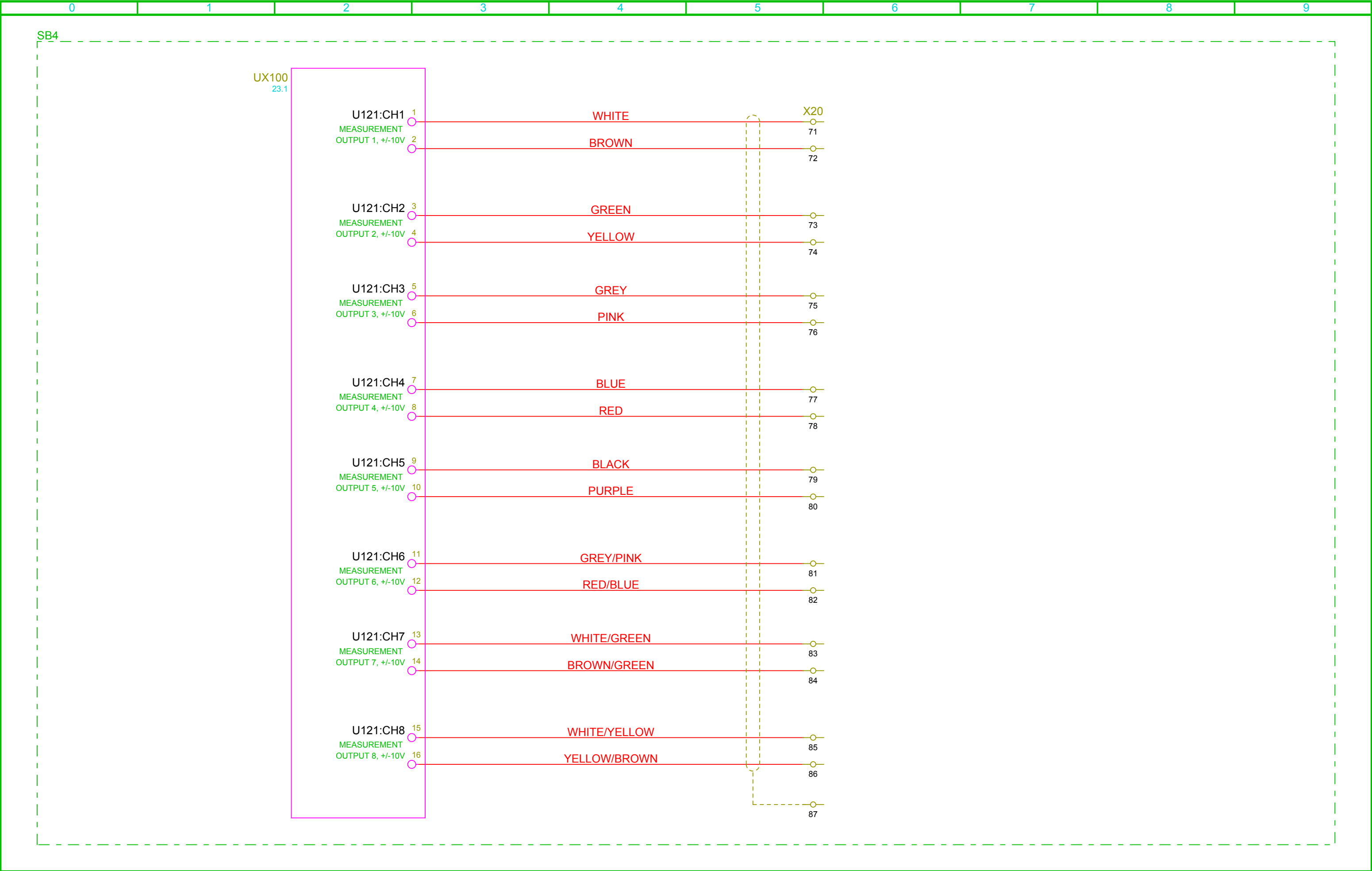












0	1	2	3	4	5	6	7	8	9
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BMH_CON_TABLE

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB1 240V CONTROL VOLTAGE		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
REV.	MODIFICATION			DATE			CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - F	
							APPR.	FILE NAME A524.DB		SHEET 1	CONT. SHEET 2

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB1 240V CONTROL VOLTAGE		DESIGN M.D.		SCALE 1 : 1		DATE 2006-10-16	
CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - F									
APPR.		FILE NAME A524.DB		SHEET 2		CONT. SHEET 3							
REV.		MODIFICATION		DATE									

0	1	2	3	4	5	6	7	8	9
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BMH_CON_TABLE

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB1 240V CONTROL VOLTAGE		DESIGN M.D.		SCALE 1 : 1		DATE 2006-10-16	
CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - F									
APPR.		FILE NAME A524.DB		SHEET 3		CONT. SHEET 4							
REV.	MODIFICATION			DATE									

0	1	2	3	4	5	6	7	8	9
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BMH_CON_TABLE

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB1 240V CONTROL VOLTAGE		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
REV.	MODIFICATION			DATE	CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - F			
					APPR.	FILE NAME A524.DB		SHEET 4	CONT. SHEET 5		

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB3 240V CONTROL VOLTAGE		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
REV.	MODIFICATION			DATE		CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - F		
						APPR.	FILE NAME A524.DB		SHEET 9	CONT. SHEET 10	

0		1		2		3		4		5		6		7		8		9	
BMH_CON_TABLE																			
TOGGLE LINK						TOGGLE LINK						TOGGLE LINK							
DEN.	ITEM	TERMINAL	SHEET	ORDER		DEN.	ITEM	TERMINAL	SHEET	ORDER		DEN.	ITEM	TERMINAL	SHEET	ORDER			
M24+4	0.2-QF1	11	3.1	1															
M24+4	0.0-K1	14	39.6	1															
M24+4	2.0-K1	14	96.6	2															
M24+4	2.3-K1	14	97.6	3															
M24+4	3.0-K1	14	92.8	4															
M24+4	3.2-K1	14	91.6	5															
M24+4	4.5-K1	14	87.5	6															
M24+4	4.6-K1	14	88.5	7															
M24+4	5.0-K3	14	51.8	8															
M24+4	5.0-K4	14	52.8	9															
M24+4	6.0-K1	14	67.6	10															
M24+4	6.2-K1	14	70.6	11															
M24+4	6.3-K1	14	71.8	12															
M24+4	8.1-K1	14	43.6	13															
M24+4	8.1-QF1	22	43.8	14															
M24+4	6.3-QF1	22	71.7	15															
M24+4	6.2-QF1	22	70.5	16															
M24+4	6.0-QF1	22	67.5	17															
M24+4	5.0-QF4	22	52.7	18															
M24+4	5.0-QF3	22	51.7	19															
M24+4	4.6-QF1	22	88.5	20															
M24+4	4.5-QF1	22	87.5	21															
M24+4	4.4-QF1	22	82.5	22															
M24+4	3.2-QF1	22	91.5	23															
M24+4	3.0-QF1	22	92.7	24															
M24+4	2.3-QF1	22	97.5	25															
M24+4	2.0-QF1	22	96.5	26															
M24+4	0.0-QF1	22	39.8	27															
M24+4	X12	11	21.3	28															
M24+4	0.3-F19	11	10.5	29															
M24+4	0.3-F20	11	10.7	30															
M24+4	0.3-QF2	43	4.1	31															
M24+4	0.3-QF3	43	4.3	32															
M24+4	U300	11	28.2	33															
M24+4	U300	33	28.2	34															
M24+4	0.3-QF1	11	3.6	35															

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB4 240V CONTROL VOLTAGE		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - F		APPR.	FILE NAME A524.DB	SHEET 13	CONT. SHEET 14				
REV.	MODIFICATION			DATE							

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB4 240V CONTROL VOLTAGE		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
REV.	MODIFICATION			DATE		CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - F		
						APPR.	FILE NAME A524.DB		SHEET 14	CONT. SHEET 15	

0		1		2		3		4		5		6		7		8		9	
BMH_CON_TABLE																			
TOGGLE LINK						TOGGLE LINK						TOGGLE LINK							
DEN.	ITEM	TERMINAL	SHEET	ORDER		DEN.	ITEM	TERMINAL	SHEET	ORDER		DEN.	ITEM	TERMINAL	SHEET	ORDER			
M24+2	0.5-F2	11	21.0	1		M24+2	X27	170	50.2	39		M24+2	2.0-S6	13	95.4	77			
M24+2	4.5-K24	11	86.4	2		M24+2	X27	167	50.2	40		M24+2	3.2-S6	11	90.3	78			
M24+2	4.5-K23	11	86.4	3		M24+2	X27	165	42.2	41		M24+2	0.7-S1	13	34.3	79			
M24+2	4.5-K22	11	86.4	4		M24+2	X27	163	42.2	42		M24+2	0.7-S2	23	32.8	80			
M24+2	4.5-K21	11	86.4	5		M24+2	X27	161	42.2	43		M24+2	0.7-S3	13	32.0	81			
M24+2	4.0-K4	11	78.3	6		M24+2	X27	159	42.2	44									
M24+2	4.0-K3	11	78.3	7		M24+2	X27	157	42.2	45									
M24+2	4.0-K2	11	78.3	8		M24+2	X27	154	42.2	46									
M24+2	4.0-K1	11	78.3	9		M24+2	X27	148	40.3	47									
M24+2	0.11-K1	11	44.5	10		M24+2	X27	135	78.2	48									
M24+2	0.4-K1	11	19.7	11		M24+2	X27	132	32.2	49									
M24+2	0.7-K1	13	32.5	12		M24+2	X27	128	32.2	50									
M24+2	0.4-F1	11	19.2	13		M24+2	X27	124	32.2	51									
M24+2	X27	299	90.2	14		M24+2	X27	120	32.2	52									
M24+2	X27	309	77B.2	15		M24+2	X27	105	7.1	53									
M24+2	X27	283	85.3	16		M24+2	X27	103	7.1	54									
M24+2	X27	278	85.3	17		M24+2	X27	101	7.1	55									
M24+2	X27	274	85.3	18		M24+2	X28	437	83.2	56									
M24+2	X27	270	82.5	19		M24+2	X28	446	84.2	57									
M24+2	X27	250	84.2	20		M24+2	0.11-U2	17	44.4	58									
M24+2	X27	247	84.2	21		M24+2	0.11-U2	9	44.4	59									
M24+2	X27	238	83.2	22		M24+2	0.11-U1	17	44.4	60									
M24+2	X27	235	83.2	23		M24+2	0.11-U1	9	44.4	61									
M24+2	X27	233	79.2	24		M24+2	0.0-U1	10	38.3	62									
M24+2	X27	230	79.2	25		M24+2	0.0-U2	10	38.3	63									
M24+2	X27	227	79.2	26		M24+2	0.0-U3	10	38.3	64									
M24+2	X27	218	59.2	27		M24+2	0.11-U3	7	44.6	65									
M24+2	X27	216	59.2	28		M24+2	0.11-U4	7	44.6	66									
M24+2	X27	213	59.2	29		M24+2	4.0-U1	7	83.3	67									
M24+2	X27	210	59.2	30		M24+2	4.0-U2	7	84.3	68									
M24+2	X27	198	52.2	31		M24+2	U121	24V	25.3	69									
M24+2	X27	193	52.2	32		M24+2	U110	24V	24.5	70									
M24+2	X27	191	51.2	33		M24+2	U109	24V	24.4	71									
M24+2	X27	186	51.2	34		M24+2	U108	24V	24.3	72									
M24+2	X27	182	50.2	35		M24+2	A2	X2: 3	34.1	73									
M24+2	X27	179	50.2	36		M24+2	A2	X3: 3	34.1	74									
M24+2	X27	176	50.2	37	M24+2	6.0-S6	13	66.3	75										
M24+2	X27	173	50.2	38	M24+2	5.0-S12	13	53.3	76										



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

SWITCHBOARD CONNECTION TABLE SB4

24VDC CONTROL VOLTAGE

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-16

DRAWING NO.
A524 - F

SHEET
15

CONT. SHEET
16

0	1	2	3	4	5	6	7	8	9
BMH_CON_TABLE									
TOGGLE LINK					TOGGLE LINK				
DEN.	ITEM	TERMINAL	SHEET	ORDER	DEN.	ITEM	TERMINAL	SHEET	ORDER
M-2	U103	COM	23.6	1	M-2	0.7-K8	A2	34.6	39
M-2	U105	COM	23.7	2	M-2	0.11-K1	A2	44.5	40
M-2	U104	COM	23.7	3	M-2	3.2-K3	A2	90.3	41
M-2	U106	COM	24.2	4	M-2	3.2-K4	A2	90.7	42
M-2	U107	COM	24.2	5	M-2	4.0-K1	A2	78.3	43
M-2	U108	COM	24.3	6	M-2	4.0-K2	A2	78.3	44
M-2	U109	COM	24.4	7	M-2	4.0-K3	A2	78.3	45
M-2	U110	COM	24.5	8	M-2	4.0-K4	A2	78.3	46
M-2	U121	0V	25.3	9	M-2	4.0-K5	A2	83.2	47
M-2	0.0-U1	11	38.3	10	M-2	4.0-K6	A2	83.6	48
M-2	0.0-U2	11	38.3	11	M-2	4.0-K7	A2	83.6	49
M-2	0.0-U3	11	38.3	12	M-2	4.0-K8	A2	84.6	50
M-2	0.11-U3	8	44.6	13	M-2	4.0-K9	A2	84.6	51
M-2	0.11-U4	8	44.6	14	M-2	4.0-K10	A2	84.6	52
M-2	4.0-U1	8	83.3	15	M-2	4.3-K3	A2	81.3	53
M-2	4.0-U2	8	84.3	16	M-2	4.3-K4	A2	81.3	54
M-2	0.11-U1	18	44.4	17	M-2	4.3-K5	A2	81.3	55
M-2	0.11-U2	18	44.4	18	M-2	4.3-K6	A2	81.3	56
M-2	X28	447	84.2	19	M-2	4.3-K7	A2	81.3	57
M-2	X28	438	83.2	20	M-2	4.3-K8	A2	81.3	58
M-2	X27	136	78.2	21	M-2	4.3-K9	A2	81.3	59
M-2	X27	168	50.2	22	M-2	4.3-K10	A2	81.3	60
M-2	X27	171	50.2	23	M-2	4.3-K11	A2	81.3	61
M-2	X27	174	50.2	24	M-2	4.3-K12	A2	81.3	62
M-2	X27	177	50.2	25	M-2	4.4-K1	A2	82.4	63
M-2	X27	180	50.2	26	M-2	4.4-K2	A2	82.4	64
M-2	X27	211	59.2	27	M-2	4.5-K10	A2	85.5	65
M-2	X27	214	59.2	28	M-2	4.5-K11	A2	86.6	66
M-2	X27	228	79.2	29	M-2	4.5-K12	A2	86.4	67
M-2	X27	231	79.2	30	M-2	4.5-K13	A2	86.6	68
M-2	X27	236	83.2	31	M-2	5.0-K9	A2	51.2	69
M-2	X27	239	83.2	32	M-2	5.0-K10	A2	52.2	70
M-2	X27	248	84.2	33	M-2	5.0-K11	A2	53.6	71
M-2	X27	251	84.2	34	M-2	5.2-K5	A2	59.3	72
M-2	X27	277	85.3	35					
M-2	X27	310	77B.2	36					
M-2	0.7-K4	A2	34.6	37					
M-2	0.7-K7	A2	34.6	38					

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB4 24VDC CONTROL VOLTAGE		DESIGN M.D.		SCALE 1 : 1		DATE 2006-10-16	
CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - F									
APPR.		FILE NAME A524.DB		SHEET 17		CONT. SHEET 18							
REV.		MODIFICATION		DATE									

[illegible]

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB4 24VDC CONTROL VOLTAGE		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
REV.	MODIFICATION			DATE		CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - F		
						APPR.	FILE NAME A524.DB		SHEET 20	CONT. SHEET 21	

[illegible]

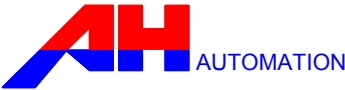
 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE SB4 24VDC CONTROL VOLTAGE		DESIGN M.D.		SCALE 1 : 1		DATE 2006-10-16	
CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - F									
APPR.		FILE NAME A524.DB		SHEET 22		CONT. SHEET 23							
REV.		MODIFICATION		DATE									

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE CABIN 240V CONTROL VOLTAGE		DESIGN M.D. SCALE 1 : 1 CHECKED PROJECT NO. 202A6086 APPR. FILE NAME A524.DB		DATE 2006-10-16 DRAWING NO. A524 - F SHEET 24 CONT. SHEET 25	
REV.	MODIFICATION	DATE									

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BMH_CON_TABLE									
TOGGLE LINK					TOGGLE LINK				
DEN.	ITEM	TERMINAL	SHEET	ORDER	DEN.	ITEM	TERMINAL	SHEET	ORDER
M24+5	11.1-X2	203	21.4	1	M24+5	6.1-U2	A1	68.5	39
M24+5	11.1-X2	211	7.1	2	M24+5	7.1-K7	11	74.5	40
M24+5	11.1-X2	217	16.1	3	M24+5	11.1-F1	11	14.2	41
M24+5	11.1-X2	232	34.7	4	M24+5	0.10-K2	11	77B.6	42
M24+5	11.1-X2	256	33.2	5					
M24+5	11.1-X2	260	33.2	6					
M24+5	11.1-X2	264	33.2	7					
M24+5	11.1-X2	267	66.2	8					
M24+5	11.1-X2	294	41.5	9					
M24+5	1.1-U1	7	104.6	10					
M24+5	1.1-U3	10	105.5	11					
M24+5	2.0-U1	10	95.4	12					
M24+5	2.0-U2	7	95.4	13					
M24+5	2.1-U3	10	99.5	14					
M24+5	2.2-U3	10	102.5	15					
M24+5	3.1-U3	10	93.5	16					
M24+5	5.0-U1	7	41.6	17					
M24+5	6.0-U1	10	66.3	18					
M24+5	6.1-U3	10	68.5	19					
M24+5	6.5-U2	7	72.3	20					
M24+5	6.5-U3	7	72.6	21					
M24+5	6.6-U2	7	73.3	22					
M24+5	6.6-U3	7	73.6	23					
M24+5	U401	11	29.2	24					
M24+5	U401	33	29.2	25					
M24+5	U411	11	30.2	26					
M24+5	U411	33	30.2	27					
M24+5	U417	11	30.7	28					
M24+5	U417	33	30.7	29					
M24+5	1.1-U2	11	105.5	30					
M24+5	1.1-U2	A1	105.5	31					
M24+5	2.1-U2	11	99.5	32					
M24+5	2.1-U2	A1	99.5	33					
M24+5	2.2-U2	11	102.5	34					
M24+5	2.2-U2	A1	102.5	35					
M24+5	3.1-U2	11	93.5	36					
M24+5	3.1-U2	A1	93.5	37					
M24+5	6.1-U2	11	68.5	38					

0	1	2	3	4	5	6	7	8	9
BMH_CON_TABLE									
TOGGLE LINK					TOGGLE LINK				
DEN.	ITEM	TERMINAL	SHEET	ORDER	DEN.	ITEM	TERMINAL	SHEET	ORDER
M-5	11.1-X2	204	21.4	1	M-5	2.1-U2	A2	99.5	39
M-5	11.1-X2	218	16.1	2	M-5	2.2-U2	A2	102.5	40
M-5	11.1-X2	233	34.7	3	M-5	3.1-U2	A2	93.5	41
M-5	11.1-X2	313	77B.6	4	M-5	6.1-U2	A2	68.5	42
M-5	11.1-X2	295	41.5	5	M-5	0.10-K2	A2	77B.6	43
M-5	2.0-U2	2	95.4	6					
M-5	1.1-U1	8	104.6	7					
M-5	1.1-U3	11	105.5	8					
M-5	2.0-U1	11	95.4	9					
M-5	2.0-U2	8	95.4	10					
M-5	2.1-U3	11	99.5	11					
M-5	2.2-U3	11	102.5	12					
M-5	3.1-U3	11	93.5	13					
M-5	5.0-U1	8	41.6	14					
M-5	6.0-U1	11	66.3	15					
M-5	6.1-U3	11	68.5	16					
M-5	6.5-U2	8	72.3	17					
M-5	6.5-U3	8	72.6	18					
M-5	6.6-U2	8	73.3	19					
M-5	6.6-U3	8	73.6	20					
M-5	U401	12	29.2	21					
M-5	U401	32	29.2	22					
M-5	U411	12	30.2	23					
M-5	U411	32	30.2	24					
M-5	U417	12	30.7	25					
M-5	U417	32	30.7	26					
M-5	1.1-K3	A2	105.5	27					
M-5	2.1-K5	A2	99.5	28					
M-5	2.2-K3	A2	102.5	29					
M-5	3.1-K3	A2	93.5	30					
M-5	6.1-K3	A2	68.5	31					
M-5	6.5-K1	A2	72.6	32					
M-5	6.6-K1	A2	73.6	33					
M-5	7.1-K7	A2	74.5	34					
M-5	11.1-K1	A2	65.3	35					
M-5	11.1-K3	A2	16.4	36					
M-5	11.1-K5	A2	17.2	37					
M-5	1.1-U2	A2	105.5	38					



SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

SWITCHBOARD CONNECTION TABLE CABIN

24VDC CONTROL VOLTAGE

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-16

DRAWING NO.
A524 - F

SHEET
26

CONT. SHEET
27

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT SWITCHBOARD CONNECTION TABLE CABIN 10VDC CONTROL VOLTAGE		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - F		APPR.	FILE NAME A524.DB	SHEET 29	CONT. SHEET 30				
REV.	MODIFICATION			DATE							

CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
0.0-W1	ÖLFLEX CLASSIC 110 3G1,5	1	MOTORSTARTER IN SB2	0.0-K16	SWITCH, COOLING FAN TRANSFORMER ROOM	0.0-Q11	15,0 m	39.8
		2		X0N		5		
		PE		PE		PE		
0.0-W2	ÖLFLEX CLASSIC 110 CY 12G0,75	1	TERMINAL IN SB4	X28401	J-BOX PT100-SENSORS ON HV-TRAFO	0.0-X101	15,0 m	38.2
		2		402		2		
		3		403		3		
		4		404		4		
		5		405		5		
		6		406		6		
		7		407		7		
		8		408		8		
		9		409		9		
		10						
		11						
		PE						
		SC						
0.0-W3	FIXED	1	J-BOX PT100-SENSORS ON HV-TRAFO	0.0-X101	Pt100-SENSOR IN TRANSFORMER	0.0-R11	1,0 m	38.1
		2		2		2		
		3		3		2		
0.0-W4	FIXED	1	J-BOX PT100-SENSORS ON HV-TRAFO	0.0-X104	Pt100-SENSOR IN TRANSFORMER	0.0-R21	1,0 m	38.1
		2		5		2		
		3		6		2		
0.0-W5	FIXED	1	J-BOX PT100-SENSORS ON HV-TRAFO	0.0-X107	Pt100-SENSOR IN TRANSFORMER	0.0-R31	1,0 m	38.1
		2		8		2		
		3		9		2		
0.0-W6	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN SB4	X27146	TEMP GUARD ON HV-TRAFO	0.0-B124	15,0 m	38.2
		2		147		21		

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		PE							
0.1-W1	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB1	0.1-X15	TERMINAL IN SB4	X413	10,0 m	1.8	
		2		6		14			
		PE							
0.1-W2	H07RN-F 4G35	BROWN	TERMINAL IN SB1	0.1-X2L1	POWER INTAKE 415V / 50Hz 100A	0.1-X10L1	25,0 m	2.8	
		BLACK		L2		L2			
		WHITE		L3		L3			
		PE							
0.3-W1	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3117	TRANSFORMER 415/240V 2,5kVA	0.4-T11	10,0 m	4.1	
		2		118		2			
		PE							
0.3-W2	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3119	TRANSFORMER 415/110V 3,5kVA	0.31-T11	10,0 m	4.3	
		2		120		2			
		PE							
0.3-W3	H07RN-F 5G16	BROWN	TERMINAL IN SB3	X3101	WELDING OUTLET ELHOUSE 415V / 50Hz 63A	0.3-X10L1	10,0 m	5.1	
		BLACK		102		L2			
		WHITE		103		L3			
		BLUE		104		N			
		PE							
0.3-W4	H07RN-F 5G16	BROWN	TERMINAL IN SB3	X3105	WELDING OUTLET SEASIDE 415V / 50Hz 63A	0.3-X11L1	28,0 m	5.3	
		BLACK		106		L2			
		WHITE		107		L3			
		BLUE		108		N			
		PE							
0.3-W5	H07RN-F 5G16	BROWN	TERMINAL IN SB3	X3109	WELDING OUTLET LANDSIDE 415V / 50Hz 63A	0.3-X12L1	25,0 m	5.6	
		BLACK		110		L2			



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - G

SHEET
2

CONT. SHEET
3

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		WHITE		111		L3			
		BLUE		112		N			
		PE							
0.3-W6	H07RN-F 5G6	BROWN	TERMINAL IN SB3	X3 113	TERMINAL IN CABIN	11.1-X1 1	65,0 m	6.1	
		BLACK		114		2			
		WHITE		115		3			
		BLUE		116		4			
		PE							
0.3-W7	ÖLFLEX CLASSIC 110 5G2,5	1	TERMINAL IN SB3	X3 147	SAFETY SWITCH, AC IN ELHOUSE	0.3-Q1 1	10,0 m	8.5	
		2		148		N			
		3							
		4							
		PE							
0.3-W8	ÖLFLEX CLASSIC 110 5G2,5	1	SAFETY SWITCH, AC IN ELHOUSE	0.3-Q1 2	AC IN ELHOUSE	0.3-AC1 L1	3,0 m	8.5	
		2		N		N			
		3							
		4							
		PE							
0.3-W9	ÖLFLEX CLASSIC 110 5G1,5	1	TERMINAL IN SB3	X3 125	SAFETY SWITCH, HEATER IN ELHOUSE	0.3-Q2 1	10,0 m	6.5	
		2		126		3			
		3		127		5			
		4		128		N			
		PE							
0.3-W10	ÖLFLEX CLASSIC 110 5G1,5	1	SAFETY SWITCH, HEATER IN ELHOUSE	0.3-Q2 2	HEATER IN ELHOUSE 3kW	0.3-E1 L1	1,0 m	6.5	
		2		4		L2			
		3		6		L3			
		4		N		N			
		PE							



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



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PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

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A524.DB

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2006-10-24

DRAWING NO.
A524 - G

SHEET
3

CONT. SHEET
4

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
0.3-W11	H07RN-F 3G1,5	BROWN	TERMINAL IN CABIN	11.1-X2211	LIGHT SWITCH CABIN	0.3-S1013	5,0 m	7.0	
		BLUE		212		14			
		PE							
0.3-W12	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27101	LIGHT SWITCH SEASIDE ENTRY	0.3-S1113	35,0 m	7.0	
		BLUE		102		14			
		PE							
0.3-W13	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27103	LIGHT SWITCH ELHOUSE	0.3-S1213	10,0 m	7.0	
		BLUE		104		14			
		PE							
0.3-W14	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3133	LIGHT 2x36W DUSTFILTER	0.3-H1L	15,0 m	7.4	
		BLUE		134		N			
		PE							
0.3-W15	H07RN-F 3G1,5	BROWN	LIGHT 2x36W DUSTFILTER	0.3-H1L	LIGHT 2x36W CABLE REEL	0.3-H2L	15,0 m	7.4	
		BLUE		N		N			
		PE							
0.3-W16	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3135	LIGHT 2x36W SEASIDE ENTRY	0.3-H3L	30,0 m	7.5	
		BLUE		136		N			
		PE							
0.3-W17	H07RN-F 3G1,5	BROWN	LIGHT 2x36W SEASIDE ENTRY	0.3-H3L	LIGHT 2x36W STAIRCASE SEASIDE	0.3-H4L	7,0 m	7.5	
		BLUE		N		N			
		PE							
0.3-W18	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3137	LIGHT 2x36W ELHOUSE	0.3-H5L	15,0 m	7.6	
		BLUE		138		N			
		PE							



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

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ELECTRICAL EQUIPMENT

CABLE LIST

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SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - G

SHEET
4

CONT. SHEET
5

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9	
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET		
0.3-W19	H07RN-F 3G1,5	BROWN	LIGHT 2x36W ELHOUSE	0.3-H5	L	LIGHT 2x36W ELHOUSE	0.3-H6	L	1,0 m	7.6
		BLUE			N		N			
		PE								
0.3-W20	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3	139	LIGHT 2x36W TURRET WALKWAY	0.3-H7	L	28,0 m	7.7
		BLUE			140		N			
		PE								
0.3-W21	H07RN-F 3G1,5	BROWN	LIGHT 2x36W TURRET WALKWAY	0.3-H7	L	LIGHT 2x36W TURRET WALKWAY	0.3-H8	L	15,0 m	7.7
		BLUE			N		N			
		PE								
0.3-W22	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27	105	LIGHT SWITCH LANDSIDE ENTRY	0.3-S13	13	28,0 m	7.0
		BLUE			106		14			
		PE								
0.3-W23	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3	181	LIGHT BELT LOADER GALLERY	0.3-H55	L	14,0 m	7.8
		BLUE			182		N			
		PE								
0.3-W24	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3	141	LIGHT SWITCH IN TURRET	0.3-S14	11	17,0 m	8.1
		BLUE			142		N			
		PE								
0.3-W25	H07RN-F 3G1,5	BROWN	LIGHT SWITCH IN TURRET	0.3-S14	14	LIGHT 2x36W IN TURRET	0.3-H9	L	1,0 m	8.1
		BLUE			N		N			
		PE								
0.3-W26	H07RN-F 3G1,5	BROWN	LIGHT 2x36W IN TURRET	0.3-H9	L	LIGHT 2x36W IN TURRET	0.3-H10	L	15,0 m	8.1
		BLUE			N		N			
		PE								



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-24	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - G	
APPR.	FILE NAME A524.DB	SHEET 5	CONT. SHEET 6

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
0.3-W27	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3143	LIGHT SWITCH IN ELHOUSE, BY DOOR 1	0.3-S1511	10,0 m	8.2	
		2		144		N			
		PE							
0.3-W28	ÖLFLEX CLASSIC 110 4G1,5	1	LIGHT SWITCH IN ELHOUSE, BY DOOR 1	0.3-S1514	LIGHT SWITCH IN ELHOUSE, BY DOOR 2	0.3-S1614	10,0 m	8.2	
		2		12		12			
		3		N		N			
		PE							
0.3-W29	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT SWITCH IN ELHOUSE, BY DOOR 2	0.3-S1611	LIGHT 2x36W IN ELHOUSE	0.3-H11L	4,0 m	8.2	
		2		N		N			
		PE							
0.3-W30	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT 2x36W IN ELHOUSE	0.3-H11L	LIGHT 2x36W IN ELHOUSE	0.3-H12L	2,0 m	8.2	
		2		N		N			
		PE							
0.3-W31	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT 2x36W IN ELHOUSE	0.3-H12L	LIGHT 2x36W IN ELHOUSE	0.3-H13L	2,0 m	8.2	
		2		N		N			
		PE							
0.3-W32	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3145	LIGHT SWITCH IN TRAF0 ROOM	0.3-S1711	10,0 m	8.3	
		2		146		N			
		PE							
0.3-W33	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT SWITCH IN TRAF0 ROOM	0.3-S1714	LIGHT 2x36W IN TRAF0 ROOM	0.3-H14L	4,0 m	8.3	
		2		N		N			
		PE							
0.3-W34	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT 2x36W IN TRAF0 ROOM	0.3-H14L	LIGHT 2x36W IN TRAF0 ROOM	0.3-H15L	4,0 m	8.3	
		2		N		N			
		PE							



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - G

SHEET
6

CONT. SHEET
7

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
0.3-W40	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3151	UPS, EMERGENCY LIGHT	0.3-U1Lin	10,0 m	9.1	
		2		152		Nin			
		PE							
0.3-W41	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3 (UPS SUPPLIED)	XU1301	UPS, EMERGENCY LIGHT	0.3-U1Lout	10,0 m	9.1	
		2		302		Nout			
		PE							
0.3-W42	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3 (UPS SUPPLIED)	XU1303	EMERGENCY LIGHT 2x36W ELHOUSE GABLE	0.3-H40L	10,0 m	9.3	
		BLUE		304		N			
		PE							
0.3-W43	H07RN-F 3G1,5	BROWN	EMERGENCY LIGHT 2x36W ELHOUSE GABLE	0.3-H40L	EMERGENCY LIGHT 2x36W STAIRCASE TOP	0.3-H41L	10,0 m	9.3	
		BLUE		N		N			
		PE							
0.3-W44	H07RN-F 3G1,5	BROWN	EMERGENCY LIGHT 2x36W STAIRCASE TOP	0.3-H41L	EMERGENCY LIGHT 2x36W TRAFO DOOR	0.3-H42L	15,0 m	9.3	
		BLUE		N		N			
		PE							
0.3-W45	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3 (UPS SUPPLIED)	XU1305	EMERGENCY LIGHT 2x36W STAIRCASE LANDSIDE	0.3-H43L	20,0 m	9.4	
		BLUE		306		N			
		PE							
0.3-W46	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3 (UPS SUPPLIED)	XU1305	EMERGENCY LIGHT 2x36W TURRET WALKWAY	0.3-H44L	20,0 m	9.5	
		BLUE		306		N			
		PE							
0.3-W47	H07RN-F 3G1,5	BROWN	EMERGENCY LIGHT 2x36W TURRET WALKWAY	0.3-H44L	EMERGENCY LIGHT 2x36W CABIN ARM	0.3-H45L	20,0 m	9.5	
		BLUE		N		N			
		PE							

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
0.3-W48	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3 (UPS SUPPLIED)	XU1307	AIRCRAFT LIGHT	0.3-H211	75,0 m	9.6	
		BLUE		308		2			
		PE							
0.3-W49	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3 (UPS SUPPLIED)	XU1309	LIGHT SENSITIVE PHOTO-CELL	0.3-B11	10,0 m	9.8	
		BLUE		310		2			
		PE							
0.3-W50	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3153	HEATER 10W CABLE REEL	8.1-R11	20,0 m	10.1	
		BLUE		154		2			
		PE							
0.3-W51	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3159	TERMINAL IN SB1	5.1-X15	6,0 m	10.5	
		2		160		6			
		PE							
0.3-W52	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3161	TERMINAL IN SB1	5.3-X15	6,0 m	10.7	
		2		162		6			
		PE							
0.3-W53	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3163	FIRE ALARM CONTROL PANEL	0.8-A1L	10,0 m	11.1	
		2		164		N			
		PE							
0.3-W60	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3171	OUTLET 240V IN ELHOUSE	0.3-X13L	10,0 m	12.1	
		2		172		N			
		PE							
0.3-W61	ÖLFLEX CLASSIC 110 3G1,5	1	OUTLET 240V IN ELHOUSE	0.3-X13L	OUTLET 240V IN ELHOUSE	0.3-X14L	4,0 m	12.1	
		2		N		N			
		PE							



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - G

SHEET
8

CONT. SHEET
9

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
0.3-W62	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3175	OUTLET 240V OUTSIDE ELHOUSE	0.3-X15L	10,0 m	12.3	
		BLUE		176		N			
		PE							
0.3-W63	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3179	OUTLET 240V STAIRCASE	0.3-X16L	30,0 m	12.6	
		BLUE		180		N			
		PE							
0.3-W64	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3177	OUTLET 240V OUTSIDE HYDRAUL	0.3-X18L	40,0 m	12.5	
		BLUE		178		N			
		PE							
0.3-W65	H07RN-F 3G1,5	BROWN	OUTLET 240V OUTSIDE HYDRAUL	0.3-X18L	OUTLET 240V ON VERTICAL PLATTFORM	0.3-X19L	50,0 m	12.5	
		BLUE		N		N			
		PE							
0.3-W71	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X511	FLOODLIGHT 400W VERTICAL PLATTFORM	0.3-H32L	45,0 m	16.5	
		BLUE		2		N			
		PE							
0.3-W72	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X513	FLOODLIGHT 400W VERTICAL PLATTFORM	0.3-H33L	45,0 m	16.7	
		BLUE		4		N			
		PE							
0.3-W73	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X515	J-BOX FLOODLIGT VERTICAL	0.3-X34L	45,0 m	16.8	
		BLUE		6		N			
		PE							
0.3-W74	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X517	LIGHT 2x36W HORIZONTAL	0.3-H47L	20,0 m	17.4	
		BLUE		8		N			
		PE							



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PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

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APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

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SHEET
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CONT. SHEET
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REV.

MODIFICATION

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0	1	2	3	4	5	6	7	8	9	
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET		
0.3-W75	H07RN-F 3G1,5	BROWN	LIGHT 2x36W HORIZONTAL	0.3-H47	L	LIGHT 2x36W STAIR TO VERTICAL	0.3-H48	L	20,0 m	17.4
		BLUE			N		N			
		PE								
0.3-W76	H07RN-F 3G1,5	BROWN	TERMINAL IN CABIN	11.1-X1	29	LIGHT 2x36W CABIN ARM	0.3-H54	L	20,0 m	17.8
		BLUE			30		N			
		PE								
0.3-W78	H07RN-F 3G1,5	BROWN	LIGHT 2x36W HORIZONTAL	0.3-H47	L	LIGHT 2x36W VERTICAL	0.3-H49	L	30,0 m	17.5
		BLUE			N		N			
		PE								
0.3-W79	H07RN-F 3G1,5	BROWN	LIGHT 2x36W VERTICAL	0.3-H49	L	LIGHT 2x36W VERTICAL	0.3-H50	L	15,0 m	17.5
		BLUE			N		N			
		PE								
0.3-W80	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X51	11	LIGHT 2x36W HOIST	0.3-H51	L	1,0 m	17.6
		BLUE			12		N			
		PE								
0.3-W81	H07RN-F 3G1,5	BROWN	LIGHT 2x36W HOIST	0.3-H51	L	LIGHT 2x36W ENTRANCE HORIZONTAL	0.3-H52	L	1,0 m	17.6
		BLUE			N		N			
		PE								
0.3-W82	H07RN-F 3G1,5	BROWN	LIGHT SWITCH IN HYDRAUL	0.3-S18	11	LIGHT 2x36W ENTRANCE HORIZONTAL	0.3-H52	L	1,0 m	17.6
		BLUE			N		N			
		PE								
0.3-W84	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X3	173	OUTLET 240V COMPUTER EQUIPMENT	0.3-X20	L	10,0 m	12.2
		2			174		N			
		PE								



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0	1	2	3	4	5	6	7	8	9	
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET		
0.3-W31.2	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT 2x36W IN ELHOUSE	0.3-H13	L	LIGHT 2x36W IN ELHOUSE	0.3-H16	L	2,0 m	8.2
		2			N			N		
		PE								
0.3-W61.2	ÖLFLEX CLASSIC 110 3G1,5	1	OUTLET 240V IN ELHOUSE	0.3-X14	L	OUTLET 240V IN TRAF0 ROOM	0.3-X5	L	10,0 m	12.1
		2			N			N		
		PE								
0.4-W1	ÖLFLEX CLASSIC 110 3G2,5	1	TRANSFORMER 415/240V 2,5kVA	0.4-T1	3	TERMINAL IN SB4	X4	1	10,0 m	4.1
		2			4			2		
		PE								
0.4-W2	ÖLFLEX CLASSIC 110 5G1,5	1	TERMINAL IN SB4	X4	3	TERMINAL IN SB1	X11	1	10,0 m	19.1
		2			4			2		
		3			5			3		
		4			6			4		
		PE								
0.4-W3	ÖLFLEX CLASSIC 110 5G1,5	1	TERMINAL IN SB4	X4	7	TERMINAL IN SB3	X12	1	8,0 m	19.4
		2			8			2		
		3			9			3		
		4			10			4		
		PE								
0.4-W4	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB4	X4	11	UPS, CONTROL VOLTAGE	0.4-U1	Lin	10,0 m	19.6
		2			12			Nin		
		PE								
0.4-W5	ÖLFLEX CLASSIC 110 3G1,5	1	UPS, CONTROL VOLTAGE	0.4-U1	Lout	TERMINAL IN SB4 (UPS SUPPLYED)	XU2	51	6,0 m	19.6
		2			Nout			52		
		PE								

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
0.4-W6	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB4 (UPS SUPPLIED)	XU253	OUTLET 240V (UPS) COMPUTER EQUIPMENT	0.4-X10L	6,0 m	20.1	
		2		54		N			
		PE							
0.4-W7	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4 (UPS SUPPLIED)	XU255	TERMINAL IN CABIN	11.1-X1101	65,0 m	20.3	
		BLUE		56		102			
		PE							
0.5-W1	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB4	X27107	TERMINAL IN SB1	X1111	10,0 m	21.2	
		2		108		12			
		PE							
0.5-W2	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB4	X27109	TERMINAL IN SB3	X1211	8,0 m	21.3	
		2		110		12			
		PE							
0.5-W3	H07RN-F 12G1,5	1	TERMINAL IN SB4	X27111	TERMINAL IN CABIN	11.1-X2203	65,0 m	21.4	
		2		112		204			
		3		113		205			
		4		114		206			
		5		115		207			
		6		116		208			
		7		117		209			
		8		118		210			
		9							
		10							
		11							
		PE							
0.7-W1	H07RN-F 5G1,5	BROWN	TERMINAL IN SB4	X27119	EMERGENCY STOP OUTSIDE EL. HOUSE	0.7-S421	10,0 m	32.1	
		BLACK		120		13			
		WHITE		121		14			



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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		BLUE		122		22			
		PE							
0.7-W2	H07RN-F 5G1,5	BROWN	TERMINAL IN SB4	X27123	EMERGENCY STOP STAIRCASE SEASIDE	0.7-S521	35,0 m	32.1	
		BLACK		124		13			
		WHITE		125		14			
		BLUE		126		22			
		PE							
0.7-W3	H07RN-F 5G1,5	BROWN	TERMINAL IN SB4	X27127	EMERGENCY STOP STAIRCASE LANDSIDE	0.7-S621	25,0 m	32.1	
		BLACK		128		13			
		WHITE		129		14			
		BLUE		130		22			
		PE							
0.7-W4	H07RN-F 5G1,5	BROWN	TERMINAL IN CABIN	11.1-X2259	EMERGENCY STOP OUTSIDE HYDRAUL	0.7-S721	40,0 m	33.1	
		BLACK		260		13			
		WHITE		261		14			
		BLUE		262		22			
		PE							
0.8-W1	ÖLFLEX CLASSIC 110 7G0,75	1	TERMINAL IN SB4	X27148	FIRE ALARM CONTROL PANEL	0.8-A1C	10,0 m	40.3	
		2		149		NO			
		3		150		NC			
		4		151		24VDC			
		5		152		0V			
		6							
		PE							
0.8-W2	ÖLFLEX CLASSIC 110 3G0,75	1	FIRE ALARM CONTROL PANEL	0.8-A11+	FIRE SIREN IN ELHOUSE	0.8-H11	4,0 m	40.6	
		2		1-		2			
		PE							



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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
0.8-W3	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN CABIN	11.1-X2 314	FIRE SIREN IN CABIN	0.8-H2 1	2,0 m	40.7	
		2		315		2			
		PE							
0.8-W4	ÖLFLEX CLASSIC 110 3G0,75	1	FIRE ALARM CONTROL PANEL	0.8-A1 1+	FIRE DETECTOR IN ELHOUSE	0.8-S1 L1	4,0 m	40.6	
		2		1-		L			
		PE							
0.8-W5	ÖLFLEX CLASSIC 110 3G0,75	1	FIRE DETECTOR IN ELHOUSE	0.8-S1 L2	FIRE DETECTOR IN TRAF0 ROOM	0.8-S2 L1	15,0 m	40.7	
		2		L		L			
		PE							
0.8-W6	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN CABIN	11.1-X2 316	FIRE DETECTOR IN CABIN	0.8-S3 L1	3,0 m	40.7	
		2		317		L			
		PE							
0.8-W7	ÖLFLEX CLASSIC 110 3G0,75	1	FIRE DETECTOR IN CABIN	0.8-S3 L2	TERMINAL IN CABIN	11.1-X2 318	3,0 m	40.8	
		2		L		319			
		PE							
0.8-W8	ÖLFLEX CLASSIC 110 5G0,75	1	TERMINAL IN SB4	X27 316	FIRE ALARM CONTROL PANEL	0.8-A1 2+	10,0 m	40.6	
		2		317		2-			
		3		318		2+			
		4		319		2-			
		PE							
0.8-W9	ÖLFLEX CLASSIC 110 BLACK 5G1,0	1	TERMINAL IN SB4	X27 316	TERMINAL IN CABIN	11.1-X2 314	60,0 m	40.6	
		2		317		315			
		3		318		316			
		4		319		317			
		PE							



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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
0.8-W10	ÖLFLEX CLASSIC 110 BLACK 5G1,0	1	TERMINAL IN CABIN	11.1-X2318	FIRE DETECTOR IN HYDRAUL	0.8-S4L1	50,0 m	40.9
		2		319		L		
		3						
		4						
		PE						
0.10-W1	H07RN-F 4G1,5	BROWN	TERMINAL IN SB4	X27309	SWITCH GATE TO UPPER PART CLOSED	0.10-S11	17,0 m	77B.1
		BLACK		311		4		
		WHITE		310		3		
		PE						
0.10-W2	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27312	MOVEMENTS ALARM 1	0.10-H11	30,0 m	77B.8
		BLUE		313		2		
		PE						
0.10-W3	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27314	MOVEMENTS ALARM 2	0.10-H21	35,0 m	77B.8
		BLUE		315		2		
		PE						
0.10-W4	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X52127	MOVEMENTS ALARM 3	0.10-H31	25,0 m	77B.8
		BLUE		126		2		
		PE						
0.11-W1	LIYCY-P 4x2x0,75	White	TENSION MONITOR IN SB4		J-BOX TENSION SENSOR LEFT SIDE	0.11-X1	100,0 m	44.2
		Brown						
		Green		0.11-U16		J106		
		Yellow		7		J108		
		Grey		4		J104		
		Pink		2		J102		
		Blue		3		J103		
		Red		1		J101		
		SC						

0	1	2	3	4	5	6	7	8	9	
CABLE	TYPE	CORE	FROM	CONNECTION		TO	CONNECTION	LENGTH	SHEET	
0.11-W2	FIXED	RED	J-BOX TENSION SENSOR LEFT SIDE	0.11-X1	J101	TENSION SENSOR LEFT SIDE	0.11-S1	10,0 m	44.1	
		BLUE			J103					
		GREEN			J106					
		YELLOW			J108					
0.11-W3	LIYCY-P 4x2x0,75	White	TENSION MONITOR IN SB4			J-BOX TENSION SENSOR RIGHT SIDE	0.11-X2	100,0 m	44.2	
		Brown								
		Green		0.11-U2	6		J106			
		Yellow			7		J108			
		Grey			4		J104			
		Pink			2		J102			
		Blue			3		J103			
		Red			1		J101			
		SC								
0.11-W4	FIXED	RED	J-BOX TENSION SENSOR RIGHT SIDE	0.11-X2	J101	TENSION SENSOR RIGHT SIDE	0.11-S2	10,0 m	44.1	
		BLUE			J103					
		GREEN			J106					
		YELLOW			J108					
0.12-W0	H07RN-F 4G25	BROWN	TERMINAL IN CABIN	11.1-X2	201	J-BOX RADAR SENSORS	0.12-X10	1	70,0 m	45.4
		BLACK			202			2		
		WHITE								
		PE								
0.12-W1	H07RN-F 3G4	BROWN	TERMINAL IN CABIN	11.1-X2	274	J-BOX RADAR SENSORS	0.12-X10	3	70,0 m	45.4
		BLUE			275			4		
		PE								
0.12-W2	ÖLFLEX 440 P 25G0,75	1	TERMINAL IN CABIN	11.1-X2	276	J-BOX RADAR SENSORS	0.12-X10	5	70,0 m	45.4
		2			277			6		
		3			278			12		



AUTOMATION

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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
		4		279		13		
		5		280		14		
		6		281		20		
		7		282		21		
		8		283		22		
		9		284		28		
		10		285		29		
		11		286		30		
		12		287		36		
		13		288		37		
		14		289		38		
		15		290		44		
		16		291		45		
		17		292		46		
		18		293		47		
		19		323		48		
		20						
		21						
		22						
		23						
		24						
		PE						
0.12-W3	H07RN-F 3G1,5	BROWN	J-BOX RADAR SENSORS	0.12-X107	RADAR SENSOR 1, FRONT SIDE	0.12-B112	10,0 m	45.1
		BLUE		8		4		
		PE						
0.12-W4	LIYCY-P 4x2x0,75	White	J-BOX RADAR SENSORS	0.12-X109	RADAR SENSOR 1, FRONT SIDE	0.12-B116	10,0 m	45.1
		Brown		10		8		
		Green		11		15		
		Yellow		12		13		
		Grey		13		14		
		Pink		14		7		

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		Blue							
		Red							
		SC							
0.12-W5	H07RN-F 3G1,5	BROWN	J-BOX RADAR SENSORS	0.12-X1015	RADAR SENSOR 2, RIGHT SIDE	0.12-B212	10,0 m	45.1	
		BLUE		16		4			
		PE							
0.12-W6	LIYCY-P 4x2x0,75	White	J-BOX RADAR SENSORS	0.12-X1017	RADAR SENSOR 2, RIGHT SIDE	0.12-B216	10,0 m	45.1	
		Brown		18		8			
		Green		19		15			
		Yellow		20		13			
		Grey		21		14			
		Pink		22		7			
		Blue							
		Red							
		SC							
0.12-W7	H07RN-F 3G1,5	BROWN	J-BOX RADAR SENSORS	0.12-X1023	RADAR SENSOR 3, REAR SIDE	0.12-B312	10,0 m	46.1	
		BLUE		24		4			
		PE							
0.12-W8	LIYCY-P 4x2x0,75	White	J-BOX RADAR SENSORS	0.12-X1025	RADAR SENSOR 3, REAR SIDE	0.12-B316	10,0 m	46.1	
		Brown		26		8			
		Green		27		15			
		Yellow		28		13			
		Grey		29		14			
		Pink		30		7			
		Blue							
		Red							
		SC							
0.12-W9	H07RN-F 3G1,5	BROWN	J-BOX RADAR SENSORS	0.12-X1031	RADAR SENSOR 4, LEFT SIDE	0.12-B412	10,0 m	46.1	



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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		BLUE		32		4			
		PE							
0.12-W10	LIYCY-P 4x2x0,75	White	J-BOX RADAR SENSORS	0.12-X10 33	RADAR SENSOR 4, LEFT SIDE	0.12-B4 16	10,0 m	46.1	
		Brown		34		8			
		Green		35		15			
		Yellow		36		13			
		Grey		37		14			
		Pink		38		7			
		Blue							
		Red							
		SC							
0.12-W11	H07RN-F 3G1,5	BROWN	J-BOX RADAR SENSORS	0.12-X10 39	RADAR SENSOR 5, HORIZONTAL	0.12-B5 12	46,0 m	47.1	
		BLUE		40		4			
		PE							
0.12-W12	LIYCY-P 4x2x0,75	White	J-BOX RADAR SENSORS	0.12-X10 41	RADAR SENSOR 5, HORIZONTAL	0.12-B5 16	46,0 m	47.1	
		Brown		42		8			
		Green		43		15			
		Yellow		44		13			
		Grey		45		14			
		Pink		46		7			
		Blue							
		Red							
		SC							
0.12-W13	H07RN-F 3G4	BROWN	J-BOX RADAR SENSORS	0.12-X10 201	TERMINAL IN A500	X51 27	40,0 m	47.4	
		BLUE		202		28			
		PE							
0.12-W15	UNITRONIC LI2YCY (TP) 10x2x0,25	WHITE	J-BOX RADAR SENSORS	0.12-X10 101	TERMINAL IN SB4	X28 410	75,0 m	48.5	
		BROWN		102		411			



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DESIGN
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CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - G

SHEET
19

CONT. SHEET
20

REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		GREEN		103		412			
		YELLOW		104		413			
		GREY		105		414			
		PINK		106		415			
		BLUE		107		416			
		RED		108		417			
		BLACK		109		418			
		VIOLET		110		419			
		GREY/PINK		111		420			
		RED/BLUE		112		421			
		WHITE/GREEN		113		422			
		BROWN/GREEN		114		423			
		WHITE/YELLOW		115		424			
		YELLOW/BROWN		116		425			
		WHITE/GREY		117		426			
		GREY/BROWN		118		427			
		WHITE/PINK		119		428			
		PINK/BROWN		120		429			
		SC							
0.12-W16	UNITRONIC LI2YCY (TP) 2x2x0,25	WHITE	J-BOX RADAR SENSORS	0.12-X10101	RADAR SENSOR 1, FRONT SIDE	0.12-B11	10,0 m	48.2	
		BROWN		102		2			
		GREEN		103		9			
		YELLOW		104		10			
		SC							
0.12-W17	UNITRONIC LI2YCY (TP) 2x2x0,25	WHITE	J-BOX RADAR SENSORS	0.12-X10105	RADAR SENSOR 2, RIGHT SIDE	0.12-B21	10,0 m	48.2	
		BROWN		106		2			
		GREEN		107		9			
		YELLOW		108		10			
		SC							
0.12-W18	UNITRONIC LI2YCY (TP) 2x2x0,25	WHITE	J-BOX RADAR SENSORS	0.12-X10109	RADAR SENSOR 3, REAR SIDE	0.12-B31	10,0 m	48.2	



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
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202A6086

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A524 - G

SHEET
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CONT. SHEET
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REV.

MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		BROWN		110		2			
		GREEN		111		9			
		YELLOW		112		10			
		SC							
0.12-W19	UNITRONIC LI2YCY (TP) 2x2x0,25	WHITE	J-BOX RADAR SENSORS	0.12-X10113	RADAR SENSOR 4, LEFT SIDE	0.12-B41	10,0 m	48.2	
		BROWN		114		2			
		GREEN		115		9			
		YELLOW		116		10			
		SC							
0.12-W20	UNITRONIC LI2YCY (TP) 2x2x0,25	WHITE	J-BOX RADAR SENSORS	0.12-X10117	RADAR SENSOR 5, HORIZONTAL	0.12-B51	46,0 m	48.2	
		BROWN		118		2			
		GREEN		119		9			
		YELLOW		120		10			
		SC							
0.31-W1	ÖLFLEX CLASSIC 110 3G6,0	1	TRANSFORMER 415/110V 3,5kVA	0.31-T13	TERMINAL IN SB3	X31201	10,0 m	4.3	
		2		4		202			
		PE							
0.31-W2	H07RN-F 3G2,5	BROWN	TERMINAL IN SB3	X31203	OUTLET 110V OUTSIDE ELHOUSE	0.31-X10L	10,0 m	13.2	
		BLUE		204		N			
		PE							
0.31-W3	H07RN-F 3G2,5	BROWN	TERMINAL IN SB3	X31205	OUTLET 110V STAIRCASE	0.31-X11L	30,0 m	13.3	
		BLUE		206		N			
		PE							
0.31-W4	H07RN-F 3G2,5	BROWN	TERMINAL IN SB3	X31207	OUTLET 110V OUTSIDE HYDRAUL	0.31-X12L	50,0 m	13.4	
		BLUE		208		N			
		PE							



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A524 - G

SHEET
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CONT. SHEET
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MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
0.31-W5	H07RN-F 3G2,5	BROWN	TERMINAL IN SB3	X31209	OUTLET 110V ON VERTICAL PLATTFORM	0.31-X13L	75,0 m	13.5	
		BLUE		210		N			
		PE							
0.31-W6	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB3	X31211	TERMINAL IN SB4	X417	10,0 m	13.7	
		2		212		18			
		PE							
1.1-W1	2YSLCYK-JB 3x50+3x10	1	MOTORSTARTER IN SB1	1.1-K12	FREQUENCY CONVERTER INLET FEEDER	1.1-A1L1	8,0 m	106.8	
		2		4		L2			
		3		6		L3			
		3xPE							
		SC							
1.1-W2	2YSLCYK-JB 3x50+3x10	1	MOTORSTARTER IN SB1	1.1-K12	FREQUENCY CONVERTER INLET FEEDER	1.1-A1L1	8,0 m	106.9	
		2		4		L2			
		3		6		L3			
		3xPE							
		SC							
1.1-W3	2YSLCYK-JB 3x50+3x10	BROWN	FREQUENCY CONVERTER INLET FEEDER	1.1-A1U	INLET FEEDER	1.1-M1U1	75,0 m	107.8	
		BLACK		V		V1			
		WHITE		W		W1			
		PE		PE		PE			
		SC							
1.1-W4	2YSLCYK-JB 3x50+3x10	BROWN	FREQUENCY CONVERTER INLET FEEDER	1.1-A1U	INLET FEEDER	1.1-M1U1	75,0 m	107.8	
		BLACK		V		V1			
		WHITE		W		W1			
		PE		PE		PE			
		SC							

CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
1.1-W5	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN SB4	X27303	FREQUENCY CONVERTER INLET FEEDER	1.1-A124V	10,0 m	107.3
		2		304		0V		
		PE						
1.1-W7	H07RN-F 5G1,5	BROWN	TERMINAL IN A500	X52155	J-BOX SPEED-GUARDS INLET FEEDER	1.1-X101	38,0 m	105.2
		BLACK		157		3		
		WHITE		158		4		
		BLUE		156		2		
		PE						
1.1-W8	FIXED	BROWN	J-BOX SPEED-GUARDS INLET FEEDER	1.1-X101	SPEED-GUARD 1 INLET FEEDER	1.1-S11	1,0 m	105.1
		BLUE		2		3		
		BLACK		3		2		
1.1-W9	FIXED	BROWN	J-BOX SPEED-GUARDS INLET FEEDER	1.1-X101	SPEED-GUARD 2 INLET FEEDER	1.1-S21	1,0 m	105.1
		BLUE		2		3		
		BLACK		4		2		
1.1-W10	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X53234	THERMISTOR INLET FEEDER	1.1-R11	38,0 m	105.1
		BLUE		235		2		
		PE						
1.1-W11	H07RN-F 4G1,5	BROWN	TERMINAL IN A500	X53236	Pt100-SENSOR INLET FEEDER	1.1-R21	38,0 m	105.1
		BLACK		237		2		
		WHITE		238		2		
		PE						
1.1-W12	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X5125	SPACE HEATER INLET FEEDER	1.1-R3L	38,0 m	105.1
		BLUE		26		N		
		PE						
2.0-W1	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	2.0-K12	HEATER 1 IN GEARBOX VERTICAL	2.0-E1L1	75,0 m	96.7

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		BLACK		4		L2			
		WHITE		6		L3			
		PE		X0 PE					
2.0-W2	H07RN-F 4G1,5	BROWN	HEATER 1 IN GEARBOX VERTICAL	2.0-E1 L1	HEATER 2 IN GEARBOX VERTICAL	2.0-E2 L1	2,0 m	96.8	
		BLACK		L2		L2			
		WHITE		L3		L3			
		PE							
2.0-W3	H07RN-F 4G1,5	BROWN	TERMINAL IN A500	X53 214	Pt100-SENSOR IN VERTICAL GEARBOX	2.0-R1 2	38,0 m	95.1	
		BLACK		215		1			
		WHITE		216		1			
		PE							
2.0-W4	ÖLFLEX 440 CP 4G0,75	1	TERMINAL IN A500	X53 217	PRESSURE GUARD VERTICAL GEARBOX	2.0-S4 1	38,0 m	95.1	
		2		218		2			
		3		219		3			
		PE							
		SC							
2.0-W5	H07RN-F 4G1,5	BROWN	TERMINAL IN A500	X52 152	J-BOX SPEED-GUARD VERTICAL CONVEYORS	2.0-X10 1	38,0 m	98.1	
		BLACK		154		3			
		WHITE		153		2			
		PE							
2.0-W6	FIXED	BROWN	J-BOX SPEED-GUARD VERTICAL CONVEYORS	2.0-X10 1	SPEED-GUARD VERTICAL CONVEYORS	2.0-S1 1	1,0 m	98.1	
		BLUE		2		3			
		BLACK		3		2			
2.0-W7	ÖLFLEX 440 CP 5G0,75	1	TERMINAL IN A500	X53 220	J-BOX VIBRATION SENSORS	2.0-X11 1	38,0 m	98.1	
		2		221		2			
		3		222		3			
		4		223		4			



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

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APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - G

SHEET
24

CONT. SHEET
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REV.

MODIFICATION

DATE

0		1		2		3		4		5		6		7		8		9	
CABLE	TYPE	CORE	FROM	CONNECTION		TO		CONNECTION		LENGTH	SHEET								
		PE																	
		SC																	
2.0-W8	FIXED	2	J-BOX VIBRATION SENSORS	2.0-X11	2	VIBRATION SENSOR 1	2.0-S7	-	1,0 m	98.1									
		1			1			+											
2.0-W9	FIXED	2	J-BOX VIBRATION SENSORS	2.0-X11	4	VIBRATION SENSOR 2	2.0-S8	-	1,0 m	98.1									
		1			3			+											
2.1-W1	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	2.1-K1	2	VERTICAL CONVEYOR 1	2.1-M1	U1	75,0 m	101.8									
		BLACK			4			V1											
		WHITE			6			W1											
		PE		X0	PE			PE											
2.1-W2	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	2.1-K1	2	VERTICAL CONVEYOR 1	2.1-M1	U1	75,0 m	101.9									
		BLACK			4			V1											
		WHITE			6			W1											
		PE		X0	PE			PE											
2.1-W3	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	2.1-K1	2	VERTICAL CONVEYOR 1	2.1-M1	U1	75,0 m	101.9									
		BLACK			4			V1											
		WHITE			6			W1											
		PE		X0	PE			PE											
2.1-W4	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	2.1-K1	2	VERTICAL CONVEYOR 1	2.1-M1	U1	75,0 m	101.9									
		BLACK			4			V1											
		WHITE			6			W1											
		PE		X0	PE			PE											
2.1-W5	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X53	224	THERMISTOR VERTICAL CONVEYOR 1	2.1-R1	1	38,0 m	99.1									
		BLUE			225			2											
		PE																	



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
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S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-24	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - G	
APPR.	FILE NAME A524.DB	SHEET 25	CONT. SHEET 26

0	1	2	3	4	5	6	7	8	9	
CABLE	TYPE	CORE	FROM	CONNECTION		TO	CONNECTION		LENGTH	SHEET
2.1-W6	H07RN-F 4G1,5	BROWN	TERMINAL IN A500	X53	226	Pt100-SENSOR VERTICAL CONVEYOR 1	2.1-R2	1	38,0 m	99.1
		BLACK			227			2		
		WHITE			228			2		
		PE								
2.1-W7	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X51	21	SPACE HEATER VERTICAL CONVEYOR 1	2.1-R3	L	38,0 m	99.1
		BLUE			22			N		
		PE								
2.2-W1	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	2.2-K1	2	VERTICAL CONVEYOR 2	2.2-M1	U1	75,0 m	103.8
		BLACK			4			V1		
		WHITE			6			W1		
		PE		X0	PE			PE		
2.2-W2	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	2.2-K1	2	VERTICAL CONVEYOR 2	2.2-M1	U1	75,0 m	103.9
		BLACK			4			V1		
		WHITE			6			W1		
		PE		X0	PE			PE		
2.2-W3	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	2.2-K1	2	VERTICAL CONVEYOR 2	2.2-M1	U1	75,0 m	103.9
		BLACK			4			V1		
		WHITE			6			W1		
		PE		X0	PE			PE		
2.2-W4	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	2.2-K1	2	VERTICAL CONVEYOR 2	2.2-M1	U1	75,0 m	103.9
		BLACK			4			V1		
		WHITE			6			W1		
		PE		X0	PE			PE		
2.2-W5	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X53	229	THERMISTOR VERTICAL CONVEYOR 2	2.2-R1	1	38,0 m	102.1
		BLUE			230			2		
		PE								



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
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SWEDEN

PROJECT 15050007
SHIPUNLOADER MERSEY DOCKS
ELECTRICAL EQUIPMENT
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1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
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DRAWING NO.
A524 - G

SHEET
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CONT. SHEET
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REV.

MODIFICATION

DATE

reproduced, or exploited in any other unauthorized way.

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
2.2-W6	H07RN-F 4G1,5	BROWN	TERMINAL IN A500	X53231	Pt100-SENSOR VERTICAL CONVEYOR 2	2.2-R21	38,0 m	102.1	
		BLACK		232		2			
		WHITE		233		2			
		PE							
2.2-W7	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X5123	SPACE HEATER VERTICAL CONVEYOR 2	2.2-R3L	38,0 m	102.1	
		BLUE		24		N			
		PE							
2.3-W1	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	2.3-K12	COOLING FAN GEARBOX VERTICAL	2.3-M1U1	70,0 m	97.8	
		BLACK		4		V1			
		WHITE		6		W1			
		PE		X0PE		PE			
3.0-W1	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	3.0-K12	HEATER IN GEARBOX HORIZONTAL	3.0-E1L1	70,0 m	92.8	
		BLACK		4		L2			
		WHITE		6		L3			
		PE		X0PE					
3.0-W2	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X52147	THERMOSTAT GEARBOX HORIZONTAL	3.0-S111	36,0 m	92.1	
		BLUE		148		12			
		PE							
3.1-W1	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	3.1-K12	HORIZONTAL CONVEYOR	3.1-M1U1	50,0 m	94.8	
		BLACK		4		V1			
		WHITE		6		W1			
		PE		X0PE		PE			
3.1-W2	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	3.1-K12	HORIZONTAL CONVEYOR	3.1-M1U1	50,0 m	94.8	
		BLACK		4		V1			
		WHITE		6		W1			
		PE		X0PE		PE			



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
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ELECTRICAL EQUIPMENT

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A524 - G

SHEET
27

CONT. SHEET
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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
3.1-W3	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	3.1-K12	HORIZONTAL CONVEYOR	3.1-M1U1	50,0 m	94.9	
		BLACK		4		V1			
		WHITE		6		W1			
		PE		X0PE		PE			
3.1-W4	H07RN-F 4G70	BROWN	MOTORSTARTER IN SB1	3.1-K12	HORIZONTAL CONVEYOR	3.1-M1U1	50,0 m	94.9	
		BLACK		4		V1			
		WHITE		6		W1			
		PE		X0PE		PE			
3.1-W5	H07RN-F 4G1,5	BROWN	TERMINAL IN A500	X52149	J-BOX SPEED-GUARD HORIZONTAL CONVEYOR	3.1-X101	40,0 m	93.1	
		BLACK		151		3			
		WHITE		150		2			
		PE							
3.1-W6	FIXED	BROWN	J-BOX SPEED-GUARD HORIZONTAL CONVEYOR	3.1-X101	SPEED-GUARD HORIZONTAL CONVEYOR	3.1-S11	2,0 m	93.1	
		BLUE		2		3			
		BLACK		3		2			
3.1-W7	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X53209	THERMISTOR HORIZONTAL CONVEYOR	3.1-R11	24,0 m	93.1	
		BLUE		210		2			
		PE							
3.1-W8	H07RN-F 4G1,5	BROWN	TERMINAL IN A500	X53211	Pt100-SENSOR HORIZONTAL CONVEYOR	3.1-R21	24,0 m	93.1	
		BLACK		212		2			
		WHITE		213		2			
		PE							
3.1-W9	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X5119	SPACE HEATER HORIZONTAL CONVEYOR	3.1-R3L	24,0 m	93.1	
		BLUE		20		N			
		PE							



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

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CONT. SHEET
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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET		
3.2-W1	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	3.2-K1	2	LUBRICATION PUMP HORIZONTAL CONVEYOR	3.2-M1	U1	50,0 m	91.8
		BLACK			4			V1		
		WHITE			6			W1		
		PE		X0	PE			PE		
3.2-W2	H07RN-F 5G1,5	BROWN	TERMINAL IN A500	X52	143	LEVEL SWITCH, GREASE CHAMBER FULL	3.2-S1	5	25,0 m	90.1
		BLACK			144			6		
		WHITE			145			4		
		BLUE			146			2		
		PE								
3.2-W3	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27	299	LEVEL SWITCH, GREASE BARREL LOW-LEVEL	3.2-S4	1	25,0 m	90.1
		BLUE			300			2		
		PE								
3.2-W4	FIXED	1	VALVE REFILL LUBR. GREASE	3.2-Y1	Y1	J-BOX LUBRICATION	5.2-X3	7	1,0 m	90.1
		2			Y2			8		
3.2-W5	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27	301	AIR-VALVE LUBR. GREASE	3.2-Y2	Y2	25,0 m	90.9
		BLUE			302			Y1		
		PE								
4.0-W1	H07RN-F 18G1,5	1	TERMINAL IN SB4	X27	131	LOCAL CONTROL BOX OUTLOADING	4.0-A1	EM-STOP	20,0 m	32.1
		2			132			EM-STOP		
		3			133			EM-STOP		
		4			134			EM-STOP		
		5			135			24V		
		6			136			0V		
		7			137			LOCAL		
		8			138			RB1 OPEN		
		9			139			OPEN RB1		
		10			140			RB1 CLOSED		



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

CHECKED

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SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
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DATE
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SHEET
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CONT. SHEET
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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		11		141		CLOSE RB1			
		12		142		RB2 OPEN			
		13		143		OPEN RB2			
		14		144		RB2 CLOSED			
		15		145		CLOSE RB2			
		16							
		17							
		PE							
4.0-W2	H07RN-F 4G1,5	BROWN	TERMINAL IN SB4	X27227	ROTARY PADDLE SWITCHMID LEVEL IN HOPPER	4.0-S1+	25,0 m	79.2	
		BLACK		229		SIG			
		WHITE		228		-			
		PE							
4.0-W3	H07RN-F 4G1,5	BROWN	TERMINAL IN SB4	X27230	ROTARY PADDLE SWITCHHIGH LEVEL IN HOPPER	4.0-S2+	25,0 m	79.2	
		BLACK		232		SIG			
		WHITE		231		-			
		PE							
4.0-W4	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27233	J-BOX OUTLOADING HATCHES	4.0-X111	16,0 m	79.2	
		BLUE		234		4			
		PE							
4.0-W5	H07RN-F 12G1,5	1	TERMINAL IN SB4	X27235	24VDC SOCKET OUTLOADING GATES	4.0-X31	15,0 m	83.2	
		2		236		2			
		3		237		3			
		4		238		4			
		5		239		5			
		6		240		6			
		7		249		7			
		8		250		8			
		9		251		9			
		10		252		10			



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AUTOMATION
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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
		11						
		PE						
4.0-W6	H07RN-F 12G1,5	1	24VDC SOCKET OUTLOADING GATES	4.0-X31	J-BOX OUTLOADING GATES	4.0-X131	5,0 m	83.2
		2		2		2		
		3		3		3		
		4		4		6		
		5		5		9		
		6		6		12		
		7		7		15		
		8		8		18		
		9		9		21		
		10		10		24		
		11						
		PE						
4.0-W7	LIYCY-P 12x2x0,75	White	TERMINAL IN SB4	X28430	ENCODER SOCKET OUTLOADING GATES	4.0-X41	15,0 m	83.2
		Brown		431		2		
		Green		432		3		
		Yellow		433		4		
		Grey						
		Pink						
		Blue		X28436		7		
		Red		437		8		
		Black		438		9		
		Purple		439		10		
		Grey/Pink		440		11		
		Red/Blue		441		12		
		White/Green		442		13		
		Brown/Green						
		White/Yellow						
		Yellow/Brown		X28445		16		
		White/Grey		446		17		

CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
		Grey/Brown		447		18		
		White/Pink						
		Pink/Brown						
		White/Blue						
		Brown/Blue						
		White/Red						
		Brown/Red						
		SC						
4.0-W8	LIYCY-P 12x2x0,75	White	ENCODER SOCKET OUTLOADING GATES	4.0-X41	J-BOX OUTLOADING GATES	4.0-X14101	5,0 m	83.2
		Brown		2		102		
		Green		3		103		
		Yellow		4		104		
		Grey						
		Pink						
		Blue		4.0-X47		107		
		Red		8		108		
		Black		9		109		
		Purple		10		110		
		Grey/Pink		11		111		
		Red/Blue		12		112		
		White/Green		13		113		
		Brown/Green						
		White/Yellow						
		Yellow/Brown		4.0-X416		116		
		White/Grey		17		117		
		Grey/Brown		18		118		
		White/Pink						
		Pink/Brown						
		White/Blue						
		Brown/Blue						
		White/Red						
		Brown/Red						

CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
		SC						
4.0-W9	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	4.0-K112	MOTOR SOCKET OUTLOADING GATE RB1	4.0-X1L1	15,0 m	83B.8
		BLACK		4		L2		
		WHITE		6		L3		
		PE		X0PE		PE		
4.0-W10	H07RN-F 4G1,5	BROWN	MOTOR SOCKET OUTLOADING GATE RB1	4.0-X1L1	MOTOR SLIDE GATE RB1	4.0-M1U1	5,0 m	83B.8
		BLACK		L2		V1		
		WHITE		L3		W1		
		PE		PE		PE		
4.0-W11	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	4.0-K152	MOTOR SOCKET OUTLOADING GATE RB2	4.0-X2L1	15,0 m	84B.8
		BLACK		4		L2		
		WHITE		6		L3		
		PE		X0PE		PE		
4.0-W12	H07RN-F 4G1,5	BROWN	MOTOR SOCKET OUTLOADING GATE RB2	4.0-X2L1	MOTOR SLIDE GATE RB2	4.0-M2U1	5,0 m	84B.8
		BLACK		L2		V1		
		WHITE		L3		W1		
		PE		PE		PE		
4.0-W13	H07RN-F 4G1,5	BROWN	J-BOX OUTLOADING GATES	4.0-X131	LIMITSWITCH SLIDE GATE RB1 OPEN	4.0-S51	5,0 m	83.1
		BLACK		3		4		
		WHITE		2		3		
		PE						
4.0-W14	H07RN-F 4G1,5	BROWN	J-BOX OUTLOADING GATES	4.0-X134	LIMITSWITCH SLIDE GATE RB1 CLOSED	4.0-S61	5,0 m	83.1
		BLACK		6		4		
		WHITE		5		3		
		PE						
4.0-W15	H07RN-F 4G1,5	BROWN	J-BOX OUTLOADING GATES	4.0-X137	LEVELSWITCH GATE RB1 HIGH LEVEL	4.0-S161	5,0 m	83.1

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		BLACK		9		4			
		WHITE		8		3			
		PE							
4.0-W17	H07RN-F 4G1,5	BROWN	J-BOX OUTLOADING GATES	4.0-X1313	LIMITSWITCH SLIDE GATE RB2 OPEN	4.0-S81	5,0 m	84.1	
		BLACK		15		4			
		WHITE		14		3			
		PE							
4.0-W18	H07RN-F 4G1,5	BROWN	J-BOX OUTLOADING GATES	4.0-X1316	LIMITSWITCH SLIDE GATE RB2 CLOSED	4.0-S91	5,0 m	84.1	
		BLACK		18		4			
		WHITE		17		3			
		PE							
4.0-W19	H07RN-F 4G1,5	BROWN	J-BOX OUTLOADING GATES	4.0-X1319	LEVELSWITCH GATE RB2 HIGH LEVEL	4.0-S181	5,0 m	84.1	
		BLACK		21		4			
		WHITE		20		3			
		PE							
4.0-W21	LIYCY-P 12x2x0,75	White	J-BOX OUTLOADING GATES	4.0-X14108	POSITION ENCODER SLIDE GATE RB1	4.0-S715	5,0 m	83.1	
		Brown		109		16			
		Green		101		1			
		Yellow		102		2			
		Grey		103		3			
		Pink		104		4			
		Blue							
		Red							
		Black							
		Purple							
		Grey/Pink		4.0-X14107		9			
		Red/Blue							
		White/Green							
		Brown/Green							



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AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

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Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

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PROJECT NO.
202A6086

FILE NAME
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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
		White/Yellow						
		Yellow/Brown						
		White/Grey						
		Grey/Brown						
		White/Pink						
		Pink/Brown						
		White/Blue						
		Brown/Blue						
		White/Red						
		Brown/Red						
		SC						
4.0-W22	LIYCY-P 12x2x0,75	White	J-BOX OUTLOADING GATES	4.0-X14117	POSITION ENCODER SLIDE GATE RB2	4.0-S1015	5,0 m	84.1
		Brown		118		16		
		Green		110		1		
		Yellow		111		2		
		Grey		112		3		
		Pink		113		4		
		Blue						
		Red						
		Black						
		Purple						
		Grey/Pink		4.0-X14116		9		
		Red/Blue						
		White/Green						
		Brown/Green						
		White/Yellow						
		Yellow/Brown						
		White/Grey						
		Grey/Brown						
		White/Pink						
		Pink/Brown						
		White/Blue						

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		Brown/Blue							
		White/Red							
		Brown/Red							
		SC							
4.0-W23	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X420	VALVE OPEN BELT RB1	4.5-Y11A1	15,0 m	86.3	
		BLUE		26		A2			
		PE							
4.0-W24	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X427	VALVE OPEN BELT RB2	4.5-Y13A1	15,0 m	86.3	
		BLUE		29		A2			
		PE							
4.0-W25	H07RN-F 3G1,5	BROWN	TERMINAL IN SB3	X3183		4.0-X5L		79.2	
		BLUE		184		N			
		PE							
4.0-W26	H07RN-F 3G1,5	BROWN	ROTARY PADDLE SWITCHMID LEVEL IN HOPPER	4.0-S1L		4.0-X5L		79.1	
		BLUE		N		N			
		PE							
4.0-W27	H07RN-F 3G1,5	BROWN	ROTARY PADDLE SWITCHHIGH LEVEL IN HOPPER	4.0-S2L		4.0-X5L		79.1	
		BLUE		N		N			
		PE							
4.3-W1	H07RN-F 4G2,5	BROWN	MOTORSTARTER IN SB1	4.3-F12	DUSTFILTER FAN ON UNLOADER	4.3-M1U1	17,0 m	80.8	
		BLACK		4		V1			
		WHITE		6		W1			
		PE		X0PE		PE			
4.3-W2	H07RN-F 12G1,5	1	TERMINAL IN SB4	X27259	DUST-CLEANING EQUIPMENT	4.3-A11	16,0 m	81.7	
		2		260		2			
		3		261		3			



AUTOMATION

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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		4		262		4			
		5		263		5			
		6		264		6			
		7		265		7			
		8		266		8			
		9		267		9			
		10		268		10			
		11		269		11			
		PE							
4.4-W1	H07RN-F 5G16 (eller 10mm2)	BROWN	MOTORSTARTER IN SB2	4.4-QF1 2	COMPRESSOR CONTROL CABINET	4.4-A1 L1	25,0 m	82.8	
		BLACK		4		L2			
		WHITE		6		L3			
		BLUE	X0	N		N			
		PE		PE		PE			
4.4-W2	FIXED	BROWN	COMPRESSOR CONTROL CABINET	4.4-A1 U	COMPRESSOR	4.4-M1 U1	1,0 m	82.7	
		BLACK		V		V1			
		WHITE		W		W1			
		PE		PE		PE			
4.4-W3	FIXED	1	COMPRESSOR CONTROL CABINET	4.4-A1 X	COMPRESSOR DRAIN VALVE	4.4-Y1 L	1,0 m	82.8	
		2		X		N			
4.4-W4	FIXED	1	COMPRESSOR CONTROL CABINET	4.4-A1 X	COMPRESSOR HEATER	4.4-R1 L	1,0 m	82.9	
		2		X		N			
4.4-W5	H07RN-F 5G1,5	BROWN	TERMINAL IN SB4	X27 270	COMPRESSOR CONTROL CABINET	4.4-A1 K06: 11	25,0 m	82.6	
		BLACK		271		K06: 12			
		WHITE		272		X108: 1			
		BLUE		273		X108: 2			
		PE							

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
4.5-W1	H07RN-F 4G2,5	BROWN	MOTORSTARTER IN SB2	4.5-K12	SOCKET & PLUG, RED 32A, DUSTFILTER BELT RB1	4.5-X1L1	15,0 m	87.7	
		BLACK		4		L2			
		WHITE		6		L3			
		PE		X0PE		PE			
4.5-W2	H07RN-F 3G1,5	BROWN	MOTORSTARTER IN SB2	4.5-K124	SOCKET & PLUG, YELLOW 16A, DUSTFILTER BELT	4.5-X2L	15,0 m	87.9	
		BLUE		34		N			
		PE							
4.5-W10	ÖLFLEX 440 P 18G0,75	1	TERMINAL IN SB4	X27274	YELLOW SOCKET & PLUG, YELLOW, OUTLOADING	4.5-X111	15,0 m	85.3	
		2		275		2			
		3		276		3			
		4		277		4			
		5		278		5			
		6		279		6			
		7		280		7			
		8		281		8			
		9		282		9			
		10		283		10			
		11		284		11			
		12		285		12			
		13		286		13			
		14		287		14			
		15		288		15			
		16		289		16			
		17							
		PE							
4.5-W11	H07RN-F 12G1,5	1	TERMINAL IN SB4	X419	10-P SOCKET OUTLOADING DUSTFILTER	4.5-X121	15,0 m	86.2	
		2							
		3		X421		3			
		4		22		4			



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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		5		23		5			
		6		24		6			
		7		25		7			
		8							
		9							
		10		X4 28		10			
		11							
		PE							
4.6-W1	H07RN-F 4G2,5	BROWN	MOTORSTARTER IN SB2	4.6-K1 2	SOCKET & PLUG, RED 32A, DUSTFILTER BELT RB2	4.6-X1 L1	15,0 m	88.7	
		BLACK		4		L2			
		WHITE		6		L3			
		PE		X0 PE		PE			
4.6-W2	H07RN-F 3G1,5	BROWN	MOTORSTARTER IN SB2	4.6-K1 24	SOCKET & PLUG, YELLOW 16A, DUSTFILTER BELT	4.6-X2 L	15,0 m	88.9	
		BLUE		34		N			
		PE							
5.0-W1	2YSLCYK-JB 3x50+3x10	1	MOTORSTARTER IN SB1	5.0-K1 2	FREQUENCY CONVERTER TRAVELLING	5.0-A1 L1	8,0 m	54.8	
		2		4		L2			
		3		6		L3			
		3xPE							
		SC							
5.0-W2	ÖLFLEX CLASSIC 110 3G6,0	1	FREQUENCY CONVERTER TRAVELLING	5.0-A1 R+	BRAKE RESISTOR TRAVELLING FREQUENCY CONV	5.0-R1 1	4,0 m	55.9	
		2		R-		2			
		PE							
5.0-W3	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN SB4	X27 208	FREQUENCY CONVERTER TRAVELLING	5.0-A1 24V	10,0 m	55.2	
		2		209		0V			
		PE							
5.0-W5	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB1	5.0-X1 41	TRANSFORMER	5.0-T1 1	6,0 m	54.7	



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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		2		42		2			
		PE							
5.0-W6	ÖLFLEX CLASSIC 110 3G10,0	1	TERMINAL IN SB1	5.0-X1 43	TRANSFORMER	5.0-T1 3	6,0 m	54.6	
		2		44		4			
		PE							
5.0-W7	ÖLFLEX CLASSIC 110 3G10,0	1	TERMINAL IN SB1	5.0-X1 45	TERMINAL IN SB1	5.1-X1 3	4,0 m	54.4	
		2		46		4			
		PE							
5.0-W8	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27 200	TRAVELLING ALARM 1	5.0-H1 1	30,0 m	53.8	
		BLUE		201		2			
		PE							
5.0-W9	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27 202	TRAVELLING ALARM 2	5.0-H2 1	30,0 m	53.8	
		BLUE		203		2			
		PE							
5.0-W10	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27 204	TRAVELLING ALARM 3	5.0-H3 1	30,0 m	53.8	
		BLUE		205		2			
		PE							
5.0-W11	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27 206	TRAVELLING ALARM 4	5.0-H4 1	29,0 m	53.8	
		BLUE		207		2			
		PE							
5.0-W12	H07RN-F 5G1,5	BROWN	TERMINAL IN SB4	X27 167	J-BOX TRAVELLING SYNC SWITCH	5.0-X11 1	25,0 m	50.1	
		BLACK							
		WHITE		X27 168		2			
		BLUE		169		3			
		PE							



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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
5.0-W13	H07RN-F 5G1,5	BROWN	TERMINAL IN SB4	X27170		5.0-X22	25,0 m	50.2	
		BLACK		164					
		WHITE		172					
		BLUE		171					
		PE							
5.0-W14	H07RN-F 5G1,5	BROWN	TERMINAL IN SB4	X27173		5.0-X23	22,0 m	50.2	
		BLACK		166					
		WHITE		174					
		BLUE		175					
		PE							
5.0-W15	H07RN-F 4G1,5	BROWN	TERMINAL IN SB4	X27176	TRAVELLING STOP LEFT	5.0-S52	20,0 m	50.8	
		BLACK		178		1			
		WHITE							
		PE							
5.0-W16	H07RN-F 4G1,5	BROWN	TERMINAL IN SB4	X27179	TRAVELLING STOP RIGHT	5.0-S62	20,0 m	50.8	
		BLACK		181		1			
		WHITE							
		PE							
5.0-W17	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN SB4	X27182	TEMP. GUARD TRAVELLING BRAKE RESISTOR	5.0-S1112	10,0 m	50.8	
		2		183		11			
		PE							
5.0-W18	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27184	ABSOLUTE ENCODER TRAVELLING	5.0-S1US	35,0 m	50.8	
		BLUE		185		GND			
		PE							
5.0-W19	ÖLFLEX 440 CP 5G0,75	1	TERMINAL IN A500	X52123	J-BOX WIND-SPEED SENSOR	5.0-X101	45,0 m	41.3	
		2		124		2			



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

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2006-10-24

DRAWING NO.
A524 - G

SHEET
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REV.

MODIFICATION

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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		3		X53201		5			
		4		202		6			
		PE							
		SC							
5.0-W20	FIXED	White	J-BOX WIND-SPEED SENSOR	5.0-X101	WIND-SPEED SENSOR	5.0-B124V	2,0 m	41.2	
		Brown		2		0V			
		Green		5		mA			
		Yellow		6		COM			
		Grey		3		HEAT+			
		Pink		4		HEAT-			
		SC							
5.0-W21	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	5.0-K32	HYDRAUL PUMP AUTOMATIC STORMLOCK 1	5.0-M1U1	25,0 m	51.8	
		BLACK		4		V1			
		WHITE		6		W1			
		PE		X0PE		PE			
5.0-W22	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27191	LIMIT SWITCH AUTO STORMLOCK 1	5.0-S81	30,0 m	51.1	
		BLUE		192		2			
		PE							
5.0-W23	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27186	AUTO STORMLOCK 1 RIGHT	5.0-S131	30,0 m	51.1	
		BLUE		187		2			
		PE							
5.0-W24	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27186	AUTO STORMLOCK 1 LEFT	5.0-S141	30,0 m	51.1	
		BLUE		188		2			
		PE							
5.0-W26	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	5.0-K42	HYDRAUL PUMP AUTOMATIC STORMLOCK 2	5.0-M2U1	25,0 m	52.8	
		BLACK		4		V1			
		WHITE		6		W1			



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0	1	2	3	4	5	6	7	8	9	
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET		
		PE		X0	PE		PE			
5.0-W27	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27	198	LIMIT SWITCH AUTO STORMLOCK 2	5.0-S10	1	30,0 m	52.1
		BLUE			199			2		
		PE								
5.0-W28	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27	193	AUTO STORMLOCK 2 RIGHT	5.0-S15	1	30,0 m	52.1
		BLUE			194			2		
		PE								
5.0-W29	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27	193	AUTO STORMLOCK 2 LEFT	5.0-S16	1	30,0 m	52.1
		BLUE			195			2		
		PE								
5.0-W31	2YSLCYK-JB 3x50+3x10	1	FREQUENCY CONVERTER TRAVELLING	5.0-A1	U	TERMINAL IN SB1	5.1-X1	L1	10,0 m	55.8
		2			V			L2		
		3			W			L3		
		3xPE								
		SC								
5.0-W4.1	LIYCY-P 4x2x0,75	White	FREQUENCY CONVERTER TRAVELLING	5.0-A1	X2 3	J-BOX TRAVELLING ENCODER	5.0-X12	1	30,0 m	55.4
		Brown			X2 4			2		
		Green								
		Yellow								
		Grey		5.0-A1	X2 6			4		
		Pink			X2 5			3		
		Blue			X1 2			5		
		Red			X1 3			6		
		SC								
5.0-W4.2	LIYCY-P 4x2x0,75	White	J-BOX TRAVELLING ENCODER	5.0-X12	1	PULSE ENCODER TRAVELLING MOTOR	5.1-S1	A+	2,0 m	55.4
		Brown			2			A-		
		Green								



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CONT. SHEET
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reproduced, or exploited in any other unauthorized way.

CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET		
		Yellow								
		Black		5.0-X12	4	B-				
		Pink			3	B+				
		Blue			5	-				
		Red			6	+				
		SC								
5.1-W1	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF1	2	TRAVELLING MOTOR 1	5.1-M1	U1	30,0 m	56.3
		2			4			V1		
		3			6			W1		
		3xPE		X0	PE			PE		
		SC								
5.1-W2	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF2	2	TRAVELLING MOTOR 2	5.1-M2	U1	30,0 m	56.4
		2			4			V1		
		3			6			W1		
		3xPE		X0	PE			PE		
		SC								
5.1-W3	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF3	2	TRAVELLING MOTOR 3	5.1-M3	U1	30,0 m	56.5
		2			4			V1		
		3			6			W1		
		3xPE		X0	PE			PE		
		SC								
5.1-W4	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF4	2	TRAVELLING MOTOR 4	5.1-M4	U1	25,0 m	56.7
		2			4			V1		
		3			6			W1		
		3xPE		X0	PE			PE		
		SC								
5.1-W5	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF5	2	TRAVELLING MOTOR 5	5.1-M5	U1	25,0 m	56.8
		2			4			V1		

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		3		6		W1			
		3xPE		X0PE		PE			
		SC							
5.1-W6	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF62	TRAVELLING MOTOR 6	5.1-M6U1	25,0 m	57.3	
		2		4		V1			
		3		6		W1			
		3xPE		X0PE		PE			
		SC							
5.1-W7	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF72	TRAVELLING MOTOR 7	5.1-M7U1	28,0 m	57.4	
		2		4		V1			
		3		6		W1			
		3xPE		X0PE		PE			
		SC							
5.1-W8	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF82	TRAVELLING MOTOR 8	5.1-M8U1	28,0 m	57.5	
		2		4		V1			
		3		6		W1			
		3xPE		X0PE		PE			
		SC							
5.1-W9	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF92	TRAVELLING MOTOR 9	5.1-M9U1	26,0 m	57.7	
		2		4		V1			
		3		6		W1			
		3xPE		X0PE		PE			
		SC							
5.1-W10	2YSLCYK-JB 3x2,5+3x0,5	1	MOTORPROTECTION IN SB1	5.1-QF102	TRAVELLING MOTOR 10	5.1-M10U1	26,0 m	57.8	
		2		4		V1			
		3		6		W1			
		3xPE		X0PE		PE			
		SC							



AUTOMATION

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202A6086

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SHEET
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CONT. SHEET
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MODIFICATION

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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
5.1-W11	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X131	TRAVELLING PARKING BRAKE 1	5.1-Y11	30,0 m	58.2	
		BLUE		32		2			
		PE							
5.1-W12	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X133	TRAVELLING PARKING BRAKE 2	5.1-Y21	30,0 m	58.3	
		BLUE		34		2			
		PE							
5.1-W13	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X135	TRAVELLING PARKING BRAKE 3	5.1-Y31	30,0 m	58.4	
		BLUE		36		2			
		PE							
5.1-W14	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X137	TRAVELLING PARKING BRAKE 4	5.1-Y41	25,0 m	58.4	
		BLUE		38		2			
		PE							
5.1-W15	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X139	TRAVELLING PARKING BRAKE 5	5.1-Y51	25,0 m	58.5	
		BLUE		40		2			
		PE							
5.1-W16	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X141	TRAVELLING PARKING BRAKE 6	5.1-Y61	25,0 m	58.6	
		BLUE		42		2			
		PE							
5.1-W17	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X143	TRAVELLING PARKING BRAKE 7	5.1-Y71	30,0 m	58.7	
		BLUE		44		2			
		PE							
5.1-W18	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X145	TRAVELLING PARKING BRAKE 8	5.1-Y81	30,0 m	58.7	
		BLUE		46		2			
		PE							



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A524 - G

SHEET
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CONT. SHEET
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MODIFICATION

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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
5.1-W19	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X147	TRAVELLING PARKING BRAKE 9	5.1-Y91	30,0 m	58.8	
		BLUE		48		2			
		PE							
5.1-W20	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.1-X149	TRAVELLING PARKING BRAKE 10	5.1-Y101	30,0 m	58.9	
		BLUE		50		2			
		PE							
5.2-W1	2YSLCYK-JB 3x16+3x2,5	1	MOTORSTARTER IN SB1	5.2-K12	FREQUENCY CONVERTER SLEWING	5.2-A1L1	10,0 m	61.8	
		2		4		L2			
		3		6		L3			
		3xPE							
		SC							
5.2-W2	ÖLFLEX CLASSIC 110 3G6,0	1	FREQUENCY CONVERTER SLEWING	5.2-A1R+	BRAKE RESISTOR SLEWING FREQUENCY CONV.	5.2-R11	4,0 m	62.9	
		2		R-		2			
		PE							
5.2-W3	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN SB4	X27225	FREQUENCY CONVERTER SLEWING	5.2-A124V	10,0 m	62.3	
		2		226		0V			
		PE							
5.2-W5	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB1	5.2-X141	TRANSFORMER	5.2-T11	6,0 m	61.7	
		2		42		2			
		PE							
5.2-W6	ÖLFLEX CLASSIC 110 3G6,0	1	TERMINAL IN SB1	5.2-X143	TRANSFORMER	5.2-T13	6,0 m	61.6	
		2		44		4			
		PE							
5.2-W7	ÖLFLEX CLASSIC 110 3G2,5	1	TERMINAL IN SB1	5.2-X145	TERMINAL IN SB1	5.3-X13	4,0 m	61.4	
		2		46		4			



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SHEET
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CONT. SHEET
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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		PE							
5.2-W8	H07RN-F 5G1,5	BROWN	TERMINAL IN SB4	X27210	J-BOX SLEWING SYNC SWITCH	5.2-X101	24,0 m	59.1	
		BLACK							
		WHITE		X27211		2			
		BLUE		212		3			
		PE							
5.2-W9	H07RN-F 5G1,5	BROWN	TERMINAL IN SB4	X27213	SLEWING LIMITSWITCH STOP BACK-UP	5.2-S31	24,0 m	59.1	
		BLACK							
		WHITE		X27214		3			
		BLUE		215		4			
		PE							
5.2-W10	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN SB4	X27216	TEMP. GUARD SLEWING BRAKE RESISTOR	5.2-S411	10,0 m	59.1	
		2		217		12			
		PE							
5.2-W11	H07RN-F 7G1,5	1	TERMINAL IN SB4	X27218	J-BOX LUBRICATION	5.2-X31	17,0 m	59.2	
		2		219		2			
		3		220		3			
		4		221		4			
		5		222		7			
		6							
		PE							
5.2-W12	FIXED	1	VALVE LUBR. SLEWING BEARING	5.2-Y1A1	J-BOX LUBRICATION	5.2-X33	1,0 m	59.0	
		2		A2		4			
5.2-W13	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27223	ABSOLUTE ENCODER SLEWING	5.2-S1US	40,0 m	59.8	
		BLUE		224		GND			
		PE							

0	1	2	3	4	5	6	7	8	9	
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET		
5.2-W14	2YSLCYK-JB 3x16+3x2,5	1	FREQUENCY CONVERTER SLEWING	5.2-A1	U	TERMINAL IN SB1	5.3-X1	L1	8,0 m	62.8
		2			V			L2		
		3			W			L3		
		3xPE								
		SC								
5.2-W4.1	LIYCY-P 4x2x0,75	White	FREQUENCY CONVERTER SLEWING	5.2-A1	X2 3	J-BOX SLEWING ENCODER	5.3-X10	1	35,0 m	62.4
		Brown			X2 4			2		
		Green								
		Yellow								
		Grey		5.2-A1	X2 6			4		
		Pink			X2 5			3		
		Blue			X1 2			5		
		Red			X1 3			6		
		SC								
5.2-W4.2	LIYCY-P 4x2x0,75	White	J-BOX SLEWING ENCODER	5.3-X10	1	PULSE ENCODER SLEWING MOTOR	5.3-S1	A+	2,0 m	62.4
		Brown			2			A-		
		Green								
		Yellow								
		Black		5.3-X10	4			B-		
		Pink			3			B+		
		Blue			5			-		
		Red			6			+		
		SC								
5.3-W1	2YSLCYK-JB 3x6,0+3x1,0	1	MOTORPROTECTION IN SB1	5.3-QF1	2	SLEWING MOTOR 1	5.3-M1	U1	35,0 m	63.3
		2			4			V1		
		3			6			W1		
		3xPE		X0	PE			PE		
		SC								



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CONT. SHEET
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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
5.3-W2	2YSLCYK-JB 3x6,0+3x1,0	1	MOTORPROTECTION IN SB1	5.3-QF22	SLEWING MOTOR 2	5.3-M2U1	35,0 m	63.5	
		2		4		V1			
		3		6		W1			
		3xPE		X0PE		PE			
		SC							
5.3-W3	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.3-X131	SLEWING PARKING BRAKE 1	5.3-Y11	35,0 m	64.6	
		BLUE		32		2			
		PE							
5.3-W4	H07RN-F 3G1,5	BROWN	TERMINAL IN SB1	5.3-X133	SLEWING PARKING BRAKE 1	5.3-Y21	35,0 m	64.7	
		BLUE		34		2			
		PE							
6.0-W1	H07RN-F 12G1,5	1	J-BOX IN HYDRAUL	6.1-X31	TERMINAL IN CABIN	11.1-X2267	50,0 m	66.2	
		2		2		268			
		3		4		269			
		4		6		270			
		5		8		271			
		6		10		272			
		7		11		273			
		8							
		9							
		10							
		11							
		PE							
6.0-W2	ÖLFLEX 440 CP 25G0,75	1	J-BOX IN HYDRAUL	6.1-X321	TERMINAL IN CABIN	11.1-X3410	50,0 m	66.2	
		2		22		411			
		3		23		412			
		4		24		413			
		5		25		414			



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APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - G

SHEET
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CONT. SHEET
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REV.

MODIFICATION

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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		6		26		415			
		7		27		416			
		8		28		417			
		9		29		418			
		10		30		419			
		11		31		420			
		12		32		421			
		13		33		422			
		14		34		423			
		15							
		16							
		17							
		18							
		19							
		20							
		21							
		22							
		23							
		24							
		PE							
		SC							
6.0-W3	H07RN-F 3G1,5	BROWN	J-BOX IN HYDRAUL	6.1-X3 51	TERMINAL IN CABIN	11.1-X1 33	50,0 m	68.2	
		BLUE		52		34			
		PE							
6.0-W4	ÖLFLEX CLASSIC 110 3G0,75	1	LEVEL GUARD HYDRAUL OIL	6.0-S1 1	J-BOX IN HYDRAUL	6.1-X3 1	5,0 m	66.1	
		2		3		2			
		PE							
6.0-W5	ÖLFLEX CLASSIC 110 3G0,75	1	FILTER GUARD TANK	6.0-S2 1	J-BOX IN HYDRAUL	6.1-X3 3	5,0 m	66.1	
		2		2		4			
		PE							



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

CHECKED

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SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
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DATE
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A524 - G

SHEET
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0	1	2	3	4	5	6	7	8	9	
CABLE	TYPE	CORE	FROM	CONNECTION		TO	CONNECTION		LENGTH	SHEET
6.0-W6	ÖLFLEX CLASSIC 110 3G0,75	1	FILTER GUARD MOVEMENTS	6.0-S3	1	J-BOX IN HYDRAUL	6.1-X3	5	3,0 m	66.1
		2			2			6		
		PE								
6.0-W7	ÖLFLEX CLASSIC 110 4G0,75	1	Pt100-SENSOR IN HYDRAUL TANK	6.0-R1	1	J-BOX IN HYDRAUL	6.1-X3	21	7,0 m	66.1
		2			2			22		
		3			2			23		
		PE								
6.0-W8	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	6.0-K1	2	HEATER IN HYDRAULIC OIL TANK	6.0-E1	L1	52,0 m	67.8
		BLACK			4			L2		
		WHITE			6			L3		
		PE		X0	PE					
6.1-W1	H07RN-F 4G35	BROWN	MOTORSTARTER IN SB1	6.1-F1	2	HYDRAULIC PUMP	6.1-M1	U1	50,0 m	69.8
		BLACK			4			V1		
		WHITE			6			W1		
		PE		X0	PE			PE		
6.1-W2	ÖLFLEX CLASSIC 110 3G0,75	1	J-BOX IN HYDRAUL	6.1-X3	24	THERMISTOR HYDRAUL PUMP	6.1-R1	1	3,0 m	68.1
		2			25			2		
		PE								
6.1-W3	ÖLFLEX CLASSIC 110 4G0,75	1	J-BOX IN HYDRAUL	6.1-X3	26	Pt100-SENSOR HYDRAUL PUMP	6.1-R2	1	3,0 m	68.1
		2			27			2		
		3			28			2		
		PE								
6.1-W4	H07RN-F 3G1,5	BROWN	J-BOX IN HYDRAUL	6.1-X3	51	SPACE HEATER HYDRAUL PUMP	6.1-R3	L	3,0 m	68.1
		BLUE			52			N		
		PE								



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
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SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

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SCALE
1 : 1

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202A6086

FILE NAME
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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
6.2-W1	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	6.2-K12	HYDRAUL COOLING FAN	6.2-M1U1	52,0 m	70.8	
		BLACK		4		V1			
		WHITE		6		W1			
		PE	X0	PE		PE			
6.3-W1	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	6.3-K12	HYDRAUL CIRCULATION PUMP	6.3-M1U1	50,0 m	71.8	
		BLACK		4		V1			
		WHITE		6		W1			
		PE	X0	PE		PE			
6.3-W2	ÖLFLEX CLASSIC 110 3G0,75	1	J-BOX IN HYDRAUL	6.1-X37	PRESSURE GUARD CIRCULATION PUMP	6.3-S11	4,0 m	71.1	
		2		8		3			
		PE							
6.3-W3	ÖLFLEX CLASSIC 110 4G0,75	1	J-BOX IN HYDRAUL	6.1-X39	LOCAL START/STOP CIRCULATION PUMP	6.3-A1COM	4,0 m	71.1	
		2		10		START			
		3		11		STOP			
		PE							
6.5-W1	ÖLFLEX CLASSIC 110 CY 4G0,75	1	J-BOX IN HYDRAUL	6.1-X329	LUFFING PROP. VALVE	6.5-Y11	8,0 m	72.8	
		2		30		GND			
		3		31		2			
		PE							
		SC							
6.5-W2	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X5115	LUFFING ANGLE SENSOR	6.5-B1L	20,0 m	72.1	
		BLUE		16		N			
		PE							
6.5-W3	ÖLFLEX 440 CP 4G0,75	1	TERMINAL IN A500	X53203	LUFFING ANGLE SENSOR	6.5-B124VDC	20,0 m	72.1	
		2		204		mA			
		3		205		0V			



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ELECTRICAL EQUIPMENT

CABLE LIST

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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		PE							
		SC							
6.6-W1	ÖLFLEX CLASSIC 110 CY 4G0,75	1	J-BOX IN HYDRAUL	6.1-X332	PENDULUM PROP. VALVE	6.6-Y11	8,0 m	73.8	
		2		33		GND			
		3		34		2			
		PE							
		SC							
6.6-W2	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X5117	PENDULUM ANGLE SENSOR	6.6-B1L	38,0 m	73.1	
		BLUE		18		N			
		PE							
6.6-W3	ÖLFLEX 440 CP 4G0,75	1	TERMINAL IN A500	X53206	PENDULUM ANGLE SENSOR	6.6-B124VDC	38,0 m	73.1	
		2		207		mA			
		3		208		0V			
		PE							
		SC							
7.1-W1	H07RN-F 4G25	BROWN	MOTORSTARTER IN SB1	7.1-F12	AUXILIARY HOIST	7.1-M1U1	45,0 m	75.8	
		BLACK		4		V1			
		WHITE		6		W1			
		PE		X0PE		PE			
7.1-W2	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN SB1	7.1-X141	TRANSFORMER	7.1-T21	4,0 m	76.8	
		2		42		2			
		PE							
7.1-W3	ÖLFLEX CLASSIC 110 3G6,0	1	TERMINAL IN SB1	7.1-X143	TRANSFORMER	7.1-T23	4,0 m	76.7	
		2		44		4			
		PE							
7.1-W4	H07RN-F 4G1,5	BROWN	TERMINAL IN A500	X52134	TELPHER TOP	7.1-S111	20,0 m	74.1	



AUTOMATION

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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		BLACK		135		12			
		WHITE		136		12			
		PE							
7.1-W5	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X52 137	J-BOX HOIST LINE-GUARD	7.1-X10 1	23,0 m	74.1	
		BLUE		138		2			
		PE							
7.1-W6	H07RN-F 3G1,5	BROWN	TELPER LINE GUARD	7.1-S4 11	J-BOX HOIST LINE-GUARD	7.1-X10 1	5,0 m	74.1	
		BLUE		12		2			
		PE							
7.1-W7	H07RN-F 3G1,5	BROWN	TERMINAL IN A500	X52 139	TELPER BRAKE SENSOR	7.1-S5 11	20,0 m	74.1	
		BLUE		140		12			
		PE							
8.1-W1	H07RN-F 4G1,5	BROWN	MOTORSTARTER IN SB2	8.1-K1 2	POWER CABLE REEL	8.1-M1 U1	20,0 m	43.8	
		BLACK		4		V1			
		WHITE		6		W1			
		PE		X0 PE		PE			
8.1-W2	H07RN-F 4G1,5	BROWN	TERMINAL IN SB4	X27 154	POWER CABLE REEL END OF CABLE	8.1-S1 NC	20,0 m	42.2	
		BLACK		155		C			
		WHITE		156		C			
		PE							
8.1-W3	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27 157	POWER CABLE REEL TO LEFT SIDE	8.1-S3 13	20,0 m	42.2	
		BLUE		158		14			
		PE							
8.1-W4	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27 159	POWER CABLE REEL SLACK LEFT SIDE	8.1-S4 13	20,0 m	42.2	
		BLUE		160		14			
		PE							



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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
8.1-W5	H07RN-F 3G1,5	BROWN	TERMINAL IN SB4	X27161	POWER CABLE REEL SLACK RIGHT SIDE	8.1-S513	20,0 m	42.2	
		BLUE		162		14			
		PE							
10.0-W1	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	RADIO REMOTE CONTROL RECIVIER	A2DP-A	ABSOLUTE ENCODER TRAVELLING	5.0-S1DP-A	35,0 m	50.8	
		GREEN		DP-B		DP-B			
		SC							
10.0-W2	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	PROFIBUS-DP CARD IN SB4	U100DP-A	RADIO REMOTE CONTROL RECIVIER	A2DP-A	2,0 m	23.3	
		GREEN		DP-B		DP-B			
		SC							
10.0-W3	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	PROFIBUS-DP CARD IN SB4	U100DP-A	I/O-NODE IN SB1	UX200DP-A	8,0 m	23.3	
		GREEN		DP-B		DP-B			
		SC							
10.0-W4	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	I/O-NODE IN SB1	UX201DP-A	I/O-NODE IN SB3	UX300DP-A	8,0 m	28.0	
		GREEN		DP-B		DP-B			
		SC							
10.0-W5	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	I/O-NODE IN SB3	UX300DP-A	FREQUENCY CONVERTER SLEWING	5.2-A1DP-A	8,0 m	28.0	
		GREEN		DP-B		DP-B			
		SC							
10.0-W6	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	FREQUENCY CONVERTER TRAVELLING	5.0-A1DP-A	FREQUENCY CONVERTER SLEWING	5.2-A1DP-A	2,0 m	62.3	
		GREEN		DP-B		DP-B			
		SC							
10.0-W7	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	FREQUENCY CONVERTER TRAVELLING	5.0-A1DP-A	FREQUENCY CONVERTER INLET FEEDER	1.1-A1DP-A	6,0 m	107.3	
		GREEN		DP-B		DP-B			
		SC							



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

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CABLE	TYPE	CORE	FROM	CONNECTION		TO	CONNECTION		LENGTH	SHEET
10.0-W8	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	ABSOLUTE ENCODER SLEWING	5.2-S1	DP-A	FREQUENCY CONVERTER INLET FEEDER	1.1-A1	DP-A	35,0 m	107.3
		GREEN			DP-B			DP-B		
		SC								
10.0-W9	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	I/O-NODE IN CABIN	UX400	DP-A	ABSOLUTE ENCODER SLEWING	5.2-S1	DP-A	25,0 m	29.0
		GREEN			DP-B			DP-B		
		SC								
10.0-W10	UNITRONIC BUS FD P L2/FIP 1x2x0,64	RED	I/O-NODE IN CABIN	UX400	DP-A	I/O-NODE IN CABIN	UX401	DP-A	5,0 m	29.0
		GREEN			DP-B			DP-B		
		SC								
10.0-W11	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN SB4	X27	305	ETHERNET SWITCH	10.0-U1	24V	4,0 m	22.5
		2			306			0V		
		PE								
10.0-W12	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN SB4	X27	307	ETHERNET SWITCH	10.0-U2	24V	4,0 m	22.6
		2			308			0V		
		PE								
10.0-W13	CAT5		OPERATOR PANEL IN CABIN	OP1	RJ45	ETHERNET SWITCH	10.0-U1	RJ45	65,0 m	22.5
11.1-W1	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN CABIN	11.1-X1	5	SWITCH FOR HEATER IN CABIN	11.1-S1	1	5,0 m	14.1
		2			6			3		
		PE								
11.1-W2	ÖLFLEX CLASSIC 110 3G1,5	1	SWITCH FOR HEATER IN CABIN	11.1-S1	2	J-BOX HEATERS IN CABIN	11.1-X20	L	5,0 m	14.1
		2			4			N		
		PE								
11.1-W3	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN CABIN	11.1-X1	9	COOLER 2,64kW IN CABIN	11.1-A1	L	5,0 m	14.3

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		2		10		N			
		PE							
11.1-W4	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN CABIN	11.1-X111	OUTLET 240V U.K. STD.	11.1-X10L	3,0 m	14.4	
		2		12		N			
		PE							
11.1-W5	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT SWITCH IN CABIN	11.1-S213	OUTLET 240V U.K. STD.	11.1-X10L	5,0 m	14.4	
		2		N		N			
		PE							
11.1-W6	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT SWITCH IN CABIN	11.1-S214	LIGHT IN CABIN 2x36W	11.1-H1L	3,0 m	14.4	
		2		N		N			
		PE							
11.1-W7	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN CABIN	11.1-X115	OUTLET 240V SWE. STD.	11.1-X12L	3,0 m	14.6	
		2		16		N			
		PE							
11.1-W8	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN CABIN	11.1-X119	SWITCH FOR POWER WINDOW	11.1-S42	5,0 m	14.8	
		2		20		N			
		PE							
11.1-W9	ÖLFLEX CLASSIC 110 3G1,5	1	SWITCH FOR POWER WINDOW	11.1-S41	POWER WINDOW	11.1-M1A1	5,0 m	14.8	
		2		N		A2			
		PE							
11.1-W10	ÖLFLEX CLASSIC 110 3G4,0	1	TERMINAL IN CABIN	11.1-X2215	WIPER CONTROL BOX	11.1-P224V	5,0 m	15.2	
		2		216		0V			
		PE							
11.1-W11	ÖLFLEX CLASSIC 110 CY 3G0,75	1	TERMINAL IN CABIN	11.1-X3424	LOAD INSTRUMENT IN CABIN	11.1-P11	5,0 m	104.8	
		2		425		2			



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT



GUNNARSTORP
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SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
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SCALE
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PROJECT NO.
202A6086

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0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		PE							
		SC							
11.1-W12	ÖLFLEX CLASSIC 110 3G0,75	1	TERMINAL IN CABIN	11.1-X2213	LOAD INSTRUMENT IN CABIN	11.1-P1X1	5,0 m	104.8	
		2		214		X2			
		PE							
11.1-W13	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN CABIN	11.1-X117	OUTLET 240V SWE. STD.	11.1-X13L	5,0 m	14.7	
		2		18		N			
		PE							
11.1-W15	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN CABIN	11.1-X1103	OUTLET 240V OP-PANEL	11.1-X15L	5,0 m	20.3	
		2		104		N			
		PE							
11.1-W16	ÖLFLEX CLASSIC 110 3G1,5	1	J-BOX HEATERS IN CABIN	11.1-X20L	HEATER 1kW IN CABIN, ON WALL	11.1-E2L	5,0 m	14.2	
		2		N		N			
		PE							
11.1-W17	ÖLFLEX CLASSIC 110 3G1,5	1	J-BOX HEATERS IN CABIN	11.1-X20L	HEATER 2kW IN CABIN, UNDER CHAIR	11.1-E1L	5,0 m	14.1	
		2		N		N			
		PE							
11.1-W18	ÖLFLEX CLASSIC 110 3G1,5	1	TERMINAL IN CABIN	11.1-X113	OUTLET 240V U.K. STD.	11.1-X11L	5,0 m	14.5	
		2		14		N			
		PE							
11.1-W19	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT SWITCH IN CABIN	11.1-S313	OUTLET 240V U.K. STD.	11.1-X11L	5,0 m	14.5	
		2		N		N			
		PE							
11.1-W20	ÖLFLEX CLASSIC 110 25G0,75	1	TERMINAL IN CABIN	11.1-X2217	TERMINAL IN CABIN CHAIR, LEFT SIDE	11.1-X41	5,0 m	16.1	
		2		218		2			



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CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
		3		219		3		
		4		220		4		
		5		221		5		
		6		222		6		
		7		223		7		
		8		224		8		
		9		225		9		
		10		226		10		
		11		227		11		
		12		228		12		
		13		229		13		
		14		230		14		
		15		231		15		
		16		320		16		
		17		321		17		
		18		322		18		
		19						
		20						
		21						
		22						
		23						
		24						
		PE						
11.1-W21	ÖLFLEX CLASSIC 110 CY 7G0,75	1	TERMINAL IN CABIN	11.1-X3401	TERMINAL IN CABIN CHAIR, LEFT SIDE	11.1-X431	5,0 m	53.2
		2		402		32		
		3		403		33		
		4		404		34		
		5						
		6						
		PE						
		SC						

CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET
11.1-W30	ÖLFLEX CLASSIC 110 25G0,75	1	TERMINAL IN CABIN	11.1-X2232	TERMINAL IN CABIN CHAIR, RIGHT SIDE	11.1-X51	5,0 m	77.7
		2		233		2		
		3		234		3		
		4		235		4		
		5		236		5		
		6		237		6		
		7		238		7		
		8		239		8		
		9		240		9		
		10		241		10		
		11		242		11		
		12		243		12		
		13		244		13		
		14		245		14		
		15		246		15		
		16		247		16		
		17		248		17		
		18		249		18		
		19		250		19		
		20		251		20		
		21		252		21		
		22						
		23						
		24						
		PE						
11.1-W31	ÖLFLEX CLASSIC 110 CY 7G0,75	1	TERMINAL IN CABIN	11.1-X3405	TERMINAL IN CABIN CHAIR, RIGHT SIDE	11.1-X531	5,0 m	104.2
		2		406		32		
		3		407		33		
		4		408		34		
		5		409		35		
		6						

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		PE							
		SC							
11.1-W32	ÖLFLEX CLASSIC 110 7G0,75	1	TERMINAL IN CABIN	11.1-X2253	TERMINAL IN CABIN CHAIR, RIGHT SIDE	11.1-X541	5,0 m	32.8	
		2		254		42			
		3		255		43			
		4		256		44			
		5		257		45			
		6		258		46			
		PE							
11.1-W19.2	ÖLFLEX CLASSIC 110 3G1,5	1	LIGHT SWITCH IN CABIN	11.1-S314	LIGHT IN CABIN 2x36W	11.1-H2L	5,0 m	14.5	
		2		N		N			
		PE							
A500-W1	H07RN-F 12G1,5	1	TERMINAL IN CABIN	11.1-X121	TERMINAL IN A500	X511	44,0 m	16.5	
		2		22		2			
		3		23		3			
		4		24		4			
		5		25		5			
		6		26		6			
		7		27		7			
		8		28		8			
		9		29		11			
		10		30		12			
		11							
		PE							
A500-W2	H07RN-F 12G1,5	1	TERMINAL IN CABIN	11.1-X135	TERMINAL IN A500	X5115	44,0 m	72.2	
		2		36		16			
		3		37		19			
		4		38		20			
		5		39		21			



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - G

SHEET
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CONT. SHEET
63

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MODIFICATION

DATE

0	1	2	3	4	5	6	7	8	9
CABLE	TYPE	CORE	FROM	CONNECTION	TO	CONNECTION	LENGTH	SHEET	
		6		40		22			
		7		41		23			
		8		42		24			
		9		43		25			
		10		44		26			
		11							
		PE							
A500-W12	ÖLFLEX 440 P 25G0,75	1	TERMINAL IN CABIN	11.1-X2 294	TERMINAL IN A500	X52 123	44,0 m	41.4	
		2		295		124			
		3		312		127			
		4		297		130			
		5		298		133			
		6		299		135			
		7		300		136			
		8		301		138			
		9		302		140			
		10		303		141			
		11		304		144			
		12		305		145			
		13		306		146			
		14		307		148			
		15		308		151			
		16		309		154			
		17		310		157			
		18		311		158			
		19							
		20							
		21							
		22							
		23							
		24							
		PE							



AUTOMATION

SMÖGEN/UDDEVALLA SWEDEN

AS BUILT

BMH

Marine

GUNNARSTORP

Box 566

S-267 25 BJUV

SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

CABLE LIST

DESIGN
M.D.

CHECKED

APPR.

SCALE
1 : 1

PROJECT NO.
202A6086

FILE NAME
A524.DB

DATE
2006-10-24

DRAWING NO.
A524 - G

SHEET
63

CONT. SHEET
64

REV.

MODIFICATION

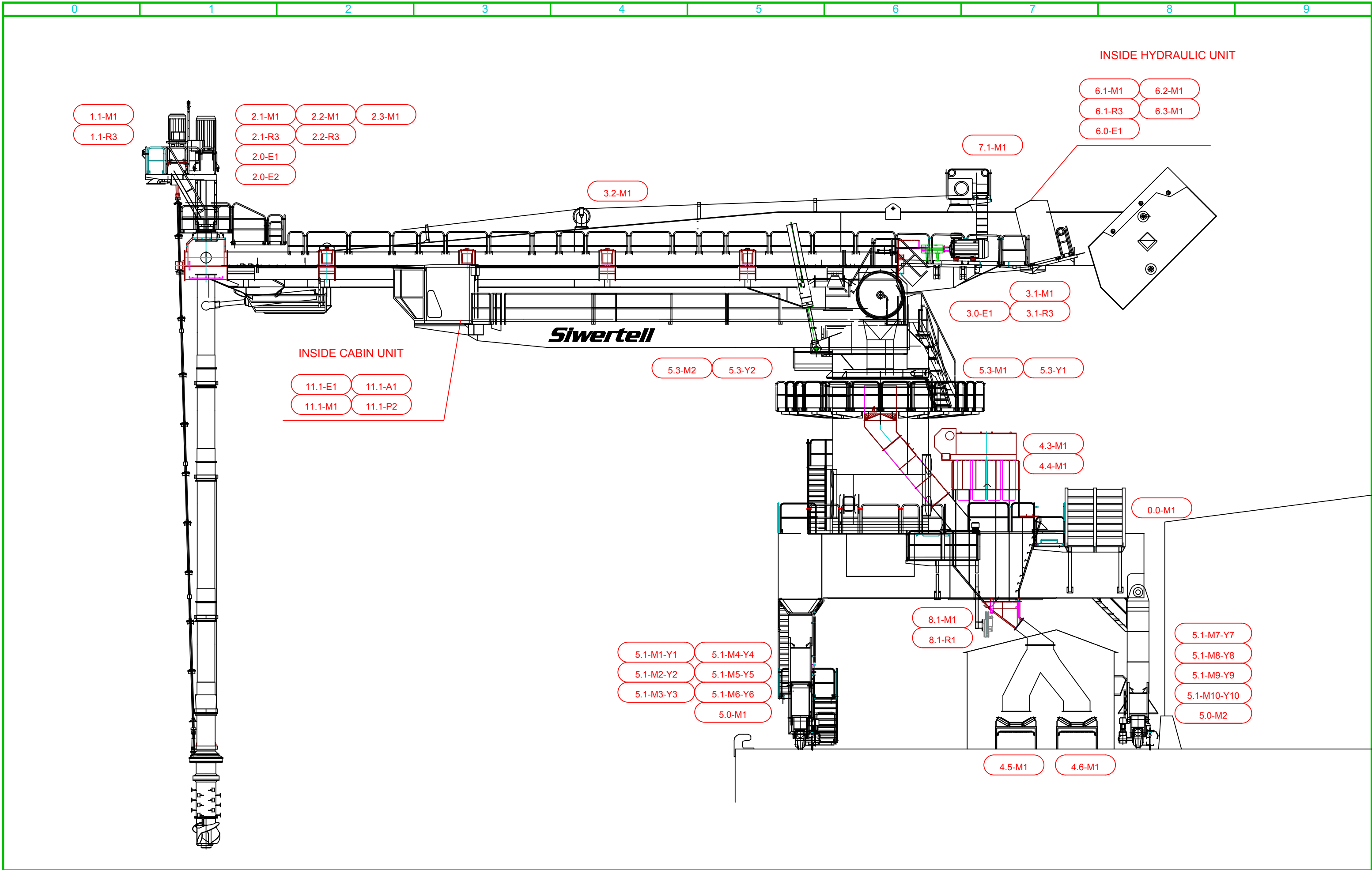
DATE

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CABLE	TYPE	CORE	FROM	CONNECTION		TO	CONNECTION		LENGTH	SHEET
A500-W15	ÖLFLEX 440 CP 25G0,75	1	TERMINAL IN CABIN	11.1-X3	426	TERMINAL IN A500	X53	201	44,0 m	41.4
		2			427			202		
		3			428			204		
		4			429			207		
		5			430			209		
		6			431			210		
		7			432			211		
		8			433			212		
		9			434			213		
		10			435			214		
		11			436			215		
		12			437			216		
		13			438			218		
		14			439			220		
		15			440			221		
		16			441			222		
		17			442			223		
		18			443			224		
		19			444			225		
		20			445			226		
		21			446			227		
		22			447			228		
		23								
		24								
		PE								
		SC								
A500-W16	ÖLFLEX 440 CP 25G0,75	1	TERMINAL IN CABIN	11.1-X3	448	TERMINAL IN A500	X53	229	44,0 m	102.3
		2			449			230		
		3			450			231		
		4			451			232		
		5			452			233		

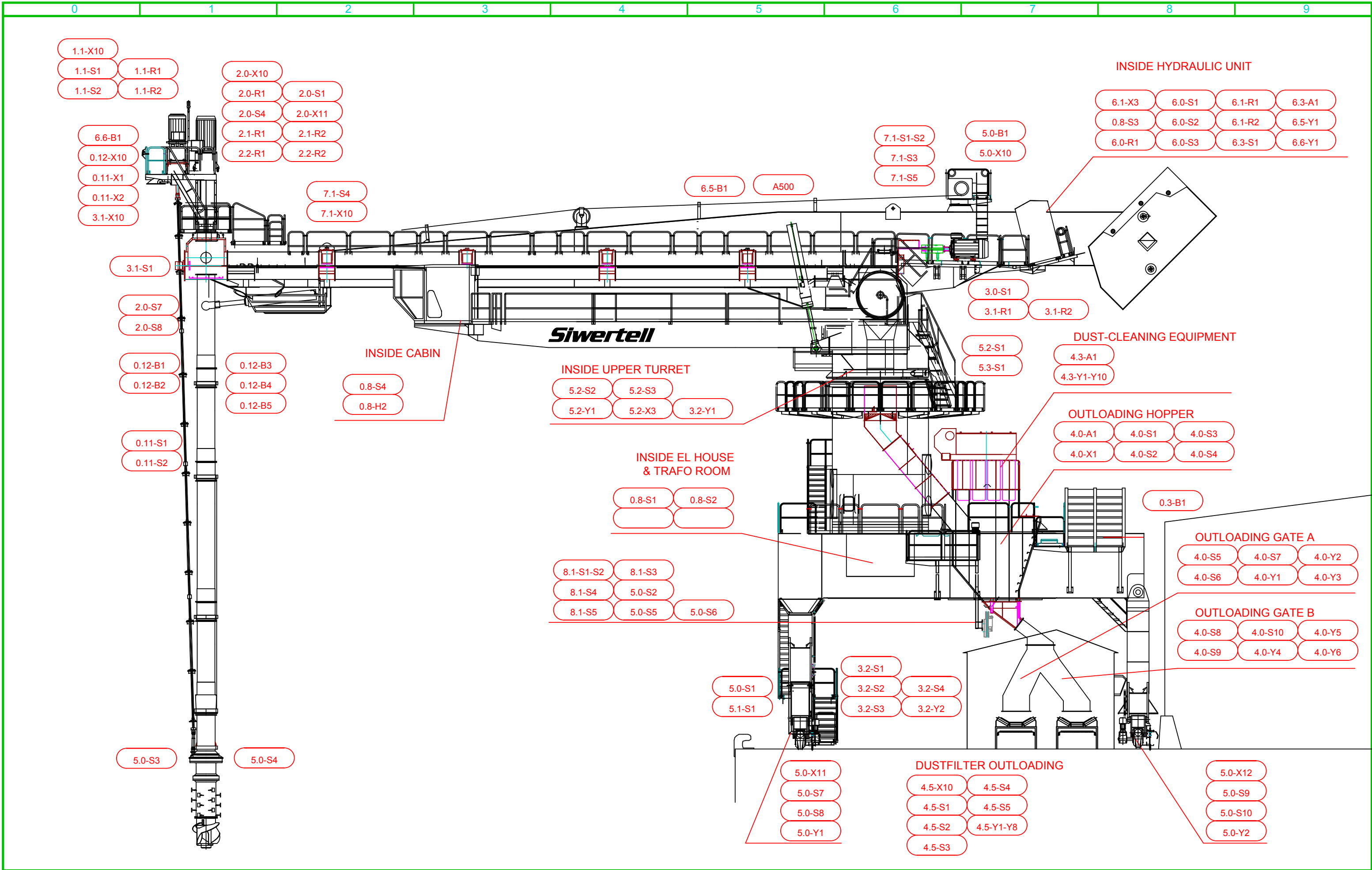
0		1		2		3		4		5		6		7		8		9	
CABLE	TYPE		CORE	FROM		CONNECTION		TO		CONNECTION		LENGTH		SHEET					
			6			453				234									
			7			454				235									
			8			455				236									
			9			456				237									
			10			457				238									
			11																
			12																
			13																
			14																
			15																
			16																
			17																
			18																
			19																
			20																
			21																
			22																
			23																
			24																
			PE																
			SC																
TP-W1	LIYCY-P 2x2x0,75		WHITE	TERMINAL FOR TELEPHONE IN SB4		TP1	3	TERMINAL FOR TELEPHONE IN CABIN		TP2	3	65,0 m		5.5					
			BROWN				4				4								
			GREEN				5				5								
			YELLOW				6				6								
			SC																
TP-W2	LIYCY-P 2x2x0,75		WHITE	TERMINAL FOR TELEPHONE IN SB4		TP1	3	TELEPHONE AT ENTRY		TP3	BLK	30,0 m		5.5					
			BROWN				4				RED								
			GREEN				5				GR								
			YELLOW				6				YO								
			SC																

reproduced, or exploited in any other unauthorized way.



AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN			AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT LOCATION OF EXTERNAL EQUIPMENT MOTORS, BRAKES & HEATERS			DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-16	
CHECKED			PROJECT NO. 202A6086		DRAWING NO. A524 - H			FILE NAME A524.DB			SHEET 1		
APPR.			SHEET 1		CONT. SHEET 2								
REV.	MODIFICATION							DATE					

reproduced, or exploited in any other unauthorized way.



PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB4

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-11-15

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
3

CONT. SHEET
4

REV.	MODIFICATION	DATE

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

SWITCHBOARD SB4

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-11-15

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
4

CONT. SHEET
5

REV.	MODIFICATION	DATE

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB4		DESIGN M.D.	SCALE 1 : 1	DATE 2006-11-15	
CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - I								
APPR.	FILE NAME A524.DB		SHEET 5	CONT. SHEET 6							
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB4		DESIGN M.D.	SCALE 1 : 1	DATE 2006-11-15	
REV.	MODIFICATION			DATE		CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - I		
						APPR.	FILE NAME A524.DB		SHEET 7	CONT. SHEET 8	

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

SWITCHBOARD SB4

DESIGN
M.D.

SCALE
1 : 1

DATE
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CHECKED

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202A6086

DRAWING NO.
A524 - I

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FILE NAME
A524.DB

SHEET
8

CONT. SHEET
9

REV.	MODIFICATION	DATE

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB4		DESIGN M.D.		SCALE 1 : 1		DATE 2006-11-15	
CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - I		APPR.		FILE NAME A524.DB		SHEET 9		CONT. SHEET 10	
REV.	MODIFICATION			DATE									

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

SWITCHBOARD SB4

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-11-15

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
10

CONT. SHEET
11

REV.

MODIFICATION

DATE _____

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB1		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
	CHECKED			PROJECT NO. 202A6086		DRAWING NO. A524 - I					
	APPR.			FILE NAME A524.DB		SHEET 13	CONT. SHEET 14				
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB1		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - I				APPR.	FILE NAME A524.DB	SHEET 14	CONT. SHEET 15		
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB1		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - I		APPR.	FILE NAME A524.DB	SHEET 15	CONT. SHEET 16				
REV.	MODIFICATION			DATE							

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

SWITCHBOARD SB1

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
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CONT. SHEET
19

REV.	MODIFICATION	DATE

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

SWITCHBOARD SB1

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
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CONT. SHEET
20

REV.	MODIFICATION	DATE

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT



PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

SWITCHBOARD SB1

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
20

CONT. SHEET
21

REV.

MODIFICATION

DATE _____

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

SWITCHBOARD SB1

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
21

CONT. SHEET
22

REV.	MODIFICATION	DATE

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB3		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - I								
APPR.	FILE NAME A524.DB		SHEET 23	CONT. SHEET 24							
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT	BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN	PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB3	DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
					CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - I	
REV.	MODIFICATION	DATE			APPR.	FILE NAME A524.DB	SHEET 24	CONT. SHEET 25

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB3		DESIGN M.D.		SCALE 1 : 1		DATE 2006-10-30	
CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - I									
APPR.		FILE NAME A524.DB		SHEET 25		CONT. SHEET 26							
REV.		MODIFICATION		DATE									

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB3		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
CHECKED	PROJECT NO. 202A6086		DRAWING NO. A524 - I			APPR.	FILE NAME A524.DB		SHEET 26	CONT. SHEET 27	
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST SWITCHBOARD SB3		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
	CHECKED			PROJECT NO. 202A6086		DRAWING NO. A524 - I					
	APPR.			FILE NAME A524.DB		SHEET 27	CONT. SHEET 28				
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST CABIN		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - I			APPR.	FILE NAME A524.DB	SHEET 28	CONT. SHEET 29	
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
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PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST CABIN		DESIGN M.D.		SCALE 1 : 1		DATE 2006-10-30	
CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - I									
APPR.		FILE NAME A524.DB		SHEET 29		CONT. SHEET 30							
REV.	MODIFICATION			DATE									

PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AS BUILT

BMH Marine
GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

CABIN

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
30

CONT. SHEET
31

REV.	MODIFICATION	DATE

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST CABIN		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
	CHECKED			PROJECT NO. 202A6086		DRAWING NO. A524 - I					
	APPR.			FILE NAME A524.DB		SHEET 31	CONT. SHEET 32				
REV.	MODIFICATION			DATE							

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

CABIN

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
33

CONT. SHEET
34

REV.	MODIFICATION	DATE

PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

CABIN

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
34

CONT. SHEET
35

REV.	MODIFICATION	DATE

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

CABIN

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
35

CONT. SHEET
36

REV.	MODIFICATION	DATE

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

 SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		 GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST CABIN		DESIGN M.D.		SCALE 1 : 1		DATE 2006-10-30	
CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - I		APPR.		FILE NAME A524.DB		SHEET 36		CONT. SHEET 37	
REV.	MODIFICATION			DATE									

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST CABIN		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
							CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - I		
APPR.	FILE NAME A524.DB	SHEET 37	CONT. SHEET 38								
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST CABIN		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
							CHECKED	PROJECT NO. 202A6086	DRAWING NO. A524 - I		
							APPR.	FILE NAME A524.DB	SHEET 38	CONT. SHEET 39	
REV.	MODIFICATION			DATE							

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

PLC INPUT/OUTPUT LIST

BMH_IO-LIST

[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN		AS BUILT		BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN		PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST CABIN		DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
	CHECKED			PROJECT NO. 202A6086		DRAWING NO. A524 - I					
	APPR.	FILE NAME A524.DB		SHEET 39	CONT. SHEET 40						
REV.	MODIFICATION			DATE							

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

CABIN

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
41

CONT. SHEET
42

REV.

MODIFICATION

DATE _____

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

CABIN

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
42

CONT. SHEET
43

REV.	MODIFICATION	DATE

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

CABIN

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
43

CONT. SHEET
44

REV.	MODIFICATION	DATE
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PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

CABIN

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
44

CONT. SHEET
45

REV.	MODIFICATION	DATE

PLC INPUT/OUTPUT LIST

BMH IO-LIST

[illegible]

AS BUILT

BMH Marine

GUNNARSTORP
Box 566
S-267 25 BJUV
SWEDEN

PROJECT 15050007

SHIPUNLOADER MERSEY DOCKS

ELECTRICAL EQUIPMENT

PLC INPUT/OUTPUT LIST

CABIN

DESIGN
M.D.

SCALE
1 : 1

DATE
2006-10-30

CHECKED

PROJECT NO.
202A6086

DRAWING NO.
A524 - I

APPR.

FILE NAME
A524.DB

SHEET
45

CONT. SHEET
46

REV.	MODIFICATION	DATE

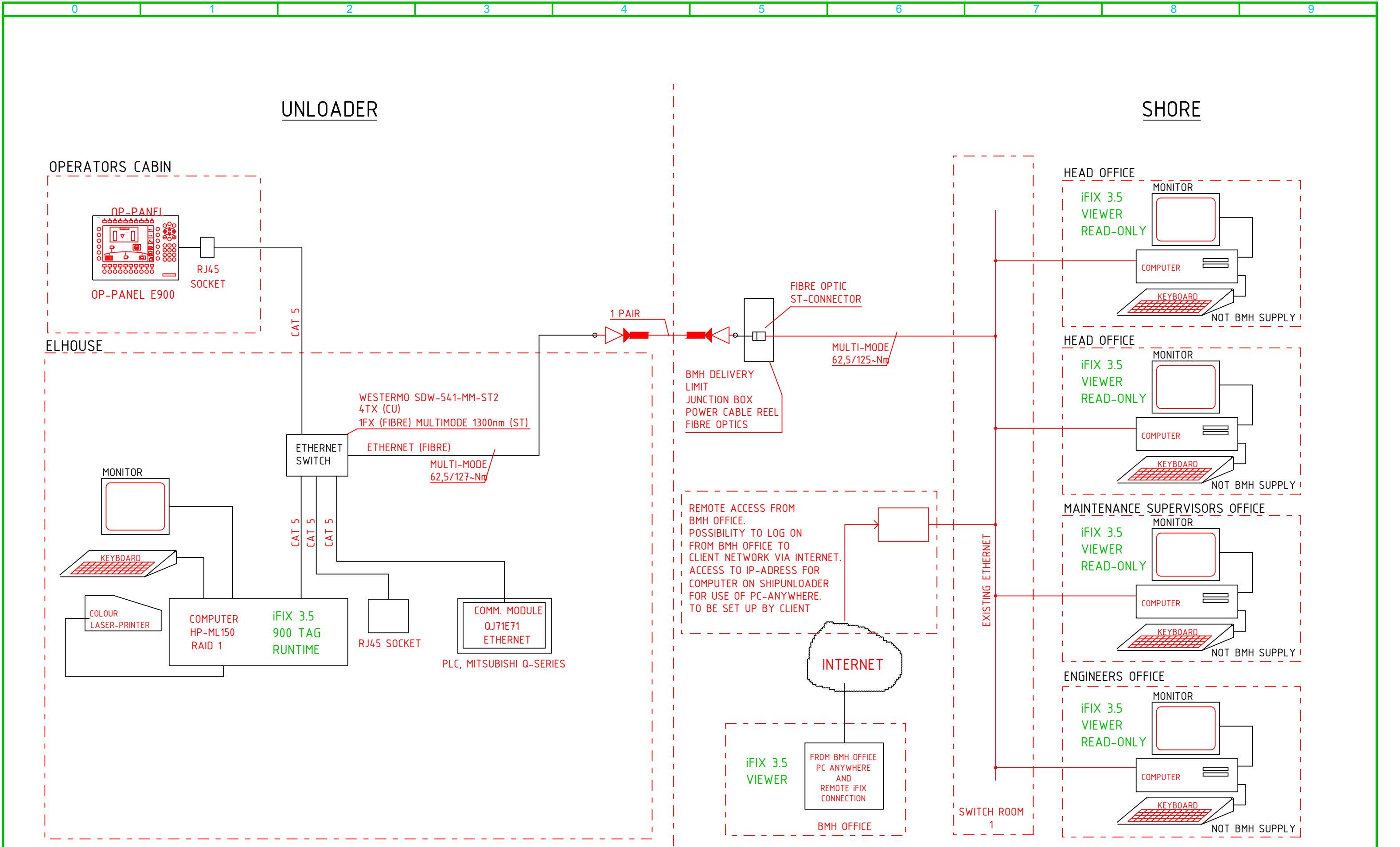
0	1	2	3	4	5	6	7	8	9
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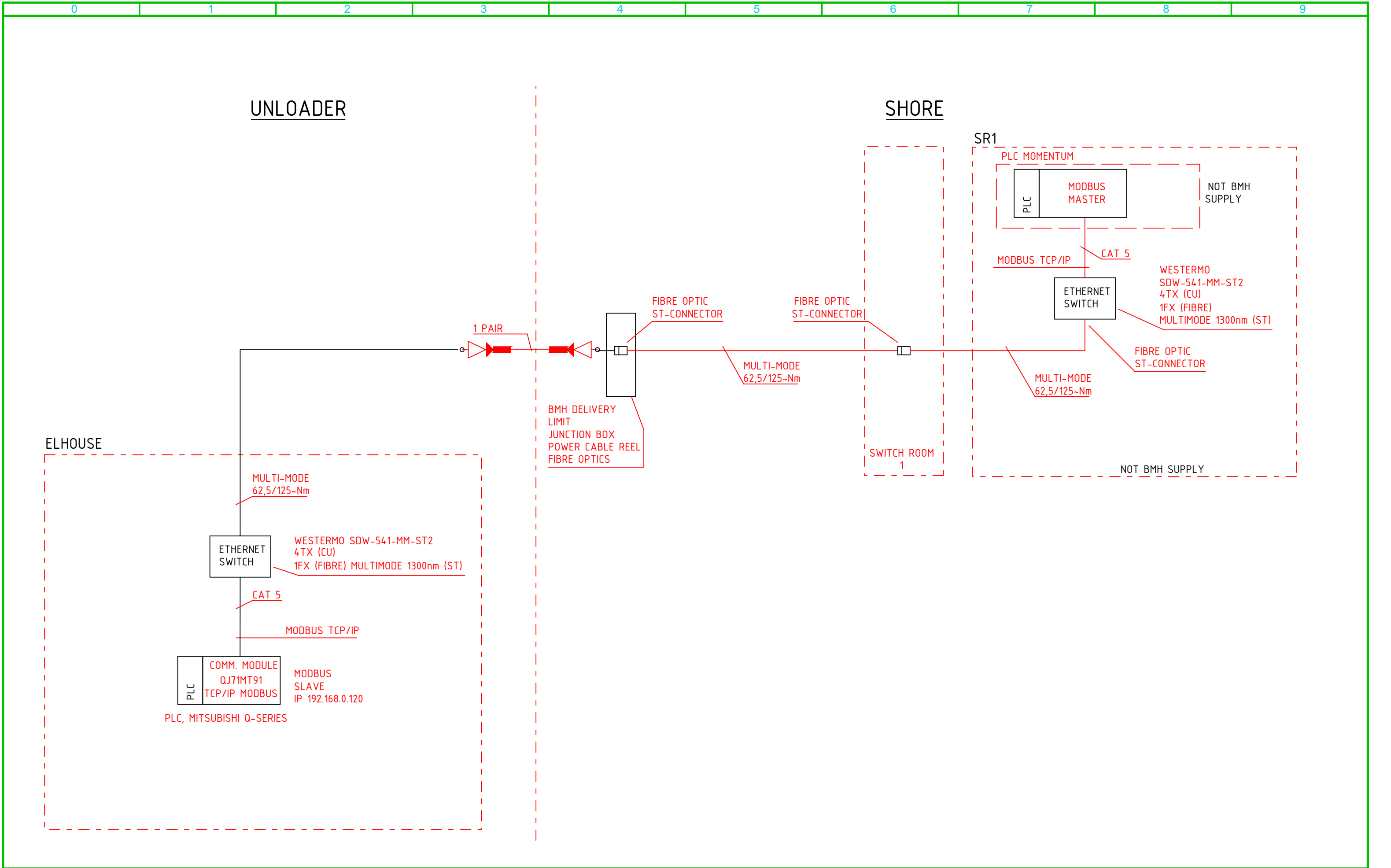
PLC INPUT/OUTPUT LIST

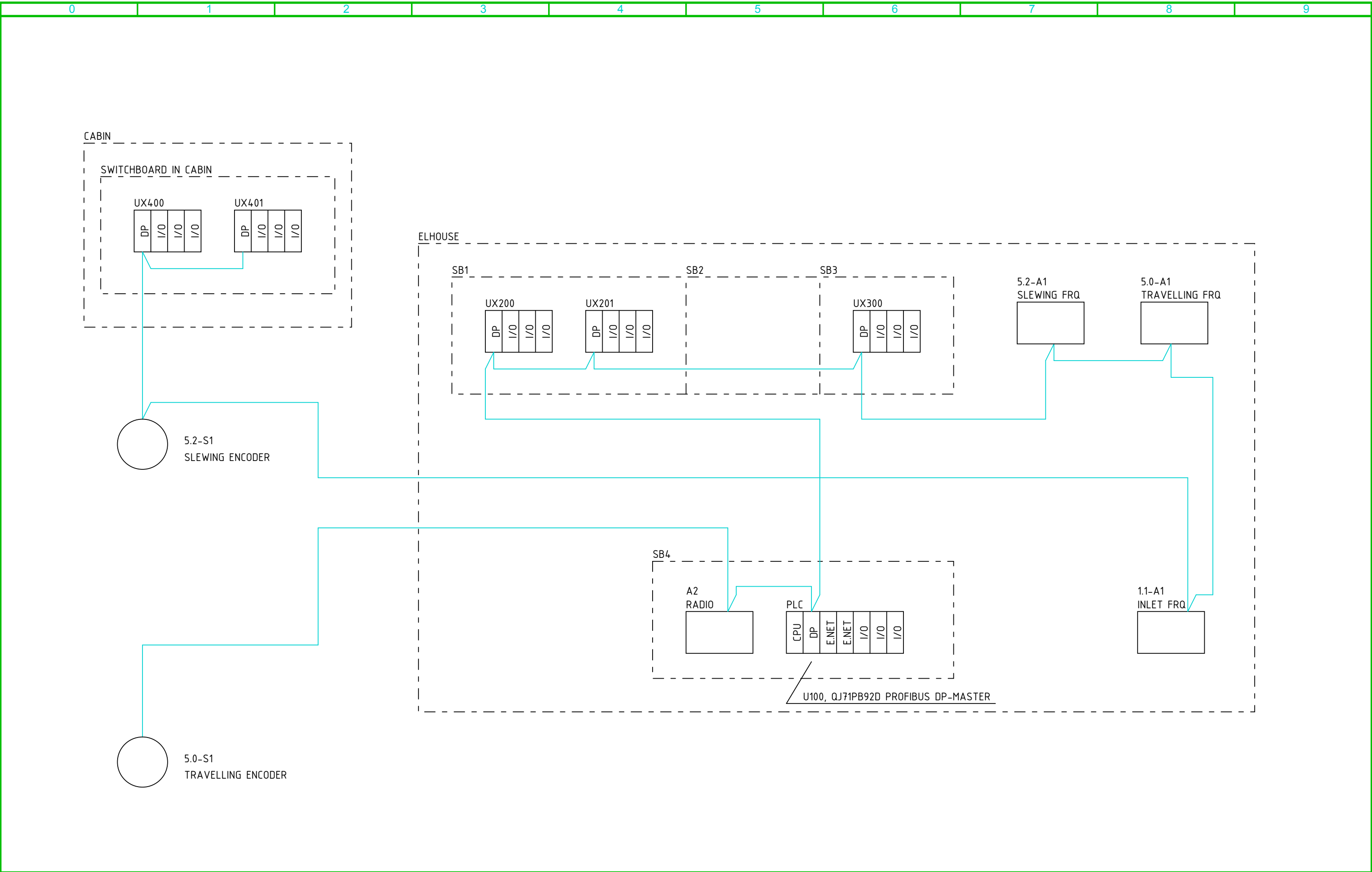
BMH_IO-LIST

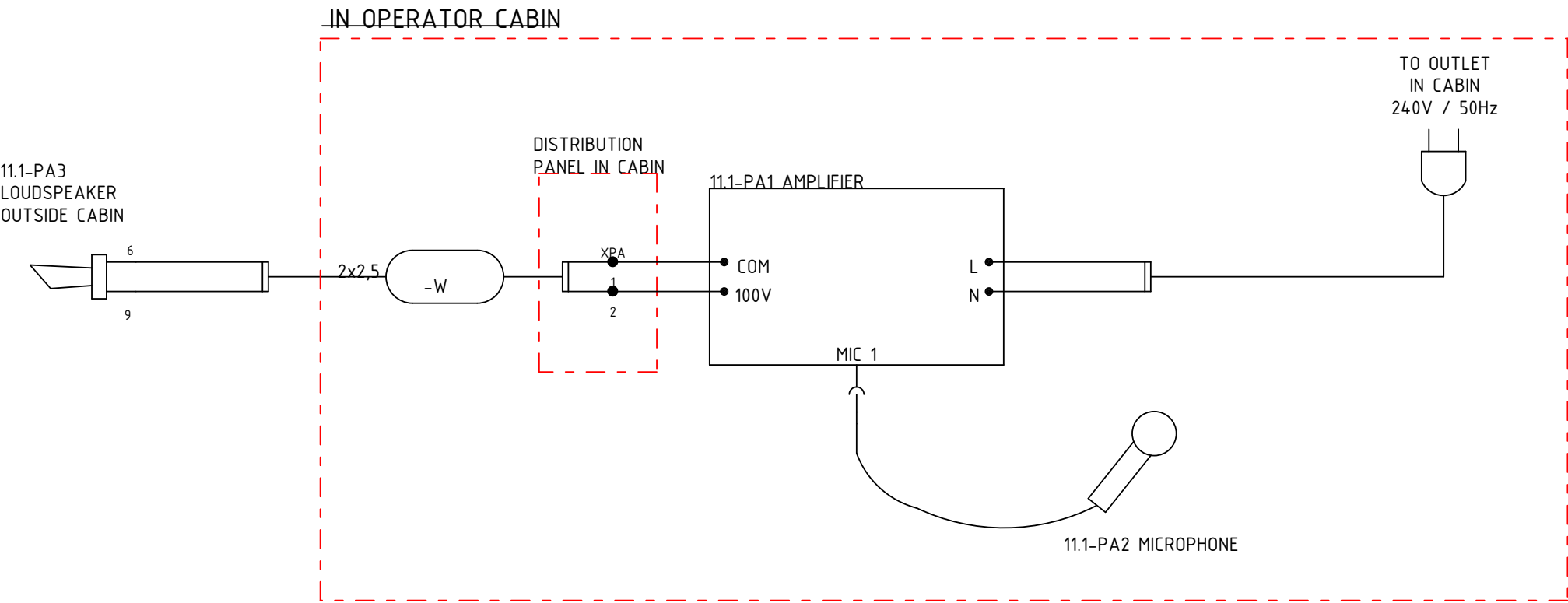
[illegible]

AH AUTOMATION SMÖGEN/UDDEVALLA SWEDEN			AS BUILT			BMH Marine GUNNARSTORP Box 566 S-267 25 BJUV SWEDEN			PROJECT 15050007 SHIPUNLOADER MERSEY DOCKS ELECTRICAL EQUIPMENT PLC INPUT/OUTPUT LIST CABIN			DESIGN M.D.	SCALE 1 : 1	DATE 2006-10-30	
	CHECKED		PROJECT NO. 202A6086		DRAWING NO. A524 - I										
	APPR.		FILE NAME A524.DB		SHEET 47		CONT. SHEET 1								
REV.	MODIFICATION				DATE										



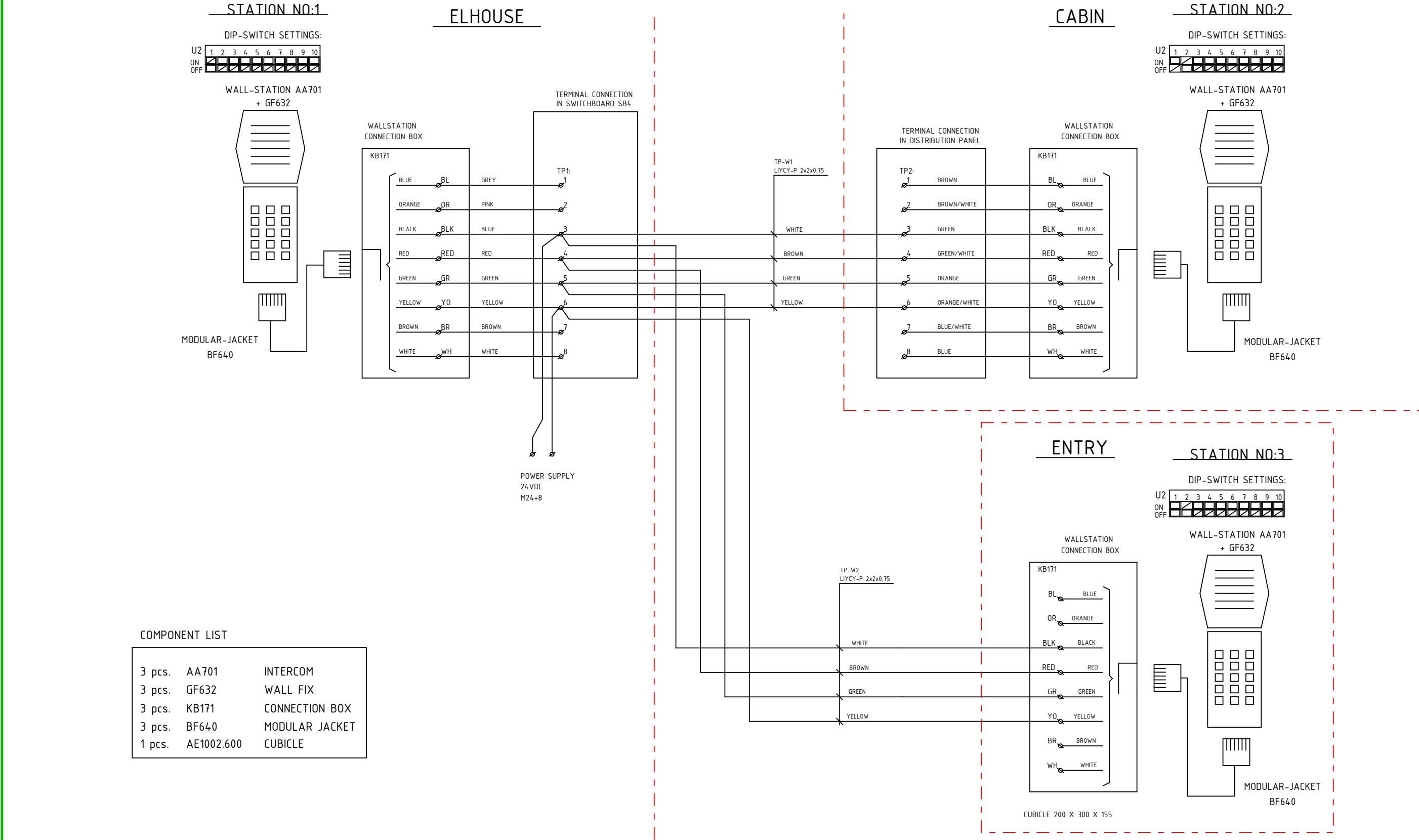




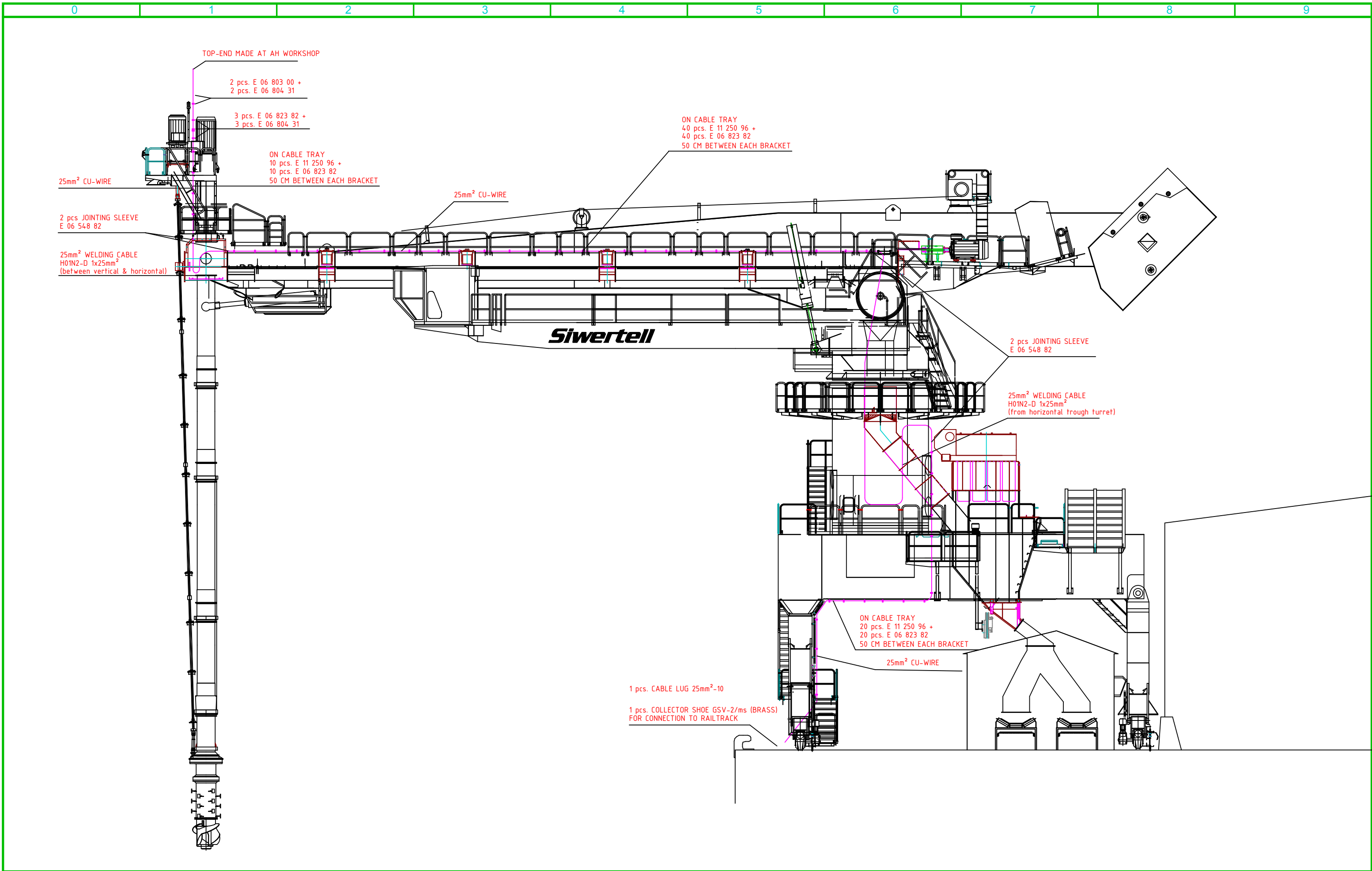


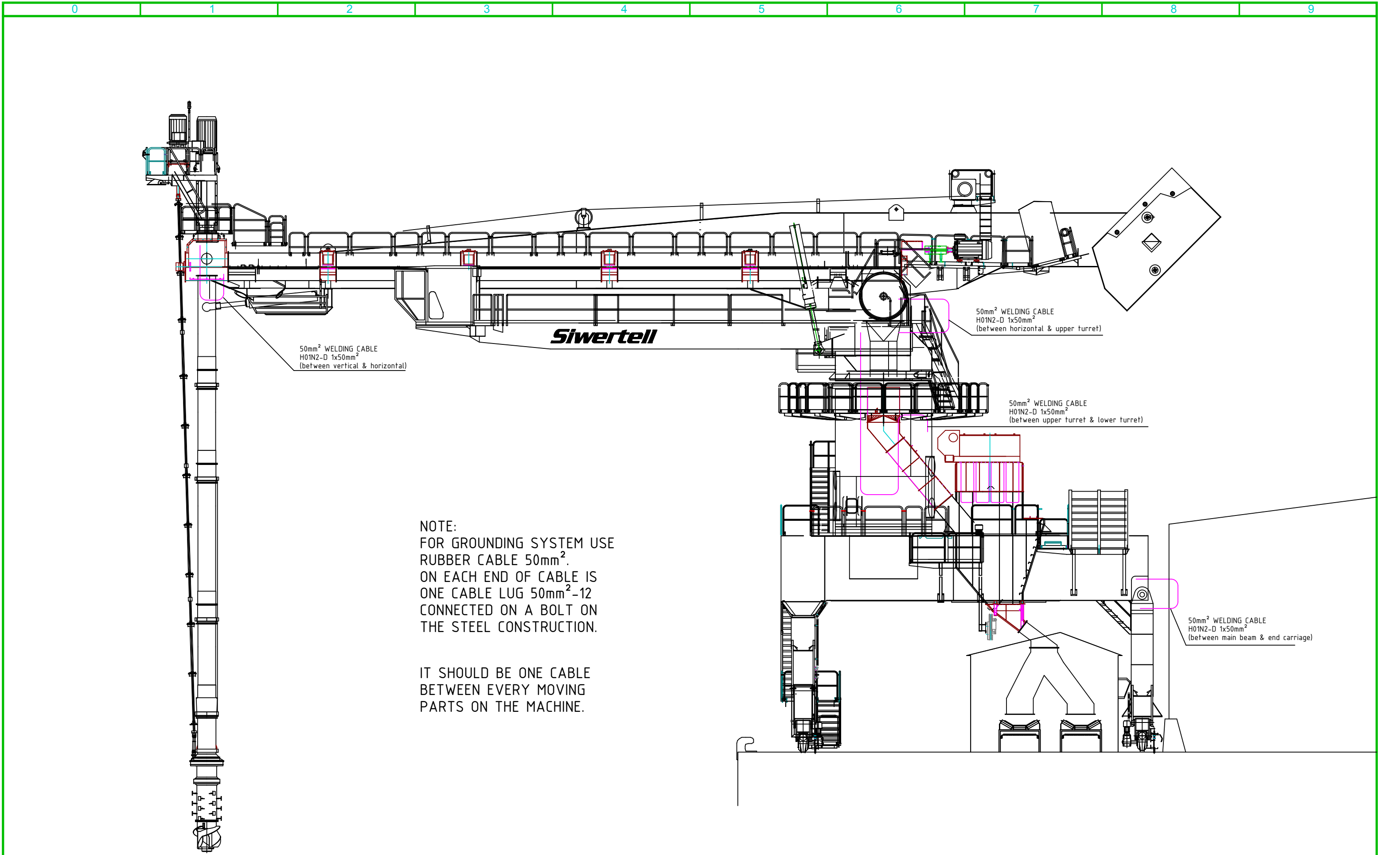
OBJEKT:	TYPE:
AMPLIFIER	INTER M PA-935
LOUDSPEAKER	DNH HP-20
MICROPHONE	PM-306D

reproduced, or exploited in any other unauthorized way.



reproduced, or exploited in any other unauthorized way.



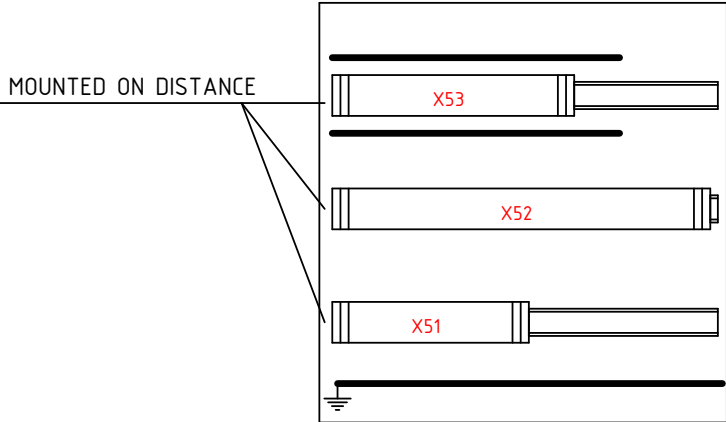
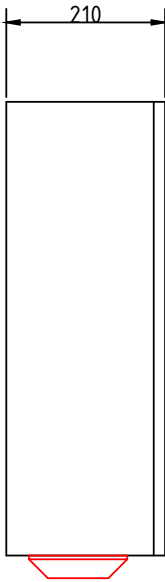
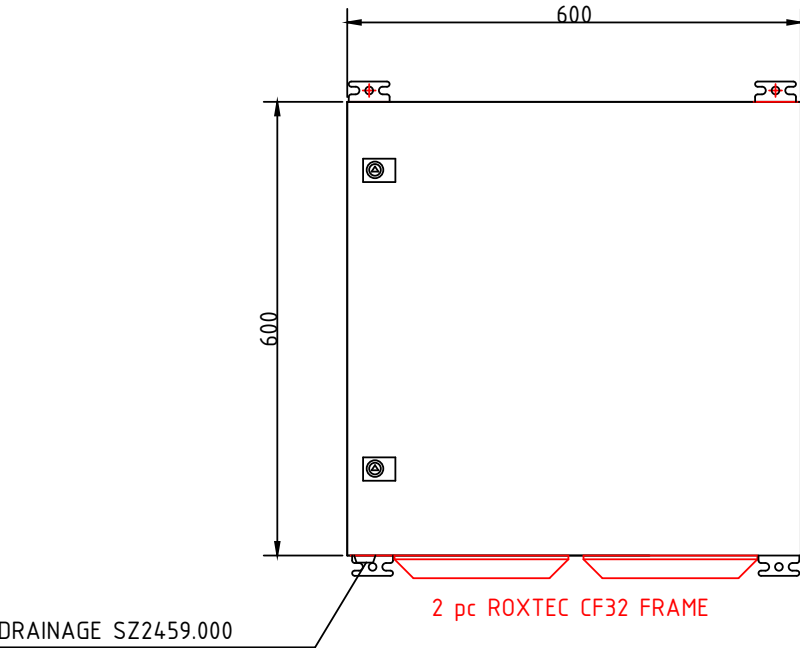


RITTAL AE 1010.600
STAINLESS STEEL

FRONT

SIDE

INTERNAL



reproduced, or exploited in any other unauthorized way.

NUMBER OF TERMINALS: 40

X51

1	16.5
2	16.5
3	16.7
4	16.7
5	16.8
6	16.9
7	17.4
8	17.4
9	17.5
10	17.5
11	17.6
12	17.6
13	17.7
14	17.7
15	72.1
16	72.1
17	73.1
18	73.1
19	93.2
20	93.2
21	99.2
22	99.2
23	102.2
24	102.2
25	105.2
26	105.2
27	47.4
28	47.4
29	
30	
31	
32	
33	
34	
35	
36	
37	
38	
39	
40	

NUMBER OF TERMINALS: 50

X53

201	41.4
202	41.4
203	72.1
204	72.1
205	72.1
206	73.1
207	73.1
208	73.1
209	93.2
210	93.2
211	93.2
212	93.2
213	93.2
214	95.1
215	95.1
216	95.1
217	95.1
218	95.1
219	95.1
220	98.2
221	98.2
222	98.2
223	98.2
224	99.2
225	99.2
226	99.2
227	99.2
228	99.2
229	102.2
230	102.2
231	102.2
232	102.2
233	102.2
234	105.2
235	105.2
236	105.2
237	105.2
238	105.2
239	
240	
241	
242	
243	
244	
245	
246	
247	
248	
249	
250	

FOR CIRCUIT DIAGRAM SEE DRAWING A524-E SHEET:

16, 17, 41, 45, 46, 47, 49, 72, 73, 74, 90, 92, 93, 95, 98, 99, 102, 105

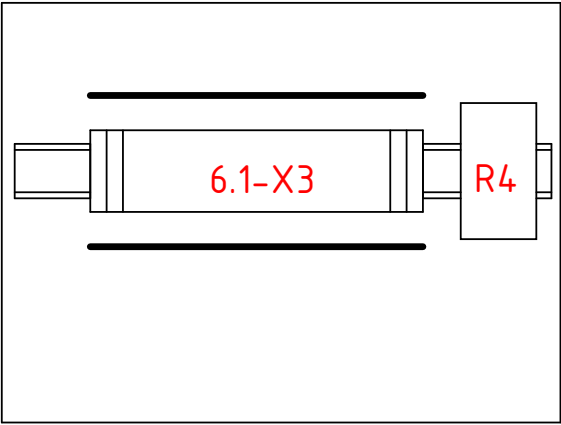
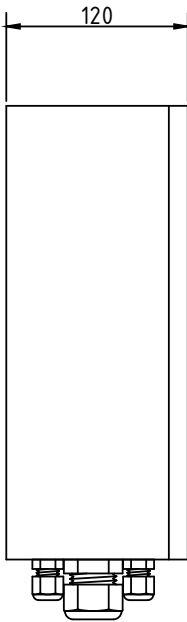
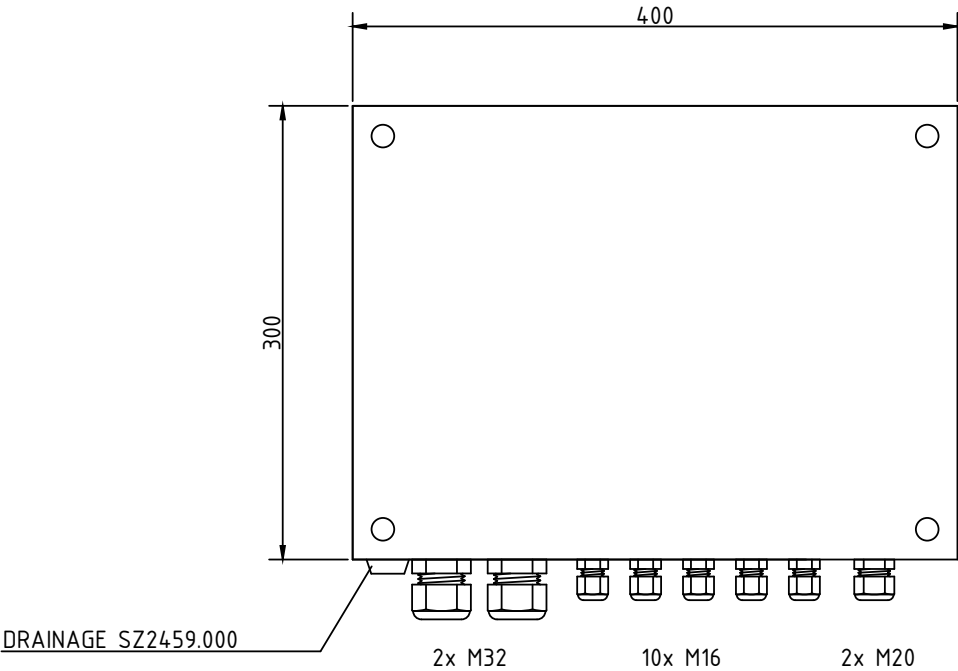
reproduced, or exploited in any other unauthorized way.

RITTAL KL 1508.510
PAINTED STEEL RAL7035

FRONT

SIDE

INTERNAL



NUMBER OF TERMINALS: 29

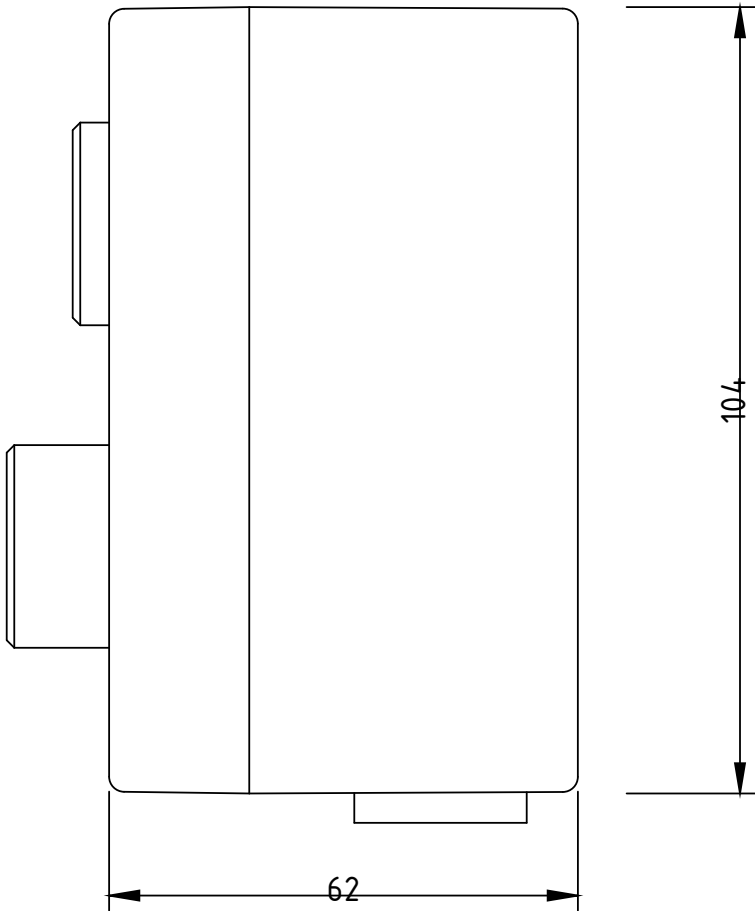
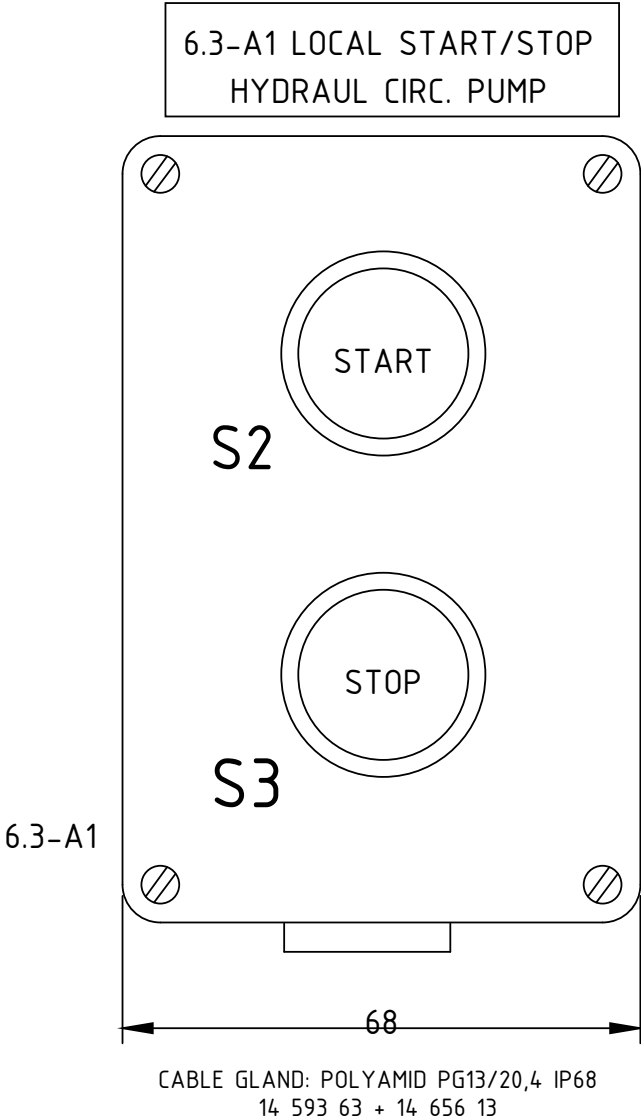
6.1-X3

1	66.1	21	66.1
2	66.1	22	66.1
3	66.1	23	66.1
4	66.1	24	68.2
5	66.1	25	68.2
6	66.1	26	68.2
7	71.1	27	68.2
8	71.1	28	68.2
9	71.1	29	72.8
10	71.1	30	72.8
11	71.1	31	72.8
		32	73.8
		33	73.8
		34	73.8
		51	68.2
		52	68.2
		53	68.2
		54	68.2

FOR CIRCUIT DIAGRAM SEE DRAWING A524-E SHEET:

66, 68, 71, 72, 73

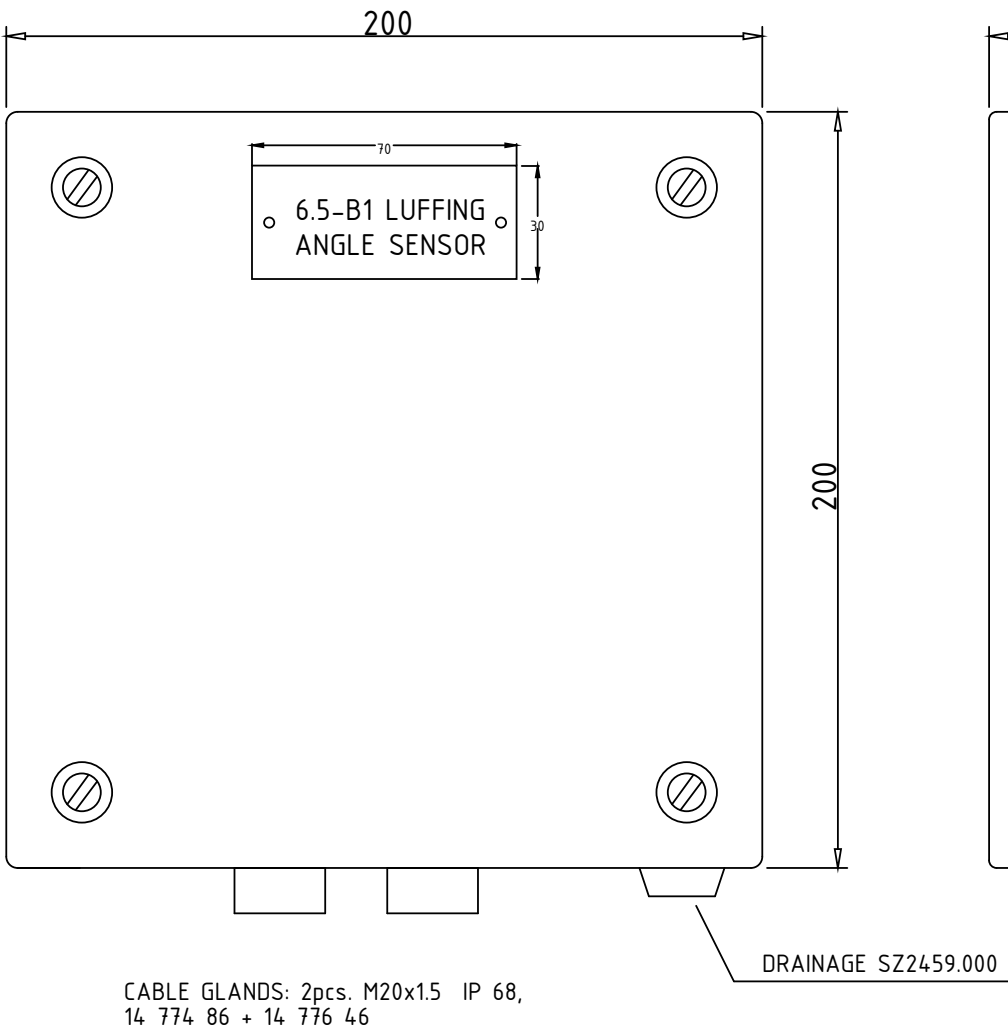
reproduced, or exploited in any other unauthorized way.



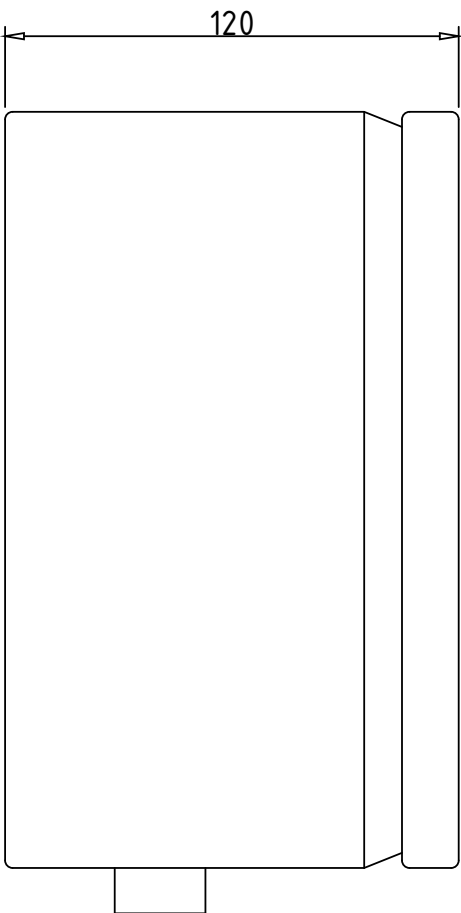
ENCLOSURE: POLYKARBONAT RAL 7035
TELEMECANIQUE XAL-B225
IP65

ITEM LIST:
PUSHBUTTON CONTROL BOX, XAL-D225

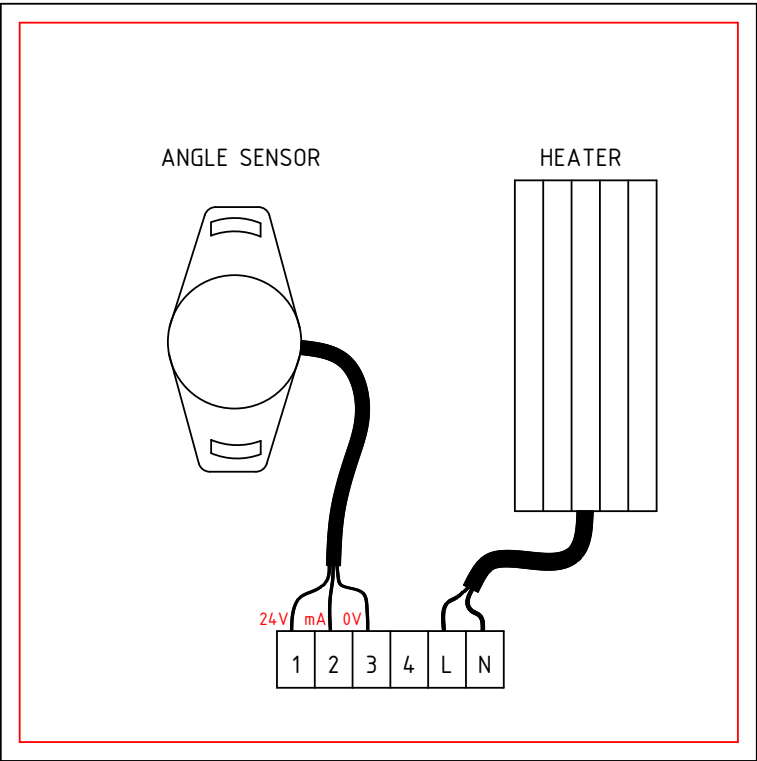
FRONT VIEW



SIDE VIEW



INTERNAL VIEW

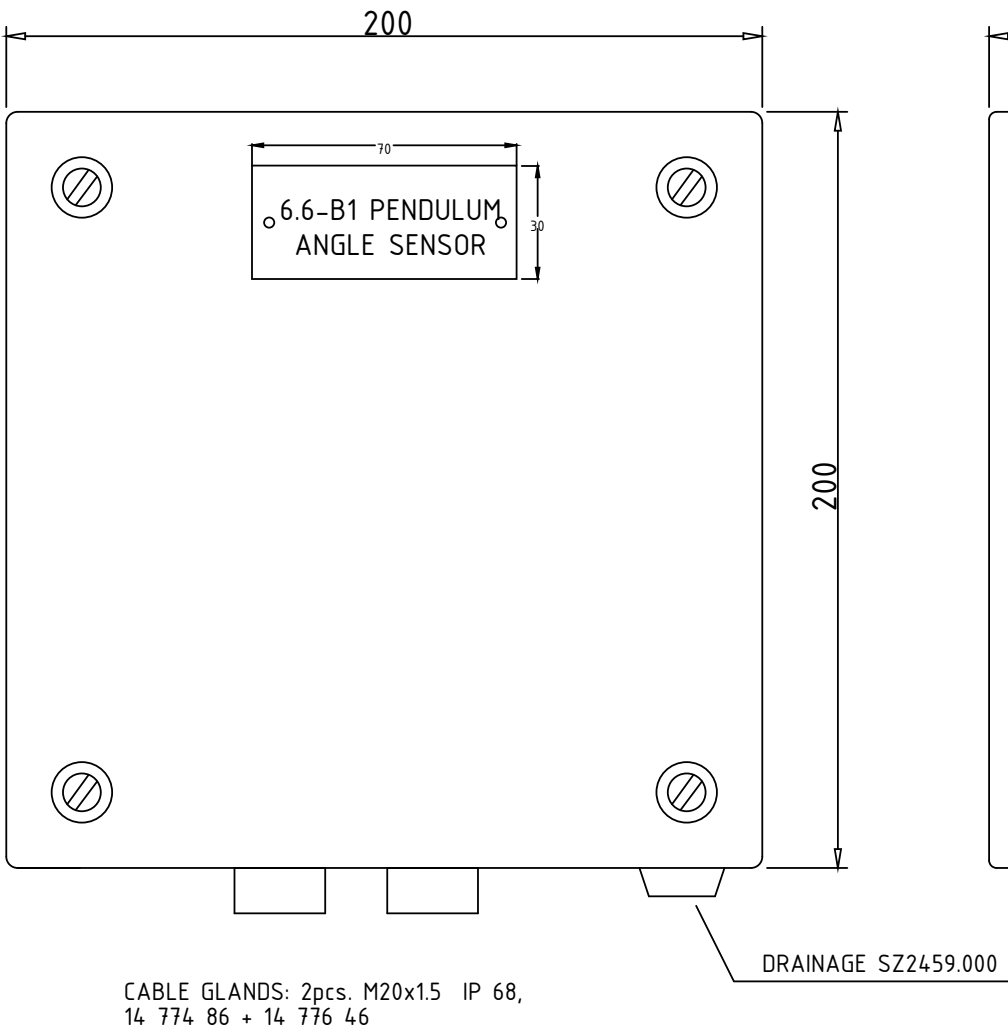


ENCLOSURE: STAINLESS STEEL IP66 / NEMA 4

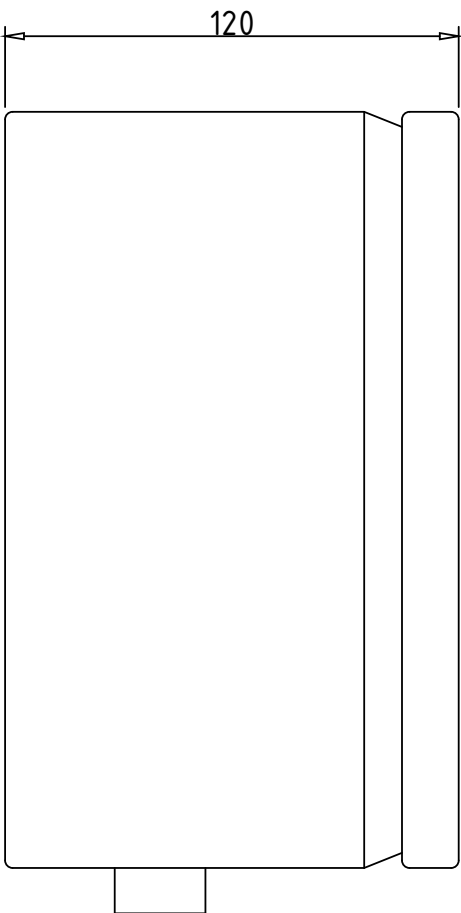
ITEM LIST:

CASING: ILS202012, ELDON
MOUNTING PLATE: IMP2020, ELDON
SENSOR: P603-90-HJ (+/-45°=>4-20mA), HERBERTEK
HEATER: SK 3105.000 (10W, 110-240VAC), RITTAL

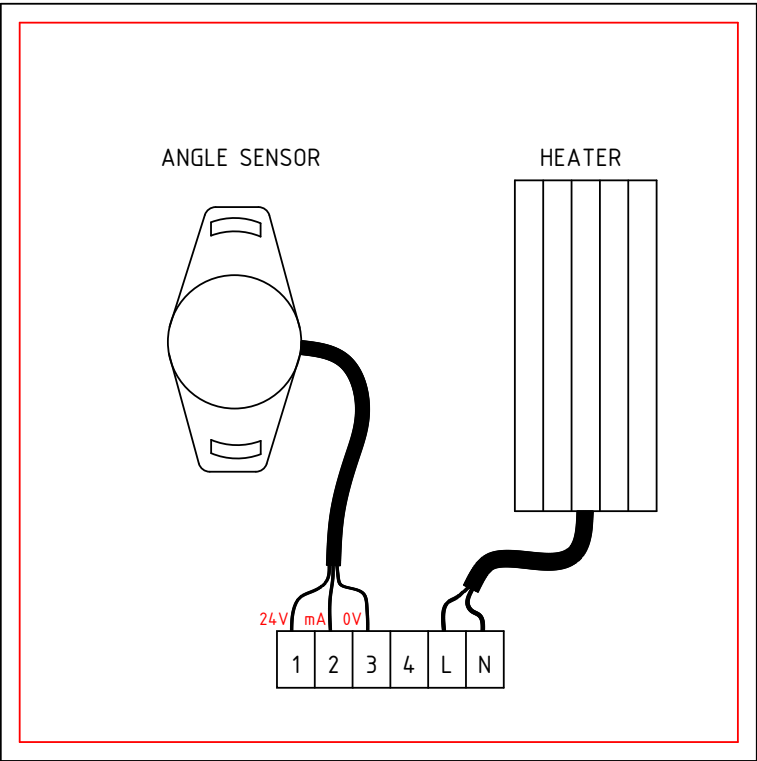
FRONT VIEW



SIDE VIEW



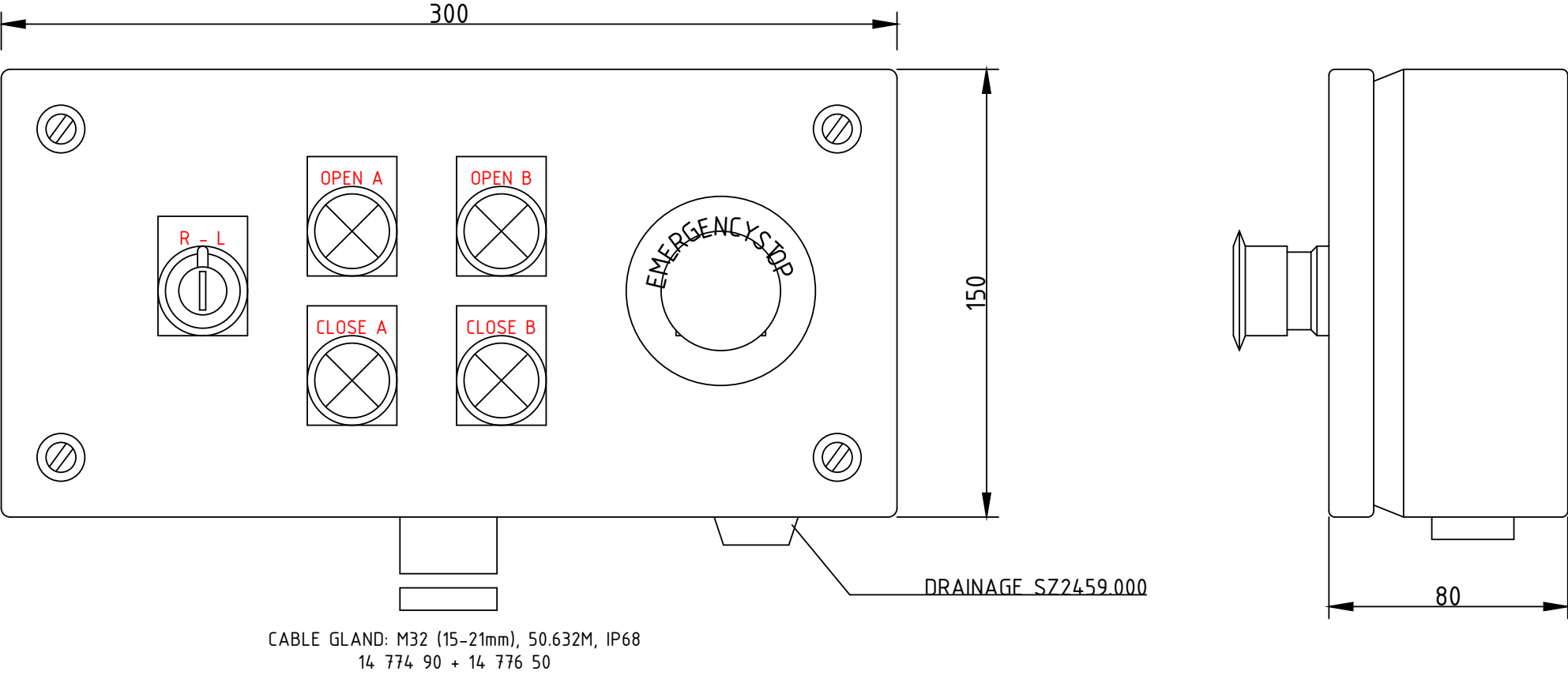
INTERNAL VIEW



ENCLOSURE: STAINLESS STEEL IP66 / NEMA 4

ITEM LIST:

CASING: ILS202012, ELDON
MOUNTING PLATE: IMP2020, ELDON
SENSOR: P603-90-HJ (+/-45°=>4-20mA), HERBERTEK
HEATER: SK 3105.000 (10W, 110-240VAC), RITTAL

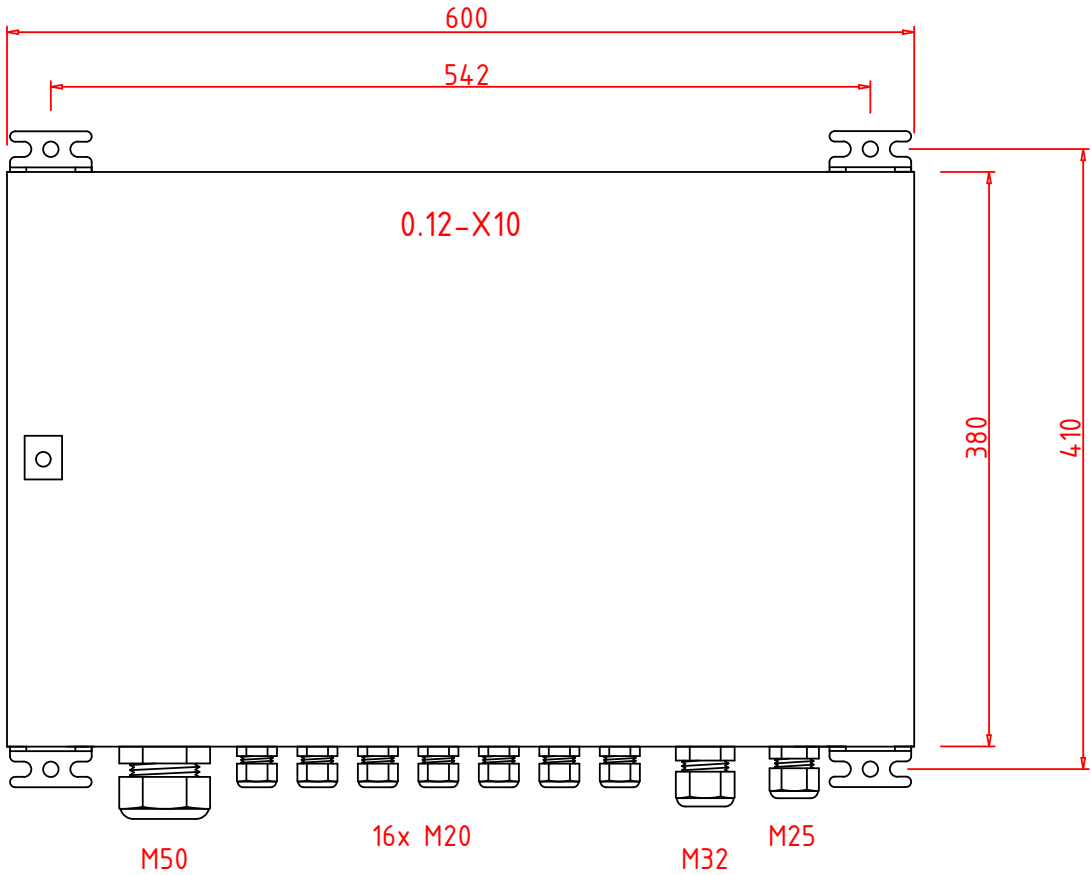


ENCLOSURE: STAINLESS STEEL

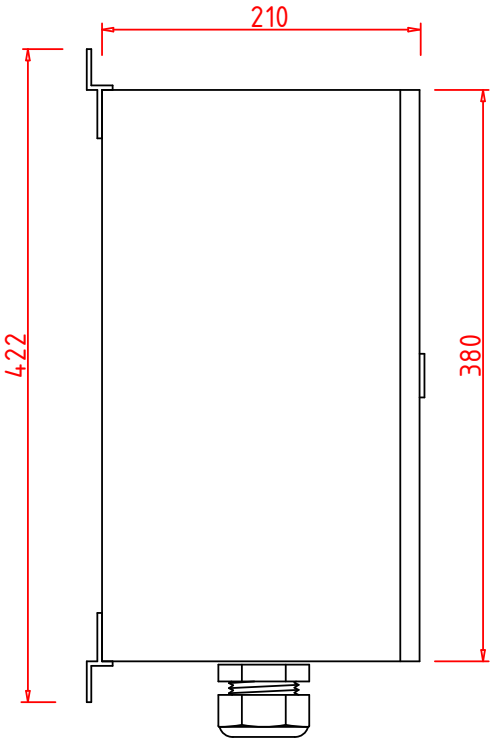
IP66

RITTAL AE 1009.600
STAINLESS STEEL

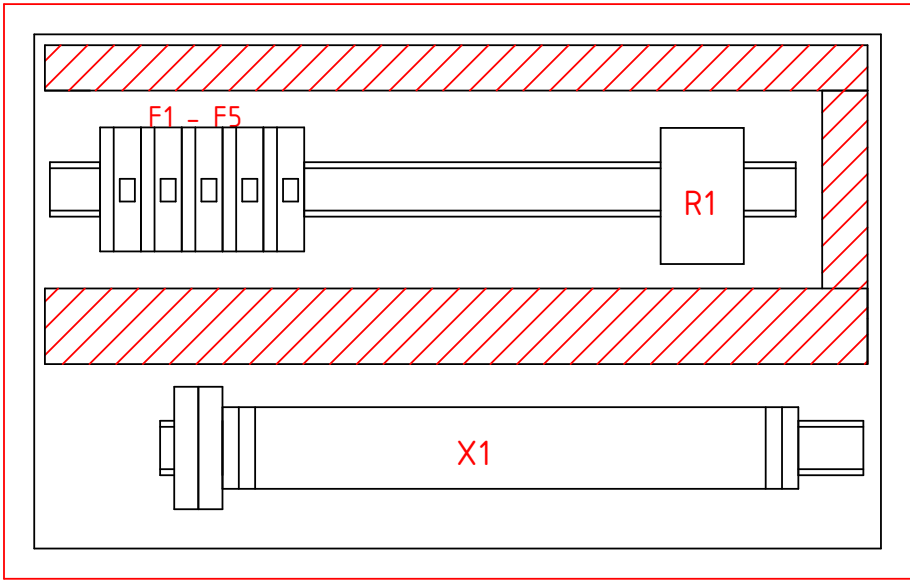
FRONT VIEW



SIDE VIEW



INTERNAL VIEW



NUMBER OF TERMINALS: 70

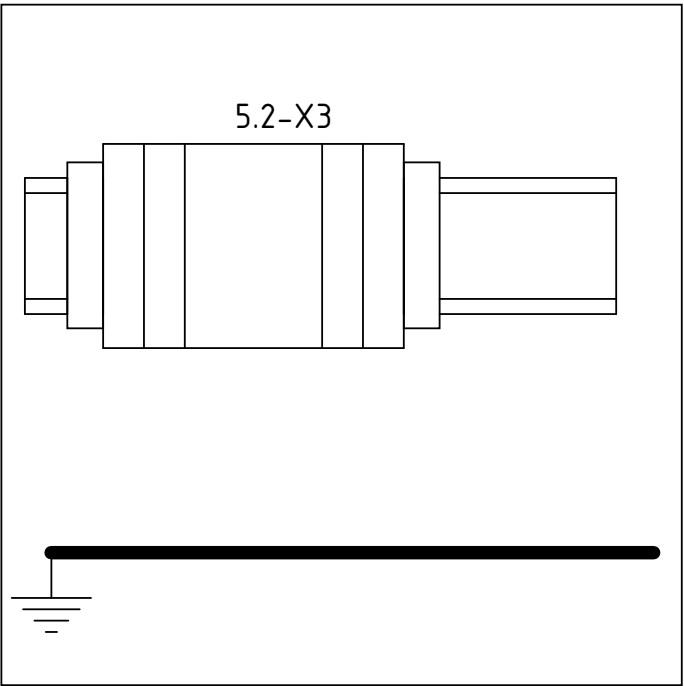
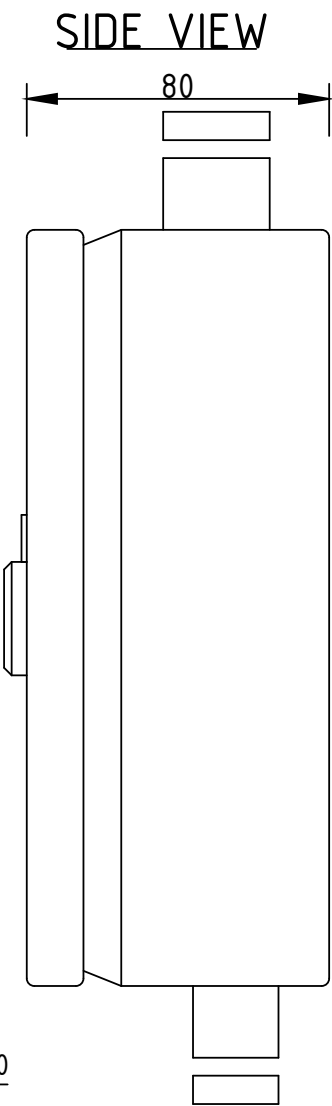
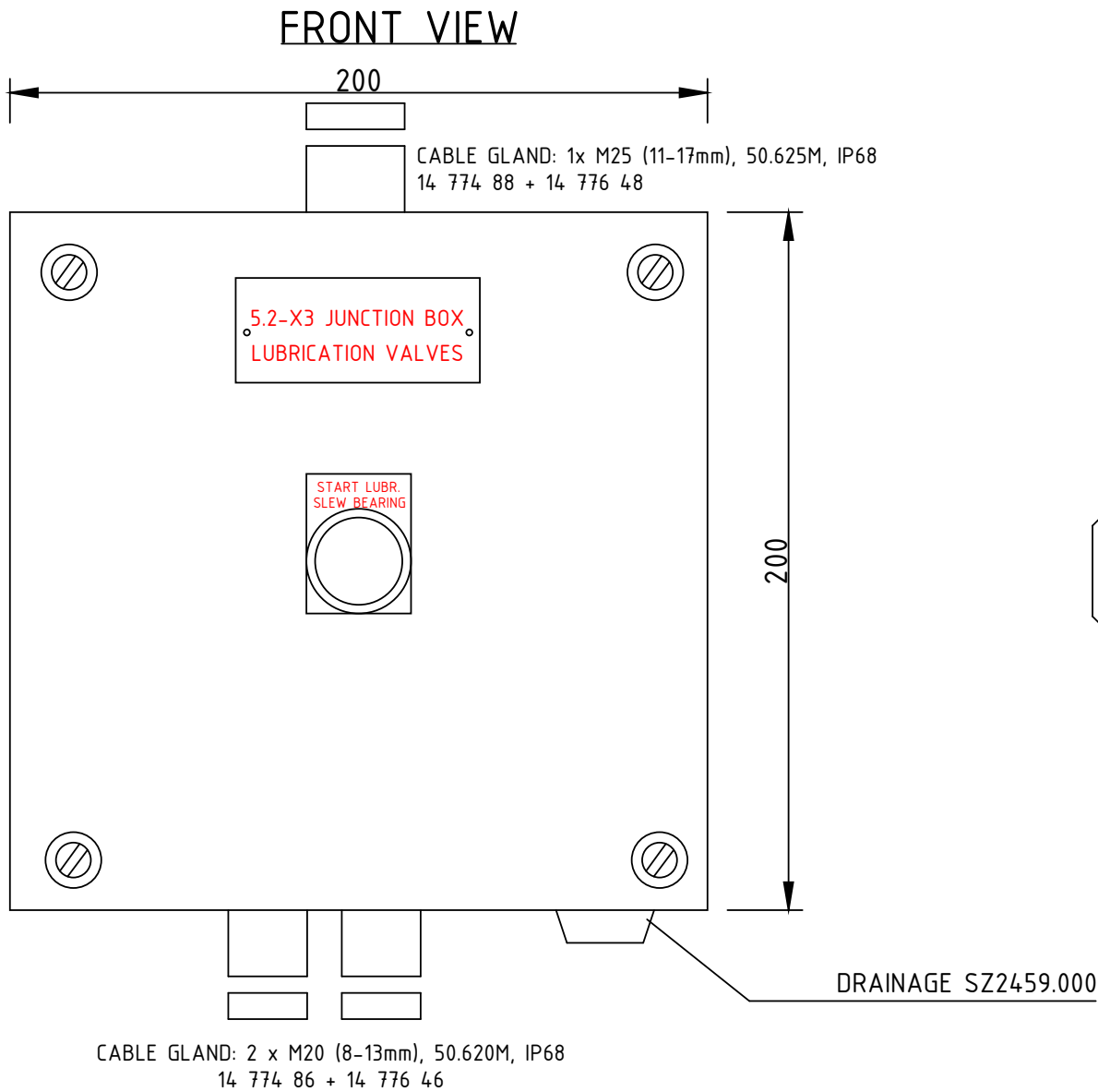
0.12-X10

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11	45.2
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14	45.2
15	45.2
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25	46.2
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27	46.2
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32	46.2
33	46.2
34	46.2
35	46.2
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38	46.2
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41	47.2
42	47.2
43	47.2
44	47.2
45	47.2
46	47.2
47	47.3
48	47.3
101	48.3
102	48.3
103	48.3
104	48.3
105	48.3
106	48.3
107	48.3
108	48.3
109	48.3
110	48.3
111	48.3
112	48.3
113	48.3
114	48.3
115	48.3
116	48.3
117	48.3
118	48.3
119	48.3
120	48.3
201	47.3
202	47.3

FOR CIRCUIT DIAGRAM SEE DRAWING A524-E SHEET:

45, 46, 47, 48

reproduced, or exploited in any other unauthorized way.



ITEM LIST:

ENCLOSURE (STAINLESS STEEL): KL 1523.010, RITTAL

PUSHBUTTON, XB4-BA21, TELEMECANIQUE

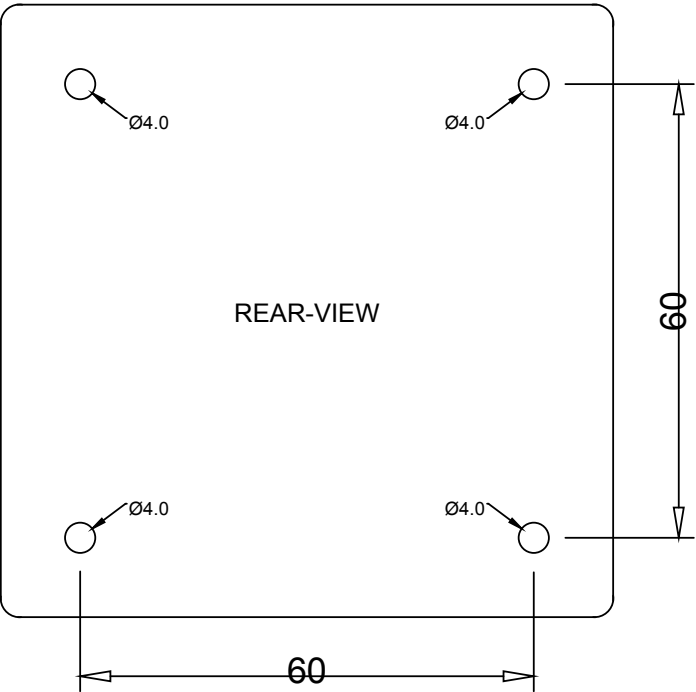
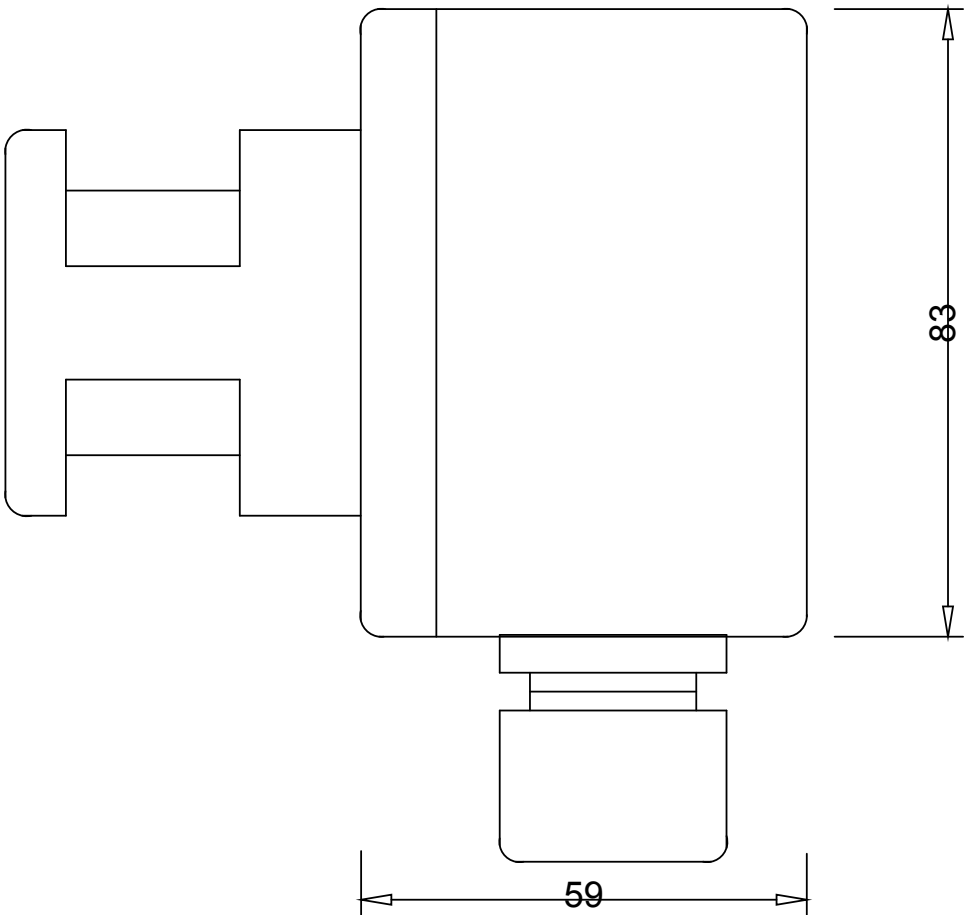
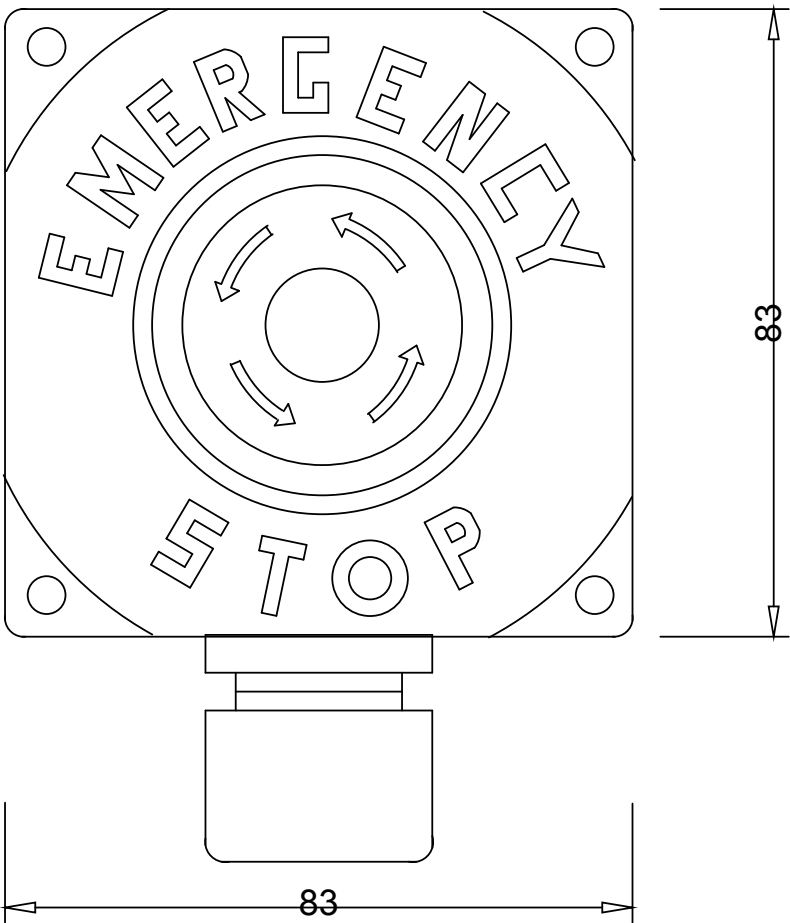
TERMINALS: WDU 4 SL, 9537440000, WEIDMULLER

CONNECTIONS, SEE DRAWINGS:

A524-E SHEET; 59, 90

ENCLOSURE: STAINLESS STEEL

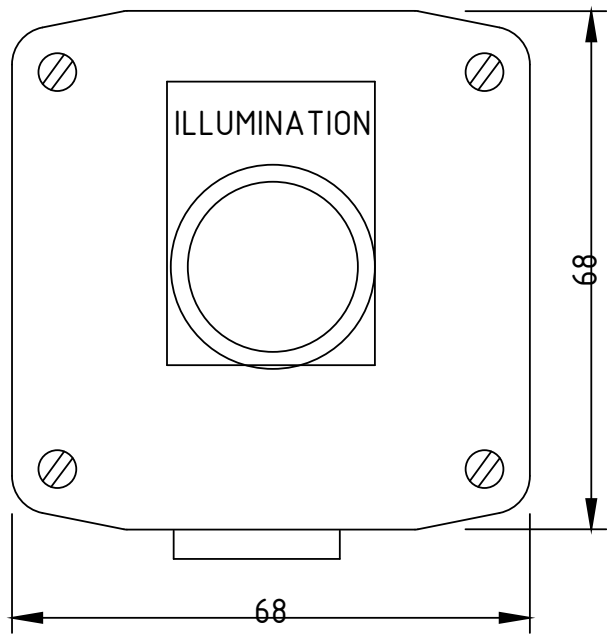
IP66



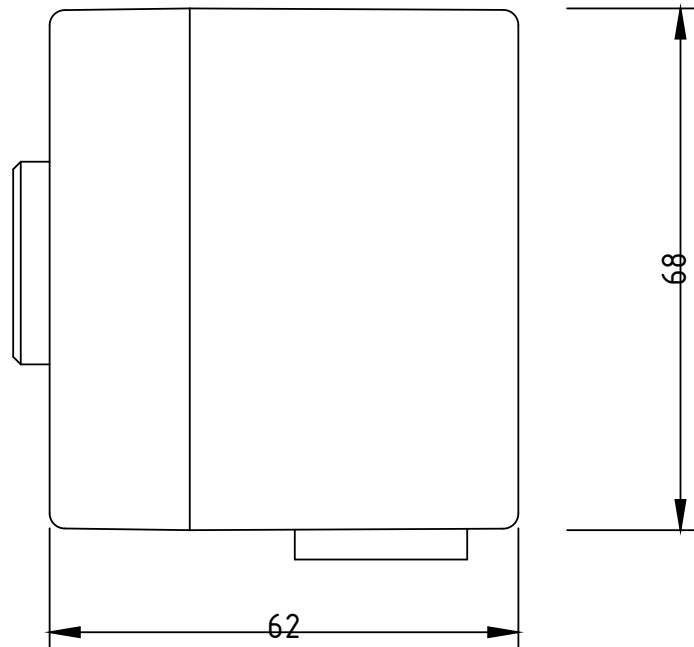
4 PCS.

EMERGENCY STOP BOX LOCATION:

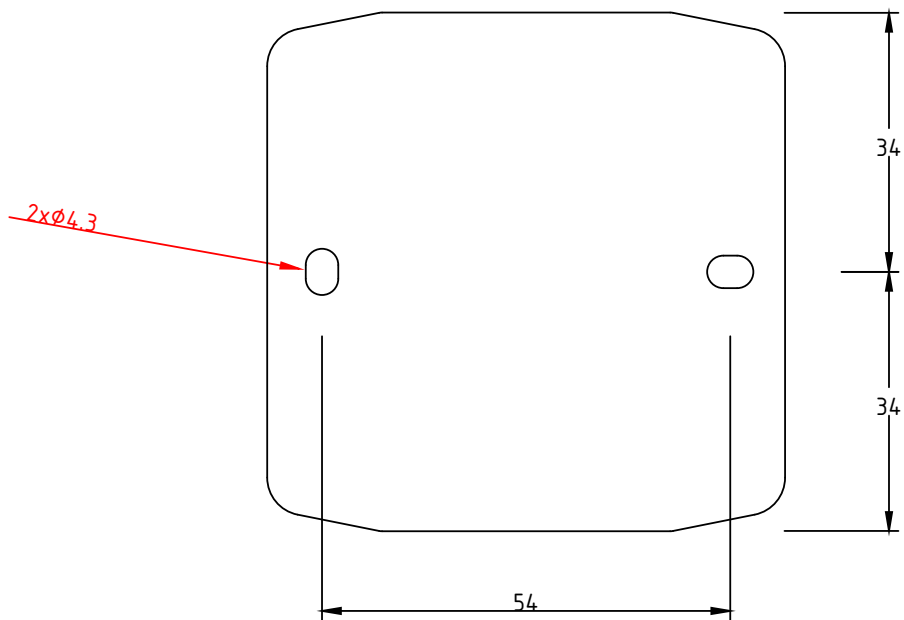
- 0.7-S4: OUTSIDE ELHOUSE
- 0.7-S5: AT STAIRCASE ENTRY
- 0.7-S6: AT STAIRCASE ENTRY
- 0.7-S7: OUTSIDE HYDRAUL



CABLE GLAND: POLYAMID PG13/20,4 IP68
14 593 63 + 14 656 13



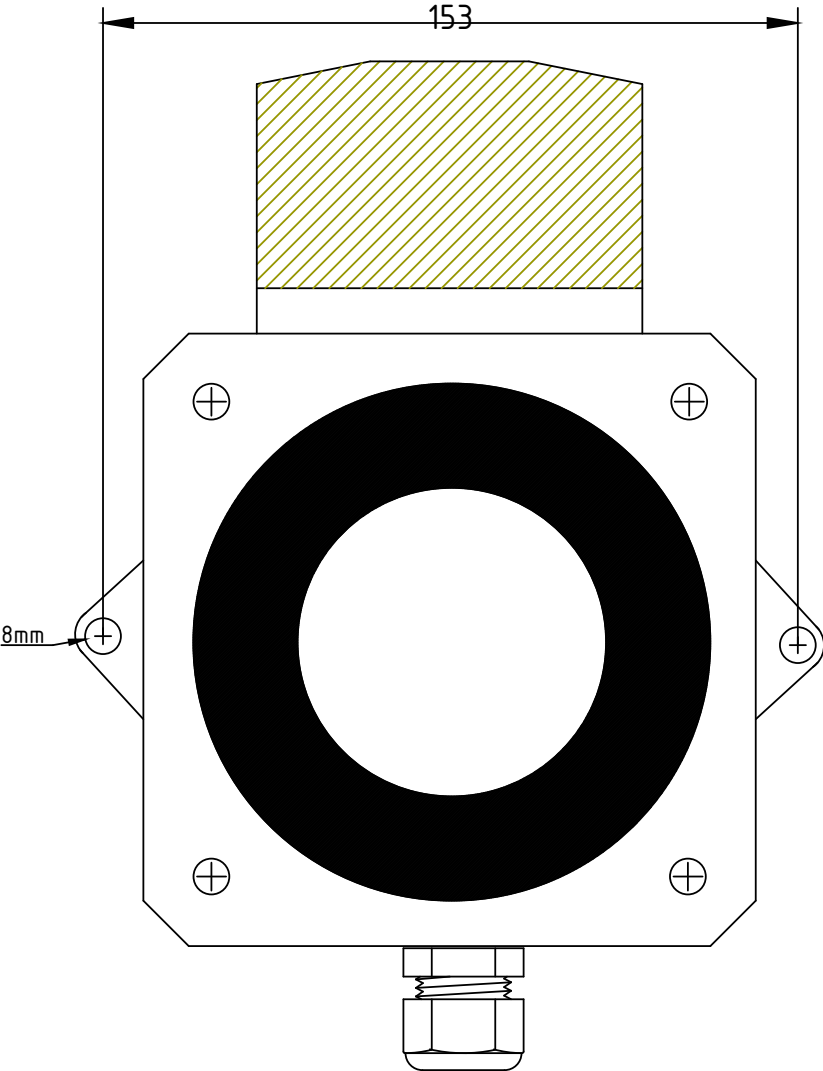
DRILL-PLANE



3 PCS.

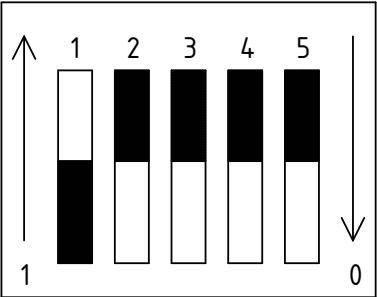
ILLUMINATION BOX LOCATION:
0.3-S10: OUTSIDE CABIN
0.3-S11: AT STAIRCASE ENTRY
0.3-S12: OUTSIDE ELHOUSE

ITEM LIST:
PUSH-BUTTON BOX: XAL-D102, TELEMECANIQUE
SIGN-HOLDER: ZBY-2101, TELEMECANIQUE
SIGN: "ILLUMINATION"



CABLE GLAND: POLYAMID PG13/20,4 IP68
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DIP-SWITCH SETTINGS



7 Pcs.

TRAVELLING ALARM LOCATION:
5.0-H1 - H4: 1pc. AT EACH ENDCARRIGE (TOT. 4pcs.)

MOVEMENTS ALARM LOCATION:
0.10-H1 - H2: 2pcs. LOWER PART OF MACHINE
0.10-H3: 1pc. UPPER PART OF MACHINE

ITEM LIST:

ALARM UNIT: YODALIGHT, YL5 (WR-IP66) 24VDC, 1YL050318

SOLDERING CONNECTIONS

- A → A WHITE
- B → B BROWN
- C → C GREEN
- D → D YELLOW

